Research record: locality, terminal information, and planar exploration laws

Autonomous mathematical investigation, with independent critical checks

29 October 2026

Abstract

The regularity bridge is now expressed through exact exit-sign criteria: finite aggregate predictable sign budget is equivalent to integrable finite-variation drift; subquadratic weighted sign energies characterize L^p absolutely continuous drift for $1 < p < 2$. A new conditional rigidity theorem identifies SLE_6 from source-linear expected L^p exit-energy bounds whose constants grow subcritically along mirrored shrinking-slit comparisons; no bounded drift density or initial tangent is assumed. Conditional crossing moments also give bounded cluster clocks along deterministic meshes and finite-mesh analytic-locality characteristic covariance without semimartingality. None of these criteria supplies the missing signed estimate from the exploration axioms. A singular analytic clock example shows why clock-path Taylor regularity alone cannot settle finite-affine clock absolute continuity; a dyadic uniform-integrability criterion identifies a precise increment-measure target. Signed-test drift energy is lower semicontinuous in the affine tensor; fixed-state affine driving laws are continuous under the standing admissibility and lifetime conditions. A compact finite-test selection principle can therefore select a common tensor jointly with signed-energy estimates, if common finite envelopes and a compact ellipticity class can be established. The 29 October priority is to understand affine-SLE_6 driving in arbitrary coordinates before imposing coordinate-sensitive stochastic regularity. Matched-prefix affine driving and capacity admit uniform first variations even for nonsimple filled hulls; the linear capacity response has every temporal exponent below one under the crossing assumptions, and the linear driver response retains every temporal exponent below one-half. The continuum crossing/locality bounds give nondegenerate subsequential adaptive quadratic-activity clocks without semimartingality. Wrong-coordinate semimartingality would force absolute continuity of the affine capacity clock, which remains open. The current priority is continuum classification under explicit RSW-type crossing assumptions, with verification for lattice limits deferred. Corrected source locality and a ray-deletion comparison give a diffusive lower driving tail; combined with the upper crossing estimates, this forces variation index two without assuming semimartingality. Once the signed restart budget is finite, the bracket is absolutely continuous and uniformly elliptic, while singular drift remains a separate issue. Analytic locality transports finite signed budgets and characteristic measures. A Schwarzian test excludes the nontrivial one-force-point SLE families without requiring initial drift tangents. None of these results proves finiteness of the signed budget or the classification. We study the corrected source-neighborhood locality axiom introduced at commit `0e47018`. The old terminal-germ counterexample is excluded; classification and critical percolation universality remain open. A conditional SLE_6 classification now allows continuous finite-variation drift with a uniform initial tangent in probability. Tightness in law along mirrored slit tests can replace deterministic initial tangents. The slit tests give inverse-linear drift and inverse-cubic variance anomalies, and a scaling argument obstructs tame driving in a wrong affine structure. Independent-color RSW shields give source-patch coalescence, an initial trace-germ zero-one law, quantitative prefix/full-curve decorrelation, and total-variation decoupling of a short prefix from a distant terminal suffix. Annular crossing statistics satisfy exponential scale mixing and a centered strong law, without assuming dilation invariance. A new conditional

signed-bias criterion supplies semimartingality and initial tangents; a discrete version passes to continuous driver limits without filtration convergence. Endpoint tensor decoding and source splicing collapse coherent affine-SLE mixtures to one deterministic common metric, without deriving that ansatz. Other results give stability criteria, anisotropy bounds, and an abstract affine-ansatz classification with one stable straight slit-tip germ.

Strong DMP in the usual natural driver filtration automatically transports canonical drift and bracket. Vanishing adapted microscopic correctors can replace raw one-step driving-bias bounds. Annular scale sums admit an exponentially integrable martingale correction and a functional Gaussian limit conditional on a deterministic variance rate. A spherical empirical-law decoder handles arbitrary sample-dependent local C^1 charts; a stationary-cylinder argument supplies dilation ergodicity for the relevant whole-plane SLE model. A canonical tube-cluster measure supplies an affine-stable intrinsic measure class. Whole-plane reversal and natural-length Palm laws then give a Borel tensor decoder from every unlabelled compact subarc of an affine simple SLE, simultaneously under all affine maps and subarc choices. Identifying such arcs on general exploration frontiers remains unresolved. Historical proofs, failed approaches, and all extra hypotheses are retained.

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1 Scope, conventions, and verification status

Current priority: continuum classification under RSW. Following the user's clarification of 28 October 2026, the immediate problem is to classify continuum exploration families satisfying strong DMP, corrected source-neighborhood locality, physical translation covariance, and explicitly

specified uniform curve-level RSW estimates. Verifying these axioms for percolation subsequential limits is deferred. The expected conclusion permits one fixed affine normalization; no conformal covariance or scale invariance is assumed.

The exploration is an ordered uniform limit of simple curves, not a simple curve. Dense self-touching, boundary contacts, and swallowing are allowed. No embedded nontrivial time interval is presumed. Ordered prefixes, filled hulls, and unlabelled accessible frontiers are distinct objects. The decoder for known simple SLE arcs with $0 < \kappa < 4$ is a separate model theorem and supplies no simple subarc of the unknown exploration. Continuous Loewner admissibility and stochastic driving regularity are separate questions. At rough restart tips, prime-end access and inherited colored boundary sides must be retained; the source-locality neighborhood convention must be stated explicitly.

Revision of the research question. The user strengthened locality in `PROJECT_DESCRIPTION.md`, commits `32f8e45` and `0e47018`. It now compares any two marked colored domains agreeing in a source neighborhood, allowing their targets to differ. Their stopped paths must agree until leaving that neighborhood. The terminal-germ counterexample in Section 3 is a historical result about the earlier fixed-target deletion axiom, and is *not* a counterexample to the corrected conjecture. Earlier references to "the proposed axioms" should be read with that historical scope. The new continuation below states its additional hypotheses explicitly.

This record starts from an otherwise empty research repository. The project specification is preserved unchanged. Its primary question is whether strong domain Markov, locality, and physical translation covariance force a domain-indexed family of exploration curves to be one fixed affine pullback of chordal SLE_6. We do not assume this conclusion or conformal covariance of an unknown family.

In the original investigation, curves are oriented, unparameterized continuous paths. After an initial exploration, the remaining domain is the component facing the terminal prime end, with the exposed tip interpreted as an appropriate prime end. The original fixed-target locality compares curves up to first encounter with a removed closed set; their respective capacity clocks need not agree. Deletions must preserve the source and terminal end neighborhoods and leave a simply connected target component. This is the customary precise setting for chordal locality. For a Jordan target and a planar closed set disjoint from that point, preservation of its terminal neighborhood is automatic.

External facts used. We use existence, conformal covariance, strong domain Markov, locality, and continuous Loewner generation of ordinary SLE_6 as established theorems; see [6]. We use elementary conformal mapping, boundary extension on Jordan domains, harmonic uniqueness, and Morera's theorem. Loewner admissibility of affine images is justified explicitly using [7]. A reversal statement below additionally uses [8]. These external results are distinguished from the arguments proved here.

Proof checks. The target-germ construction was found independently in two investigations. The boundary-race computation and the single-domain metric-identification argument were checked independently, including the sign of the diffusion, the exclusion of simultaneous swallowing, and the boundary-extension step. The global-homeomorphism rigidity argument and the local boundary-twist test were also independently checked. These checks are evidence of correctness, not a replacement for expert human review. Nothing here proves universality for general critical percolation.

2 An affine metric is identifiable from one chordal law

For a bounded Jordan domain D, distinct $p, q \in \partial D$, and an orientation-preserving invertible real-linear map A, put

$$\nu^A_{D,p,q} = (A^{-1})_* \operatorname{Law}\big(\mathrm{SLE}_6(AD; Ap, Aq)\big). \tag{1}$$

Translations in an affine normalizer are immaterial. The tensor $A^{\mathsf{T}}A$ is its Euclidean conformal metric, determined only up to a positive scalar.

2.1 A boundary-swallowing race

Lemma 2.1 (Strict boundary-race probability)**.** *Let η be chordal SLE_6 in $\mathbb{H}$ from 0 to ∞. For $x < 0 < y$, let τ_x, τ_y denote their Loewner swallowing times. Then*

$$\mathbb{P}(\tau_y < \tau_x) = F\left(\frac{-x}{y-x}\right), \qquad F(r) = \frac{\int_0^r [u(1-u)]^{-2/3}\,du}{\int_0^1 [u(1-u)]^{-2/3}\,du}. \tag{2}$$

In particular F is strictly increasing on $(0,1)$. The event in (2) is invariant under orientation-preserving homeomorphisms of marked Jordan domains, when formulated as the order of separation from the terminal prime end by the filled exploration hull.

Proof. Use $\partial_t g_t(z) = 2/(g_t(z) - W_t)$ with $W_t = \sqrt{6}B_t$. Until the first swallowing time put

$$X_t = g_t(x) - W_t < 0, \quad Y_t = g_t(y) - W_t > 0, \quad T_t = Y_t - X_t, \quad R_t = -X_t/T_t \in (0,1).$$

Then

$$dX_t = \frac{2}{X_t}\,dt - \sqrt{6}\,dB_t, \qquad dY_t = \frac{2}{Y_t}\,dt - \sqrt{6}\,dB_t,$$
$$dT_t = \frac{2T_t}{-X_tY_t}\,dt > 0, \qquad dR_t = \frac{2(1-2R_t)}{R_t(1-R_t)T_t^2}\,dt + \frac{\sqrt{6}}{T_t}\,dB_t.$$

With $ds = dt/T_t^2$, the generator of R is

$$\mathcal{L} = 3\frac{d^2}{dr^2} + \frac{2(1-2r)}{r(1-r)}\frac{d}{dr}.$$

The increasing solution of $\mathcal{L}F = 0$, $F(0) = 0$, $F(1) = 1$ has $F'(r) = C[r(1-r)]^{-2/3}$, giving (2) by bounded optional stopping. To justify the terminal values, $|X|/\sqrt{6}$ and $Y/\sqrt{6}$ are Bessel processes of dimension $1 + 4/6 = 5/3$ up to their first zero; thus the first swallowing time is finite almost surely. Alternatively, the accessible endpoints of the displayed diffusion give the same stopping argument. There is no simultaneous swallowing, since $T_t \geq y - x > 0$, whereas both $X_t, Y_t \to 0$ would force $T_t \to 0$. At the first swallowing, $R = 1$ exactly when y is swallowed first. ODE solutions and the continuous driver stay finite up to that first time.

Swallowing means that the marked boundary end has been cut off from the target by the explored filled hull (including the boundary swallowed at first contact). This definition uses only the domain, the oriented path, and its order of traversal. A homeomorphism carries filled target components to filled target components. Capacity parametrization therefore has no role in the comparison event. □

2.2 Boundary rigidity and affine uniqueness

Lemma 2.2 (A single SLE law identifies its conformal boundary map)**.** *Let $h : \overline{D} \to \overline{D'}$ be an orientation-preserving homeomorphism between closed bounded Jordan domains, with $p' = h(p)$, $q' = h(q)$. If $h_*\mathrm{SLE}_6(D; p, q) = \mathrm{SLE}_6(D'; p', q')$, then $h|_{\partial D}$ agrees with the boundary extension of a conformal map $f : D \to D'$.*

Proof. Choose conformal maps $\phi : D \to \mathbb{H}$ and $\phi' : D' \to \mathbb{H}$ sending their marked endpoints to $0, \infty$. Their Jordan boundary extensions give an increasing homeomorphism

$$H = \phi' \circ h \circ \phi^{-1} : \overline{\mathbb{R}} \to \overline{\mathbb{R}}$$

fixing $0, \infty$. For every $x < 0 < y$, equality of the corresponding boundary-race probabilities and strictness in Lemma 2.1 give

$$\frac{-H(x)}{H(y) - H(x)} = \frac{-x}{y - x}, \quad \text{hence} \quad \frac{H(x)}{H(y)} = \frac{x}{y}.$$

Fixing one $x < 0$ and one $y > 0$ shows $H(s) = cs$ for all real s, with $c > 0$. Consequently $f = (\phi')^{-1} \circ (c\phi)$ has exactly the required boundary values. Equality of interior maps is not asserted for an arbitrary h. □

Theorem 2.3 (Single-domain affine identifiability)**.** *For even one bounded Jordan D and one pair of distinct boundary points p, q, the equality $\nu^A_{D,p,q} = \nu^B_{D,p,q}$ holds if and only if BA^{-1} is a Euclidean similarity. Equivalently,*

$$B^\mathsf{T} B = cA^\mathsf{T} A \qquad (c > 0).$$

The statement also allows orientation-reversing normalizers by composing one map with a reflection.

Proof. The equality of laws means $M = BA^{-1}$ carries the SLE law in AD to the SLE law in BD, at the corresponding endpoints. By Lemma 2.2, a conformal map $f : AD \to BD$ has boundary values equal to M. Both real coordinate functions of the real-linear map M are harmonic. The real and imaginary parts of f are also harmonic, continuous on the closure, with the same boundary values. The maximum principle on the bounded domain gives $f = M$ throughout AD. Thus M is complex-linear and invertible, hence a similarity. The converse is ordinary SLE similarity covariance. Reflection covariance reduces the remaining orientation case to this one. □

3 A counterexample to the original fixed-target locality

3.1 The conserved information and the admissible domain class

Let $G(D, q)$ be the embedded germ of D at its terminal prime end. Concretely, target neighborhoods may be taken as

$$f^{-1}\big(\mathbb{H} \setminus \overline{B(0, R)}\big), \qquad R \to \infty,$$

for any conformal $f : D \to \mathbb{H}$ sending q to infinity. The embedded germ records the planar domain in a full neighborhood of that end; one may shrink neighborhoods without changing it. A full neighborhood contains a neighborhood in the prime-end compactification, rather than merely an arbitrarily narrow approach sector. Germ equivalence means agreement on such neighborhoods with the same inherited marked end.

At a Jordan boundary point this is the ordinary local boundary germ. At a slit tip one must retain which prime end is intended. A germ is called C^1-regular here when it has a singleton

endpoint and, in a full end neighborhood, a single C^1 boundary arc with a nonzero tangent and the domain locally on one side. The tangent line $\ell(G)$ is unoriented. All other germs receive the default label below. Regularity and tangent are properties of the germ itself.

Finite chordal Loewner hulls in a target-normalized half-plane chart are bounded. They therefore miss some target neighborhood. At every finite stopping time prior to reaching the target, the remaining marked domain D_t satisfies

$$G(D_t, q) = G(D, q). \tag{3}$$

Likewise locality deletions away from the terminal end preserve this germ. Thus the following proof already works on domains with regular targets, together with their slit domains, a class closed under all operations needed in the argument. Assigning defaults to nonregular end germs extends it to arbitrary prime ends under the stated end-neighborhood convention. Locality for sets accumulating at a nonsingleton impression must always be formulated in prime-end terms; it is not defined by pretending the end is a single physical point.

Proposition 3.1 (Conserved-label construction)**.** *Assign to each embedded terminal germ an orientation-preserving invertible real-linear map $A(G)$, measurably and in a translation-covariant way: $A(G(D + v, q + v)) = A(G(D, q))$. Define*

$$P_{D,p,q} = (A(G(D,q))^{-1})_* \mathrm{SLE}_6\big(A(G(D,q))D; A(G(D,q))p, A(G(D,q))q\big). \tag{4}$$

Then this family has strong domain Markov, locality, and translation covariance as unparameterized paths.

Proof. Write $A = A(G(D, q))$. By (3) the same deterministic A is prescribed after any finite initial exploration. Apply strong domain Markov to SLE_6 in AD, then pull the tail back by A^{-1}. This gives exactly (4) in the remaining target component, at its exposed tip and target. Stopping times are carried to stopping times because the invertible map preserves the explored filtration; the capacity clock is an adapted increasing time change. For deletion of K away from both ends, the two domains use the same A. SLE_6 locality in AD and $AD \setminus AK$, pulled back by A^{-1}, couples their paths identically until first encounter with K. Finally the displayed covariance of the label and translation covariance of SLE_6 prove physical translation covariance. No conditioning on a hidden random parameter occurs: the label is a deterministic conserved function of the current marked domain. □

3.2 A concrete example with full similarity covariance

Fix $a > 0$, $a \neq 1$, and set

$$A(G) = \begin{cases} \Pi_{\ell(G)} + a\Pi_{\ell(G)^\perp}, & G \text{ is } C^1\text{-regular}, \\ I, & \text{otherwise}. \end{cases} \tag{5}$$

Here Π is orthogonal projection. On smooth-target domains this is a continuous function of the tangent in the topology controlling that tangent; on arbitrary germs it is a deterministic local geometric rule. If desired, the class can first be restricted to piecewise smooth domains and their finite slit descendants, avoiding any measurable-space conventions for the class of all planar domains.

For a similarity $S(x) = cRx + v$, $c > 0$, R orthogonal, one has

$$A(G(SD, Sq)) = RA(G(D, q))R^{-1}.$$

Combining this with SLE_6 similarity and reflection covariance proves $P_{SD,Sp,Sq} = S_* P_{D,p,q}$. Thus adding scale invariance, reflection covariance, or covariance under every rotation does not exclude this construction.

Theorem 3.2 (The proposed fixed-affine classification is false)**.** *The ensemble (4)–(5) satisfies the three proposed axioms, and full Euclidean similarity covariance, but there is no one fixed affine transformation sending all its laws to ordinary chordal SLE_6. This failure already occurs for smooth targets on one disk.*

Proof. Take D to be the unit disk and fix $p = -1$. At $q_h = i$ the tangent is horizontal, while at $q_v = 1$ it is vertical. The respective matrices are

$$A_h = \begin{pmatrix} 1 & 0 \\ 0 & a \end{pmatrix}, \qquad A_v = \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}.$$

A fixed affine normalizer with linear part B would imply $\nu^B_{D,p,q_h} = \nu^{A_h}_{D,p,q_h}$ and $\nu^B_{D,p,q_v} = \nu^{A_v}_{D,p,q_v}$. Theorem 2.3 makes both BA_h^{-1} and BA_v^{-1} similarities. Hence $A_v A_h^{-1} = \mathrm{diag}(a, a^{-1})$ would be a similarity, forcing $a^2 = 1$, contrary to $a > 0$, $a \neq 1$. □

3.3 Why these really are continuous-driving Loewner curves

Continuity and the absence of transverse crossings alone would not suffice to settle this point. Instead use the deterministic characterization in [7, Theorem 1.1 and Remark 1.4]: a continuous half-plane path with strictly increasing filled hulls is a continuous-driving Loewner trace if and only if it is a locally uniform limit of simple interior paths. The condition is independent of parametrization.

An ordinary SLE_6 trace has this property. Conjugating an invertible affine map by the conformal charts of its source and image domains gives a quasiconformal self-map of $\mathbb{H}$. It extends to a homeomorphism of the closed half-plane with its point at infinity, fixing the two marked ends after normalization. It carries simple interior approximants to simple interior approximants; uniform convergence survives on compact time intervals because the extended map is uniformly continuous on relevant compact sets. Filled hulls and their strict growth are topologically preserved. Yuan's theorem gives a continuous driving function after reparametrizing by capacity. This also treats nonsmooth slit domains in the Markov property without assuming Euclidean continuity of their uniformizing maps on every boundary point. The counterexample therefore applies to the driving-function formulation as well as to curves.

3.4 Quasiconformal images have infinite capacity lifetime

Yuan's characterization gives a continuous Loewner driver after recoding an affine image by its capacity, but this alone does not assert that the capacity lifetime is infinite. A continuous curve tending to infinity need not have infinite half-plane capacity. The following argument proves the required assertion for the quasiconformal changes arising here.

Lemma 3.3. *Let $F : \mathbb{H} \to \mathbb{H}$ be a fixed K-quasiconformal self-homeomorphism whose boundary extension fixes 0 and ∞. If η is ordinary chordal SLE_6 from 0 to ∞, then $F\eta$ has unbounded imaginary part almost surely. Consequently its half-plane-capacity lifetime is infinite.*

Proof. Put $C = \overline{B(i, 1/4)} \subset \mathbb{H}$. The boundary map of F is increasing, so $F(R) > 0$ for $R > 0$ and $F(R) \to +\infty$ as $R \to \infty$. The normalized maps

$$F_R(z) = \frac{F(Rz)}{F(R)}$$

are K-quasiconformal half-plane maps fixing $0, 1, \infty$. Reflecting them across the real line gives normalized quasiconformal sphere maps. Compactness of this normalized family, including its inverses, implies

$$\inf_{R>0} \inf_{z\in C} \operatorname{Im} F_R(z) = c_K > 0.$$

Indeed, a sequence violating positivity would have a subsequence converging uniformly on the sphere to a quasiconformal homeomorphism preserving the half-plane, and a subsequence of points in C would then be mapped to its real boundary, an impossibility. Hence

$$\inf_{z\in RC} \operatorname{Im} F(z) \geq c_K F(R) \longrightarrow \infty. \tag{QC1}$$

The event $E = \{\eta \cap C \neq \varnothing\}$ has positive probability. One elementary justification avoids invoking a support theorem. In the normalization $\partial_t g_t = 2/(g_t - W_t)$, the Brownian event

$$\sup_{0\leq t\leq 1} |W_t| < \varepsilon, \qquad 0 < \varepsilon < 1/4,$$

has positive probability. On this event the Loewner hull stays within the vertical strip $|\operatorname{Re} z| \leq \varepsilon$ up to time 1. For example, a point initially to the right of that strip has increasing real part under the Loewner ODE and retains a positive real gap from the driver, so it cannot be swallowed in finite time; the other side is identical. If the entire trace up to time 1 had height at most 1, its filled hull would lie in the half-disk of radius $\sqrt{1+\varepsilon^2}$. By capacity monotonicity its capacity would be at most $1 + \varepsilon^2 < 2$, whereas its actual capacity is 2. Here a half-disk of radius R has hydrodynamic map $z + R^2/z$ and capacity R^2. Thus the continuous trace crosses height 1 in the strip, and that crossing lies in C.

Fix $r > 1$. Brownian scaling, and therefore scalar dilation of ordinary SLE_6, is mixing. Indeed $Z_s = e^{-s/2} B_{e^s}$, $s \in \mathbb{R}$, is a stationary Gaussian process with covariance $e^{-|s-u|/2}$; scalar Brownian scaling becomes a log-time shift. Decay of these covariances gives mixing first for Gaussian cylinder functions and then for the whole path sigma-field. Its inverse dilation is ergodic as well. Birkhoff's theorem applied to 1_E gives

$$\eta \cap r^n C \neq \varnothing \qquad \text{for infinitely many positive integers } n, \quad \text{almost surely.}$$

In fact these visits have limiting positive frequency, since $1_E(S_{r^{-n}}\eta) = 1_{\{\eta\cap r^n C\neq\varnothing\}}$. Equation (QC1) then shows that $F\eta$ has unbounded height.

Every finite initial image trace is compact away from infinity and has finite capacity. The standard Loewner height estimate

$$\max\{\operatorname{Im} z : z \in K\}^2 \leq 2 \operatorname{hcap}(K)$$

for a half-plane hull, or equivalently $\max \operatorname{Im} K_u \leq 2\sqrt{u}$ when $\operatorname{hcap}(K_u) = 2u$, now forces the capacities of these image prefixes to tend to infinity. Thus the capacity-recoded image has lifetime $[0, \infty)$. □

For an affine image in a bounded marked domain, conjugating the affine map by its source and image conformal charts gives exactly such a fixed quasiconformal F fixing the marked ends. This lemma therefore completes the infinite-time driving-function assertion for the target-germ counterexample. It uses properties of SLE_6, rather than an invalid general assertion about arbitrary curves tending to infinity.

4 A positive classification within the global-deformation class

The preceding example does not arise from one deterministic global change of spatial coordinates. Within that narrower class the proposed affine conclusion does follow, and no differentiability of the change of coordinates is required.

Theorem 4.1 (Full-family homeomorphism symmetries are conformal)**.** *Let $h : \mathbb{C} \to \mathbb{C}$ be an orientation-preserving homeomorphism such that, for every bounded Jordan domain and marked boundary pair,*

$$h_*\mathrm{SLE}_6(D; p, q) = \mathrm{SLE}_6(hD; hp, hq).$$

Then h is an entire conformal automorphism, hence $h(z) = \alpha z + \beta$ for complex $\alpha \neq 0$. For an orientation-reversing homeomorphism, the corresponding conclusion is anticonformality.

Proof. Take any nondegenerate closed Euclidean triangle T and distinct $p, q \in \partial T$. Lemma 2.2 gives a conformal $f_T : T^\circ \to h(T^\circ)$ whose continuous boundary extension equals h on ∂T. Cauchy's theorem, applied on interior triangles and then passed to the boundary by continuity, yields

$$\int_{\partial T} h(z)\, dz = \int_{\partial T} f_T(z)\, dz = 0.$$

Since h is continuous and this holds for every triangle, Morera's theorem makes h entire holomorphic. Its injectivity makes it conformal, and an entire conformal automorphism of the plane is complex affine. Composition with a reflection proves the other orientation case. □

Theorem 4.2 (Stationary global deformations are real affine)**.** *Let $h : \mathbb{C} \to \mathbb{C}$ be a homeomorphism and define, on all bounded Jordan domains,*

$$P^h_{D,p,q} = (h^{-1})_*\mathrm{SLE}_6(hD; hp, hq).$$

If this ensemble is covariant under every physical translation, then h is a real-affine homeomorphism.

Proof. Reflection covariance lets us assume h preserves orientation. For every $v \in \mathbb{C}$, physical translation covariance says that

$$\psi_v = h \circ T_v \circ h^{-1}, \qquad T_v(z) = z + v,$$

preserves the full chordal SLE_6 family. All bounded Jordan domains are available, since inverse images under a plane homeomorphism are bounded Jordan domains. Theorem 4.1 gives $\psi_v(z) = \alpha_v z + \beta_v$. For $v \neq 0$, the conjugate of a nonzero translation has no fixed point. A complex affine map with $\alpha_v \neq 1$ has a fixed point, so $\alpha_v = 1$ for all v. It follows that $h(x + v) = h(x) + \beta_v$, and putting $x = 0$ gives

$$h(x + v) - h(0) = (h(x) - h(0)) + (h(v) - h(0)).$$

The continuous additive map $h - h(0)$ is real-linear. It is invertible because h is a homeomorphism. □

What this does not prove. The original axioms do not give a representation by one global h. The counterexample shows that replacing a domain-dependent geometry by one fixed spatial geometry is a substantive missing step, not an innocent coordinate choice.

5 Boundary stability and other ways to remove terminal dependence

5.1 An explicit failure of continuity under small target perturbations

The tangent rule is continuous when target tangents converge, but not under ordinary uniform approximation of the domains and their boundaries. This defect persists even when every domain is smooth and the approximating domains are uniform quasidisks.

Proposition 5.1 (Local rotation of a terminal germ)**.** *For the ensemble of Theorem 3.2, there are smooth Jordan $D_n \to D$, with uniformly converging boundary parametrizations and fixed distinct marked points p, q, for which $P_{D_n,p,q}$ does not converge to $P_{D,p,q}$ in the uniform unparameterized-curve topology.*

Proof. Let D be the unit disk, $q = i$, $p = -1$. Choose a smooth radial cutoff χ equal to 1 near 0 and 0 outside the unit disk, and define the orientation-preserving smooth diffeomorphism

$$F(re^{i\theta}) = r\exp(i\theta + i(\pi/2)\chi(r)).$$

It is a rotation by $\pi/2$ near 0 and the identity outside the disk. Its quasiconformal distortion is finite, since $r\chi'(r)$ is bounded. Put $F_\epsilon(z) = q + \epsilon F((z-q)/\epsilon)$ and $D_\epsilon = F_\epsilon(D)$. Then $F_\epsilon(q) = q$, and $F_\epsilon(p) = p$ for small ϵ, while $\sup_z |F_\epsilon(z) - z| \le 2\epsilon$. All F_ϵ have one common quasiconformal bound. At q the target tangent has rotated by $\pi/2$, so the matrix used for every D_ϵ is A_v, whereas the matrix for D is A_h.

The domains $A_v D_\epsilon$ are uniform quasidisks converging to $A_v D$, with the marked boundary points converging as well. Kernel convergence and uniform boundary equicontinuity of normalized Riemann maps on uniform quasidisks give uniform convergence of these maps on the closed disk. Coupling their chordal SLE_6 paths with one common disk SLE path therefore gives convergence of the entire unparameterized curves. Thus

$$P_{D_\epsilon,p,q} \implies \nu^{A_v}_{D,p,q}.$$

By Theorem 2.3, this limit differs from $P_{D,p,q} = \nu^{A_h}_{D,p,q}$ because $A_v A_h^{-1} = \mathrm{diag}(a, a^{-1})$ is not a similarity. □

This proves that an appropriate boundary-stability requirement excludes the example. It does not prove that boundary stability plus the three original axioms suffices for classification.

Corollary 5.2 (Boundary stability classifies tangent-dependent affine laws)**.** *Suppose, on smooth bounded Jordan domains with smooth targets, an ensemble has the form $P_{D,p,q} = \nu^{A_{\ell(D,q)}}_{D,p,q}$, where A_ℓ is an arbitrary orientation-preserving invertible linear map depending only on the unoriented target tangent line. If these laws are continuous under the uniform smooth-boundary approximations used in Proposition 5.1, then one fixed affine map normalizes all of them. If the ensemble is also covariant under all rotations, that fixed map is a similarity.*

Proof. Given any two tangent lines ℓ_1, ℓ_2, take a disk having tangent ℓ_1 at q and apply the preceding local twist with the rotation carrying ℓ_1 to ℓ_2. Every approximating domain uses the constant matrix A_{ℓ_2}. SLE continuity in that fixed metric shows that the limiting law is $\nu^{A_{\ell_2}}_{D,p,q}$, whereas the assumed boundary stability makes it $\nu^{A_{\ell_1}}_{D,p,q}$. Single-pair identifiability implies $A_{\ell_2} A_{\ell_1}^{-1}$ is a similarity. Thus all $A_\ell^{\mathsf{T}} A_\ell$ are proportional, which is exactly one fixed affine family. For rotation covariance its fixed tensor Q satisfies $R^{\mathsf{T}} Q R = Q$ for every rotation R, by the same metric uniqueness, hence Q is scalar. □

5.2 Reversibility eliminates this mechanism inside the affine-germ ansatz

Here is a rigorous limitation of the constructed counterexamples, useful for formulating a more faithful percolation classification. Suppose

$$P_{D,p,q} = A(D,q)_*^{-1}\mathrm{SLE}_6(AD; Ap, Aq)$$

on bounded Jordan domains, and suppose this family is reversible: reversing $P_{D,p,q}$ gives $P_{D,q,p}$. Then, for all p, q,

$$A(D,p)^T A(D,p) = c(D,p,q)A(D,q)^T A(D,q), \qquad c(D,p,q) > 0.$$

Use here the external reversibility theorem of J. Miller and S. Sheffield, *Imaginary geometry III: reversibility of SLE$_\kappa$ for $\kappa \in (4,8)$*, Annals of Mathematics 184 (2016), 455–486, https://annals.math.princeton.edu/2016/184-2/p03. Indeed, ordinary SLE_6 reversibility says that reversing the first curve uses the same matrix $A(D,q)$, now with endpoints q, p. By assumed reversibility this is equal to the law constructed with $A(D,p)$. Single-pair affine identifiability therefore says that $A(D,p)A(D,q)^{-1}$ is a similarity, which is equivalent to the displayed matrix relation. For a rectangle with one endpoint in a horizontal side and the other in a vertical side, the proposed matrices $\mathrm{diag}(1,a)$ and $\mathrm{diag}(a,1)$ violate this relation when $a \neq 1$. Thus the construction explicitly fails reversibility.

This result only eliminates endpoint-dependent anisotropy *within this ansatz*; it does not prove the desired classification after reversibility is added. Near-critical percolation suggests a separate criticality condition remains necessary.

6 Further affine rigidity and target consistency

The following argument concerns ordinary chordal SLE_6; it does not assume conformal invariance of an unknown exploration family. For SLE_6 in $\mathbb{H}$ from 0 to ∞, the left-passage event $L(z)$ is defined by the eventual side of the oriented trace containing z. For a nonsimple trace this is also its eventual Loewner swallowing side. It is invariant under orientation-preserving homeomorphisms of the marked half-plane. A fixed interior point belongs to the trace with probability zero.

Lemma 6.1 (Strict left-passage function)**.** *For $z = x + iy$, $y > 0$,*

$$\mathbb{P}[L(z)] = F(x/y), \qquad F(u) = \frac{\int_{-\infty}^{u}(1+v^2)^{-2/3}\,dv}{\int_{-\infty}^{\infty}(1+v^2)^{-2/3}\,dv}.$$

In particular, F is strictly increasing.

Proof. This is the standard SLE left-passage formula; its short diffusion derivation explains why strict monotonicity holds even for the nonsimple SLE_6 trace. Write $g_t(z) - \sqrt{\kappa}B_t = X_t + iY_t$. Then

$$dX_t = \frac{2X_t}{X_t^2+Y_t^2}\,dt - \sqrt{\kappa}\,dB_t, \qquad dY_t = -\frac{2Y_t}{X_t^2+Y_t^2}\,dt.$$

For $U = X/Y$ and $ds = dt/Y_t^2$, Itô's formula gives

$$dU_s = \frac{4U_s}{1+U_s^2}\,ds - \sqrt{\kappa}\,d\widetilde{B}_s.$$

The increasing scale function with limits $0, 1$ solves

$$\frac{\kappa}{2}F'' + \frac{4u}{1+u^2}F' = 0, \qquad F'(u) = C_\kappa(1+u^2)^{-4/\kappa}.$$

The normalizing integral is finite for $\kappa < 8$. The clock runs to infinity up to the point's maximal Loewner lifetime: otherwise $d\log Y_s = -2(1+U_s^2)^{-1}ds$ keeps Y positive and the diffusion extends. The bounded martingale $F(U_s)$ converges. A limit in $(0,1)$ would force U_s to converge to a finite value and its martingale quadratic variation to be infinite, a contradiction. Thus $U_s \to -\infty$ or $+\infty$. Optional stopping and the usual topological side identification for chordal Loewner chains give the formula at $\kappa = 6$. □

Proposition 6.2 (Linear symmetries of half-plane SLE_6)**.** *If an invertible real-linear map M carries $SLE_6(\mathbb{H}, 0, \infty)$ to $SLE_6(M\mathbb{H}, 0, \infty)$, then M is a Euclidean similarity, possibly reversing orientation.*

Proof. Reflection invariance reduces to $\det M > 0$. Rotate $M\mathbb{H}$ back to $\mathbb{H}$. The resulting map is $T(x,y) = (ax+by, cy)$ with $a, c > 0$. By topological preservation of the side event and Lemma 6.1,

$$F(u) = F((au+b)/c) \qquad (u \in \mathbb{R}).$$

Strict monotonicity implies $a = c$, $b = 0$. Undo the rotation. □

6.1 Target consistency is a substantive additional condition

Additional axiom; not a consequence proved here. For two targets q_1, q_2 on a boundary arc J whose closure avoids p, require explorations towards them to couple identically until first hitting J (or use the usual stopping formulation before target separation). A common percolation coloring gives this: the two boundary conditions disagree only on J, so the exploration algorithms see identical colors until reaching it. The stated locality axiom fixes q and does not assert this property.

Proposition 6.3 (Target consistency removes variable affine target metrics)**.** *Suppose D is a bounded Jordan domain analytic near p, and*

$$\mathcal{P}_{D,p,q} = (A_q^{-1})_* \,\mathrm{SLE}_6(A_q D, A_q p, A_q q).$$

If target consistency holds for q_1, q_2 on an arc avoiding p, then $A_{q_2}A_{q_1}^{-1}$ is a similarity. In particular, their metric tensors $A_q^{\mathsf{T}}A_q$ are proportional.

Proof. Translate p to zero and dilate by $r \to \infty$. The domains converge locally to the tangent half-plane H_p, and targets tend to infinity. For each fixed q, conformal kernel convergence at the analytic start and the usual initial-segment construction of SLE give limiting law

$$(A_q^{-1})_* \,\mathrm{SLE}_6(A_q H_p, 0, \infty).$$

Convergence can be tested on paths stopped on crosscuts strictly inside fixed half-disks, followed by exhaustion. The dilated disagreement arc escapes every fixed neighborhood, so target consistency says that the two limiting half-plane laws coincide. Apply A_{q_1}, rotate the image half-plane, and use Proposition 6.2. □

This eliminates the specific target-tangent obstruction, not every possible counterexample. The proof uses initial-segment convergence for ordinary SLE_6, not for an unknown general ensemble.

6.2 A compatibility graph for all target-germ affine decorations

The previous target-consistency proposition also gives a useful classification within the entire class of affine decorations, without a claim about arbitrary exploration ensembles. Define a graph whose vertices are admissible embedded target germs, modulo translations, and join G_1, G_2 when they can both occur on one bounded Jordan domain with an analytic starting point p elsewhere. The targets must lie on a boundary arc whose closure avoids p. Assign to each vertex the projective positive-definite tensor $[Q(G)] = [A(G)^{\mathsf{T}} A(G)]$.

Corollary 6.4 (Target-germ metrics are constant on compatibility components)**.** *For a target-germ affine SLE_6 decoration satisfying target consistency, $[Q(G)]$ is constant on every connected component of this graph. If the graph is connected, the decoration is one fixed-affine SLE_6 family.*

Proof. Proposition 6.3 makes the tensors proportional across every edge. Propagate along finite paths. If a component has representative matrix A_0, proportionality implies $A(G) = c_G R_G A_0$ with $c_G > 0$ and R_G orthogonal. Similarity covariance of SLE_6 then gives

$$A(G)_*^{-1}\,\mathrm{SLE}_6(A(G)D, A(G)p, A(G)q) = (A_0^{-1})_*\,\mathrm{SLE}_6(A_0 D, A_0 p, A_0 q),$$

independently of G. □

For ordinary smooth local boundary germs, connectivity is suggested by a simple planar surgery: place two small representatives far apart, retain their marked local boundary arcs, join the remaining endpoints with disjoint arcs to form one Jordan boundary, and place an analytic start on a joining arc. A fully general surgery statement for nonlocally connected prime-end impressions is not asserted here; the graph formulation keeps this issue explicit. For the straight-line tangent germs in the counterexample, joint realization on an ordinary rounded rectangle already connects horizontal and vertical targets and excludes their distinct tensors.

6.3 Finite rotations and a fixed anisotropy

Proposition 6.5. *Suppose an ensemble has one fixed affine SLE_6 normalization with linear part A, and is covariant under rotation R_θ, where $\theta \notin \pi\mathbb{Z}$. Then $A^{\mathsf{T}}A$ is scalar, so A is a similarity. A half-turn alone does not suffice.*

Proof. The map $AR_\theta A^{-1}$ preserves the SLE_6 family and hence is a similarity. Its determinant is 1, so it is orthogonal. For $Q = A^{\mathsf{T}}A$ this says $R_\theta^{\mathsf{T}} Q R_\theta = Q$. A symmetric real 2×2 matrix commuting with a rotation other than 0 or π is scalar. For $R_\pi = -I$, every Q satisfies the relation. □

The fixed-normalizer hypothesis is essential: the target-tangent construction can be covariant under every rotation and still have no fixed normalizer. Rotation symmetry by itself does not repair the conjecture.

7 What locality actually says about a semimartingale driving process

Status. The coordinate-change identities below are proved, under explicitly stated continuous-semimartingale assumptions. The conditional martingale theorem does not require absolute continuity of quadratic variation. They give a conditional SLE classification. The original three axioms do not currently imply these semimartingale or local-martingale assumptions, and no such implication is claimed. In particular the domain Markov property does not make a driving process Markov in its single real coordinate.

7.1 Deletion of a hull and the drift anomaly

Use chordal Loewner normalization

$$\partial_t g_t(z) = \frac{2}{g_t(z) - W_t}, \qquad g_t(z) = z + \frac{2t}{z} + O(z^{-2}).$$

Suppose on an interval before collision with a fixed hull that

$$dW_t = dM_t + b_t\,dt, \qquad d\langle M\rangle_t = v_t\,dt, \tag{L1}$$

where M is a continuous local martingale and $v_t \geq 0$. Let A be a bounded half-plane hull away from the initial tip, and let $h_0 : \mathbb{H} \setminus A \to \mathbb{H}$ have hydrodynamic normalization at infinity. Grow the same trace in the deleted domain, mapping by h_0 first, and denote its Loewner maps by $\widehat{g}_u$. Up to the first encounter with A, write

$$h_t = \widehat{g}_{u(t)} \circ h_0 \circ g_t^{-1}, \qquad \widehat{W}_{u(t)} = h_t(W_t), \qquad \dot{u}(t) = h_t'(W_t)^2. \tag{L2}$$

The map h_t is analytic across the real line near W_t, by Schwarz reflection, and $h_t'(W_t) > 0$. These formulas are local near the driving point; no differentiability of the trace is used.

Differentiating the composition in (L2) gives

$$\partial_t h_t(z) = \frac{2h_t'(W_t)^2}{h_t(z) - h_t(W_t)} - \frac{2h_t'(z)}{z - W_t}. \tag{L3}$$

For $z = W_t + \epsilon$ the first term is $2h_t'(W_t)/\epsilon - h_t''(W_t) + O(\epsilon)$, while the second term is $2h_t'(W_t)/\epsilon + 2h_t''(W_t) + O(\epsilon)$. Consequently

$$(\partial_t h_t)(W_t) = -3h_t''(W_t). \tag{L4}$$

The random field $h_t(z)$ has finite variation in time locally before collision, so ordinary Itô's formula (equivalently the finite-variation case of Itô–Wentzell) applies to $h_t(W_t)$. Thus

$$d\widehat{W}_{u(t)} = h_t'(W_t)\,dM_t + \left[h_t'(W_t)b_t + \left(\frac{v_t}{2} - 3\right)h_t''(W_t)\right]dt. \tag{L5}$$

After the capacity time change, its characteristics are therefore

$$\boxed{\widehat{v}_{u(t)} = v_t, \qquad \widehat{b}_{u(t)} = \frac{b_t}{h_t'(W_t)} + \left(\frac{v_t}{2} - 3\right)\frac{h_t''(W_t)}{h_t'(W_t)^2}.} \tag{L6}$$

The coefficient -3, and hence the exceptional variance 6, follow from the normalization $\partial_t g_t = 2/(g_t - W_t)$. A different capacity normalization changes both constants consistently.

Logical content. Locality identifies the law of the transformed driving process with the law prescribed for the deleted physical domain in its own conformal chart. It does *not* identify that law with the original driving law in its original chart. In particular the two drifts in (L6) need not vanish, or be equal. A geometry-dependent drift can transform with the second term in (L6); deleting that term without an additional hypothesis is exactly where an invalid proof of $v = 6$ would enter. Physical translation invariance acts by translating the physical domain and trace. It supplies no invariance under the nonlinear map h_t and cannot by itself justify identifying these two characteristics.

7.2 A precise conditional classification by driving martingales

Conditional theorem (proved).
Suppose an ensemble of continuous Loewner-generating chordal traces satisfies strong domain Markov and locality, with the conventional endpoint-preserving hull deletions. Suppose also that, for every deterministic marked domain, its capacity driving process in a chordal conformal chart is a continuous local martingale in its natural filtration. Assume the usual measurability and restart consistency of this family, so that the strong domain Markov property applies in the chart at a random slit domain. Then

$$W_t - W_0 = \sqrt{6}\, B_t,$$

and the ensemble is ordinary chordal SLE_6. No absolute continuity or differentiability assumption on quadratic variation is needed. The chart premise is independent of which chordal chart is selected, since two such charts differ by $z \mapsto az + b$, $a > 0$, and the corresponding deterministic capacity time change preserves local martingales.

Proof. Fix an initial marked domain and map it to $(\mathbb{H}, 0, \infty)$. Delete the inverse image of a vertical slit at $x \neq 0$, of height $2\sqrt{s}$ for $s > 0$. The hydrodynamic map of this slit is

$$h_0(z) = x + \sqrt{(z-x)^2 + 4s}, \tag{L7}$$

with the branch asymptotic to $z - x$ at infinity. At zero,

$$h_0'(0) = \frac{|x|}{\sqrt{x^2+4s}}, \qquad h_0''(0) = -\frac{4s\ \mathrm{sgn}(x)}{(x^2+4s)^{3/2}} \neq 0. \tag{L8}$$

Let $C_t = \langle W \rangle_t$, initially with no regularity beyond the continuity and monotonicity of quadratic variation. The general Itô version of (L5) is

$$d\widehat{W}_{u(t)} = h_t'(W_t)\, dW_t + \frac{1}{2} h_t''(W_t)\, dC_t - 3h_t''(W_t)\, dt. \tag{L8a}$$

By locality and the hypothesis on the deleted domain, $\widehat{W}_u$ is a continuous local martingale; its time change $\widehat{W}_{u(t)}$ is a local martingale in the same trace filtration. Indeed, the inverse clock $t(u)$ is the original-domain capacity of the initial trace reconstructed from the deleted-domain curve. It is an adapted, continuous increasing functional of that trace, so $u(t)$ is a stopping time in the deleted-driver filtration. The usual stopping-time change theorem, after localization, therefore applies. That filtration is also the one generated by W_t, since each driving function determines its initial hull and trace. Localizing before collision and before the chart derivatives become unbounded, the stochastic integral in (L8a) is a local martingale. Its remaining continuous finite-variation term must therefore vanish identically. As signed measures,

$$h_t''(W_t)(dC_t - 6\, dt) = 0. \tag{L8b}$$

The second derivative starts nonzero by (L8) and remains so on a positive, possibly random, initial time interval. Equation (L8b) proves $C_t = 6t$ on that interval.

This conclusion applies at the start of every deterministic marked domain. To propagate it without any density assumption, set

$$T = \inf\{t \geq 0 : C_t - 6t \neq 0\}, \qquad \inf \varnothing = \infty.$$

In the usual right-continuous filtration this is a stopping time: the first entrance time of the continuous adapted process $C_t - 6t$ into the open set $\mathbb{R} \setminus \{0\}$. The positive initial zero interval

shows $T > 0$ almost surely. If $T < \infty$, continuity implies $C_T - 6T = 0$, and by the definition of T there must be nonzero values arbitrarily soon after T. Restarting the exploration at T using the strong domain Markov property gives a positive interval on which $C_{T+s} - C_T = 6s$, a contradiction. (One may localize T before deterministic bounds and before any terminal time.) Hence $C_t = 6t$ throughout. Lévy's characterization now gives the asserted Brownian driving law, and the Loewner correspondence identifies the trace as chordal SLE_6. □

Translation invariance was not needed in this conditional theorem. The additional condition is substantial: it asks for local martingales *in the conformal charts of all physical domains*, not just one chosen half-plane. A possible route toward universality would be to derive such a martingale condition from suitable physical observables. Nothing in this calculation constructs those observables.

7.3 A constant-characteristic version and an explicit two-slit test

A related sufficient assumption is that every marked-domain driving process has the same deterministic constant variance κ and drift b, in its capacity chart. Under locality (L6), a deletion with initial derivatives $d = h_0'(0)$ and $e = h_0''(0)$ must obey

$$b = \frac{b}{d} + \left(\frac{\kappa}{2} - 3\right)\frac{e}{d^2}. \tag{L9}$$

Choose the two vertical slits in (L7) at x and $-x$ with the same height. Their derivatives have the same $d \in (0, 1)$ and opposite nonzero e. Subtracting their instances of (L9) yields $\kappa = 6$; either equation then gives $b = b/d$, hence $b = 0$. Thus the only constant-characteristic ensemble in this class is SLE_6.

This two-slit computation is a useful check on the signs and constants, but its universality premise is much stronger than physical translation invariance. The current bottleneck is to prove an adequate chart-martingale or characteristic condition from genuinely coarse properties of critical percolation, rather than to manipulate (L6) further.

8 Dilation symmetry and criticality

8.1 Dilation covariance and subsequential limits

Lemma 8.1. *Every family with a fixed affine SLE_6 normalization is covariant under all scalar dilations: $\mathcal{P}_{rD,rp,rq} = r_*\mathcal{P}_{D,p,q}$ for $r > 0$.*

Proof. Dilations commute with the linear part of the normalizer; conjugating by its full affine map gives a Euclidean similarity. Use SLE similarity covariance. □

Lemma 8.2 (The cluster set, not each cluster point, inherits dilations)**.** *Let X be a topological space of ensembles with continuous dilation actions T_r. Suppose $P_\delta \in X$ has exact mesh covariance $T_r P_\delta = P_{r\delta}$ and is relatively compact as $\delta \downarrow 0$. Its cluster set $\mathcal{C}$ satisfies $T_r\mathcal{C} = \mathcal{C}$. If $\mathcal{C} = \{P\}$, then $T_r P = P$ for every $r > 0$.*

Proof. If $P_{\delta_n} \to P$, continuity gives $P_{r\delta_n} = T_r P_{\delta_n} \to T_r P$. Thus $T_r\mathcal{C} \subset \mathcal{C}$; use $T_{1/r}$ for equality. □

For laws indexed by marked domains the action is $(T_r P)_{D,p,q} = r_* P_{r^{-1}D, r^{-1}p, r^{-1}q}$. One must choose a topology in which the actions are continuous; this is not silently supplied by abstract tightness. The distinction is real: on $X = S^1$, take $P_\delta = \exp(i\omega \log \delta)$ and $T_r z = \exp(i\omega \log r)z$, $\omega \neq 0$. The cluster set is all of S^1 but has no member fixed by every dilation. Uniform nondegenerate

observable bounds do not remove this possibility: $\frac{1}{2} + \varepsilon \sin(\omega \log(R/\delta))$, $0 < \varepsilon < 1/2$, remains uniformly bounded while oscillating in scale phase. This is a topological example, not a percolation counterexample. It nevertheless explains why deriving dilation invariance of every subsequential limit from tightness or RSW alone would be circular.

8.2 All-scale RSW and a nonzero near-critical parameter

Proposition 8.3 (A finite correlation scale conflicts with all-scale RSW)**.** *Suppose continuum colorings P^λ obey $T_r P^\lambda = P^{r^{-\alpha}\lambda}$ for some $\alpha > 0$. For a fixed rectangle shape let $c(\lambda)$ be its unit-scale white crossing probability. Assume $c(\lambda) \to 1$ as $\lambda \to +\infty$ and $c(\lambda) \to 0$ as $\lambda \to -\infty$. If one P^λ has white crossing probabilities in $[\varepsilon, 1-\varepsilon]$ at every scale $R > 0$, with fixed $\varepsilon > 0$, then $\lambda = 0$.*

Proof. Scaling a size-R rectangle down to unit size changes the parameter to $R^\alpha \lambda$. Its crossing probability is $c(R^\alpha \lambda)$, which tends to 1 or 0 for positive or negative λ. □

This is a conditional calculation, separate from existence and regularity of near-critical continuum percolation. For standard triangular-lattice near-critical percolation $\alpha = 3/4$. RSW merely on each bounded scale range, with constants depending on that range, does not exclude nonzero mass. Moreover the target-tangent construction has bounded anisotropy and no preferred scale; removing a near-critical parameter does not address the target-consistency obstruction.

9 Near-critical obstruction: what is rigorous and what is not yet transferred

The missing criticality assumption is a separate obstruction from terminal germ dependence. On triangular mesh η, write

$$r(\eta) = \frac{\eta^2}{\alpha_4^\eta(\eta, 1)}, \qquad p_\eta = \frac{1}{2} + \lambda r(\eta), \qquad \lambda \neq 0,$$

where α_4 is the critical alternating four-arm probability. The normalization is $r(\eta) = \eta^{3/4+o(1)}$. At every mesh the exploration has exact spatial domain Markov and locality properties for every p_η. These two properties alone cannot express criticality.

Established external facts (not reproved here). Nolin–Werner, *Asymmetry of near-critical percolation interfaces*, JAMS 22 (2009), 797–819, https://arxiv.org/abs/0710.1470, establish subsequential limits of near-critical interfaces and their singularity with respect to ordinary SLE_6. Garban–Pete–Schramm, *The scaling limits of near-critical and dynamical percolation*, JEMS 20 (2018), 1195–1268, https://arxiv.org/abs/1305.5526, construct the configuration-valued near-critical scaling limit and establish its conformal covariance. Their Section 13.2 explicitly distinguishes the measurable trace of an interface from its order as a curve: the latter uniqueness is not proved there. Their Section 11 Markov property concerns change of the near-critical parameter, not the spatial exploration stopping-time property needed here. Thus neither statement may silently be substituted for full convergence of ordered exploration kernels in random slit domains. A 2025 Durham thesis of Léonie Papon, introduction, https://etheses.durham.ac.uk/id/eprint/16026/1/Papon-thesis-v2.pdf, also records ordered near-critical percolation interface uniqueness as a remaining challenge. These observations are limits of the sources checked, not a proof that no later theorem exists.

A simple necessary test for fixed affine SLE. Every fixed affine image of the ordinary SLE ensemble is covariant under all scalar dilations. Indeed, if the linear part of the affine map is L, then $L(rx) = rLx$; the translation part is absorbed by translation covariance. Consequently a genuine failure of dilation covariance excludes *every* fixed affine normalization. Singularity relative to ordinary SLE alone does not have this implication: an anisotropic affine image may also be singular.

Under a physical dilation by $s > 0$, the near-critical parameter changes as

$$s\,\omega^{\lambda} \overset{\text{law}}{=} \omega^{s^{-3/4}\lambda}.$$

The sign of the exponent follows already on the lattice: the original mesh η becomes $s\eta$ while the same occupation probability has effective parameter $\lambda r(\eta)/r(s\eta) \to s^{-3/4}\lambda$. Known nonconstant crossing probabilities as functions of λ therefore witness a macroscopic scale in the configuration law. Transferring this witness to a full domain-indexed curve ensemble requires the ordered-kernel step just identified. This investigation therefore records near-critical interfaces as a *conditional additional counterexample*, not as the proof of the counterexample in the main text.

9.1 Near-critical germ singularity and the energy of a driving drift

A stronger external obstruction. Aumann [9], Theorem 2.1 and Corollary 2.2, proves mutual singularity of distinct near-critical triangular-percolation interface limits already on the raw initial ordered-germ sigma-field. The theorem assumes a common subsequence with the stated quad-crossing convergence; for comparison with the critical parameter zero, that condition follows from Cardy's formula. Corollary 5.1 gives singularity under nontrivial scalar dilations at a fixed nonzero parameter, again with its convergence hypotheses. Thus a curve-level dilation obstruction is available, without identifying random slit-domain continuation kernels. It does not construct the coherent Borel strong-DMP family still needed for a counterexample to the corrected three-axiom conjecture.

The following consequence concerns a proposed driving representation, not a claim of semi-martingality of a near-critical interface.

Proposition 9.1 (Finite driving energy preserves the Brownian initial germ)**.** *Let $X_0 = 0$ be continuous on $[0,T]$, and let B be Brownian motion in a usual filtration to which X is adapted. Suppose $\kappa > 0$, $\rho > 0$ is a stopping time almost surely, and, for $t \le \rho \wedge T$,*

$$X_t = \sqrt{\kappa}B_t + \int_0^t b_s\,ds, \qquad \int_0^{\rho\wedge T} b_s^2\,ds < \infty \quad \textit{almost surely,} \tag{GE1}$$

where b is progressively measurable. On the canonical continuous-path space put $\mathcal{H}_{0+} = \bigcap_{t>0}\sigma(x_s : 0 \le s \le t)$, using the raw sigma-fields. Then, for every $E \in \mathcal{H}_{0+}$,

$$\mathbb{P}\{X \in E\} = \mathbb{W}_\kappa(E) \in \{0,1\}, \tag{GE2}$$

where $\mathbb{W}_\kappa$ is Wiener measure of variance rate κ.

Proof. Set

$$\sigma = \rho \wedge T \wedge \inf\left\{t : \int_0^{t\wedge\rho} b_s^2 ds \ge \kappa\right\}.$$

The finite energy in (GE1) implies $\sigma > 0$ almost surely. Define $\theta_s = \kappa^{-1/2} b_s \mathbf{1}_{\{s\le\sigma\}}$. Its integrated square is at most one. Novikov's criterion and Girsanov's theorem (for example [10], Theorems 1–2) therefore make

$$Z_T = \exp\left\{-\int_0^T \theta_s\,dB_s - \frac{1}{2}\int_0^T \theta_s^2 ds\right\}$$

a strictly positive density of an equivalent probability Q. Under Q,

$$\widetilde{X}_t = \sqrt{\kappa} B_t + \int_0^{t\wedge\sigma} b_s ds$$

is Brownian motion of variance rate κ on $[0,T]$. The paths X and $\widetilde{X}$ agree on the strictly positive interval $[0,\sigma]$. Every raw germ event has the same membership for two paths agreeing on some positive initial interval: choose any deterministic $1/n$ inside that interval and use measurability in $\sigma(x_s : s \le 1/n)$. Consequently $Q\{X \in E\} = \mathbb{W}_\kappa(E)$. Blumenthal's zero–one law makes the latter probability zero or one; equivalence of P and Q gives (GE2). There is no global Novikov assumption on the original drift. □

Corollary 9.2 (A near-critical limit cannot be a finite-energy drift of SLE)**.** *Consider a nonzero near-critical interface limit covered by Aumann's comparison with the critical limit. Assume it has a continuous Loewner-generating parametrization, with an adapted Borel correspondence between driver prefixes and ordered trace prefixes. Its driver in a capacity chart where the critical comparison has law $\sqrt{6}B$ cannot satisfy (GE1) with $\kappa = 6$.*

Proof. A raw ordered trace-germ event is unchanged when two nonconstant Loewner traces agree for a positive capacity interval. In the preceding proof the original and the Q-Brownian drivers have precisely such an interval in common. The corresponding traces therefore have the same raw germ-event membership. Under Q the comparison trace is SLE_6; its raw initial germ is zero–one (equivalently use the Brownian germ and the adapted correspondence). Equivalence then gives the same germ probabilities under the original law. This contradicts the germ event separating the nonzero near-critical and critical laws in Aumann's Corollary 2.2. □

The energy must diverge immediately if such an AC representation exists. More precisely, suppose $X = \sqrt{\kappa}B + \int_0^{\cdot} b_s ds$ on $[0,T]$, with b progressively measurable and integrable in absolute value, and suppose a raw germ event E has probability one for X and zero under $\mathbb{W}_\kappa$. Then

$$\int_0^t b_s^2 ds = \infty \quad \text{for every } 0 < t \le T, \quad \text{almost surely.} \tag{GE3}$$

Indeed set $\sigma = T \wedge \inf\{t : \int_0^t b_s^2 ds \ge \kappa\}$, allowing $\sigma = 0$ on instantaneous infinite energy. This is a stopping time in the usual filtration. In all cases $\int_0^\sigma b_s^2 ds \le \kappa$, by monotone convergence from times strictly below σ when $\sigma > 0$. The same bounded-energy change of measure as above makes $\widetilde{X} = \sqrt{\kappa}B + \int_0^{\cdot\wedge\sigma} b_s ds$ Brownian under an equivalent probability. Thus $E(\widetilde{X})$ has probability zero under both probabilities. On $\{\sigma > 0\}$, however, X and $\widetilde{X}$ have the same germ, so $E(X) = 1$ forces $\mathbb{P}(\sigma > 0) = 0$. Finite energy on any positive initial interval would imply $\sigma > 0$, proving (GE3). The identical argument uses the trace-germ separator for the near-critical comparison.

Why square integrability cannot be weakened to absolute integrability. For $0 < \alpha < 1/2$, the process $X_t = \sqrt{\kappa}B_t + t^\alpha$ has a continuous absolutely continuous finite-variation drift, with density $\alpha t^{\alpha-1}$ integrable at zero but with infinite square integral on every positive interval. Brownian small-time Hölder regularity gives $B_t/t^\alpha \to 0$ almost surely. Hence $X_t/t^\alpha \to 1$ almost surely, while the corresponding Wiener limit is zero. This explicitly separates their raw initial germs and shows why the energy hypothesis in Proposition 9.1 is substantive.

Scope and compatibility with source shielding. This excludes Brownian driving with a locally bounded, or more generally locally square-integrable, absolutely continuous drift. It does not prove that a near-critical driver is not a semimartingale: a singular finite-variation part, infinite drift energy at every positive initial time, or a different martingale bracket is not excluded. Neither the bracket-six premise nor Loewner generation is supplied by the singularity theorem. Raw germ singularity between two laws is compatible with an individual germ zero–one law for each: different laws may assign different deterministic values to the same germ event. Source shielding alone therefore cannot identify the universal germ law.

9.2 An explicit limit-transfer lemma

Here is a precise sufficient condition for the commonly asserted passage of the spatial Markov property to scaling limits. It isolates the missing assumption rather than labeling it automatic.

Let (I_n, D_n, Y_n) denote respectively the explored initial segment, remaining marked domain, and future curve at a stopping time. Suppose these random objects take values in Polish spaces and converge jointly in distribution to (I, D, Y). Let K_n be the future-curve kernels, so

$$\mathbb{E}[f(Y_n) \mid I_n, D_n] = K_n f(D_n).$$

Assume there is a Borel probability kernel K such that, for every bounded continuous f and every sequence $d_n \to d$ with d in a full-law set for D, one has $K_n f(d_n) \to K f(d)$. Then

$$\mathbb{E}[f(Y) \mid I, D] = K f(D).$$

Proof. By a Skorokhod representation the triples may be assumed to converge almost surely. For any bounded continuous test function h of (I, D), the exact finite-n identity gives

$$\mathbb{E}[h(I_n, D_n) f(Y_n)] = \mathbb{E}[h(I_n, D_n) K_n f(D_n)].$$

Bounded convergence and the sequential-kernel hypothesis give the same identity at the limit, with $Kf(D)$ on the right. The monotone class theorem extends the identity to bounded measurable h, and a countable convergence-determining class of f identifies the conditional kernel. □

The decisive hypothesis controls kernels at *random slit domains*; convergence for one deterministic Jordan domain is insufficient. Analogous coupled convergence can transfer locality before first approach to a removed closed set: stop first at positive-distance neighborhoods, where uniform curve convergence leaves the two coupled initial segments identical, then decrease the distance to zero. Exact first-hit stopping needs a separate continuity argument and is not implicit in this observation.

10 Affine identifiability without Jordan boundary regularity

Status and interpretation. The extension proved here uses only standard quasiconformal compactness, prime-end extension for quasiconformal disk maps, Fatou's theorem for bounded analytic functions, and uniqueness of bounded harmonic functions from their almost-everywhere disk boundary values. It removes the Jordan assumption from affine metric identifiability. At completely arbitrary prime ends a conformal image of SLE need not have a Euclidean-continuous endpoint; its law is therefore understood in the prime-end compactification, or as its conformal hull exploration. The result applies in particular whenever the physical curves are continuous, as in the exploration setting of interest. No classification of arbitrary exploration ensembles is asserted.

10.1 A quasiconformal disk map fixing the boundary stays close to the identity

Lemma (proved). For each finite $K \geq 1$ there exists $C(K) < \infty$ such that every orientation-preserving K-quasiconformal self-homeomorphism $h : \mathbb{D} \to \mathbb{D}$ whose boundary extension fixes every point of $\partial\mathbb{D}$ satisfies

$$d_{\mathbb{D}}(h(z), z) \leq C(K) \qquad (z \in \mathbb{D}). \tag{RA1}$$

Here $d_{\mathbb{D}}$ is hyperbolic distance.

Proof. First there is $r(K) < 1$ with $|h(0)| \leq r(K)$ for all such maps. To see this, extend h to the sphere by reflection in the unit circle:

$$\widetilde{h}(z) = \frac{1}{\overline{h(1/\bar{z})}} \quad (|z| > 1).$$

This is a K-quasiconformal sphere map, and fixes three prescribed distinct points of the unit circle. The standard compactness theorem for normalized K-quasiconformal sphere maps, including compactness of their inverses, precludes a sequence with $h_n(0) \to \zeta \in \partial\mathbb{D}$: a limiting quasiconformal homeomorphism would send both 0 and the fixed boundary point ζ to ζ. Thus such an $r(K)$ exists. For an arbitrary z choose a disk automorphism a_z sending z to 0. The map $a_z \circ h \circ a_z^{-1}$ is again K-quasiconformal with identity boundary map, so

$$|a_z(h(z))| \leq r(K).$$

Conformal invariance of the hyperbolic metric gives (RA1), for example with $C(K) = 2 \operatorname{arctanh} r(K)$ in the usual normalization. □

Consequence (proved). For every $\zeta \in \partial\mathbb{D}$, the image of the radial approach $r\zeta$ under h is a nontangential approach to the same ζ, with aperture bounded solely in terms of K. Indeed rotate so $\zeta = 1$ and write

$$h(r) = \frac{r + u_r}{1 + r u_r}, \qquad |u_r| \leq r(K).$$

Then

$$|1 - h(r)| \leq (1 - r)\frac{1 + r(K)}{1 - r(K)},$$

while

$$1 - |h(r)| \geq \frac{1 - r}{2}\frac{1 - r(K)}{1 + r(K)}.$$

The ratio $|1 - h(r)|/(1 - |h(r)|)$ is therefore uniformly bounded. In particular $h(r\zeta) \to \zeta$ within a fixed Stolz region.

10.2 The identifiability theorem on any bounded simply connected domain

Theorem (proved). Let D be any bounded simply connected planar domain and let p, q be distinct prime ends. For orientation-preserving invertible real-linear maps A, B, define

$$\nu^A_{D,p,q} = (A^{-1})_* \operatorname{Law}(\mathrm{SLE}_6(AD; Ap, Aq)).$$

Then

$$\nu^A_{D,p,q} = \nu^B_{D,p,q} \iff BA^{-1} \text{ is a Euclidean similarity} \iff B^{\mathsf{T}}B = cA^{\mathsf{T}}A \quad (c > 0). \tag{RA2}$$

The statement holds for exploration laws in the prime-end compactification, and hence for their physical curves whenever defined continuously.

Proof. Set $M = BA^{-1} : AD \to BD$. This bilipschitz map carries null crosscut chains to null crosscut chains, with equivalent chains remaining equivalent; thus it induces the natural homeomorphism of prime-end compactifications. Choose conformal charts $\phi_A : AD \to \mathbb{H}$ and $\phi_B : BD \to \mathbb{H}$ sending the corresponding source and target to $0, \infty$. The interior map

$$H = \phi_B \circ M \circ \phi_A^{-1} : \mathbb{H} \to \mathbb{H}$$

is quasiconformal with a boundary extension to $\mathbb{R} \cup \{\infty\}$, fixing $0, \infty$. The boundary map is increasing. The boundary-swallowing race of the main record remains a prime-end topological event: an affine homeomorphism preserves the order in which boundary ends are cut off from the target by the explored filled hull. Equality of the two path laws, followed by strictness of that race probability, therefore gives, for every $x < 0 < y$,

$$\frac{-H(x)}{H(y) - H(x)} = \frac{-x}{y - x}.$$

Consequently $H(x) = cx$ on the entire extended real boundary, for some $c > 0$. This step uses prime-end boundary values of H, not Euclidean boundary values of either Riemann map.

Pass to disk charts, absorbing the conformal map $z \mapsto cz$ into the second chart. We obtain bounded conformal maps

$$\psi_A : \mathbb{D} \to AD, \qquad \psi_B : \mathbb{D} \to BD$$

such that

$$h = \psi_B^{-1} \circ M \circ \psi_A$$

is a K-quasiconformal disk self-homeomorphism with identity boundary map. Thus pointwise in the disk,

$$M\psi_A = \psi_B \circ h. \tag{RA3}$$

Fatou's theorem gives radial limits $\psi_A^*(\zeta)$ and nontangential limits $\psi_B^*(\zeta)$ for almost every circle point ζ. The preceding lemma shows that $h(r\zeta)$ approaches that same ζ nontangentially. Taking $r \uparrow 1$ in (RA3) yields

$$M\psi_A^*(\zeta) = \psi_B^*(\zeta) \quad \text{for almost every } \zeta \in \partial\mathbb{D}. \tag{RA4}$$

Both real coordinates of $M\psi_A$ are bounded harmonic functions on the disk, since M is real-linear and ψ_A is bounded analytic. The coordinates of ψ_B are also bounded harmonic. Uniqueness of the Poisson representation for bounded harmonic functions now gives

$$M\psi_A = \psi_B \qquad \text{throughout } \mathbb{D}.$$

Thus M is conformal on the nonempty open set AD. Writing $M(z) = \alpha z + \beta \bar{z}$, holomorphicity of $M\psi_A$ gives $\beta\overline{\psi_A'(z)} = 0$. Since ψ_A' never vanishes, $\beta = 0$. Invertibility and orientation imply that M is a direct Euclidean similarity. The converse is SLE similarity covariance. Reflections reduce orientation-reversing normalizers to this case. □

The boundedness assumption was used exactly when invoking bounded analytic Fatou limits and bounded harmonic uniqueness. No assertion about arbitrary unbounded domains follows without a growth or uniqueness condition.

10.3 A necessary and sufficient DMP condition inside the affine ansatz

Suppose, as an explicit additional ansatz, that an ensemble has the form

$$P_{D,p,q} = \nu_{D,p,q}^{A(D,p,q)} \tag{RA5}$$

for a deterministic measurable affine label on each marked bounded simply connected domain. Remove the inessential positive scalar from its metric by putting

$$T(D,p,q) = \frac{A(D,p,q)^{\mathsf{T}} A(D,p,q)}{\sqrt{\det(A(D,p,q)^{\mathsf{T}} A(D,p,q))}}. \tag{RA6}$$

Thus T is a positive-definite symmetric matrix of determinant one. Assume the usual Borel family and strong conditional-kernel formulation, so that equality of conditional path laws at a stopping time is an equality of probability kernels almost surely.

Corollary (proved). Within ansatz (RA5), the strong domain Markov property holds if and only if, for every starting marked domain and every admissible stopping time τ,

$$T(D_\tau, p_\tau, q) = T(D,p,q) \qquad \text{almost surely.} \tag{RA7}$$

Here D_τ is the target component of the remaining domain. It need not have Jordan boundary.

Proof. Fix the starting label $A = A(D,p,q)$. Under the fixed-A affine SLE law, ordinary SLE's strong domain Markov property gives, conditional on the explored initial segment, the remaining law

$$\nu_{D_\tau,p_\tau,q}^{A}.$$

Strong DMP for (RA5) demands that this be $\nu_{D_\tau,p_\tau,q}^{A(D_\tau,p_\tau,q)}$ almost surely. The rough-domain theorem (RA2), applied at each realized slit domain, is equivalent to equality of their normalized metric tensors, namely (RA7). Conversely (RA7) and similarity covariance identify these two conditional laws, proving strong DMP. □

This corollary classifies the DMP constraint only *within the stated ansatz*: the affine tensor must be a conserved quantity of the exploration state. A target-germ tensor supplies exactly such a conserved quantity. The corollary neither proves that a general ensemble has affine-SLE form nor that every conserved tensor must arise from a target germ. Locality, translation covariance, target consistency, and boundary continuity impose further constraints which are not characterized by (RA7) alone.

11 Local target consistency passes to marginal curve limits

Scope. This lemma transfers an exploration coupling only inside a fixed neighborhood of the common start. It needs convergence of the two marginal curve laws, but neither convergence of random slit-domain kernels nor a limiting strong domain Markov theorem. It does not assert a coupling of the limiting curves all the way to a distant boundary arc.

11.1 Stopping an ordered unparameterized curve on a circle

Fix $p \in \mathbb{R}^2$ and $R_0 > 0$. Work in the usual metric space of oriented continuous curves modulo increasing reparameterization, with the uniform Fréchet topology. Curves start at p and end outside $\overline{B(p, R_0)}$. Small changes of the starting points are harmless; for the sequence version they need only lie inside every fixed smaller ball for sufficiently large indices. For a continuous representative $\gamma : [0, 1] \to \mathbb{R}^2$ and $0 < r < R_0$, put

$$\tau_\gamma(r) = \inf\{t : |\gamma(t) - p| \geq r\}, \qquad E_r\gamma = \gamma|_{[0,\tau_\gamma(r)]}. \tag{LT1}$$

The stopped path is regarded as an ordered unparameterized curve, so this definition is independent of pauses or changes of clock.

Lemma 11.1 (Almost every circle is a continuity circle)**.** *For each fixed continuous curve γ, the map $(\eta, s) \mapsto E_s\eta$ is continuous at (γ, r) for all but at most countably many radii $r \in (0, R_0)$. The stopping map is Borel. Consequently, for any probability law μ on these curves, E_r is μ-almost surely continuous for Lebesgue-almost every deterministic r.*

Proof. Choose any continuous representative of γ. The function $r \mapsto \tau_\gamma(r)$ is nondecreasing and hence has at most countably many discontinuities. Suppose it is continuous at r. If $(\gamma_n, r_n) \to (\gamma, r)$ in the product of Fréchet and ordinary metrics, choose increasing reparameterizations for which representatives $\widetilde{\gamma}_n$ converge uniformly to γ. Writing $\varepsilon_n = \|\widetilde{\gamma}_n - \gamma\|_\infty + |r_n - r|$, first-hit times obey, for every slightly larger positive error tending to zero,

$$\tau_\gamma(r - 2\varepsilon_n) \;\leq\; \tau_{\widetilde{\gamma}_n}(r_n) \;\leq\; \tau_\gamma(r + 2\varepsilon_n). \tag{LT2}$$

If $\varepsilon_n = 0$ the assertion is immediate; using a strictly larger error avoids any immaterial equality convention. Continuity of τ_γ at r gives convergence of the middle times to $\tau_\gamma(r)$. Uniform continuity of γ then makes the stopped representatives, linearly recoded onto $[0, 1]$, converge uniformly. This proves joint continuity at (γ, r).

The stopping map is understood on all curves reaching the selected radius, including already stopped curves reaching a larger radius. For parameterized curves the stopping time is Borel, since

$$\{\tau_\gamma(r) \leq t\} = \left\{ \max_{0 \leq u \leq t} |\gamma(u) - p| \geq r \right\}.$$

Stopping and linear recoding are therefore Borel path operations. They commute with reparameterization and give the usual Borel stopping map on ordered unparameterized curves.

For completeness, the almost-every-radius assertion does not require a measurable choice of representatives. For a map between metric spaces its set of continuity points is a G_δ set, obtained from its oscillations on neighborhoods. Apply this to the joint stopping map. Its discontinuity set in the product of curve space and radius space is Borel, and every vertical section has Lebesgue measure zero by the countability argument above. Fubini's theorem now gives the assertion under μ. □

Why a bad radius can occur. A path may first touch a circle, then spend a nontrivial segment inside it, and only later cross it. A small outward perturbation can postpone its first exit until that later crossing. This is why stopping continuity cannot simply be claimed at every deterministic radius for every path. The preceding lemma handles these tangencies without excluding them by an unproved regularity assumption.

11.2 Transfer using only two marginal limits

Theorem 11.2 (Transfer of local target consistency)**.** *For $i = 1, 2$, let γ_n^i be oriented continuous random curves whose marginal laws converge weakly in the uniform Fréchet topology to laws μ^i. Assume the limiting curves start at p and end outside $\overline{B(p, R_0)}$. Suppose that for every fixed $r \in (0, R_0)$ and all sufficiently large n,*

$$\mathrm{Law}(E_r\gamma_n^1) = \mathrm{Law}(E_r\gamma_n^2). \tag{LT3}$$

Then the same equality holds for the limiting curves at every *$r \in (0, R_0)$:*

$$(E_r)_*\mu^1 = (E_r)_*\mu^2. \tag{LT4}$$

The hypothesis (LT3) may equivalently be replaced by a discrepancy tending to zero in any metric inducing weak convergence of the stopped laws.

Proof. Lemma 11.1 gives a full-Lebesgue-measure set of radii at which E_r is almost surely continuous under both limiting laws. At each such radius the continuous mapping theorem and (LT3) give (LT4). This part needs no joint convergence of (γ_n^1, γ_n^2).

To obtain every smaller radius, fix arbitrary $r < R_0$ and choose a good $s \in (r, R_0)$. Equality of the stopped-at-s laws remains equality after applying the Borel map E_r. Since $E_r(E_s\gamma) = E_r\gamma$ for $r < s$, this proves (LT4) at r without requiring any continuity of E_r there. □

11.3 Application to two percolation targets

Let D be a Jordan domain, p a common boundary start, and let the targets q_1, q_2 lie on a boundary arc J with $p \notin \overline{J}$. The boundary colors for the two Dobrushin explorations differ only on the arc between the targets. Use identical independent bulk colors and identical boundary colors outside a lattice approximation J_n of this disagreement arc, with a common discrete starting edge. The discrete exploration rules produce exactly the same ordered initial path until the exploration first encounters J_n. Choose

$$0 < R_0 < \mathrm{dist}(p, \overline{J}).$$

For all large n, the approximating disagreement arc stays outside the closed ball $\overline{B(p, R_0)}$. Thus (LT3) holds inside every smaller ball. If a common subsequence makes each marginal exploration converge, Theorem 11.2 supplies local target consistency for their limiting initial segments. The lattice approximations must in fact have the stated common-start/common-color coupling; a small displacement of two unrelated explorations alone would not justify pathwise closeness.

Consequence for affine decorations. The affine-metric elimination argument needs only equality of these local prefix laws, not equality until the full distant arc. Suppose D is analytic near p and the two limiting marginals have laws

$$(A_i^{-1})_*\,\mathrm{SLE}_6(A_iD, A_ip, A_iq_i), \qquad i = 1, 2.$$

Translate p to zero and blow up space. By conformal kernel convergence at the smooth start, the local initial laws tend to

$$(A_i^{-1})_*\,\mathrm{SLE}_6(A_iH_p, 0, \infty),$$

where H_p is the tangent half-plane. Equality in every sufficiently small physical ball transfers after dilation to equality of the limiting stopped laws in every fixed half-disk (first at common continuity

radii, then all smaller radii by the same composition argument). These local restrictions determine the entire half-plane law. Strict left-passage identifiability therefore makes $A_2A_1^{-1}$ a similarity.

Consequently the target-tangent affine counterexample cannot be the family of simultaneous two-target subsequential percolation limits on a fixed smooth domain with targets carrying different tangent directions. This exclusion is independent of the more difficult transfer of a strong domain Markov property to random limiting slit domains.

12 Affine metrics encoded in initial path germs

Status and motivation. The result below is proved for the *oriented initial trace germ* of an affine image of half-plane SLE_6 at its smooth starting boundary point. It does not construct a source-dependent domain Markov family. In particular, it does not establish that the metric can be recovered from the unlabelled boundary germ at a later random exploration tip. Those distinctions are important when assessing whether target consistency alone could repair the classification conjecture.

12.1 State space and the meaning of a trace germ

We retain the chronological order of a path but forget its original clock. For the half-plane curves used here, the usual half-plane-capacity clock provides a canonical representative: the filled hull at time t has hydrodynamic map $g_t(z) = z + 2t/z + o(1/z)$. This clock is determined by the ordered unparameterized trace. Section 3.3 justifies continuous Loewner driving of affine images; hence they admit this clock. The clock has infinite total lifetime almost surely: its total half-plane capacity T is positive, while spatial scale invariance makes its law equal to that of r^2T for every $r > 0$; the only such probability laws on $[0, \infty]$ are supported on $\{0, \infty\}$. Thus $T = \infty$. All functions below are functions of the ordered unparameterized trace.

Use the standard Borel space of these capacity-parameterized continuous curves, with the sigma-field generated by restrictions to finite time intervals. Equivalently, this can be regarded as a Borel realization of the usual ordered unparameterized path space. For $t > 0$, let

$$\mathcal{F}_t = \sigma\{\gamma(s) : 0 \le s \le t\}, \qquad \mathcal{F}_{0+} = \bigcap_{t>0} \mathcal{F}_t.$$

The latter is the initial trace-germ sigma-field. It is unaffected by changing any part of the curve after a positive capacity time. The canonical time is used only to give an explicit measurable coding of this germ.

For $r > 0$, let $S_r\gamma$ be the spatially dilated trace, again put in the canonical clock. Capacity scaling gives the exact identity

$$(S_r\gamma)(t) = r\gamma(t/r^2). \tag{SG1}$$

In particular S_r is invertible with inverse $S_{1/r}$.

12.2 Dilation ergodicity

Lemma 12.1 (Brownian scaling is mixing)**.** *For a one-sided Brownian motion $B = (B_t)_{t\ge0}$ and any fixed $r > 1$, the measure-preserving transformation*

$$(\mathcal{S}_rB)_t = rB_{t/r^2}$$

is mixing and hence ergodic.

Proof. Set $Z_s = e^{-s/2} B_{e^s}$ for $s \in \mathbb{R}$. This is a stationary centered Gaussian process with

$$\mathbb{E}[Z_s Z_u] = e^{-|s-u|/2}.$$

It is the stationary Ornstein–Uhlenbeck process. Under $\mathcal{S}_r$, it becomes $Z_{s-2\log r}$. Covariances between finite collections of coordinates separated by $n(2\log r)$ tend to zero, so the corresponding jointly Gaussian vectors become independent in the limit. Approximation by finite-coordinate cylinder functions gives mixing on the full path sigma-field. The correspondence is invertible almost surely: $B_t = \sqrt{t}\, Z_{\log t}$ for $t > 0$, with $B_0 = 0$. Thus mixing transfers back to Brownian scaling. The two-sided log-time process here is obtained from a one-sided Brownian path at times $t > 0$; no independent Brownian past is being added. □

Lemma 12.2 (Affine SLE_6 is dilation mixing)**.** *Let A be any invertible linear map carrying $\mathbb{H}$ onto itself and fixing its positive boundary orientation. Define*

$$\mu_A = \mathrm{Law}(A^{-1}\eta), \qquad \eta \sim \mathrm{SLE}_6(\mathbb{H}, 0, \infty),$$

with the physical half-plane-capacity representative understood. For every fixed $r > 1$, S_r preserves μ_A and is mixing under μ_A.

Proof. The SLE trace is a measurable, scaling-equivariant function of its Brownian Loewner driver $\sqrt{6}B$. The trace exists almost surely. Therefore its dilation action is a factor of the Brownian scaling action and is mixing by Lemma 12.1. The map on ordered traces $\eta \mapsto A^{-1}\eta$ commutes with scalar spatial dilations, and is an invertible measurable map after recoding by capacity. Mixing and invariance therefore transfer to μ_A. □

12.3 Separating distinct metrics from arbitrarily short initial segments

Normalize the maps preserving $\mathbb{H}$ by

$$A_{b,c} = \begin{pmatrix} 1 & b \\ 0 & c \end{pmatrix}, \qquad b \in \mathbb{R},\ c > 0. \tag{SG2}$$

This parametrizes all constant positive-definite conformal metrics, up to a positive scalar, in a fixed half-plane frame: the tensor is $A_{b,c}^{\mathsf{T}} A_{b,c}$ with its $(1,1)$ entry normalized to 1. The laws $\mu_{A_{b,c}}$ are distinct. Indeed, the strictly increasing SLE_6 left-passage function F gives

$$\mu_{A_{b,c}}[L(x+iy)] = F\left(\frac{x+by}{cy}\right). \tag{SG3}$$

Equality of the probabilities at $z = i$ and $z = 1 + i$ first identifies b/c and then $1/c$, hence both parameters.

Lemma 12.3 (Unit-time restrictions distinguish the metrics)**.** *If $A \neq A'$ are two normalized maps in (SG2), the laws of their traces restricted to capacity time $[0,1]$ are distinct.*

Proof. If the restrictions agreed, dilation invariance and (SG1) would make the restrictions to $[0,T]$ agree for every $T > 0$. Those restrictions generate the full path sigma-field, so the entire laws would agree, contrary to (SG3). □

Theorem 12.4 (Metric decoding from an initial trace germ)**.** *Let $\{A_j : j \in J\}$ be any finite or countable collection of distinct normalized maps (SG2). There exists an $\mathcal{F}_{0+}$-measurable function D with values in $J \cup \{\dagger\}$ such that*

$$\mu_{A_j}[D(\gamma) = j] = 1 \qquad \text{for every } j \in J. \tag{SG4}$$

In particular, restrictions of these affine SLE_6 measures to the initial trace-germ sigma-field are mutually singular, even though each separately has its usual almost-sure germ zero-one property.

Proof. For each pair $j \neq k$, Lemma 12.3 supplies a Borel set of unit-time traces whose indicator f_{jk} has distinct expectations under μ_{A_j} and μ_{A_k}. There are only countably many such pairs; list all the resulting bounded unit-time observables as $(f_m)_{m\geq 1}$. Write $v_{j,m} = \int f_m \, d\mu_{A_j}$. The vectors $v_j = (v_{j,m})_m$ are pairwise distinct.

Fix $r > 1$. By Lemma 12.2 and Birkhoff's ergodic theorem, for each m and each j,

$$\frac{1}{N}\sum_{n=1}^{N} f_m(S_{r^n}\gamma) \longrightarrow v_{j,m} \qquad \mu_{A_j}\text{-almost surely.} \tag{SG5}$$

The intersection of these events over m still has full μ_{A_j} measure. Every term $f_m(S_{r^n}\gamma)$ depends only on the trace up to time r^{-2n}. For any $t > 0$, all sufficiently late terms are consequently $\mathcal{F}_t$-measurable. Omitting finitely many initial terms does not change a Cesàro limit. Thus every limit in (SG5), and the event that the limit exists, is $\mathcal{F}_{0+}$-measurable.

Define $D = j$ if all limits exist and their vector equals v_j; use $D = \dagger$ otherwise. Countability of J and pairwise distinctness of the v_j make this a measurable unambiguous definition. Formula (SG5) proves (SG4). □

12.4 A common decoder for the continuum of normalized metrics

The countable proof extends to all parameters $(b, c) \in \Theta = \mathbb{R} \times (0, \infty)$, provided the standard path coding is used jointly in the affine parameter. We state this coding point explicitly.

Joint Borel capacity coding. Let η be an ordinary capacity-parameterized SLE_6 trace and K_s its filled hull. Before recoding, its affine image is the continuous path $A_{b,c}^{-1}\eta(s)$. Its physical capacity clock is

$$u_{b,c,\eta}(s) = \tfrac{1}{2}\,\mathrm{hcap}(A_{b,c}^{-1}K_s). \tag{SG6}$$

Use the standard measurable coding of compact half-plane hulls, in which filling a continuous path prefix and taking its half-plane capacity are Borel operations. Linear transformation is jointly continuous on compact sets. Consequently $(b, c, \eta, s) \mapsto u_{b,c,\eta}(s)$ is Borel. The inverse clock is measurable because, on the set where the clock is continuous and strictly increasing,

$$u_{b,c,\eta}^{-1}(t) = \inf\{s : u_{b,c,\eta}(s) > t\}$$

can be computed using rational s (with equality interpreted by continuity). Thus evaluation of the recoded image on $[0, 1]$ is jointly Borel in (b, c, η). The good-clock conditions are Borel using rational tests; on their complement use a cemetery path. For each fixed (b, c) this complement has SLE probability zero by the affine local-growth and lifetime arguments above. No single full-probability set valid simultaneously for all affine parameters is required.

This paragraph uses the usual deterministic measurability of hull capacity as part of the standard Borel curve coding. If that coding fact is not taken as established, the next corollary should be read conditionally on joint Borel recoding. The preceding countable theorem needs only measurable recoding for each of its countably many fixed maps.

Corollary 12.5 (Continuum metric decoder)**.** *With the joint Borel coding just specified, there exists one $\mathcal{F}_{0+}$-measurable map*

$$D_{\mathrm{all}} : \mathcal{C} \longrightarrow \Theta \cup \{\dagger\}$$

such that for every $(b, c) \in \Theta$,

$$\mu_{A_{b,c}}[D_{\mathrm{all}}(\gamma) = (b, c)] = 1. \tag{SG7}$$

The same single measurable rule therefore recovers every normalized affine metric, with almost-sure correctness under each metric law separately.

Proof. Choose a countable generating π-system for the standard Borel space of unit-capacity path segments, including the cemetery state. For example, use finite intersections of rational-time evaluation cylinders in rational rectangles, together with the cemetery singleton. Let $(f_m)_{m\geq 1}$ be their indicator functions. They determine probability laws on unit segments. Joint Borel recoding and integration against the fixed ordinary SLE law imply that

$$v : \Theta \to [0, 1]^{\mathbb{N}}, \qquad v_m(b, c) = \int f_m \, d\mu_{A_{b,c}}, \tag{SG8}$$

is Borel. Lemma 12.3 and the determining property of (f_m) imply that v is injective. Both parameter and sequence spaces are standard Borel, so the Lusin–Souslin theorem makes $v(\Theta)$ Borel and $v^{-1} : v(\Theta) \to \Theta$ Borel.

Fix $r > 1$. Form the vector $L(\gamma)$ of Cesàro limits

$$L_m(\gamma) = \lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} f_m(S_{r^n}\gamma),$$

using a cemetery value if any limit fails to exist. The proof of Theorem 12.4 shows that L is $\mathcal{F}_{0+}$-measurable. Define $D_{\mathrm{all}} = v^{-1}(L)$ when $L \in v(\Theta)$ and set it to $\dagger$ otherwise. This is a measurable definition; no measurable selection theorem is needed. For each fixed (b, c), ergodicity and Birkhoff applied to the countable family (f_m) give $L = v(b, c)$ almost surely under $\mu_{A_{b,c}}$. This proves (SG7). □

Quantifiers and limitations. The statement is one common decoder with correctness under each law: $\forall(b, c)\ \mu_{A_{b,c}}(D_{\mathrm{all}} = (b, c)) = 1$. It does not assert one common full-probability set of Brownian seeds on which correctness holds simultaneously after every uncountable collection of affine transformations. More importantly, the decoded object is still an ordered initial trace germ at a smooth start, not the bare boundary germ of a random slit-domain tip.

What has and has not been proved. The theorem demonstrates an exact mechanism by which a microscopic ordered path germ can store a constant affine metric. It is more than identifiability of the law: a single arbitrarily short trace germ distinguishes any prescribed countable collection of metrics almost surely, and the continuum corollary extends this under the stated joint Borel coding. The result does not contradict a germ zero-one law, since that law concerns one probability measure, while the distinguishing full-measure germ sets differ between measures.

A hypothetical source-germ decoration could choose a metric from the smooth initial source geometry, then attempt to recover the same metric from the slit boundary at all later tips. That recovery does *not* follow from Theorem 12.4. Three obstacles remain:

1. A later tip is not a smooth half-plane starting boundary point; the nearby past trace can have a different two-sided or conditioned local law.

2. The remaining marked domain gives an embedded boundary germ, which may lose chronological information when the trace has touched itself. The theorem uses an oriented initial trace germ and does not prove that the needed data are recoverable from that boundary alone.

3. A domain Markov construction would need one measurable rule valid at all relevant stopping times and compatible with locality deletions, not merely a decoder at a deterministic initial point under each fixed metric law.

Extending local metric decoding to random interior or tip germs, or proving that such decoding is impossible from the domain state, is a concrete next research question. No full counterexample satisfying target consistency is claimed from this mechanism.

13 A fixed metric follows inside the affine-SLE ansatz with target consistency and reversal

Status. This theorem assumes that every marginal is already an affine image of SLE_6. It does not derive that ansatz from coarse percolation properties. It shows that quite weak physical coherence conditions then eliminate all variation of the affine conformal structure between domains and marked points. Geometric reversal is an explicit additional hypothesis; it is not asserted here to follow from RSW or from the original three axioms.

Let $\mathcal{D}$ be the class of bounded Jordan domains whose boundaries consist of finitely many regular real-analytic arcs meeting at corners. Each analytic arc extends to a neighborhood of its closed parameter interval. Restrict marks to regular boundary points, meaning interior points of these arcs. Suppose

$$P_{D,p,q} = (A_{D,p,q}^{-1})_* \,\mathrm{SLE}_6(A_{D,p,q}D, A_{D,p,q}p, A_{D,p,q}q), \tag{AA1}$$

where $A_{D,p,q}$ is an invertible orientation-preserving linear map. Write $[Q_{D,p,q}]$ for the projective positive-definite tensor $[A_{D,p,q}^{\mathsf{T}} A_{D,p,q}]$; brackets identify positive scalar multiples.

The following elementary local identifiability principle will be used: if two laws of the form (AA1), on domains agreeing near the same regular start, have identical initial paths stopped in every sufficiently small ball, then their projective tensors are equal. Indeed blow up at the regular start. Conformal kernel convergence for ordinary SLE gives half-plane limits $A_i^{-1}\,\mathrm{SLE}_6(A_iH, 0, \infty)$. Equality of local prefixes gives equality of these half-plane laws; the strictly monotone SLE_6 left-passage function makes $A_2A_1^{-1}$ a similarity. This argument uses only local SLE convergence and the affine ansatz.

Theorem 13.1 (Conditional global metric rigidity)**.** *Assume (AA1), together with the following properties on $\mathcal{D}$:*

1. Translation covariance.

2. Locality for nested domains*: if $D' \subset D$ agrees with D near regular marks p, q, the two initial laws agree until they encounter the removed set.*

3. Local target consistency*: for fixed D, p and any two regular targets q_1, q_2, the two initial laws agree when stopped on every sufficiently small circle centered at p.*

4. Geometric reversibility*: reversing an ordered path with law $P_{D,p,q}$ gives law $P_{D,q,p}$.*

Then there is one fixed positive-definite tensor Q_0, up to scalar, with $[Q_{D,p,q}] = [Q_0]$ for every admissible triple. Equivalently, one fixed linear map transforms the entire family into ordinary chordal SLE_6. No domain Markov hypothesis is needed for this conditional conclusion.

Proof. First, remove dependence on marks inside one domain. Fix D and a regular start p. Local target consistency and local identifiability make $[Q_{D,p,q}]$ independent of q. Ordinary SLE_6 is geometrically reversible by [8]. Therefore reversal of (AA1) retains its same tensor, and the assumed reversal of P implies

$$[Q_{D,p,q}] = [Q_{D,q,p}].$$

Given regular p, p', choose a third regular point q. Changing the target with fixed start q and reversing at the two ends shows

$$[Q_{D,p,q}] = [Q_{D,q,p}] = [Q_{D,q,p'}] = [Q_{D,p',q}].$$

Thus there is a tensor $[Q(D)]$ independent of both marks.

Second, compare nested domains with retained regular boundary arcs. If $D' \subset D$ agrees with D near two distinct regular boundary points, choose them as p, q. Locality makes the initial laws agree in some ball about p, and local identifiability gives $[Q(D')] = [Q(D)]$. The removed set may meet other parts of the boundary; it stays a positive distance from the selected marks.

Third, create a horizontal boundary segment without changing the metric. For any $D \in \mathcal{D}$, choose a horizontal line $\{y = c\}$ with $\min_{\partial D} \operatorname{Im} z < c < \max_{\partial D} \operatorname{Im} z$, at a regular value of the boundary height and not through a corner, and intersect D with $\{y > c\}$. Such a regular value with nonempty intersection exists: on each nonhorizontal analytic boundary arc the critical points are isolated, and horizontal arcs contribute only finitely many excluded height values. Choose the level also away from all corner heights. The analytic arcs are understood to extend regularly to neighborhoods of their closed parameter intervals, so there are only finitely many critical points unless an arc is horizontal. There are finitely many transverse crossings, and a component U of the intersection is again a piecewise real-analytic Jordan domain. It contains an original analytic boundary arc above the cutting line and also an open horizontal segment of the cutting line on its boundary. Choose two regular marks on the retained original arc. The second step gives $[Q(U)] = [Q(D)]$.

Choose a sufficiently short subsegment of the horizontal boundary segment. An upper half-disk V on that diameter lies in U, agrees with U near two distinct interior diameter points, and belongs to $\mathcal{D}$. The second step again gives $[Q(V)] = [Q(U)]$.

Finally, compare all horizontal upper half-disks. Translation covariance identifies their tensors for equal radii and arbitrary centers. Two upper half-disks with the same center and radii $r < R$ are nested and agree near any two distinct points on the smaller open diameter. The second step makes their tensors equal. Thus all such half-disks have one tensor $[Q_0]$, and the third step shows $[Q(D)] = [Q_0]$ for every D.

Choose a fixed A_0 with $A_0^{\mathsf{T}} A_0 = Q_0$. Equality of projective tensors gives $A_{D,p,q} = c_{D,p,q} R_{D,p,q} A_0$ with $c_{D,p,q} > 0$ and $R_{D,p,q}$ orthogonal. Similarity covariance of SLE_6 in (AA1) removes those extra factors, leaving exactly $(A_0^{-1})_* \, \mathrm{SLE}_6(A_0 D, A_0 p, A_0 q)$. □

Boundary regularity in the cutting step. A transverse horizontal cut of a piecewise real-analytic Jordan domain has finitely many intersections with its boundary. Each component is simply connected: a Jordan loop in the intersection has its interior in both the original Jordan domain and the upper half-plane. Its boundary consists of finitely many original arcs and cutting-line segments meeting at transverse corners, so the component is Jordan with piecewise real-analytic boundary. At least one retained original arc and one cutting segment occur in every nontrivial component chosen in the proof. This also makes all marked-point comparisons legitimate within the stated class.

Use for the percolation program. Local target consistency can be transferred from common-color discrete couplings to simultaneous marginal curve limits without random-domain kernel

convergence, as established separately. Reversibility is a further assumption. For a general lattice model with critical density different from 1/2, reversing the color-oriented exploration is not automatically the same as swapping its start and target in a fixed convention: it also changes which boundary arc has each color. No exact discrete color symmetry, nor limiting reversal theorem, is supplied by the present argument. The main unsolved analytic step remains establishing (AA1), or an alternative local structure theorem, from the original axioms plus appropriate coherence.

14 Continuous affine labels: deterministic support and source independence

Status. The result here is conditional on the deterministic affine-SLE ansatz and an explicit continuity condition on its metric labels. It does not apply to a general exploration ensemble. It gives a further restriction inside that ansatz: a continuous conserved label cannot encode information in the source prime end. Source independence alone gives no target or domain independence; the final corollary adds explicit target-consistency and locality hypotheses to obtain a global metric.

14.1 The precise continuity premise

Let $T(D, p, q)$ be the determinant-one conformal metric tensor of the affine label, as in (RA6), on all bounded simply connected domains with two distinct prime ends. We use the following sufficient premise, stated in a form that is stable under the fixed affine change of coordinates needed for each starting law.

Affine-stable marked-chart continuity. For every fixed invertible real-linear map L, suppose $f_n : \mathbb{H} \to E_n$ are conformal maps onto bounded simply connected domains, with a nondegenerate conformal limit $f_n \to f : \mathbb{H} \to E$ locally uniformly in $\mathbb{H}$. The marked prime ends are $f_n(0), f_n(\infty)$ and $f(0), f(\infty)$, respectively; these are prime-end extensions, whether or not they have Euclidean point limits. Require

$$T(LE_n, Lf_n(0), Lf_n(\infty)) \longrightarrow T(LE, Lf(0), Lf(\infty)). \tag{SS1}$$

Thus the topology is locally uniform convergence of inverse conformal charts with the two designated boundary preimages $0, \infty$, and the premise also applies after each fixed affine map. This is an explicit additional hypothesis; it is not inferred merely from DMP, locality, or physical translation invariance. An equivalent affine-stable marked kernel notion may be used instead, provided it supplies (SS1).

Why fixed affine maps preserve this marked topology (proved using standard Beltrami stability). In fact the displayed premise for all L follows from its case $L = I$. Here is the needed topological fact. Write an orientation-preserving real-linear map as $L(z) = \alpha z + \beta \bar{z}$, with $|\beta/\alpha| < 1$. The Beltrami coefficients of $L \circ f_n$ are

$$\mu_n(z) = \frac{\beta}{\alpha} \frac{\overline{f_n'(z)}}{f_n'(z)}.$$

Since the nonvanishing derivatives f_n' converge locally uniformly to f', these coefficients converge pointwise in $\mathbb{H}$ to the analogous μ, all with the same bound $k < 1$. Extend them to the lower half-plane by reflection, $\mu_n(\bar{z}) = \overline{\mu_n(z)}$. Let h_n be the normalized quasiconformal sphere solutions

fixing $0, 1, \infty$. The standard stability theorem for the measurable Riemann mapping theorem (uniform bound $k < 1$ and almost-everywhere convergence of coefficients) gives locally uniform convergence of h_n and their inverses to h and h^{-1}. Reflection symmetry and normalized uniqueness imply that every h_n maps $\mathbb{H}$ onto itself and fixes the two marked boundary points $0, \infty$.

The maps

$$\chi_n = L \circ f_n \circ h_n^{-1} : \mathbb{H} \longrightarrow LE_n$$

are conformal by Beltrami factorization. They converge locally uniformly to the nondegenerate conformal map $\chi = L \circ f \circ h^{-1}$, and their marked endpoints are precisely $Lf_n(0), Lf_n(\infty)$. This proves continuity of a fixed orientation- preserving affine change of the physical domains in the marked conformal- chart topology. A reflection handles the other orientation. Consequently ordinary marked-chart continuity of T implies (SS1). Keeping (SS1) explicit in the proof below makes clear that its charts are holomorphic in the affine SLE coordinates, even when their physical pullbacks are only quasiconformal. The additional analytical input in this paragraph is the standard normalized Beltrami stability theorem; no claim about Euclidean uniform boundary convergence is needed.

14.2 Full support transfers conservation to deterministic Loewner controls

Fix a starting label $A = A(D, p, q)$ and a conformal map $f : \mathbb{H} \to AD$ with $f(0) = Ap$, $f(\infty) = Aq$. For any continuous real function w on $[0, t]$ with $w(0) = 0$, let g_t^w denote the associated hydrodynamic Loewner map on its surviving half-plane domain. Define the marked target state by the inverse conformal chart

$$f_t^w(z) = f\big((g_t^w)^{-1}(z + w(t))\big), \qquad z \in \mathbb{H}. \tag{SS2}$$

This state is a simply connected domain with two designated prime ends for *every* continuous w. The assertion does not say that every such w generates a trace. No trace is needed to define the domain state in (SS2). For the Brownian driving paths of SLE the state is its actual remaining target domain.

The map $w \mapsto f_t^w$ is continuous from the uniform norm on $C[0, t]$ to locally uniform convergence of conformal maps on $\mathbb{H}$. For example, the inverse map can be represented by the reverse ODE

$$\dot{v}_s = -\frac{2}{v_s - w(t - s)}, \quad v_0 = z, \quad (g_t^w)^{-1}(z) = v_t. \tag{SS3}$$

Its imaginary part is nondecreasing. On a compact set with $\operatorname{Im} z \geq \delta > 0$, denominators stay away from zero and the usual ODE stability estimate gives locally uniform continuity in w. Translations by $w(t)$ and composition with f preserve this continuity. The corresponding metric functional

$$\mathcal{T}_t(w) = T\big(A^{-1} f_t^w(\mathbb{H}), A^{-1} f_t^w(0), A^{-1} f_t^w(\infty)\big) \tag{SS4}$$

is therefore continuous by (SS1), with $L = A^{-1}$.

Deterministic-support lemma (proved). If the affine-label ensemble has strong DMP and satisfies (SS1), then for every $t > 0$ and every continuous control $w \in C[0, t]$ with $w(0) = 0$,

$$\mathcal{T}_t(w) = T(D, p, q). \tag{SS5}$$

For the source-independence conclusion below it suffices to apply this to smooth linear ramps.

Proof. Under the original affine SLE law the driving process in AD is $\sqrt{6}B$. The DMP conservation criterion (RA7) gives $\mathcal{T}_t(\sqrt{6}B) = T(D, p, q)$ almost surely. Brownian measure has full support on

the space of continuous functions starting at zero in the uniform topology. If (SS5) failed at one control w, continuity of $\mathcal{T}_t$ would give an open uniform neighborhood on which it differs from the fixed tensor. Brownian full support gives that neighborhood positive probability, contradicting almost-sure conservation. □

One could avoid defining the functional at non-trace-generating controls by restricting the statement to controls at which the state functional has a continuous extension, and applying full support there. In particular smooth ramps suffice. The all-prime-end ansatz already makes (SS4) defined on every state, so the formulation above is simpler.

14.3 Rapid driving ramps move the source at vanishing capacity

Theorem (proved, within the stated ansatz). Under strong DMP and (SS1), the tensor $T(D, p, q)$ is independent of the source prime end: for every bounded simply connected D and every three prime ends with $p, p' \neq q$,

$$T(D, p, q) = T(D, p', q). \tag{SS6}$$

It may still depend on the target or on the domain; these are unresolved by this argument.

Proof. Use the notation above and write $Ap' = f(a)$ for the unique finite $a \in \mathbb{R}$ corresponding to the new source. Choose $t_n \downarrow 0$ and use the smooth control

$$w_n(s) = \frac{as}{t_n}, \qquad 0 \leq s \leq t_n.$$

The deterministic-support lemma says that every resulting metric equals $T(D, p, q)$. From the reverse equation (SS3),

$$|(g_{t_n}^{w_n})^{-1}(z) - z| \leq \frac{2t_n}{\operatorname{Im} z}. \tag{SS7}$$

Consequently, locally uniformly in $\mathbb{H}$,

$$f_{t_n}^{w_n}(z) = f\big((g_{t_n}^{w_n})^{-1}(z + a)\big) \longrightarrow f(z + a).$$

The limiting chart has image AD, source $f(a) = Ap'$, and the same target $f(\infty) = Aq$. Applying (SS1) with $L = A^{-1}$ gives (SS6). □

The controls may have very large slopes as $t_n \to 0$; this is harmless. Each fixed ramp lies in the support of Brownian driving on its own finite interval. The argument uses the possibility of rare events, not their uniform probability as $n \to \infty$. The limiting source change is a statement about marked prime-end charts, not Euclidean convergence of a curve of vanishing diameter: the capacity vanishes while the tip may travel far along a thin boundary region.

14.4 A fixed metric in the affine ansatz without a reversal hypothesis

Corollary (proved, conditional on the global affine ansatz). Suppose the ensemble has a deterministic affine-SLE label on *all* bounded simply connected domains and distinct prime ends, has strong DMP, and its normalized tensor T is continuous for marked inverse-conformal- chart convergence. Suppose in addition that its piecewise real-analytic Jordan subfamily has translation covariance, locality for nested domains retaining neighborhoods of regular marks, and local target consistency in the sense used by the conditional global metric-rigidity theorem. Then there is one fixed determinant-one tensor Q_0 such that

$$T(D, p, q) = Q_0$$

for every bounded simply connected marked domain. No geometric reversal hypothesis is needed for this conditional conclusion.

Proof. The source-independence theorem removes dependence on p. On a regular piecewise real-analytic Jordan domain, local target consistency together with the local half-plane identifiability argument removes dependence on q as well. Thus these domains have a tensor $Q(D)$ independent of both marks. The nested-cut argument in the preceding conditional global metric-rigidity theorem now applies verbatim, without its reversal step: nested locality identifies tensors of domains agreeing near two retained regular marks; a transverse horizontal cut creates a horizontal segment; a small upper half-disk on that segment retains the metric; translation and nesting identify all such half-disks. Hence all piecewise real-analytic Jordan domains have the same tensor Q_0.

To extend to an arbitrary bounded simply connected D, choose a conformal $f : \mathbb{D} \to D$ and distinct circle points ζ_p, ζ_q representing p, q. For $r_n \uparrow 1$, set

$$D_n = f(r_n\mathbb{D}), \qquad p_n = f(r_n\zeta_p), \quad q_n = f(r_n\zeta_q).$$

Each D_n is an analytic Jordan domain: f extends holomorphically and univalently across the compact circle $|z| = r_n$, with nonzero derivative. Choose a conformal $m : \mathbb{H} \to \mathbb{D}$ with $m(0) = \zeta_p$, $m(\infty) = \zeta_q$. The marked inverse charts

$$f_n(z) = f(r_n m(z))$$

converge locally uniformly to $f(m(z))$, giving exactly the desired marked prime-end limit, even if f has no Euclidean limit at one of the selected circle points. Tensor continuity and $T(D_n, p_n, q_n) = Q_0$ give $T(D, p, q) = Q_0$. The fixed-affine normalizer follows from SLE similarity covariance as in the preceding theorem. □

This corollary removes a reversal assumption *inside the global affine ansatz*, at the price of strong DMP and a precise continuity premise. It still supplies no argument that the original three axioms, RSW, or local target consistency force a marginal exploration law to belong to that ansatz.

15 Continuation under the corrected source-neighborhood locality

The specification at commit `0e47018` requires equality of initial curve laws whenever the physical domains and their boundary colors agree near the source; the remote target and remote domain may differ. We call this strong source locality below. We use one fixed boundary-color orientation, so two distinct color-orientation sectors are not an additional hidden label. No counterexample to the corrected three axioms in that convention is established in this continuation. Neither is the full classification proved.

There are three main positive advances: a quantitative driving-drift rigidity theorem needing no assumed regularity of quadratic variation; a common-metric theorem inside the affine-SLE ansatz requiring stability only at one straight slit-tip germ; and target-patch stability from a precise conditional unforced-crossing estimate. A coarse nonconcentration estimate also makes affine anisotropies compact. A common-color RSW shield additionally proves quantitative source-patch stability in flat and straight-slit geometries; it is not asserted for arbitrary curve laws or arbitrary domain approximations. Every analytical or coupling premise added to the project axioms is identified explicitly. The stronger wrong-coordinate obstruction concerns the known affine-SLE family, not an unknown general scaling limit.

Independent audits checked the signed-measure characteristic identities, the mirrored-slit coefficients, analytic-germ covariance, the narrow straight-tip support argument, the rough domain-control extension, and the target-patch and rank-collapse estimates. In particular, Brownian full support was not used to extend almost-sure identities to null deterministic histories without a stated stability premise.

16 A near-source drift-boundedness criterion

Status. This is a new conditional rigidity result. It does not assume conformal invariance, Brownian driving, or local-martingale driving. Instead it shows that a locally bounded *initial driving drift*, together with an absolutely continuous drift part and locality, already forces zero drift and variance six. No regularity of quadratic variation is assumed. Strong DMP then identifies the whole driving law. The boundedness and characteristic hypotheses are not derived from RSW or from the three project axioms. A finite-dimensional bulk-jet closure is given as one explicit sufficient hypothesis.

Only a particular deletion consequence of the corrected source-neighborhood locality is needed. Arbitrary target changes and the full force of corrected locality are not used in this theorem. Thus the result is compatible with, but does not exhaust, the corrected research objective.

16.1 Chart-dependent initial characteristics and their covariance

Let $f : \mathbb{H} \to D$ be an inverse chordal conformal chart, with source $f(0) = p$ and target $f(\infty) = q$, both interpreted as prime ends. For the physical ensemble in D, let W^f be its driving process in this chart, with half-plane capacity normalization

$$\partial_t g_t(z) = \frac{2}{g_t(z) - W_t^f}, \qquad W_0^f = 0.$$

Assume that it is a continuous semimartingale in its natural trace filtration, with decomposition

$$W_t^f = M_t^f + \int_0^t b_s^f \, ds, \qquad C_t^f = \langle M^f \rangle_t. \tag{DR1}$$

The drift-density paths b^f are assumed finite and right-continuous, after local stopping if necessary. No absolute continuity or regularity of C^f is assumed. At the initial time suppose

$$b_0^f = B(f) \in \mathbb{R} \tag{DR2}$$

is deterministic. It may depend on the entire physical marked domain and its conformal chart. In particular no Markov property of the single real driver is assumed. A finite deterministic initial variance rate $V(f)$ will be *derived* below from locality and (DR1)–(DR2).

Let $A \subset \overline{\mathbb{H}}$ be a half-plane hull disjoint from 0, let $h : \mathbb{H}\setminus A \to \mathbb{H}$ be hydrodynamically normalized, and set

$$\bar{h}(z) = h(z) - h(0), \qquad f_A = f \circ \bar{h}^{-1}. \tag{DR3}$$

The source is still the same physical prime end, since $\bar{h}(0) = 0$; the target is also unchanged. Put $d = h'(0) > 0$, $e = h''(0)$. Source locality, together with the usual deletion-chart calculation, will give a finite deterministic initial variance rate $V(f)$ and the covariance

$$\boxed{V(f_A) = V(f), \qquad B(f_A) = \frac{B(f)}{d} + \left(\frac{V(f)}{2} - 3\right)\frac{e}{d^2}.} \tag{DR4}$$

These are identities for *initial* characteristics and do not identify the two complete driving laws.

Here are the details without assuming a quadratic-variation density. If h_t is the evolving deletion map, then $du = h_t'(W_t)^2 dt$, and Itô's formula gives

$$d\widehat{W}_{u(t)} = h_t'(W_t)\, dM_t + h_t'(W_t) b_t\, dt + \frac{1}{2} h_t''(W_t)\, dC_t - 3h_t''(W_t)\, dt. \tag{DR5}$$

The -3 comes from

$$\partial_t h_t(z) = \frac{2h_t'(W_t)^2}{h_t(z) - h_t(W_t)} - \frac{2h_t'(z)}{z - W_t}, \qquad (\partial_t h_t)(W_t) = -3h_t''(W_t).$$

The comparison of filtrations is justified by geometric invertibility: from a deleted-domain initial trace one reconstructs its original trace and capacity, so the inverse capacity clock is adapted and its inverse is a stopping-time clock. Locality therefore compares the same trace filtration, not an unrelated enlargement. Adaptedness of an arbitrary increasing clock would not suffice for this step.

Local bracket-regularization lemma (proved). Suppose $h''(0) \neq 0$. Then on a positive initial interval, C_t is absolutely continuous and has a finite right-continuous density. In particular its initial right derivative $V(f)$ exists and is deterministic.

Proof. The deleted-domain driver also has an absolutely continuous drift, $\int \widehat{b}_u\, du$, by (DR1). Uniqueness of the continuous semimartingale decomposition applied to (DR5), after the justified capacity time change, gives the signed-measure identity

$$h_t''(W_t)\big(dC_t - 6\, dt\big) = 2\big[h_t'(W_t)^2 \widehat{b}_{u(t)} - h_t'(W_t) b_t\big] dt. \tag{DR5a}$$

The second derivative starts nonzero and remains so on a positive initial interval. On that interval the bracket is absolutely continuous, with

$$v_t = 6 + 2\frac{h_t'(W_t)^2 \widehat{b}_{u(t)} - h_t'(W_t) b_t}{h_t''(W_t)}. \tag{DR5b}$$

The derivative paths of h_t along the continuous driving point are continuous, while the two drift-density paths are right-continuous and finite. Thus v is right-continuous and finite there. In particular

$$V(f) = v_0 = 6 + 2\frac{d^2B(f_A) - dB(f)}{e} \tag{DR5c}$$

is deterministic. This is an intrinsic initial rate of C, so it does not depend on which nonzero-e deletion was used to derive it. Applying the same time change to bracket gives $V(f_A) = V(f)$. Rearranging (DR5c) gives the drift identity (DR4). □

There is always an available nonzero-e test deletion, for example one vertical slit at any fixed real point different from 0. Right-continuity makes the initial drift and variance rates intrinsic, rather than arbitrary versions of densities at a single time. The bracket density has only been proved locally at the start; its global regularity is not being presumed.

Source-neighborhood convention. In half-plane coordinates these slit deletions preserve a full neighborhood of the source end. Under f this is a prime-end neighborhood. Thus (DR4) is justified on arbitrary rough domains if the corrected locality assumption is formulated for such neighborhoods, or if these particular deletions are explicitly admissible. For regular analytic source boundaries it is ordinary Euclidean neighborhood locality. If locality is instead stated only using Euclidean balls at non-singleton source impressions, the all-rough-domain extension must remain an explicit additional hypothesis.

16.2 Two vanishing slits force both characteristics

For $\epsilon > 0$ take the reflected vertical slits

$$A_\epsilon^\pm = \{\, \pm\epsilon^2 + iy : 0 \le y \le 2\epsilon \,\}. \tag{DR6}$$

Their hydrodynamic maps are

$$h_\epsilon^\pm(z) = \pm\epsilon^2 + \sqrt{(z \mp \epsilon^2)^2 + 4\epsilon^2}, \tag{DR7}$$

with the square-root branch asymptotic to $z \mp \epsilon^2$ at infinity. Let $f_\epsilon^\pm = f \circ (h_\epsilon^\pm - h_\epsilon^\pm(0))^{-1}$. Both sequences are conformal charts of bounded simply connected domains and

$$f_\epsilon^\pm \longrightarrow f \quad \text{locally uniformly in } \mathbb{H}. \tag{DR8}$$

Indeed the slits have diameter 2ϵ and approach 0; directly from (DR7), their maps and inverse maps converge to the identity on compact subsets of $\mathbb{H}$, while $h_\epsilon^\pm(0) \to 0$. Crucially their boundary derivatives do *not* converge to the identity jet:

$$d_\epsilon = (h_\epsilon^\pm)'(0) = \frac{\epsilon}{\sqrt{\epsilon^2 + 4}} \longrightarrow 0, \qquad e_\epsilon^\pm = (h_\epsilon^\pm)''(0) = \mp\frac{4}{\epsilon(\epsilon^2 + 4)^{3/2}}. \tag{DR9}$$

In particular

$$\frac{e_\epsilon^\pm}{d_\epsilon^2} = \mp\frac{4}{\epsilon^3\sqrt{\epsilon^2 + 4}}. \tag{DR10}$$

Thus compact convergence of inverse charts permits very large second-jet changes precisely at the source boundary.

Initial rigidity theorem (proved). Fix f and assume (DR1)–(DR4) for f and the charts $f_\epsilon^\pm$. If

$$\sup_{0<\epsilon<\epsilon_0} \max\{|B(f_\epsilon^+)|, |B(f_\epsilon^-)|\} < \infty \tag{DR11}$$

for some $\epsilon_0 > 0$, then

$$B(f) = 0, \qquad V(f) = 6. \tag{DR12}$$

No continuity of B is required.

Proof. Write $b = B(f)$, $v = V(f)$ and $b_\epsilon^\pm = B(f_\epsilon^\pm)$. Equation (DR4) gives

$$b_\epsilon^\pm = \frac{b}{d_\epsilon} + \left(\frac{v}{2} - 3\right)\frac{e_\epsilon^\pm}{d_\epsilon^2}.$$

Since $e_\epsilon^- = -e_\epsilon^+$,

$$b_\epsilon^+ + b_\epsilon^- = \frac{2b}{d_\epsilon}, \qquad b_\epsilon^+ - b_\epsilon^- = (v-6)\frac{e_\epsilon^+}{d_\epsilon^2}. \tag{DR13}$$

The first right side is bounded only if $b = 0$, since $d_\epsilon \to 0$. The second is bounded only if $v = 6$, since its multiplier has magnitude asymptotic to $2\epsilon^{-3}$. This proves (DR12). □

Exact instability and weaker rate conditions. For real numbers U, C one has $\max\{|U + C|, |U - C|\} = |U| + |C|$. Hence the two drift covariances give the exact identity

$$\boxed{\max\{|b_\epsilon^+|, |b_\epsilon^-|\} = \frac{|b|\sqrt{\epsilon^2+4}}{\epsilon} + \frac{4|v/2-3|}{\epsilon^3\sqrt{\epsilon^2+4}}.} \tag{DR14}$$

There is no possible cancellation between the two reflected tests. In particular, if the displayed maximum is $o(\epsilon^{-3})$, then $v = 6$. If it is $o(\epsilon^{-1})$, then $b = 0$ and $v = 6$. Boundedness is more than sufficient. Conversely, if $v \neq 6$, its leading blow-up is $|v-6|\epsilon^{-3}$; if $v = 6$ and $b \neq 0$, its leading blow-up is $2|b|\epsilon^{-1}$. This is a concrete, quantitative obstruction to any alternative semimartingale driving law with tame initial drift under source perturbations.

16.3 Local boundedness in marked-chart topology suffices

Equip inverse marked charts with locally uniform convergence on $\mathbb{H}$, retaining the fixed preimages $0, \infty$ of source and target. If B is locally bounded at every f in this topology, then (DR8) implies (DR11). In particular continuity of B is sufficient, but unnecessarily strong. This is a hypothesis about the driving characteristic itself; weak convergence of curve laws alone does not imply it.

Conditional classification theorem (proved). Suppose a Borel family of continuous Loewner-generating exploration laws on all bounded simply connected marked prime-end domains has strong DMP and source-neighborhood locality (including the test deletions (DR6) in each chart). Suppose its drivers satisfy (DR1)–(DR2), with finite right-continuous drift-density paths and the usual restart compatibility of characteristic kernels, with no initial assumption on quadratic variation. If the initial drift map B is locally bounded in marked inverse- chart topology, then every driving process is

$$W_t^f = \sqrt{6}\, B_t,$$

and the family is chordal SLE_6 in these conformal coordinates.

Proof. The initial rigidity theorem gives $B(f) = 0$ and the derived $V(f) = 6$ for every marked inverse chart. At any deterministic rational time r, condition on the initial exploration and use strong DMP in its remaining marked domain and current conformal chart. The centered future driver is exactly the restarted driver; its initial drift is b^f_r, by right-continuity. The restarted law has initial drift zero, so $b^f_r = 0$ almost surely. Countably many rational times and right-continuity give $b^f_t = 0$ throughout each localized lifetime. Therefore all domain chart drivers are continuous local martingales.

For completeness the no-density local-martingale locality argument then identifies their brackets. With both drifts zero, (DR5a) becomes $h''_t(W_t)(dC_t - 6\,dt) = 0$. A nonzero-$e$ vertical deletion yields $C_t = 6t$ on a positive initial interval. Let

$$T = \inf\{t \geq 0 : C_t - 6t \neq 0\}.$$

This is an entrance stopping time of a continuous adapted process into an open set, in the usual right-continuous filtration. The initial zero interval gives $T > 0$. If $T < \infty$, continuity gives $C_T = 6T$ and the first-deviation definition gives nonzero values arbitrarily soon after T. Strong DMP restarted at T gives another positive zero interval, a contradiction. Thus $C_t = 6t$ throughout. Lévy's characterization gives Brownian driving, and Loewner generation identifies SLE_6. □

The Borel-kernel condition means that the family admits the standard strong conditional-kernel interpretation on random slit domains, and that the characteristics can be chosen consistently on that family. It is not a claim that deterministic-domain convergence of discrete explorations proves this property. Translation invariance is not needed in this conditional theorem. If its hypotheses hold only after one fixed physical affine normalization, the conclusion is the corresponding fixed-affine SLE_6 family; an affine change of physical coordinates is not presumed to preserve semimartingale driving in an arbitrary preselected complex structure.

16.4 A finite-dimensional closure that is completely classified

For example, fix finitely many interior points $z_j \in \mathbb{H}$ and finitely many nonnegative derivative orders m_j. Let

$$J(f) = \big(f^{(k)}(z_j) : 1 \leq j \leq N, 0 \leq k \leq m_j\big).$$

Suppose the initial drift has a representation

$$B(f) = \mathcal{B}(J(f)), \tag{DR15}$$

where $\mathcal{B}$ is locally bounded at every realized jet vector. The bracket has no assumed density or finite-dimensional representation; its initial rate is supplied by the local regularization lemma. Locally uniform convergence of conformal maps implies convergence of all these interior jets by Cauchy's formula, so (DR15) implies local boundedness of B in marked-chart topology. The conditional classification theorem therefore applies: within (DR1)–(DR2), every source-local strong-DMP family with such a locally bounded finite-dimensional *bulk-jet drift closure* is SLE_6.

This statement does not cover a closure depending on unbounded source boundary derivatives such as $f'(0)$ or a boundary pre-Schwarzian. Those quantities can change without control under (DR8), and can carry exactly the singular terms in (DR13). Nor does it show that true percolation limits have semimartingale driving in a fixed physical complex structure. The new rigorous reduction is that any non-SLE candidate with regular characteristic rates must violate the displayed near-source drift bound, in the explicit ϵ^{-1} or ϵ^{-3} manner above.

16.5 The criterion also detects a wrong affine coordinate system

Corollary (proved). Let A be an orientation-preserving real-linear map which is not a Euclidean similarity, and let the physical ensemble be the fixed-affine pullback

$$P_{D,p,q} = (A^{-1})_* \operatorname{Law}(\mathrm{SLE}_6(AD; Ap, Aq)).$$

Although this family satisfies corrected source-neighborhood locality and strong DMP, in the original Euclidean complex structure it cannot satisfy both the regular semimartingale characteristic assumptions (DR1)–(DR2) in all marked domains and the locally bounded initial-drift condition above.

Proof. If both properties held, the conditional classification theorem would make this ensemble ordinary SLE_6 in the original complex structure. The single-domain affine identifiability theorem would then make A a similarity, a contradiction. □

Thus the drift criterion selects a conformal structure. It is not a coordinate-free consequence of coarse crossing bounds: bounded affine changes preserve such bounds, whereas a wrong affine structure necessarily violates at least one of the criterion's additional analytic hypotheses. The corollary does not decide whether the failure is semimartingality, right-continuity of the drift density, deterministic initial drift, or boundedness of the initial drift; distinguishing these failures is a separate question.

16.6 What corrected locality says at a regular analytic source germ

Here is a separate necessary local structure statement that uses the full corrected locality, including changes of the distant target. It concerns initial characteristics only, not existence of a process from arbitrary coefficient choices.

Suppose the physical boundary is analytic near p and the source colors are fixed there. Let G denote this embedded colored source germ. For a chart f extending conformally across 0, define

$$a(f) = |f'(0)| > 0, \qquad P(f) = \operatorname{Re} \frac{f''(0)}{f'(0)}. \tag{DR16}$$

If another physical marked domain has the same source germ but any distant target, let k be its inverse chordal chart with $k(0) = p$. The local map $h = k^{-1} \circ f$ is analytic across 0, takes the real boundary interval to a real interval, and has $h(0) = 0$, $d = h'(0) > 0$, $e = h''(0) \in \mathbb{R}$. The two initial physical traces agree by corrected locality. The local Loewner coordinate comparison is exactly (DR4), with the same derivation from pole cancellation and Itô's formula; a globally defined deletion map is unnecessary. Consequently $V(f) = V(k)$ depends only on G.

Proposition (proved). Under the semimartingale assumptions above, corrected locality at analytic source germs is equivalent at the level of initial characteristic covariance to the existence of two source-germ quantities $V(G)$ and $\beta(G)$ such that

$$\boxed{V(f) = V(G), \qquad B(f) = a(f)\,\beta(G) + \left(3 - \frac{V(G)}{2}\right) P(f).} \tag{DR17}$$

Here "equivalent at the level of covariance" means precisely that every covariant initial-characteristic assignment has this form, and every assignment of this form obeys the local chart covariance identity. It does not mean that arbitrary V, β yield a domain Markov exploration family.

Proof. Since $k = f \circ h^{-1}$, the chain rule gives

$$a(k) = \frac{a(f)}{d}, \qquad P(k) = \frac{P(f)}{d} - \frac{e}{d^2}.$$

Write $c = V(G)/2 - 3$. Combining these transformations with $B(k) = B(f)/d + ce/d^2$ yields

$$\frac{B(k) + cP(k)}{a(k)} = \frac{B(f) + cP(f)}{a(f)}.$$

This common value is the source-germ invariant $\beta(G)$, proving (DR17). Conversely direct substitution gives the required covariance. □

Physical translation invariance makes $V(G)$ and $\beta(G)$ invariant under translations of the colored germ. If dilation covariance is separately assumed, then $V(\lambda G) = V(G)$ and $\beta(\lambda G) = \lambda^{-1}\beta(G)$. For a straight source-boundary germ, which is unchanged up to translation by dilation, this would force $\beta(G) = 0$; the dimensionless variance can still depend on the boundary orientation before an affine metric has been identified.

The boundary pre-Schwarzian term in (DR17) explains the quantitative anomaly in the vanishing-slit test. Even when the physical source germ is unchanged, $P(f_\epsilon)$ may grow as ϵ^{-3} while $f_\epsilon \to f$ on every interior compact set. Thus source locality by itself gives a connection-type transformation law, rather than vanishing drift. The locally bounded bulk closure theorem removes exactly this uncontrolled boundary dependence. The missing global step is to understand how DMP constrains these local assignments at rough, dynamically created source germs, where the analytic boundary jet in (DR16) no longer exists.

17 Rigidity from an averaged initial finite-variation characteristic

Scope. This conditional theorem weakens the preceding driving-regularity premise. The canonical finite-variation characteristic need not be absolutely continuous and need not possess a right-continuous density. We assume only a deterministic first-order tangent in probability, uniformly on shrinking initial intervals. No such property is derived from RSW or the three axioms.

Let the driver in inverse chart f have canonical decomposition

$$W^f = M^f + A^f, \qquad W_0^f = M_0^f = A_0^f = 0, \tag{FV1}$$

where M^f is a continuous local martingale and A^f is continuous, adapted, and locally of finite variation. Write $C^f = \langle M^f \rangle$. Assume for every deterministic state and inherited chordal chart that a finite deterministic number $B(f)$ satisfies

$$\frac{1}{t} \sup_{0 \le s \le t} |A_s^f - B(f)s| \xrightarrow[t \downarrow 0]{\mathbb{P}} 0. \tag{FV2}$$

An almost-sure derivative $A_t^f/t \to B(f)$ implies (FV2), but (FV2) is weaker. No bound on $\mathrm{TV}(A^f; [0,t])/t$ is assumed.

Lemma 17.1 (A small finite-variation displacement in an analytic field)**.** *Let $W = M + A$ as in (FV1), let $C = \langle M \rangle$, and suppose $\sup_{s \le t} |A_s| = O_{\mathbb{P}}(t)$ and $A_t/t \to B$ in probability. Let $H_t(x)$ be a random field analytic in x on a fixed neighborhood of zero for a positive initial time interval. Suppose it has absolutely continuous finite variation in t, and, after localization, its first three*

spatial derivatives and the first spatial derivative of $\partial_t H_t$ are bounded there. Assume $H_t'(x) \to d$ as $(t,x) \to (0,0)$. Then

$$\frac{1}{t}\int_0^t H_s'(W_s)\,dA_s \longrightarrow dB \quad \textit{in probability.} \tag{FV3}$$

Proof. Localize so M, A, W stay in the spatial neighborhood. Apply the finite-variation Itô–Wentzell formula to $F_t(x) = H_t(x + A_t) - H_t(x)$ at $x = M_t$. It gives

$$\begin{aligned}\int_0^t H_s'(W_s)\,dA_s = {} & H_t(M_t + A_t) - H_t(M_t) \\ & - \int_0^t [H_s'(M_s + A_s) - H_s'(M_s)]\,dM_s \\ & - \frac{1}{2}\int_0^t [H_s''(M_s + A_s) - H_s''(M_s)]\,dC_s \\ & - \int_0^t [(\partial_s H_s)(M_s + A_s) - (\partial_s H_s)(M_s)]\,ds.\end{aligned}$$

The first term divided by t converges to dB by the mean-value formula. The second term is a continuous local martingale N_t with bracket at most $K^2 \sup_{s\le t} |A_s|^2 C_t$. Thus $N_t/t \to 0$ in probability: stop when $|A_s| > Lt$ or $C_s > \eta$, apply the martingale maximal inequality, then let $t \downarrow 0$, $\eta \downarrow 0$, and $L \uparrow \infty$. Here $C_t \to 0$ almost surely and $\sup |A_s|/t$ is tight. The third term divided by t is bounded by $K \sup |A_s| C_t/t = o_{\mathbb{P}}(1)$; the fourth is bounded by $K \sup |A_s| = o_{\mathbb{P}}(1)$. No total-variation estimate was used. □

Proposition 17.2 (Initial covariance without a drift density)**.** *Suppose source locality allows one test deletion with hydrodynamic map h, $d = h'(0) > 0$, $e = h''(0) \neq 0$, and both original and deleted drivers obey (FV1)–(FV2), with their canonical characteristics in the same trace filtration. Then*

$$C_t^f/t \longrightarrow V(f) \in [0,\infty) \quad \textit{in probability,}$$

where $V(f)$ is deterministic, and

$$V(f_A) = V(f), \qquad B(f_A) = \frac{B(f)}{d} + \left(\frac{V(f)}{2} - 3\right)\frac{e}{d^2}. \tag{FV4}$$

Proof. Use the same evolving deletion maps and stopping-time capacity clocks as in (DR5), now without assuming a drift density. Put $p_t = h_t'(W_t)$, $q_t = h_t''(W_t)$ and $u(t) = \int_0^t p_s^2 ds$. Uniqueness of the canonical semimartingale decomposition gives

$$\widehat{A}_{u(t)} = \int_0^t p_s\,dA_s + \frac{1}{2}\int_0^t q_s\,dC_s - 3\int_0^t q_s\,ds. \tag{FV5}$$

The deletion field satisfies the analytic-field lemma after localization; its Loewner field derivative and spatial derivatives are bounded near the uncollided source. Therefore the first integral divided by t tends to $dB(f)$, without assuming C_t/t is tight. Also $u(t)/t \to d^2$. Condition (FV2) for the deleted state implies $\widehat{A}_{u(t)}/t \to d^2 B(f_A)$: on the event $u(t) \le Kt$, bound the error by $t^{-1}\sup_{s\le Kt} |\widehat{A}_s - B(f_A)s|$, and choose fixed $K > d^2$.

Since $q_t \to e \neq 0$, (FV5) first implies C_t/t is tight: with probability tending to one q_s has fixed sign and magnitude at least $|e|/2$ on $[0,t]$, so $|\int q_s dC_s| \ge |e| C_t/2$. Consequently

$$\frac{1}{t}\int_0^t q_s\,dC_s - e\frac{C_t}{t} \longrightarrow 0 \quad \text{in probability.}$$

Taking (FV5) divided by t now gives

$$V(f) = 6 + \frac{2\{d^2B(f_A) - dB(f)\}}{e}.$$

This is intrinsic, nonnegative, and deterministic. The bracket relation $\widehat{C}_{u(t)} = \int_0^t p_s^2 dC_s$ gives $V(f_A) = V(f)$. For this last step the tangents of $C, \widehat{C}$ may be evaluated at asymptotically deterministic clocks by monotonic squeezing; convergence in probability of an arbitrary signed process would not justify that step. □

Lemma 17.3 (A finite-variation process with zero right tangent at every stopping time)**.** *Let $A_0 = 0$ be a continuous adapted locally finite-variation process. If for every bounded stopping time τ before lifetime*

$$\frac{A_{\tau+h} - A_\tau}{h} \longrightarrow 0 \quad \text{in probability as } h \downarrow 0, \tag{FV6}$$

then A is indistinguishably zero.

Proof. Work on a compact localized interval. Pathwise put $V_t = \mathrm{TV}(A; [0,t])$, and use its generalized inverse $\theta(u) = \inf\{t : V_t \geq u\}$ for $0 < u < V_T$. The usual arclength parametrization of a continuous finite-variation path is $G(u) = A_{\theta(u)}$: it is 1-Lipschitz, satisfies $A_t = G(V_t)$, and has $|G'(u)| = 1$ for Lebesgue almost every u. Indeed the variation of G on every interval is its interval length. The monotone function θ has a finite derivative $\theta'(u) \in [0, \infty)$ for almost every u. Exclude also the countably many levels corresponding to nontrivial flat intervals of V.

At a level where these properties hold, let $s = \theta(u)$ and $v(h) = V_{s+h} - u$. Then $v(h) > 0$ and $v(h) \to 0$ as $h \downarrow 0$. The generalized-inverse bounds $\theta((u + v(h))-) \leq s + h \leq \theta((u + v(h))+)$ and differentiability of θ at u give $h/v(h) \to \theta'(u)$. Combining this with differentiability of G gives

$$\lim_{h \downarrow 0} \frac{A_{s+h} - A_s}{h} = \begin{cases} G'(u)/\theta'(u), & \theta'(u) > 0, \\ \mathrm{sgn}(G'(u))\,\infty, & \theta'(u) = 0. \end{cases} \tag{FV7}$$

Thus the extended right derivative is nonzero at almost every variation level, including levels whose inverse belongs to a singular part of dV.

For deterministic $u > 0$ use the stopping times $\tau_u = \inf\{t : V_t \geq u\}$, truncated at a deterministic endpoint T. Fubini and the preceding pathwise statement show that for Lebesgue almost every deterministic u, on $\{\tau_u < T\}$ the quotient in (FV6) has a nonzero or infinite pathwise limit almost surely. But (FV6) gives convergence in probability to zero at this stopping time, which forbids such an event of positive probability. Thus $\mathbb{P}(\tau_u < T) = 0$ for Lebesgue almost every u, and $\mathbb{E}[V_T] = \int_0^\infty \mathbb{P}(\tau_u < T) du = 0$ after localizing V_T to be integrable. Hence $V_T = 0$ almost surely. Localization and a countable exhaustion of the lifetime prove the claim. □

Theorem 17.4 (Conditional classification from a finite-variation initial tangent)**.** *Suppose a Borel family on all relevant marked slit states has strong DMP and source locality, including the same mirrored test deletions (DR6). Suppose canonical decompositions (FV1), compatible with conditional restart in the inherited chordal charts, obey (FV2). If $B(f)$ is locally bounded in inverse-chart topology, or just satisfies the paired slit bound (DR11) at every state, the family is chordal SLE_6 in these coordinates.*

Proof. Proposition 17.2 gives exactly (DR4), hence the reflected-slit sum and difference argument gives $B(f) = 0$, $V(f) = 6$ at every deterministic state.

At a stopping time τ, the centered future driver is the driver of the remaining physical state in the inherited conformal chart. Its capacity parameter is exactly $s = t - \tau$ in the original half-plane coordinate: recentering by W_τ introduces no additional scale clock. Strong DMP and conditional-characteristic compatibility identify its canonical finite-variation part with $A_{\tau+s} - A_\tau$ under the regular conditional law. Applying (FV2) with $B = 0$, and integrating these conditional probabilities over the random state, gives (FV6). Lemma 17.3 therefore makes $A = 0$.

The same conditional restart argument gives $(C_{\tau+h} - C_\tau)/h \to 6$ in probability at every stopping time. Apply Lemma 17.3 to the continuous finite-variation process $C_t - 6t$: it is identically zero. Thus $W = M$ and $\langle M \rangle_t = 6t$ globally. Lévy's characterization and continuous Loewner generation identify SLE_6. □

Strict weakening and remaining limitation. Even an absolutely continuous drift with density $b_s = B + \sin(1/s)$, $s > 0$, has no right limit at zero, whereas $\int_0^t b_s ds = Bt + O(t^2)$ and therefore satisfies (FV2). The theorem also allows an a priori singular finite-variation drift; its exclusion is a conclusion, not an assumption. A claim that an actual exploration family has these initial characteristic tangents remains unproved. The theorem still depends on choosing one correct physical conformal structure and on tame behavior under the source slit tests.

17.1 The wrong-coordinate obstruction also only needs finite-variation tangents

Scaling lemma with probability tangents. The exact-scaling lemma (SM1)–(SM4) remains true if its drift assumption is replaced by continuous finite variation and (FV2), and its bracket assumption by $C_t/t \to v$ in probability for deterministic v. Indeed (FV2) makes $\sup_{s \le T} |c^{-1/2} A_{cs}| = O_{\mathbb{P}}(\sqrt{c}) \to 0$. Since C is increasing, a finite deterministic mesh and monotonic squeezing show

$$\sup_{s \le T} |C_{cs}/c - vs| \longrightarrow 0 \quad \text{in probability.}$$

The continuous-martingale functional central limit argument in the earlier scaling lemma then applies unchanged. Exact scaling makes the whole law Brownian with variance v.

Local transfer of the tangent. In Proposition 17.2, the analytic-field argument actually gives the uniform version

$$\frac{1}{t} \sup_{s \le t} \left| \int_0^s h_r'(W_r)\, dA_r - dB(f)s \right| \longrightarrow 0 \quad \text{in probability.}$$

Use the martingale maximal inequality in its proof, and the same uniform initial bound for A. Monotonic squeezing similarly makes $C_{ts}/t \to V(f)s$ uniformly on compact s intervals in probability. Equation (FV5) therefore transfers (FV2) through a local analytic chart change, once the initial bracket tangent has been obtained; it also transfers its inverse, since the first chart derivative is positive. No total-variation rate bound is introduced by this transfer.

Corollary 17.5 (Wrong-affine obstruction under finite-variation tangents)**.** *Consider the canonical fixed-affine SLE_6 family with matrix A on all bounded marked domains. If every Euclidean chart driver admits (FV1)–(FV2), then A is a Euclidean similarity. No local boundedness of $B(f)$ is needed.*

Proof. Choose the bounded flat-source domain and one test deletion used in the bounded-domain obstruction theorem. Proposition 17.2 gives its deterministic bracket tangent in probability. Couple

its initial physical trace by known affine-SLE locality to the canonical half-plane affine-SLE trace. The local analytic chart transfer just proved gives a finite-variation decomposition and uniform deterministic drift tangent for the half-plane driver, and the same deterministic bracket tangent. Its exact Brownian scaling and the extended scaling lemma make its law ordinary Brownian driving. The boundary-swallowing-race and affine identification arguments in the half-plane obstruction theorem then give variance six and force A to be a similarity. □

17.2 A tightness-in-law variant without deterministic initial tangents

Proposition (proved). There is a useful alternative formulation of the preceding theorem. Allow (FV2) to have a finite random tangent β^f rather than a deterministic $B(f)$. Assume that for every fixed chart f the family of initial tangents in its two shrinking slit domains is uniformly tight:

$$\lim_{K\to\infty} \sup_{0<\epsilon<\epsilon_0} \max_{\sigma\in\{+,-\}} \mathbb{P}\{|\beta^{f_\epsilon^\sigma}| > K\} = 0. \tag{FV8}$$

Then the same classification follows, with all other family and restart hypotheses unchanged.

The analytic-field lemma and initial covariance proof apply verbatim to finite random tangents, producing an intrinsic finite random bracket tangent V^f. For each fixed ϵ, source locality and the invertible local coordinate changes put the original and both deleted initial traces on one probability space. Only their initial germs are needed, so no simultaneous extension of this coupling past their exit times is required. On that space the covariances give

$$\beta^{f_\epsilon^+} + \beta^{f_\epsilon^-} = 2\beta^f/d_\epsilon, \qquad \beta^{f_\epsilon^+} - \beta^{f_\epsilon^-} = (V^f - 6)e_\epsilon^+/d_\epsilon^2.$$

Uniform tightness of the two marginals makes their sum and difference uniformly tight under any coupling. Since $d_\epsilon \to 0$ and $|e_\epsilon^+|/d_\epsilon^2 \to \infty$, these identities force $\beta^f = 0$ and $V^f = 6$ almost surely. Formally, for $a > 0$,

$$\mathbb{P}\{|\beta^f| > a\} \le \sum_{\sigma=\pm} \mathbb{P}\{|\beta^{f_\epsilon^\sigma}| > a/d_\epsilon\} \longrightarrow 0,$$

and the identical argument with threshold $a|e_\epsilon^+|/(2d_\epsilon^2)$ gives $V^f = 6$. The stopping-time lemma then kills the finite-variation drift and identifies the bracket as before. Local boundedness of a deterministic drift tangent is one special case of (FV8); no deterministic initial-tangent premise is otherwise necessary for this version.

18 Canonical characteristics under a genuine strong restart law

The strong DMP required above is substantive. Once it is an identity of regular conditional laws in the usual natural driver filtration, however, compatibility of the continuous canonical characteristics follows from uniqueness. It is useful to separate these two issues.

Proposition 18.1 (Strong DMP transports continuous characteristics)**.** *Let $\{P_f\}$ be a Borel family of laws on $C_0([0,\infty),\mathbb{R})$. In each state the coordinate process has a continuous canonical decomposition $W = M^f + A^f$ in its usual natural filtration, where A^f has finite variation on compact intervals. Write $C^f = \langle M^f\rangle$. Fix an initial state f and a bounded stopping time τ. Suppose f_τ is measurable at τ and the strong restart identity is literally*

$$\mathcal{L}_{P_f}\big((W_{\tau+t} - W_\tau)_{t\ge 0} \mid \mathcal{F}_\tau^f\big) = P_{f_\tau}. \tag{KR1}$$

Then, for almost every conditioned history, the shifted finite-variation process $A^f_{\tau+t} - A^f_\tau$ and bracket $C^f_{\tau+t} - C^f_\tau$ are the canonical characteristics of the restarted law. This is equality up to indistinguishability under that conditional law. No continuity of $f \mapsto P_f$ is required.

The same conclusion applies to an inherited Loewner chart when the future coordinate in (KR1) is exactly that inherited driving process. Changing to a different chart or clock still requires the appropriate invertible adapted coordinate transformation.

Proof. Write $P = P_f$, $M = M^f$, $A = A^f$, and $C = C^f$. The continuous adapted processes M, A, C have raw Borel adapted versions, up to one P-evanescent set. Indeed, at a positive time approximate from strictly earlier rational times; a variable measurable in the usual filtration at $r < t$ has a version measurable in the raw filtration at some time strictly between r and t. Make these choices on the countable rational approximation grid and take the continuous limits. Their values at zero are fixed. On exceptional paths retain the raw adapted pathwise-limit definitions, rather than zeroing a nonadapted exceptional set. After conditioning, freeze the stopped past, τ, and any additional information in $\mathcal{F}_\tau$. Thus the shifted processes are adapted Borel functionals of the future coordinate $U_t = W_{\tau+t} - W_\tau$ alone.

Choose stopping times $\sigma_n \uparrow \infty$ such that M^{σ_n} and $(M^2 - C)^{\sigma_n}$ are bounded martingales; for example stop at time n or at the first time $|M| + C$ reaches n. For rational $0 \le r < s$ and a bounded cylinder test H of U at rational times at most r, optional sampling and the tower property give

$$\mathbb{E}_P\Big[(M_{(\tau+s)\wedge\sigma_n} - M_{(\tau+r)\wedge\sigma_n})H \mid \mathcal{F}^f_\tau\Big] = 0. \tag{KR2}$$

Use a countable cylinder class, for example indicators of finite intersections of rational half-lines. Disintegrating all these identities at once gives, outside one null set of histories, their martingale identities under the conditional future law. Monotone classes and continuity extend them to every time and bounded past test. The bounded martingales remain martingales after usual augmentation by conditioning at times decreasing to the desired time.

Under any such conditional law put $T_n = (\sigma_n - \tau)^+$. The stopped shifted martingale $M_{\tau+t\wedge T_n} - M_\tau$ is the process in (KR2) with its initial value subtracted: if $\sigma_n \le \tau$ both are identically zero; otherwise they coincide. The T_n are stopping times in the conditional future natural filtration, increase to infinity, and hence $M_{\tau+t} - M_\tau$ is a continuous local martingale there. The same argument applied to $M^2 - C$, subtracting the frozen cross term $2M_\tau(M_{\tau+t} - M_\tau)$, identifies its bracket as $C_{\tau+t} - C_\tau$.

The shifted A is adapted, continuous, and of finite variation, so it is predictable. Consequently

$$U_t = (M_{\tau+t} - M_\tau) + (A_{\tau+t} - A_\tau)$$

is a canonical decomposition in the conditional future natural filtration. By (KR1) this is the usual natural filtration of the coordinate process under P_{f_τ}. Uniqueness of the continuous canonical decomposition and bracket proves the claim. A separate jointly measurable choice of characteristics over every deterministic state is unnecessary for this almost-every-history conclusion. □

Consequence for the driving criteria. Under the genuine strong restart identity (KR1), the explicit canonical-restart-compatibility premises in the earlier conditional driving theorems are redundant. The locality identification of natural trace filtrations and the existence of the strong conditional kernels remain necessary premises of those arguments. This reduction does not deduce strong DMP from deterministic-domain convergence.

18.1 Pointwise convergence, even of continuous kernels, is insufficient

Here is a small counterexample to an otherwise tempting kernel-transfer argument. Let V take the values $-1, 1$ with equal probability and put $I_n = V$, $D_n = V/n$, and $Y_n = V$. Choose continuous functions $g_n : \mathbb{R} \to [-1, 1]$ supported in the union of intervals of radius $1/(4n^2)$ around $-1/n$ and $1/n$, with $g_n(1/n) = 1$ and $g_n(-1/n) = -1$. Define $K_n(d) = \delta_{g_n(d)}$. These are weakly continuous probability kernels, and for every fixed deterministic d,

$$K_n(d) \longrightarrow \delta_0 =: K(d). \tag{KR3}$$

The limiting kernel K is continuous. Moreover the exact conditional identity $\mathcal{L}(Y_n \mid I_n, D_n) = K_n(D_n)$ holds. Nevertheless $(I_n, D_n, Y_n) \to (V, 0, V)$ almost surely, and $\mathcal{L}(V \mid V, 0) = \delta_V \neq K(0)$. Thus pointwise convergence at every deterministic state, and continuity of every kernel including the limit, do not justify a conditional limit at random states. Uniform equicontinuity, sequential convergence along the actually visited states, or another quantitative substitute is essential.

Lemma 18.2 (An annealed substitute for sequential kernel convergence)**.** *Let the past, state, and future variables take values in fixed Polish spaces. Suppose $(I_n, D_n, Y_n) \Rightarrow (I, D, Y)$ and $\mathcal{L}(Y_n \mid I_n, D_n) = K_n(D_n)$. Suppose K is a Borel kernel such that $d \mapsto K\varphi(d)$ is continuous at D almost surely, for each member φ of a countable bounded continuous convergence-determining class on the future space. It suffices that*

$$\mathbb{E}\big|K_n\varphi(D_n) - K\varphi(D_n)\big| \longrightarrow 0 \tag{KR4}$$

for each such φ. Then $\mathcal{L}(Y \mid I, D) = K(D)$.

Proof. For any bounded continuous test h of (I_n, D_n), replace $K_n\varphi(D_n)$ in the exact finite-n conditional identity by $K\varphi(D_n)$. The error is bounded by $\|h\|_\infty$ times (KR4). The continuous mapping theorem, with its almost-sure-continuity version, then passes the resulting identity to the limit. A monotone class in h, followed by the countable measure-determining class in φ, identifies the conditional law. □

Condition (KR4) only asks for a stability estimate on domains sampled by the exploration. It can therefore be more economical than a theorem uniform over every rough slit domain. It has not been proved here for percolation explorations. For strong DMP it must hold for the relevant stopping times, or be combined with an independent strong-Markov extension theorem.

19 Rigidity from conditional first-moment bounds

Scope. The next criterion replaces the a priori semimartingale and initial-tangent premises by conditional first-moment bounds. It does not derive those bounds from RSW. They must hold in one fixed physical conformal structure, with the stated control along the specified shrinking slit tests. No quadratic-moment bound or initial drift derivative is assumed.

Lemma 19.1 (Conditional bias produces a bounded drift density)**.** *Let X be a bounded continuous adapted real process, with $X_0 = 0$, in a usual filtration. Suppose a finite deterministic K satisfies*

$$|\mathbb{E}[X_t - X_s \mid \mathcal{F}_s]| \le K(t - s) \quad \text{almost surely,} \qquad 0 \le s \le t. \tag{BI1}$$

Then $X = M + \int_0^{\cdot} b_r dr$, where M is a continuous local martingale and b is predictable with $|b| \le K$ almost everywhere.

Proof. On a fixed finite time interval, X_t+Kt and $-X_t+Kt$ are class-D submartingales. Their Doob–Meyer decompositions have increasing predictable parts A^+ and A^-. Adding the decompositions and using uniqueness gives $A^+ + A^- = 2Kt$. Thus A^+ and A^- are absolutely continuous, with densities in $[0, 2K]$. The decomposition of X has drift density $b = dA^+/dt - K \in [-K, K]$. Since X and its drift are continuous, its martingale part is continuous. Uniqueness patches the decompositions over finite time intervals. □

Proposition 19.2 (Controlled rate bounds force the tangents to exist)**.** *Use the mirrored shrinking slits $A^{\pm}_{\epsilon}$ in (DR6), with initial derivatives d_ϵ and $e^{\pm}_{\epsilon}$ in (DR9). Assume the driver in a state f, and the drivers in all these deleted states, have canonical continuous-semimartingale decompositions with absolutely continuous drift. In each state the drift density is bounded on some strictly positive initial stopping interval. The interval may depend on the state and sample. Let $K_\epsilon < \infty$ be a deterministic bound common to the two deleted states. If $d_\epsilon K_\epsilon \to 0$ along a sequence tending to zero, then in the original state*

$$\frac{1}{t}\sup_{s\le t}|A_s| \longrightarrow 0, \qquad \frac{C_t}{t} \longrightarrow 6 \quad \text{almost surely as } t \downarrow 0. \tag{BI2}$$

Here A is canonical drift and C is the martingale bracket. No right continuity or initial tangent of the drift densities is presumed.

Proof. By locality, compare the original and each deleted driver using the same trace filtration and the stopping-time capacity clocks. With $p^\sigma_t = (h^\sigma_t)'(W_t)$ and $q^\sigma_t = (h^\sigma_t)''(W_t)$, uniqueness of canonical characteristics gives

$$\frac{1}{2}q^\sigma_t\,(dC_t - 6dt) = \big[(p^\sigma_t)^2\widehat{b}^\sigma_{u_\sigma(t)} - p^\sigma_t b_t\big]dt. \tag{BI3}$$

One fixed nonzero-q_0 test makes C absolutely continuous on a positive initial interval; write $dC_t = v_t dt$ there. For each fixed ϵ, both mirrored identities hold almost everywhere on a positive initial interval, as a two-by-two system in b_t and $(v_t - 6)/2$. Its coefficients converge as $t \downarrow 0$ to

$$\begin{pmatrix} d_\epsilon & e^+_\epsilon \\ d_\epsilon & -e^+_\epsilon \end{pmatrix},$$

whose determinant is nonzero. The right sides are $(p^\sigma_t)^2\widehat{b}^\sigma_{u_\sigma(t)}$, with each transformed density bounded by K_ϵ. Continuity of the coefficients and matrix inversion therefore give the pathwise bounds

$$\operatorname*{ess\,limsup}_{t\downarrow 0}|b_t| \le d_\epsilon K_\epsilon, \qquad \operatorname*{ess\,limsup}_{t\downarrow 0}|v_t - 6| \le \frac{2d^2_\epsilon K_\epsilon}{|e^+_\epsilon|}. \tag{BI4}$$

These assertions concern the same original densities. Intersect the probability-one sets for a countable sequence $\epsilon \downarrow 0$. By assumption $d_\epsilon K_\epsilon \to 0$; also $d_\epsilon/|e^+_\epsilon| \to 0$, so both right sides tend to zero. Integrating the resulting essential right limits proves (BI2). The positive comparison intervals may shrink with ϵ: the essential limsup is taken at fixed ϵ first. □

Theorem 19.3 (SLE from controlled conditional driving bias)**.** *Let a Borel exploration family have continuous Loewner drivers on all $t \ge 0$, source locality including the test deletions, and strong DMP. Require locality to identify the completed natural trace filtrations via the invertible capacity clocks. Require strong DMP to be the regular-conditional-law identity (KR1) for the inherited driver in its usual natural filtration. Proposition 18.1 then supplies canonical characteristic compatibility; it is not a separate premise. Suppose that, in every deterministic state f, there is an increasing*

sequence of stopping times $\rho_{f,m}$ increasing almost surely to infinity, with $\rho_{f,1} > 0$ almost surely, such that each $X_t^{f,m} = W^f_{t\wedge\rho_{f,m}}$ is bounded and satisfies (BI1), with a finite deterministic bound $K(f,m)$. Require

$$\lim_{n\to\infty} \epsilon_n \max_{\sigma=\pm} K(f^\sigma_{\epsilon_n}, 1) = 0 \tag{BI5}$$

at every state, for some sequence $\epsilon_n \downarrow 0$. In particular uniformly bounded constants suffice; growth $o(\epsilon_n^{-1})$ is allowed. Then the family is chordal SLE_6 in these coordinates.

Proof. Lemma 19.1, applied to the localizing sequence, gives global continuous-semimartingale decompositions with locally bounded absolutely continuous drift. Uniqueness patches the decompositions on overlapping stopped intervals. Proposition 19.2 then gives the zero drift tangent and bracket tangent six in every state.

Use the conditional-kernel compatibility of canonical decompositions in the inherited natural trace filtration, as in the earlier conditional characteristic theorems. At every bounded stopping time τ, conditional DMP and (BI2) give

$$\frac{A_{\tau+h} - A_\tau}{h} \longrightarrow 0, \qquad \frac{C_{\tau+h} - C_\tau}{h} \longrightarrow 6 \quad \text{in probability.}$$

The zero-stopping-tangent lemma (Lemma 17.3) applied to A and $C - 6t$ gives $A = 0$ and $C_t = 6t$. Lévy's characterization identifies the driver as $\sqrt{6}B$, hence the trace as SLE_6. □

Rates in the slit test. Since $d_\epsilon \sim \epsilon/2$ and $|e_\epsilon^+| \sim 1/(2\epsilon)$, the two bounds in (BI4) are of orders ϵK_ϵ and $\epsilon^3 K_\epsilon$. The weaker condition $\epsilon^3 K_\epsilon \to 0$ alone forces the initial bracket tangent to be six. The full classification above also needs the stronger drift condition in (BI5).

What this reduction gains and what it leaves open. Conditional bias bounds concern first moments of observable driver increments, rather than unknown canonical decompositions. They produce semimartingality, bounded drift densities, and the missing initial tangents without assuming quadratic variation estimates. They are nevertheless substantial cancellation estimates: a bound on $\mathbb{E}|W_{t+h} - W_t|$ of Brownian order $\sqrt{h}$, or tightness from RSW, does not imply (BI1). The rate control in (BI5) is coordinate sensitive and is not justified by convergence of curve laws in the shrinking slit domains. Thus this is a conditional route toward the desired regularity, not a proof of percolation universality.

19.1 An observable discrete criterion that passes to curve limits

There is no need to identify the discrete or limiting canonical characteristics in order to test the preceding criterion. Signed conditional first moments on a discrete exploration clock suffice. The following elementary transfer also avoids convergence of filtrations. It does not establish the separate strong-DMP premise of the rigidity theorem.

Proposition 19.4 (Discrete conditional bias passes to the natural limit filtration)**.** *Fix $T > 0$. For each n, let $0 = t_0^n < t_1^n < \cdots < t_{N_n}^n = T$ be a deterministic partition with maximum spacing tending to zero. Let X^n be its continuous linear interpolation of grid values, all bounded by one deterministic constant L. Suppose the grid values are adapted to filtrations $(\mathcal{G}_j^n)_j$ and*

$$\left|\mathbb{E}\left[X^n_{t^n_{j+1}} - X^n_{t^n_j} \mid \mathcal{G}^n_j\right]\right| \le (K + \delta_n)(t^n_{j+1} - t^n_j), \qquad \delta_n \downarrow 0. \tag{DB1}$$

Suppose X^n converges in distribution in $C([0,T])$ to X. Then in the usual augmentation of the natural filtration of X,

$$|\mathbb{E}[X_t - X_s \mid \mathcal{F}_s]| \le K(t-s), \qquad 0 \le s \le t \le T. \tag{DB2}$$

Consequently X has a continuous local-martingale part and an absolutely continuous canonical drift with density bounded by K.

Proof. Summing the conditional inequalities and using the tower property shows that the grid processes $X^n_{t^n_j} \pm (K+\delta_n)t^n_j$ are respectively a submartingale and a supermartingale. In particular, for grid points $s_n \le t_n$ and every bounded nonnegative $\mathcal{G}^n_{j(s_n)}$-measurable H_n,

$$\left|\mathbb{E}[(X^n_{t_n} - X^n_{s_n})H_n]\right| \le (K+\delta_n)(t_n - s_n)\mathbb{E}H_n. \tag{DB3}$$

Choose grid points $s_n \to s$ and $t_n \to t$, with $s_n \le t_n$, and set $H_n = H(X^n_{\wedge s_n})$, where H is any bounded nonnegative continuous function on $C([0,T])$. The stopped interpolated path is measurable at the grid time s_n. Convergence in $C([0,T])$ and uniform continuity of each limiting path give joint convergence of the evaluations and stopped paths. The deterministic bound L permits passage of their expectations to the limit in (DB3).

Bounded continuous tests determine finite measures on the stopped-path space. Thus the resulting two inequalities extend to all bounded nonnegative tests measurable in the raw natural filtration at s. This proves (DB2) there. Equivalently $X_t + Kt$ and $-X_t + Kt$ are continuous class-D submartingales. These remain submartingales under the usual augmentation: for $u \downarrow s$, apply the inequality at $u < t$, then use reverse-martingale convergence and continuity of X to condition on $\bigcap_{u>s} \mathcal{F}^{\mathrm{raw}}_u$. Completion changes only null sets. Lemma 19.1 now supplies the asserted canonical decomposition. □

Exploration-time variant. An equivalent usable sufficient hypothesis allows a random discrete clock. Let the successive capacity times t^n_j and grid values X^n_j be adapted to the discrete revealed-color filtration, with uniform deterministic bounds on the drivers and clock horizons. Assume the times are nondecreasing, interpolation is well defined, and $J_n(s) = \inf\{j : t^n_j \ge s\} < \infty$ almost surely for each $s < T$. Suppose their largest spacing tends to zero in probability and

$$\left|\mathbb{E}[X^n_{j+1} - X^n_j \mid \mathcal{G}^n_j]\right| \le (K+\delta_n)\mathbb{E}[t^n_{j+1} - t^n_j \mid \mathcal{G}^n_j]. \tag{DB4}$$

After continuous interpolation, assume the same uniform-path convergence. Then (DB2) again follows. Indeed $X^n_j \pm (K+\delta_n)t^n_j$ are bounded discrete sub/supermartingales. Use optional sampling at $J_n(s) = \inf\{j : t^n_j \ge s\}$ and $J_n(t)$, with a test of the path stopped at $t^n_{J_n(s)}$. The indices are stopping times because $\{J_n(s) \le j\} = \{t^n_j \ge s\}$. The clock overshoots tend to zero in probability; tightness in the continuous-path topology makes the stopped paths and sampled values converge to their deterministic-time counterparts. Boundedness then passes the inequalities to the limit as above. At T, use continuity from times below T.

Fixed-interval block estimates also suffice. The per-reveal condition (DB4) is sufficient, not necessary for this transfer argument. It is enough that for each fixed $s < t$ and every bounded nonnegative continuous stopped-path test H, the tested inequalities (DB3), using approximations of s, t, hold up to an error $o(1)\|H\|_\infty$. The same weak-convergence proof then gives (DB2). Thus one may instead seek signed cancellation over mesoscopic blocks, allowing microscopic biases to cancel before a fixed capacity interval. No per-reveal estimate is asserted for lattice percolation.

Thus a possible lattice-level research target is the signed cancellation estimate (DB4), after suitable bounded stopping, in an unknown but fixed affine normalization. The constants must satisfy the shrinking-slit control in Theorem 19.3, and the limit must separately have its stated strong-DMP and conditional-characteristic compatibility. Neither a Brownian-size absolute increment estimate nor RSW alone is being substituted for this signed conditional estimate.

20 Microscopic correctors for signed driving cancellation

The per-reveal condition (DB4) is sensitive to bounded microscopic oscillations which disappear in the continuum limit. The following criterion permits exactly this kind of cancellation. It supplies neither a corrector for a percolation exploration nor the required slit-uniform bounds; those remain mathematical tasks.

Proposition 20.1 (Vanishing-corrector transfer on an exploration clock)**.** *Fix a deterministic horizon T. In the random-clock setting of (DB4), assume $|X_j^n| \le L$, $t_0^n = 0$, $0 \le t_j^n \le T_*$, every level $s < T$ is reached almost surely, and the maximum clock spacing tends to zero in probability. Require zero clock increments to have zero driver increments, so continuous interpolation is well defined. Suppose the interpolated X^n converge in law in $C([0,T])$ to X. Let R_j^n be adapted real correctors with $\mathbb{E}\sup_j |R_j^n| < \infty$ for every n, such that*

$$\mathbb{E}\sup_j |R_j^n| \longrightarrow 0. \tag{MC1}$$

Assume also zero clock increments have zero corrector increments. Set $Y_j^n = X_j^n + R_j^n$ and suppose for $K < \infty$ and $\delta_n \downarrow 0$ that

$$\left|\mathbb{E}[Y_{j+1}^n - Y_j^n \mid \mathcal{G}_j^n]\right| \le (K+\delta_n)\mathbb{E}[t_{j+1}^n - t_j^n \mid \mathcal{G}_j^n]. \tag{MC2}$$

Then X satisfies (DB2) in its usual natural filtration. In particular its canonical drift is absolutely continuous with density bounded by K. If $K = 0$, X is a continuous martingale.

Proof. The grid processes $Y_j^n + (K+\delta_n)t_j^n$ and $-Y_j^n + (K+\delta_n)t_j^n$ are submartingales. For each fixed n, their absolute values at all stopping indices are dominated by the integrable variable $L + \sup_j |R_j^n| + (K+\delta_n)T_*$. Consequently optional sampling applies to the almost surely finite indices, for $0 \le s < t < T$, $J_n(s) = \inf\{j : t_j^n \ge s\}$ and $J_n(t)$, even when they have no deterministic upper bound. For every bounded nonnegative $\mathcal{G}_{J_n(s)}^n$-measurable test H_n, it gives

$$\left|\mathbb{E}[(Y_{J_n(t)}^n - Y_{J_n(s)}^n)H_n]\right| \le (K+\delta_n)\mathbb{E}[(t_{J_n(t)}^n - t_{J_n(s)}^n)H_n].$$

Replace Y^n by X^n. The resulting error is at most $2\|H_n\|_\infty \mathbb{E}\sup_j |R_j^n| = o(1)\|H_n\|_\infty$. Take $H_n = H(X^n_{\cdot\wedge t^n_{J_n(s)}})$ for a bounded nonnegative continuous stopped-path test H. Clock overshoots vanish in probability, and tightness in $C([0,T])$ identifies the limiting sampled values and stopped paths. The bound on X^n and on the clock then passes their expectations to the limit and yields

$$|\mathbb{E}[(X_t - X_s)H(X_{\cdot\wedge s})]| \le K(t-s)\mathbb{E}H(X_{\cdot\wedge s}).$$

As in Proposition 19.4, bounded continuous tests determine the corresponding finite signed measures, giving (DB2) in the raw natural filtration. Continuity and reverse-martingale convergence extend it to the usual augmentation. The endpoint T follows by continuity, and Lemma 19.1 identifies the canonical drift. □

A finite-state cell problem. There is a useful sufficient construction of such a corrector. For simplicity let $t_j^n = jh_n^2$ and let E_j^n be an adapted process in a fixed finite set $\mathcal{E}$ whose conditional transition matrix given $\mathcal{G}_j^n$ is a fixed irreducible stochastic matrix P. Suppose

$$\mathbb{E}[X_{j+1}^n - X_j^n \mid \mathcal{G}_j^n] = h_n b(E_j^n) + h_n^2 r_j^n, \qquad |r_j^n| \le K. \tag{MC3}$$

Let π be the unique invariant probability of P. If $\sum_e \pi(e)b(e) = 0$, the finite linear system

$$(I - P)\chi = b \tag{MC4}$$

has a solution χ, unique up to an additive constant. This follows because the kernel of $I - P$ is the constants and the image is the π-mean-zero hyperplane; no aperiodicity assumption is needed. The adapted corrector $R_j^n = h_n \chi(E_j^n)$ then obeys (MC1), and

$$\mathbb{E}[\Delta(X^n + R^n)_j \mid \mathcal{G}_j^n] = h_n\{b + P\chi - \chi\}(E_j^n) + h_n^2 r_j^n = h_n^2 r_j^n.$$

Thus (MC2) holds. The same computation remains valid before an adapted stopping index, provided both the driver and corrector are frozen there. The capacity clock may continue after this stopping index, with zero corrected driver increments. For a random clock the right side of (MC3) can instead be a residual bounded by $K\mathbb{E}[\Delta t_j^n \mid \mathcal{G}_j^n]$.

This is only a finite-state sufficient model. The local conformal geometry of an explored percolation boundary is not represented by the tip orientation alone. No closure to a fixed finite P, and no solution of a percolation cell problem, is asserted.

Remark 20.2 (Raw one-step bias may diverge for Brownian approximants). Let $Z_t = \sqrt{6}B_{t\wedge\tau}$, where $\tau = \inf\{t : |\sqrt{6}B_t| = L\}$, and let $h_n \downarrow 0$. At the deterministic grid times $t_j^n = jh_n^2$, put

$$X_j^n = Z_{t_j^n} + h_n\{(-1)^j - 1\}. \tag{MC5}$$

With the Brownian grid filtration, $X_0^n = 0$ and the conditional signed increment is exactly $-2h_n(-1)^j$. Hence any (DB4) constant must satisfy $K_n \ge 2/h_n \to \infty$. Nevertheless $|X_j^n| \le L + 2h_n$ and continuous interpolation converges uniformly on compact time intervals, almost surely, to Z. The corrector $R_j^n = h_n\{1 - (-1)^j\}$ has supremum at most $2h_n$ and makes $X^n + R^n$ an exact grid martingale. On a fixed positive horizon the sum of the absolute one-step predictable biases diverges like h_n^{-1}, while their cumulative signed sum stays bounded by $2h_n$.

Implication for the universality reduction. A percolation proof need not bound the raw per-reveal conditional bias. It would suffice to construct an adapted corrector satisfying (MC1) and the corrected estimate (MC2), in one fixed affine normalization. The resulting constants must still satisfy the shrinking-slit condition (BI5), and strong continuation and characteristic compatibility remain separate requirements of Theorem 19.3. The example proves only a logical obstruction to treating (DB4) as a necessary property of approximating drivers; it is not a counterexample to (DB4) for a particular percolation lattice.

21 Strong source locality inside the affine-SLE ansatz

Scope and revised axiom. Here locality means agreement of exploration laws up to exiting any common source neighborhood, including agreement of the boundary colors there; the targets and the domains outside the neighborhood may differ. This is stronger than the earlier target-preserving

deletion axiom. All statements below use the revised version. We investigate the deterministic affine-SLE marginal ansatz, not arbitrary exploration laws.

For each marked domain $S = (D, p, q)$ suppose

$$P_S = (A_S^{-1})_* \, \mathrm{SLE}_6(A_S D, A_S p, A_S q), \qquad A_S \in \mathrm{GL}^+(2, \mathbb{R}). \tag{AF1}$$

Let $T(S)$ be the determinant-one representative of $[A_S^{\mathsf{T}} A_S]$. The metric-identification results elsewhere in the record make this independent of the choice of representing matrix. When invoking the domain Markov property we assume (AF1) on the resulting rough slit states as well, not merely on initially regular domains.

21.1 Exact freedom remaining without the domain Markov property

An embedded colored source germ is the germ of (D, p) at the source, including the white and black boundary sides. At a regular source it may be represented by the physical boundary curve and its interior side in an arbitrarily small disk. Denote it by $G_s(D, p)$.

Proposition 21.1 (Strong locality leaves arbitrary regular-source germ metrics)**.** *On bounded piecewise real-analytic Jordan domains with regular source, strong locality forces $T(D, p, q)$ to depend only on the colored embedded source germ. Translation covariance makes this germ function invariant under translation. Conversely, every measurable translation-invariant choice of a positive-definite projective tensor from the source germ produces, through (AF1), a translation-covariant affine-SLE family with strong locality. No domain Markov assertion is part of this converse.*

Proof. If two triples have the same regular source germ, strong locality identifies the initial laws stopped inside a common source disk. Blow up at the regular source. The limiting laws are affine images of half-plane SLE_6 from 0 to ∞. Strict monotonicity of the left-passage probability makes the two affine tensors proportional, hence their determinant-one forms equal. In particular the target and the remote domain cannot change T. Translation covariance gives the corresponding equality for translated germs, by the same local identifiability.

Conversely, choose a canonical matrix from the tensor, for example its positive-definite square root after determinant normalization. Equal source germs use the same matrix. Applying it to two domains that agree near their source gives transformed domains with the same colored source neighborhood. The usual strong locality/target-independence property of SLE_6 couples its initial paths there. Pull back the coupling. Translation covariance follows from the assumed invariance of the tensor assignment. □

Thus strong locality and translations alone do *not* give a fixed metric, even inside the affine-SLE ansatz. This is an exact obstruction, rather than a gap in a proposed proof.

21.2 An explicit source-germ example, and a direct proof that DMP excludes it

Fix $a > 0$, $a \neq 1$. At a source germ locally equal to a half-plane with boundary line ℓ, assign

$$A_s = \Pi_\ell + a\Pi_{\ell^\perp}; \tag{AF2}$$

at every other source germ assign $A_s = I$. Here locally equal means a genuine full embedded end neighborhood with straight boundary, not an artificial crosscut drawn through the interior. This rule is translation covariant and is even covariant under Euclidean similarities. It yields strong locality by the preceding proposition and the identical-germ argument on arbitrary admissible source ends.

It has no fixed affine normalizer: sources on horizontal and vertical flat arcs have tensors $\mathrm{diag}(1, a^2)$ and $\mathrm{diag}(a^2, 1)$, which are not proportional. Regular-source half-plane identifiability already proves this on two sources of a rectangle.

The failure of DMP can be proved without any unestablished claim about the fractal regularity of a later hull boundary.

Lemma 21.2 (A fresh horizontal-level tip is not a flat source germ)**.** *Let a continuous curve in a rectangular domain start below a horizontal line $L = \{y = c\}$ and stop at its first hit z of L, with z in the interior of the rectangle. Suppose the target is above L. Then the target component of the slit domain is not locally equal to a half-plane at z.*

Proof. The pre-stopping trace lies below L and meets L only at z. The entire upper part of the rectangle is connected to the target and avoids the trace, so it survives in the target component. Every point of L other than z also survives: the compact trace stays a positive distance from each such point, which can be joined vertically to the upper part. Thus the removed filled hull meets L only at z.

In a sufficiently small ball about z, the full open upper half-ball belongs to the target component. Any putative straight boundary line through z would therefore have to be horizontal, since every other line enters this upper half-ball. But a horizontal boundary segment would contain other points of L, all of which are interior surviving points. Contradiction. The same argument rules out a flat full prime-end germ at this tip. □

Proposition 21.3 (The source-flat/default-identity rule violates DMP)**.** *The family (AF1)–(AF2) does not satisfy the domain Markov property.*

Proof. Take a rectangle, source at the midpoint of its bottom side, and target at the midpoint of its top side. The initial matrix is $A_h = \mathrm{diag}(1, a)$, a nonsimilarity. Stop at first hit of an intermediate horizontal line. The hit is almost surely interior: the two possible boundary endpoints of the line are fixed boundary points, and SLE_6 hits each fixed boundary point other than its marks with probability zero; an affine map preserves this statement. The path must reach the line before reaching its target.

By Lemma 21.2, the prescribed matrix at the restarted source is I. But ordinary SLE_6 DMP in the fixed A_h coordinates says that the true conditional continuation has matrix A_h. Identifiability of the affine metric for one marked bounded prime-end domain makes these continuation laws unequal, since A_h is not a similarity. Thus DMP fails. □

This example distinguishes two separate conclusions. Revised strong locality eliminates the earlier *target*-germ construction. It still permits arbitrary *source*-germ metrics before DMP is imposed; the simple flat source rule above is then genuinely excluded by DMP.

21.3 The exact DMP restriction and why full support is not enough by itself

Proposition 21.4 (Conservation of the affine metric)**.** *Within the all-state affine-SLE ansatz, DMP is equivalent to*

$$T(D_t, p_t, q) = T(D, p, q) \quad \textit{almost surely} \tag{AF3}$$

at every relevant stopping time. This is in addition to the ordinary SLE conditional kernel in the current fixed affine coordinates.

Proof. In the original affine coordinates $A = A_{D,p,q}$, ordinary SLE_6 DMP gives an affine-SLE continuation with the same matrix A. The family prescribes matrix $A_{D_t,p_t,q}$. The two laws coincide exactly when their tensors are proportional, by the one-pair metric-identification theorem. This proves both implications, including the stopping-time version when the kernels are measurable. □

Strong locality on regular starts and conservation do not immediately erase information stored in rough restart germs. Full support of Brownian driving alone cannot extend an almost-sure identity to deterministic analytic curves. For example, bounded-variation controls are dense in $C_0[0,t]$, have Brownian probability zero, and form a Borel set (total variation at most N is closed). The Borel function equal to 0 on bounded-variation controls and 1 elsewhere is almost surely constant under Brownian measure, yet differs on every smooth control. Some stability at the deterministic restart states is essential to that particular support argument. It is not silently implied by full support.

The initial-trace-germ metric decoder proved elsewhere gives a concrete reason to take this obstruction seriously: different affine metrics can already be distinguished from arbitrarily short ordered trace germs. It does not yet provide a decoder from the unlabelled boundary germ of the current target component, and therefore does not construct a DMP counterexample here.

A probabilistic alternative to topological stability. Within the more specific class of source-germ decorations, suppose a measurable metric label $t(G)$ is defined on every reachable colored source germ and DMP gives $t(G_\tau) = t(G_0)$ almost surely. If the laws ν_1, ν_2 of two restart germs are not mutually singular, their initial metrics must coincide: otherwise ν_i is supported on the disjoint measurable fiber $t^{-1}(t(G_0^i))$, which makes the laws mutually singular. Consequently a connected graph of such nonsingular restart-law comparisons would force a common tensor without any continuity assumption. This is a conditional communication criterion, not a proof that the required overlap occurs. Brownian full support supplies topological support only; mutually singular measures can have identical full topological support. Initial-germ metric decoding gives an explicit instance of that distinction. For the unrestricted affine ansatz, even expressing T as a function of a rough source germ rather than the full marked state needs local metric identifiability at that rough germ; only regular and straight-tip germs have been proved sufficient in this fragment.

21.4 Straight slit-tip germs identify the metric locally

Let $\mathcal{S} = \mathbb{C} \setminus (-\infty, 0]$, with source the slit tip 0 and target the end at infinity. The two sides of the slit are distinct colored boundary arcs. A canonical straight-tip germ in physical space agrees, up to translation, with $\mathcal{S}$ near 0, with fixed colors on its two sides.

Lemma 21.5 (Local metric identification at a straight slit tip)**.** *If two affine-SLE laws have identical initial prefixes in a common canonical straight-tip neighborhood, their projective affine tensors are equal. Consequently strong locality and translations assign one tensor T_* to all states having this canonical colored source germ.*

Proof. Blow up at the tip. The straight-boundary corner of opening 2π has a square-root uniformizing coordinate, so ordinary conformal kernel convergence identifies the two local limits as affine-SLE laws in $\mathcal{S}$ from 0 to ∞. Equality of prefixes in every fixed disk determines equality of these complete limiting laws. Stopping continuity may first be used at almost every radius and then extended to all smaller radii by composition.

It remains to identify the metric in $\mathcal{S}$. If a linear map M carries ordinary SLE_6 in $\mathcal{S}$ to SLE_6 in $M\mathcal{S}$, rotate the image slit to the negative real axis. The resulting map N preserves $\mathcal{S}$ and its

slit orientation. The conformal map $\psi(z) = i\sqrt{z}$ takes $\mathcal{S}$ to $\mathbb{H}$ with marks $0, \infty$. For $z = \rho e^{i\theta}$, $-\pi < \theta < \pi$, the left-passage probability is

$$F\big(-\tan(\theta/2)\big),$$

where F is strictly increasing. Topological side preservation forces $\arg Nz = \arg z$ for every $z \in \mathcal{S}$. In particular, writing $N(x, y) = (\alpha x + \beta y, \delta y)$ with $\alpha, \delta > 0$, tests at $z = i$ and $z = 1 + i$ give $\beta = 0$ and $\alpha = \delta$. Thus M is a similarity. Apply this to the two limiting affine laws. Translations and strong locality then identify all canonical straight-tip states. □

21.5 A narrower sufficient stability premise than full domain continuity

The preceding lemma permits a support argument requiring continuity only at one kind of source germ, rather than at every marked domain.

Straight-tip stability (additional hypothesis). Suppose $f_n : \mathbb{H} \to E_n$ and $f : \mathbb{H} \to E$ are conformal maps onto bounded simply connected domains with $f_n \to f$ locally uniformly and nondegenerate limit. Mark the ends by $0, \infty$. For each fixed invertible linear L, require

$$T(LE_n, Lf_n(0), Lf_n(\infty)) \longrightarrow T(LE, Lf(0), Lf(\infty)) \tag{AF4}$$

only when the limiting physical state has a canonical horizontal straight-tip source germ, up to translation and with the prescribed colors. The approximating states may have rough boundaries. No continuity is assumed at other limiting states. This is a condition at one local germ type across its different global representatives, not a derived consequence of strong locality or DMP.

Theorem 21.6 (One stable slit-germ type forces a common regular-domain metric)**.** *Assume the all-state affine-SLE ansatz, strong DMP, revised strong locality, translation covariance, and straight-tip stability (AF4). Then one fixed matrix normalizes every marginal on bounded piecewise real-analytic Jordan domains with regular source and target. Neither geometric reversal nor full marked-domain continuity is required.*

Proof. By Lemma 21.5, every state with the canonical straight-tip germ has the same determinant-one tensor T_*. Fix a regular initial triple and its matrix A. Choose a simple deterministic arc from its source into its interior that ends with a horizontal straight segment traversed from left to right. The arc stays away from the target and can be chosen piecewise analytic, smooth at its finitely many joins. Its final point lies in an interior disk and no earlier part of the arc enters a smaller disk about that point. The resulting slit source therefore has exactly the canonical germ, with the same fixed boundary colors.

In A coordinates, uniformize the initial domain by $f : \mathbb{H} \to AD$, source 0, target ∞. The chosen simple arc has a continuous Loewner control w_* on a finite capacity interval $[0, t]$. For every continuous control w on this interval define the inverse chart of the remaining state by

$$f_t^w(z) = f\big((g_t^w)^{-1}(z + w(t))\big). \tag{AF5}$$

The inverse Loewner ODE shows $w \mapsto f_t^w$ continuous from the uniform norm to locally uniform convergence on $\mathbb{H}$. No trace-continuity theorem is needed for general approximating controls. At $w = w_*$ the physical state $A^{-1} f_t^{w_*}(\mathbb{H})$ has the canonical germ. Thus (AF4), with $L = A^{-1}$, makes the metric functional of (AF5) continuous at w_*.

Under the original affine-SLE law, the driving control is $\sqrt{6}B$. Conservation (AF3) says its restarted metric is the original tensor almost surely. Brownian measure has full uniform support on

$C_0[0,t]$. If the metric at w_* differed from the original tensor, continuity at w_* would give a positive-Brownian-probability neighborhood with different metrics, contradicting conservation. Hence the original tensor equals the metric at the deterministic slit state, which is T_*. The argument applies to every regular initial triple. Similarity covariance removes the residual scalar and orthogonal factors from the representing matrices. □

Extension at the marked-domain-control level. If the all-state ansatz and (AF4) are formulated in inverse conformal charts, the same proof gives a common tensor on *all bounded simply connected prime-end states*, without assuming an initially regular Euclidean boundary. Here are the extra deterministic details. Given $f : \mathbb{H} \to AD$ with source $f(0)$, first use the zero driving control for a positive time. Its vertical chart slit has a positive-time endpoint in AD. The physical image of the slit is analytic near that endpoint, because f is conformal at its interior preimage. Earlier parts of the slit cannot accumulate at this endpoint: that would contradict continuity of f^{-1} at an interior point. Its local source domain is therefore a straightenable analytic slit of opening 2π.

Continue from this tip along a simple physical arc which initially avoids the old slit, stays in the remaining domain, and ends with an isolated horizontal straight segment traversed from left to right. Such an arc exists by polygonal connectivity of an open planar domain, with smoothing away from the final straight segment. In the current normalized half-plane chart it is a simple curve starting at the accessible tip and otherwise interior, hence has a continuous Loewner driver starting at zero. Concatenating the initial zero driver and this continuation driver is a continuous control on a finite interval. Its final physical state has the canonical source germ, exactly as in the regular proof. Conservation, continuity at this one limiting germ type, and Brownian full support again force the initial tensor to be T_*.

This argument constructs Loewner domain states; it does not assert that an initial non-singleton prime-end impression has a Euclidean landing point. If the model is restricted to Euclidean continuous curves including their initial endpoints, retain Theorem 21.6 on the regular class. The extension is valid in the inverse-chart/prime-end formulation of the all-state affine ansatz used in (AF4).

Proposition 21.7 (Pointwise analytic-history consistency is another sufficient premise)**.** *In Theorem 21.6, straight-tip stability may be replaced by this explicit condition: for every regular initial state and every deterministic simple piecewise analytic initial arc of the kind used in the proof, the family's law in the resulting slit state is the ordinary SLE conditional continuation in the initial fixed affine coordinates. Under this condition, one fixed metric again normalizes all regular-domain marginals.*

Proof. Choose the arc ending in a canonical horizontal slit germ. The pointwise condition gives its tensor equal to the original tensor. Strong locality and translations make the slit state's tensor T_*. No support or continuity argument is needed. □

This condition is not automatically part of ordinary stochastic DMP. Analytic deterministic histories have probability zero under Brownian driving, and a regular conditional kernel is determined only almost surely. Requiring its specified affine-SLE extension to agree at those histories is extra pointwise consistency. The proposition is therefore a conditional theorem, not a proof that the revised three axioms already force one metric.

Precise remaining issue. This theorem substantially narrows the stability needed by the support method, but it does not remove stability altogether. Without (AF4), the support argument cannot rule out a conserved label encoded in rough source germs. Neither the existence of such a label

nor its impossibility under strong locality and DMP has been proved. The task of deriving the affine-SLE ansatz from arbitrary exploration laws remains separate and open.

22 A nonconcentration estimate bounds affine anisotropy

This result is a concrete use of a coarse curve estimate. It does not identify a metric, but makes the possible normalized metrics compact. It concerns the affine-SLE ansatz in a flat source half-plane, not an unspecified coloring associated to arbitrary curve laws.

For an orientation-preserving real-linear map B carrying $\mathbb{H}$ onto itself, let μ_B be the law of $B\eta$, where $\eta \sim \mathrm{SLE}_6(\mathbb{H}; 0, \infty)$. Suppose a collection $\mathcal{B}$ obeys a uniform small-ball estimate

$$\sup_{B\in\mathcal{B}} \sup_{z\in\overline{\mathbb{H}}\setminus\{0\}} \mu_B\{\gamma \cap B(z, \rho|z|) \neq \varnothing\} \le u(\rho), \qquad \lim_{\rho\downarrow 0} u(\rho) = 0. \tag{6}$$

An example of the required numerical form is $u(\rho) = C\rho^\alpha$. A sufficient input is a first-entry annulus bound which explicitly includes interior balls when both marked ends are outside the outer ball. The boundary-unforced G2/C2 estimate used for driver tightness alone does not supply this: its unforced annulus part is empty if the inner circle misses the current domain boundary. Thus the displayed nonconcentration estimate remains an explicit hypothesis here. The later ray-locality comparison controls a particular driving-confinement event, not this full interior hitting estimate. No coloring is presumed for an abstract family.

Proposition 22.1 (Compactness of normalized metrics)**.** *Under (6), the condition numbers $\|B\|\,\|B^{-1}\|$ for $B \in \mathcal{B}$ are uniformly bounded. Equivalently, the determinant-one conformal tensors of B^{-1} stay in a compact subset of the positive-definite determinant-one matrices.*

Proof. Suppose otherwise, and choose a sequence with diverging condition number. Ordinary SLE_6 is invariant under scalar spatial dilations, so replacing B_n by $B_n/\|B_n\|$ does not change μ_{B_n} or its condition number. Pass to a subsequence with $B_n \to M$, $\|M\| = 1$, $\det M = 0$. Thus M has rank one and $M\overline{\mathbb{H}} \subset \overline{\mathbb{H}}$. Write $Mz = v\ell(z)$ with $|v| = 1$ and a nonzero real-linear functional ℓ.

An SLE_6 trace on any positive capacity interval is almost surely not contained in a prescribed straight line. One can use the standard nondegeneracy of Brownian driving here: a strictly growing straight-ray Loewner trace has a deterministic driver of the form $c\sqrt{t}$, whereas Brownian motion almost surely has no such trajectory. Therefore $\max_{0\le t\le 1} |\ell(\eta(t))| > 0$ almost surely. Choose $\delta > 0$ so that this maximum exceeds 2δ with probability greater than $3/4$. At least one choice of sign $\sigma \in \{+1, -1\}$ has

$$\mathbb{P}\{\max_{0\le t\le 1} \sigma\ell(\eta(t)) > 2\delta\} > 3/8.$$

Fix that sign and put $z_0 = \sigma\delta v \neq 0$. By continuity and $\eta(0) = 0$, $M\eta$ hits z_0 on the displayed event. In particular $z_0 \in \overline{\mathbb{H}}$.

On each sample the compact trace $\eta[0, 1]$ is bounded, so $B_n\eta \to M\eta$ uniformly on this interval. Consequently, for every fixed $\rho > 0$,

$$\liminf_n \mu_{B_n}\{\gamma \cap B(z_0, \rho|z_0|) \neq \varnothing\} \ge \mathbb{P}\{M\eta[0, 1] \text{ hits } z_0\} > 3/8.$$

Choosing ρ with $u(\rho) < 3/8$ contradicts (6). Uniform boundedness of the condition numbers follows. The tensor conclusion is elementary singular-value normalization. □

The proposition does not force equality of different metrics: every uniformly bounded collection of affine distortions can still contain many members. Its use is compactness in attempts to recover metric continuity from convergence of exploration laws, rather than postulating continuity of a hidden metric label. Establishing the necessary convergence at rough slit tips remains separate.

23 Source locality, microscopic memory, and quantitative target stability

Current status under the revised axiom. The present project specification compares any two colored marked domains which agree near the source, including different targets. The earlier terminal-tangent construction does not satisfy this stronger axiom. This investigation has not produced an unconditional counterexample to the revised single-color-convention conjecture. It does prove a quantitative boundary stability statement from a precise conditional crossing estimate, and it isolates the exact additional state-recovery step needed for a source-germ counterexample. Near-critical configuration convergence is not used as a substitute for ordered exploration-kernel convergence.

23.1 What an affine source-germ counterexample would require

Let $\mathfrak{G}_s$ be the space of embedded colored source prime-end germs, with full end neighborhoods retained. Fix finitely or countably many orientation-preserving invertible matrices A_i and a deterministic Borel label $\pi : \mathfrak{G}_s \to I$. Consider

$$P_{D,p,q} = \nu_{D,p,q}^{A_{\pi(G_s(D,p))}}. \tag{SO1}$$

Unlike a target label, this label is automatically unchanged when two marked domains agree in a source neighborhood. It need not be unchanged after an exploration step.

Proposition 23.1 (A precise conditional source-label construction)**.** *Suppose π is translation covariant, and suppose that for every initial marked domain and every admissible finite stopping time of the fixed-label affine SLE exploration,*

$$[A_{\pi(G_s(D_t,p_t))}^{\mathsf{T}} A_{\pi(G_s(D_t,p_t))}] = [A_{\pi(G_s(D,p))}^{\mathsf{T}} A_{\pi(G_s(D,p))}] \qquad \textit{almost surely.} \tag{SO2}$$

Then (SO1) has strong DMP, the revised source-neighborhood locality, and translation covariance. If two regular source germs receive genuinely different projective tensors, the ensemble cannot have a single fixed affine SLE normalization.

Proof. The rough-domain affine identifiability and DMP criterion in the research record say exactly that (SO2) is the DMP condition within this deterministic affine ansatz. When two source neighborhoods agree, their labels agree; the two laws therefore use the same fixed matrix. Ordinary SLE_6 source locality, with the matching boundary colors, couples their prefixes through that neighborhood; pulling back the coupling proves the revised locality. Translation covariance follows from that of the label and SLE. Finally, local affine identifiability at a regular start identifies the projective tensor from its initial prefix law, excluding a common normalizer when the assigned tensors differ. □

Condition (SO2) is not established for a nonconstant source-germ label. The proved initial ordered-trace-germ decoder does not establish it. A remaining slit domain has an *unlabelled accessible frontier*, while the decoder uses the chronological order of a trace started from a smooth boundary. Filling swallowed bubbles can erase parts of that order. Furthermore a usable label must work at every admissible stopping time. Decoding at each separate deterministic time would not by itself supply that stronger assertion.

One possible route is to show that a constant affine metric can be recovered from every nonempty open piece of an SLE accessible frontier. A simultaneous statement on a countable spatial basis

could then handle all tip neighborhoods, avoiding the unjustified intersection of uncountably many probability-one events. No such frontier theorem is proved here; SLE duality alone does not give the required universal measurable decoder. Therefore (SO1)–(SO2) is a useful construction criterion, not a completed counterexample.

For a finite collection of matrices with bounded condition numbers, ordinary affine SLE marginals retain uniform versions of the standard unforced-quad crossing bounds, by quasiconformal modulus distortion. Consequently marginal crossing bounds would not on their own contradict a hypothetical conserved-label construction. This does not assert that the resulting ensemble comes from a single coloring or satisfies a configuration-level RSW statement.

23.2 An independent-metric-noise construction degenerates

The tempting alternative of putting independent bounded conformal structures at continuum points cannot produce a measurable random Beltrami field. Here is a slightly stronger formulation.

Lemma 23.2 (No independently scattered bounded random density)**.** *Let $X(\omega, z)$ be a jointly measurable complex random function on $\mathbb{R}^2$ with $|X| \le K$ almost surely. Suppose the random variables*

$$X(B) = \int_B X(z)\, dz$$

are independent for disjoint bounded Borel sets B. Then $X(\omega, z) = \mathbb{E}[X(z)]$ for Lebesgue-almost every z, almost surely. In particular a stationary such field is almost surely a deterministic constant almost everywhere.

Proof. Partition a bounded rectangle B into rectangles B_j of maximum area at most δ. Independence and boundedness give

$$\operatorname{Var} X(B) = \sum_j \operatorname{Var} X(B_j) \le 4K^2 \sum_j |B_j|^2 \le 4K^2 |B| \delta.$$

Letting $\delta \downarrow 0$ proves that $X(B)$ is deterministic. Intersecting the full-probability events for rectangles with rational vertices, and using uniqueness of locally finite absolutely continuous measures, gives the first assertion. Under stationarity the mean function is constant almost everywhere. □

For a bounded Beltrami coefficient $|\mu| \le k < 1$, this rules out nontrivial independently scattered metric noise. Nontrivial stationary coefficients with spatial correlation are possible, but exploration then reveals information about coefficient values away from the revealed trace. Annealed DMP in an undecorated slit domain is not automatic. That latter statement is an obstruction to the proposed proof strategy, not a nonexistence theorem for all random-metric models.

23.3 A quantitative stability consequence of source locality

We state the crossing input precisely rather than call it "RSW" without specifying a path implication. Assume constants $C < \infty$ and $\alpha > 0$ such that, conditional on any explored initial segment, the probability that the remaining curve makes an *unforced* crossing of an annulus $A(z, r, R)$ is at most

$$C(r/R)^{\alpha}. \tag{SO3}$$

Here a component of the annulus in the remaining target domain is unforced when it does not separate the current source from the target. In particular, if those ends can be joined inside $B(z, r)$ in the remaining domain, every crossing of this annulus is unforced. The conditional formulation

is crucial. It is a standard type of hypothesis in the Kemppainen–Smirnov framework; see A. Kemppainen and S. Smirnov, *Random curves, scaling limits and Loewner evolutions*, Ann. Probab. 45 (2017), 698–779, https://arxiv.org/abs/1212.6215. The argument below uses (SO3) directly; it does not claim to derive it from bare rectangle crossing bounds for an arbitrary path ensemble.

Theorem 23.3 (Localized target perturbations are small in curve law)**.** *Consider two colored marked Jordan domains (D_i, p, q_i), $i = 1, 2$, whose colored domains agree outside $B(z, \epsilon)$, with common source $p \notin \overline{B(z, R_0)}$ and targets $q_i \in B(z, \epsilon)$. Suppose the revised source locality holds and both exploration laws satisfy (SO3) with the same constants. Couple the common prefixes through first entry of $\overline{B(z, 2\epsilon)}$. Assume that, at this entry, the current source and target can be joined in their respective remaining domains inside $B(z, 4\epsilon)$, almost surely. Then, for $4\epsilon < R < R_0$, there is a coupling of the full curves with*

$$\mathbb{P}\{d_{\mathrm{F}}(\gamma_1, \gamma_2) > 2R\} \le 2C(4\epsilon/R)^{\alpha}, \tag{SO4}$$

where d_{F} is oriented uniform Fréchet distance. Consequently the distances between the two curve laws tend to zero as $\epsilon \downarrow 0$; convergence to a common limit follows if either family has a limit. After fixing a spatial unit, their Prokhorov distance, and their bounded Lipschitz distance for the truncated Fréchet metric, are $O(\epsilon^{\alpha/(\alpha+1)})$.

Proof. The two colored domains coincide in the source neighborhood $\mathbb{R}^2 \setminus \overline{B(z, 2\epsilon)}$. The revised locality therefore identifies the laws of their prefixes stopped on the circle of radius 2ϵ. Use a common prefix and the regular conditional tail kernels to obtain a full coupling with the correct marginals. No agreement of the two tails is asserted.

At this stopping time the connection hypothesis makes every crossing of $A(z, 4\epsilon, R)$ unforced in either remaining domain. Estimate (SO3) therefore bounds the conditional probability that either tail exits $B(z, R)$ by $C(4\epsilon/R)^{\alpha}$. The union bound gives twice this quantity for the probability that at least one tail exits.

On the complementary event, the two curves have a common ordered prefix and both remaining tails stay in a ball of diameter $2R$. Aligning the common prefix and then reparameterizing their tails gives $d_{\mathrm{F}}(\gamma_1, \gamma_2) \le 2R$. This proves (SO4). The coupling definition of Prokhorov distance gives a bound by $\max\{2R, 2C(4\epsilon/R)^{\alpha}\}$. Bounded Lipschitz test functions give, for example, $2R + 4C(4\epsilon/R)^{\alpha}$. Taking R proportional to $\epsilon^{\alpha/(\alpha+1)}$ proves the stated rate. □

A sufficient geometric situation for the connection hypothesis. Suppose the original domain is a half-plane in $B(z, R_0)$, each modified domain is Jordan, and all modifications and their target points are inside $B(z, \epsilon)$. The half-annulus $B(z, 2\epsilon) \setminus \overline{B(z, \epsilon)}$ in that half-plane is connected and unchanged. A target access arc reaches this half-annulus from inside the smaller ball. At first entry of the radius-2ϵ ball the earlier trace has no point in its interior. Thus the entering target-side prime end connects through the unchanged half-annulus to the target access arc inside $B(z, 4\epsilon)$, without crossing the explored trace. Filling non-target components does not remove this connection. This verifies the hypothesis in that locally flat situation. For a wild target patch one must check the connection condition; proximity of the two Euclidean endpoints alone does not establish it.

What this rules out and what it leaves open. The theorem proves a genuine boundary-continuity consequence of strengthened locality plus a specified conditional unforced-crossing bound. Remote localized target perturbations cannot change a macroscopic path law by a fixed amount under these hypotheses. This is relevant to schemes that store information at a distant target. It does not establish continuity under perturbations of the source itself: when two source germs differ,

the revised locality supplies no common initial prefix, and the preceding coupling argument does not start. Thus it does not yet eliminate (SO1)–(SO2) or prove the marked-chart continuity needed by the conditional affine-ansatz classification.

23.4 Color orientation is a separate formulation issue

If a state space includes *both* boundary-color orientations as separate sectors, black-on-left and black-on-right relative to the oriented exploration are conserved by DMP, and colored source locality never compares those two sectors. Assigning one fixed affine matrix to one sector and a different fixed affine matrix to the other then gives an exact example with strong locality, DMP, and translations, with no common matrix across the two sectors. Local affine identifiability proves non-equivalence when the tensors are not proportional.

This is *not* a counterexample when the project notation fixes the usual one-color convention for every (D, p, q), as we assume in the current core investigation. In that convention only one of the sectors is present. Likewise a two-ended exploration or reversal argument must track the fact that reversal exchanges the color orientation; no color-exchange symmetry at a critical density different from $1/2$ is supplied by the three axioms.

24 A wrong affine metric cannot have tame driving drift in every bounded domain

Status. The conclusion here is stronger than the earlier wrong-coordinate consequence of drift boundedness. For a fixed non-similarity affine deformation of SLE_6, regular semimartingale driving with absolutely continuous right-continuous drift cannot hold in every bounded domain in the wrong Euclidean complex structure. No local boundedness of the initial drift functional is needed. The proof uses exact scale invariance only for the *known fixed-affine SLE_6 half-plane law*; it does not assume that arbitrary critical subsequential limits are scale invariant. All characteristic regularity assumptions remain explicit.

24.1 Exact Brownian scaling and a regular initial bracket

Scaling lemma (proved). Let W be a continuous real process on $[0, \infty)$ with $W_0 = 0$, whose law has exact Brownian scaling:

$$(c^{-1/2} W_{ct})_{t \geq 0} \overset{\text{law}}{=} (W_t)_{t \geq 0} \qquad (c > 0). \tag{SM1}$$

Suppose up to a strictly positive stopping time it is a continuous semimartingale

$$W_t = M_t + \int_0^t b_s \, ds, \tag{SM2}$$

where $M_0 = 0$ is a continuous local martingale and b is finite and right-continuous at 0. Suppose its bracket has a finite deterministic initial right derivative

$$\lim_{t \downarrow 0} \frac{\langle M \rangle_t}{t} = v \in [0, \infty) \quad \text{almost surely.} \tag{SM3}$$

Then the entire law of W is that of $\sqrt{v} B$. This includes the identically zero process if $v = 0$.

Proof. Fix $T < \infty$ and stop the local decomposition at a positive initial localizing time. As $c \downarrow 0$, with probability tending to one the whole interval $[0, cT]$ lies before that time. Right-continuity of b gives

$$\sup_{t \le T} \left| c^{-1/2} \int_0^{ct} b_s \, ds \right| \le \sqrt{c}\, T \sup_{s \le cT} |b_s| \longrightarrow 0$$

almost surely on the localized event. Writing $C_t = \langle M \rangle_t$, (SM3) implies

$$\sup_{t \le T} |C_{ct}/c - vt| \longrightarrow 0. \tag{SM4}$$

Indeed $|C_{ct}/c - vt| \le t \sup_{0 < s \le cT} |C_s/s - v|$. The continuous-martingale functional central limit theorem now gives $c^{-1/2} M_{c\cdot} \Rightarrow \sqrt{v} B$ on $C[0, T]$. One may see this directly from Dambis–Dubins–Schwarz: extend its Brownian motion if necessary and write

$$c^{-1/2} M_{ct} = B^{(c)}_{C_{ct}/c}, \qquad B^{(c)}_u = c^{-1/2} \widetilde{B}_{cu}.$$

Each $B^{(c)}$ has standard Brownian law. Uniform bracket convergence to the deterministic clock and the Brownian modulus of continuity imply the stated limit; independence between $B^{(c)}$ and the bracket is unnecessary. Adding the negligible drift gives the same limit for W. By exact scaling, however, each rescaled process already has the law of W on $[0, T]$. That law is therefore Brownian with variance v. Arbitrariness of T proves the assertion. □

The regular initial bracket is not automatic for a general self-similar continuous process. In the affine-SLE application below it follows from locality and absolutely continuous drift in two domain charts, using the local bracket-regularization identity.

24.2 The half-plane affine-SLE obstruction

Let A be an invertible orientation-preserving real-linear map and let η be chordal SLE_6 in $A\mathbb{H}$ from 0 to ∞. Put $\gamma = A^{-1}\eta \subset \overline{\mathbb{H}}$ and let W be its half-plane-capacity driving function. Its continuous Loewner generation and infinite capacity lifetime are supplied by the affine-admissibility result already proved in the research record. Its law has exact Brownian scaling (SM1): ordinary SLE is dilation covariant, A commutes with positive scalar dilations, and capacity scales by the square of physical length.

Proposition (proved). If this W satisfies (SM2)–(SM3), then A is a Euclidean similarity.

Proof. The scaling lemma makes W ordinary Brownian driving of variance v. Thus the trace is ordinary chordal SLE_v in $\mathbb{H}$. We identify v without a Hausdorff-dimension theorem.

For $\kappa \le 4$ the nonzero real boundary points are never swallowed by SLE_κ: their Loewner distances divided by $\sqrt{\kappa}$ are Bessel processes of dimension $1 + 4/\kappa \ge 2$ (the case $\kappa = 0$ is immediate). An affine image of SLE_6 swallows boundary points, so $v > 4$. For any $\kappa > 4$, the boundary-race calculation gives

$$\mathbb{P}_\kappa(\tau_y < \tau_x) = F_\kappa\!\left(\frac{-x}{y - x}\right), \qquad F_\kappa(r) = \frac{\displaystyle\int_0^r [u(1-u)]^{-4/\kappa}\, du}{\displaystyle\int_0^1 [u(1-u)]^{-4/\kappa}\, du}, \quad x < 0 < y. \tag{SM5}$$

This includes $\kappa \geq 8$. To check it, the ratio $R = -X/(Y - X)$ has generator

$$\frac{\kappa}{2}\frac{d^2}{dr^2} + \frac{2(1-2r)}{r(1-r)}\frac{d}{dr}$$

after $ds = dt/(Y - X)^2$. Its scale density is $[r(1-r)]^{-4/\kappa}$, integrable at both ends exactly when $\kappa > 4$. The first swallowing is finite almost surely by the Bessel description; simultaneous swallowing is impossible since $Y - X$ is positive and nondecreasing. Bounded optional stopping gives (SM5).

Rotate $A\mathbb{H}$ conformally back to $\mathbb{H}$. The resulting map has matrix

$$T = \begin{pmatrix} \alpha & \beta \\ 0 & \delta \end{pmatrix}, \qquad \alpha, \delta > 0.$$

It acts on the real boundary by $x \mapsto \alpha x$, so it preserves the ratio $-x/(y - x)$ and the topological swallowing order. The law of γ therefore has boundary-race function F_6, while its Brownian driving gives F_v. Equality for every $r \in (0, 1)$ forces $v = 6$: compare the derivatives in (SM5), proportional respectively to $[r(1-r)]^{-4/v}$ and $[r(1-r)]^{-2/3}$.

Finally $T^{-1}\mathrm{SLE}_6(\mathbb{H}; 0, \infty)$ is ordinary SLE_6. Half-plane affine identifiability now implies T is a similarity. For completeness, the strictly increasing SLE_6 left-passage function

$$L(u) = \frac{\int_{-\infty}^{u}(1+s^2)^{-2/3}\,ds}{\int_{-\infty}^{\infty}(1+s^2)^{-2/3}\,ds}$$

and preservation of topological side give $L(u) = L((\alpha u + \beta)/\delta)$ for all real u. Thus $\alpha = \delta$ and $\beta = 0$. Rotation does not affect whether A is a similarity, proving the proposition. □

24.3 Transfer from bounded-domain regularity to the half-plane germ

The preceding proposition is already a complete unbounded-domain statement. The following additional argument shows why requiring the tame drift structure in *every bounded domain* is enough to trigger it. We spell out the local coordinate transfer, rather than silently assume semimartingality is preserved by a general physical affine map.

Bounded-domain obstruction theorem (proved). Consider the canonical fixed-affine SLE_6 family

$$P_{D,p,q} = (A^{-1})_* \,\mathrm{Law}(\mathrm{SLE}_6(AD; Ap, Aq))$$

on bounded simply connected marked domains. Suppose in the original Euclidean complex structure every domain-chart driver is a continuous semimartingale whose finite-variation part is $\int_0^t b_s\,ds$, with finite right-continuous drift density and deterministic initial value $B(f)$, consistently under the usual conditional kernel interpretation. Then A is a Euclidean similarity. No boundedness or continuity of the map $f \mapsto B(f)$ is assumed.

Proof. Choose a bounded half-disk $D \subset \mathbb{H}$ with 0 in the interior of its real diameter, and a distinct target on that diameter away from 0. Choose the boundary colors so they agree near 0 with the half-plane problem $(\mathbb{H}, 0, \infty)$. Let $\phi : D \to \mathbb{H}$ be its chordal chart sending these two marks to $0, \infty$. It extends analytically across 0, with $\phi'(0) > 0$.

The local bracket-regularization lemma applies to this bounded-domain driver: delete one further vertical slit in its ϕ chart, away from its marks, with nonzero $h''(0)$. Both domains have tame absolutely continuous drift by hypothesis. The signed-measure identity

$$h_t''(W_t)(dC_t - 6\,dt) = 2\big[h_t'(W_t)^2\widehat{b}_{u(t)} - h_t'(W_t)b_t\big]dt$$

therefore gives a finite right-continuous initial bracket density in D, with deterministic initial value v_D.

Corrected source-neighborhood locality of ordinary SLE_6, preserved by the fixed map A^{-1}, couples this bounded-domain curve to the canonical affine-SLE_6 half-plane curve so that their physical initial segments are identical until leaving a sufficiently small half-neighborhood of 0. This uses ordinary SLE_6 locality including distant target changes; it is not a new hypothesis about an unknown nonconformal family. On that common initial trace, the coordinate change is the analytic map ϕ at the source.

A local analytic coordinate change and its inverse preserve the tame semimartingale structure just obtained. More explicitly, if t is physical half-plane capacity and u is the bounded-chart capacity, the evolving local map satisfies

$$W_D(u(t)) = h_t(W_H(t)), \qquad \dot{u}(t) = h_t'(W_H(t))^2.$$

Initially $h_0 = \phi$, so $h_0'(0) > 0$ and all required local derivatives are finite. Before the trace leaves the common neighborhood, localize so the first derivative stays bounded away from zero and the derivatives stay bounded. Both capacity clocks are adapted functions of the same initial trace; their inverses are stopping-time clocks. The inverse field h_t^{-1} has finite variation in time, as follows by differentiating the Loewner composition. Applying the finite-variation Itô–Wentzell formula to the semimartingale W_D, and changing the capacity clock, produces a continuous-semimartingale decomposition of W_H on a positive initial interval. Its drift is absolutely continuous and right-continuous, and its initial bracket density is the same scalar v_D. These conclusions use only a *conformal local chart transition*, not the nonconformal map A.

For an explicit initial check, writing $d = \phi'(0)$, $e = \phi''(0)$, the transferred rates obey

$$V_H = v_D, \qquad B_H = dB_D - \left(\frac{v_D}{2} - 3\right)\frac{e}{d}, \tag{SM6}$$

so both are finite and deterministic. Thus the canonical half-plane driver satisfies (SM2)–(SM3). Its exact scaling and the half-plane proposition force A to be a similarity. □

Only a bounded domain flat at its source and one of its source-preserving slit deletions are needed for the bracket part of this contradiction. The full all-bounded-domain assumption is a convenient sufficient formulation. The obstruction does not identify the precise failure for a particular wrong affine metric: semimartingality, absolute continuity or right-continuity of the drift, or deterministic initial drift may fail. It does show that at least one must fail somewhere. It supplies no argument that an unknown exploration family admits a correct affine normalization with tame drift.

25 A discrete source shield and stability of two exploration starts

Status and model. The deterministic shielding statement and the ensuing probabilistic estimate are proved here for independent site percolation on a triangular-lattice triangulation of a Jordan disk. Boundary vertices are wired black or white, with exactly two color changes, the marked source and target. The exploration is the nonbranching face-dual black–white interface between those changes. The same argument applies to a cellular model with the corresponding nonbranching primal/dual geometry and the stated white RSW bounds. It is an actual common-color coupling argument. It is *not* inferred for an arbitrary random-curve family from abstract locality or G2 alone.

Work at mesh δ. Two triangulated domains, their source positions, and their boundary colors may differ inside $B(p, \epsilon)$, but agree outside it. Their common target lies outside $B(p, R_0)$. Outside

the source patch and inside this latter ball, assume the common geometry is either a lattice half-plane with straight boundary, or a canonical thin straight slit with its two boundary sides. The lattice approximations are chosen with free first rows adjacent to the wired boundary rows; slit width is $O(\delta)$. The black and white wired sides are the same for the two configurations outside the patch. Couple all free colors outside the patch identically. Those shared colors have the independent percolation law. The colors inside the patch may be coupled arbitrarily, with their correct marginal laws.

25.1 The deterministic one-edge crosscut

In a triangular face, either zero or two of its primal edges have endpoints of different colors. Hence the dual interface graph has degree zero or two at every interior dual vertex. In the Dobrushin disk it has a unique open simple path between the source and target; other components are loops.

Lemma 25.1 (A one-heterochromatic-edge shield)**.** *Let a simple primal crosscut c of a triangulated disk connect a white wired boundary vertex to a black wired boundary vertex. Suppose every edge of c is white–white except its final edge e, which joins a white* interior *vertex to the black boundary vertex. Suppose c separates the entire source patch from the target. Then the exploration must use the dual edge e^*, with the direction determined by its black and white sides. After using e^* it cannot reenter the source side of the crosscut.*

Consequently, if the same such crosscut and its colors occur in both configurations above, their explorations have identical ordered suffixes from e^* to the common target.

Proof. The source–target dual path must cross the primal crosscut. Primal and dual vertices are distinct, and dual edges cross primal edges only at their matched transverse intersections. No interface edge can cross a white–white primal edge. Thus the crossing must be e^*, and matching colors give the same direction in both configurations.

The dual interface path is simple, so it cannot traverse e^* twice. All other primal edges of c remain monochromatic, so they cannot be crossed later either. The path therefore stays on the target side after using e^*. In particular it never returns to the altered source patch. Outside the patch the two triangulations and all colors coincide. At each subsequent dual vertex the outgoing edge is uniquely determined by the local colors and the incoming edge. Starting at the same directed edge e^*, the two paths consequently have the same entire suffix. □

Using boundary *vertices* in this formulation avoids an ambiguity of cell-exterior pictures: e is an interior primal edge incident to a boundary vertex, so its dual edge is an ordinary interior edge. Neither a crossing through a dual/primal common vertex nor an unexplained passage around the end of a physical boundary edge is being used.

25.2 Uniform probability of a white arch

Assume a uniform white RSW lower bound for fixed-aspect rectangles of free vertices. No connection to a prescribed individual vertex is assumed. We give the geometry for a horizontal half-plane source, translated to $p = 0$, with black boundary on the negative real side and white boundary on the positive side. Let $r \gg \epsilon + \delta$.

Use a white vertical crossing in each of

$$[-3r, -2r] \times [0, 3r], \qquad [2r, 3r] \times [0, 3r],$$

and a white horizontal crossing of

$$[-3r, 3r] \times [r, 2r].$$

At the bottom of a stem, start the rectangle in the first free vertex row, not in the fixed wired row. Rounding by $O(\delta)$ changes the aspect ratios by a uniformly bounded amount. The crossings in the overlapping squares intersect by planarity. Thus their union contains a white path from a free vertex adjacent to the black boundary to a free vertex adjacent to the white boundary. Add a white–white edge at the white endpoint and choose an edge e to a black boundary vertex at the other endpoint. Trim the white path at its last white-wired-boundary contact before reaching the left endpoint, so that all its subsequent white vertices are interior; then erase loops. This gives a proper simple primal crosscut with just the one heterochromatic edge e.

All these vertices lie, allowing lattice rounding, in $A(0, r/2, 6r)$. The crosscut and the boundary arc through the source enclose the source patch. Its source-side component is inside $B(0, 6r)$: the crosscut and that local boundary arc form a Jordan polygon contained there, so their bounded interior is also contained there. The common half-annulus between the patch and crosscut shows that the entire altered domain portion of the source patch is on that side.

Each rectangle event is increasing in white colors of independent free vertices. RSW and FKG therefore give a constant $\vartheta > 0$, independent of r, δ and the contents of the source patch, for this white-arch event. Importantly, the left endpoint is *some* first-row white vertex adjacent to the black boundary, selected after the crossing occurs. Requiring connection to one predetermined microscopic vertex would not have this uniform lower bound.

Straight slit geometry. The same construction works for a finite-width lattice slit along the negative real axis, with black upper lip and white lower lip, by using five crossings around a square annulus: an upper-left vertical stem, a top horizontal crossing, a right vertical crossing, a bottom horizontal crossing, and a lower-left vertical stem. The stems occupy $[-3r, -2r] \times [0, 3r]$ and $[-3r, -2r] \times [-3r, 0]$ on their respective sides of the slit. The other three rectangles may be

$$[-3r, 3r] \times [2r, 3r], \quad [2r, 3r] \times [-3r, 3r], \quad [-3r, 3r] \times [-3r, -2r].$$

The first free upper row is adjacent to the black wired lip; the first free lower row is adjacent to the white wired lip. The resulting crosscut goes around the right side of the slit and separates its tip patch from the distant target. The boundary arc through the tip, including the two separate lattice lips, stays inside $B(0, 6r)$. The finite-width triangulated domain is Jordan; one need not identify its two physical boundary rows. RSW and FKG again give a uniform $\vartheta > 0$ after adjusting constants.

25.3 Polynomial source-patch stability

Theorem 25.2 (Common-color coalescence outside a source patch)**.** *Under the preceding geometric and percolation hypotheses, there are constants $C < \infty$ and $\alpha > 0$ depending only on the RSW bounds and the fixed geometry such that, for $0 < R < R_0/6$, the common-color coupling of the two full exploration curves satisfies*

$$\mathbb{P}\{d_{\mathrm{F}}(\gamma_1, \gamma_2) > 2R\} \le C \left(\frac{\epsilon + \delta}{R}\right)^{\alpha}. \tag{SH1}$$

More concretely, outside an event of the displayed probability the curves coalesce at a directed dual edge inside $B(p, R)$; both prefixes before that edge stay inside this ball, and the entire suffixes are identical.

Proof. Put $r_0 = 16(\epsilon + \delta)$ and $r_j = 16^j r_0$. For every j with $6r_j \le R$, the corresponding white-arch event depends only on shared free vertices in $A(p, r_j/2, 6r_j)$. These vertex sets are disjoint for different j, up to an immaterial lattice enlargement absorbed by increasing r_0. The events are

therefore independent and each has probability at least ϑ. If there are N available bands, the probability of no shield is at most $(1-\vartheta)^N$.

On any shield event select a white crosscut by a deterministic finite-graph rule using only shared colors, for example a lexicographic simple path in the crossing cluster. The two configurations select the same crosscut and the same single heterochromatic edge. Lemma 25.1 gives a common suffix. Before crossing the shield, each path stays in its source-side component, contained in $B(p, 6r_j) \subset B(p, R)$. Synchronize the two prefixes by any increasing reparameterization and then align their common suffix. The prefixes are in a ball of diameter $2R$, so their oriented Fréchet distance is at most $2R$.

The number of bands is a logarithmic function of $R/(\epsilon+\delta)$. For instance with $\alpha = -\log(1-\vartheta)/\log 16 > 0$, enlarging a fixed constant C gives $(1-\vartheta)^N \le C((\epsilon+\delta)/R)^\alpha$. This also covers the case with no available band by making its right-hand side at least one. □

Passage to marginal curve limits. Suppose along a common mesh subsequence the two exploration laws converge in the usual Polish oriented uniform-Fréchet curve space. The coupled pairs are tight because their two marginals are tight. Choose a joint subsequential limit. The set $\{d_{\mathrm{F}} > 2R\}$ is open in the product curve space; the Portmanteau inequality transfers (SH1), with $\delta \downarrow 0$, to this limiting coupling. Thus the same quantitative source-patch stability holds for the two marginal curve limits. It requires neither convergence of random slit-domain kernels nor a limiting strong DMP theorem.

If source patches themselves shrink and the mesh is negligible relative to their radii, this gives continuity in law for these particular local source perturbations. One may first fix R and let $\epsilon \downarrow 0$, then let $R \downarrow 0$, or balance the coupling estimate to obtain an explicit Prokhorov rate.

What is not proved. This is not a consequence of abstract unforced-crossing estimates for one curve. It uses a common independent coloring and the planar nonbranching interface to force *coalescence at one edge*. It also is not the full marked inverse-conformal-chart stability condition used in the affine-ansatz support theorems: arbitrary converging charts may create forced thin fjords, noncommon microscopic boundaries, or source approaches outside this shared half-plane/slit geometry. Extending the shield coupling to such approximations requires additional work. No classification of arbitrary source-local exploration ensembles follows from (SH1) alone.

25.4 Common curved source annuli and local flattening

The source-shield argument extends to a common curved source neighborhood. Importantly, the two boundaries must still agree outside the altered source patch. This is sufficient to compare a regular source with its local flattening; it is not a comparison of two unrelated approximations of a whole slit.

Here is a precise geometric hypothesis. Let H be an orientation-preserving ambient K-bi-Lipschitz map on a disk about 0, with $H(0) = p$. In that disk the reference domain is either the upper half-plane, with its two colored boundary rays, or the complement of the negative real ray, with distinct colored lips. Outside $B(p, \epsilon)$, the two discrete domains have the same graph, colors, and free vertices, and their common source neighborhood is a compatible lattice approximation of this H-image. By compatible we mean that all ambiguity in discretizing each smooth boundary/lip lies in a fixed $c\delta$ tube, vertices a distance greater than $c\delta$ on the interior side are free, and each boundary contact has the prescribed color of that boundary side. There are no additional boundary components or fjords outside that tube. The domains inside the altered patch can be arbitrary Dobrushin disks, with their sole source color change in the patch. Their common target is outside

the source neighborhood. We use independent Bernoulli colors and uniform two-color RSW for all fixed-aspect rectangles needed below.

Proposition 25.3 (A shield in common bi-Lipschitz source annuli)**.** *Under these hypotheses there are $C_K < \infty$ and $\alpha_K > 0$ such that, for R sufficiently small relative to the common source neighborhood, the common-color coupling satisfies*

$$\mathbb{P}\{d_{\mathrm{F}}(\gamma_1, \gamma_2) > 2R\} \le C_K \left(\frac{\epsilon + \delta}{R}\right)^{\alpha_K}. \tag{SH2}$$

On its good event the two prefixes are contained in $B(p, R)$ and the curves have an exactly common directed suffix beginning at an edge in that ball. Constants also depend on the fixed discretization constant and RSW bounds. The same statement holds for two marginal scaling limits through a joint subsequential coupling, in the precise sense proved below.

Proof. Take the three half-plane crossing corridors, or the five slit crossing corridors, used for the straight-source shield, and apply H. Their dimensions remain comparable to r, their overlap regions contain disks of radius comparable to r with constants depending only on K, and their nonterminal portions stay a distance comparable to r from the boundary. A finite chain of overlapping axis-aligned rectangles of bounded aspect forces each required crossing and connection. The number and aspect bounds depend only on K: follow the image of a corridor's central segment with balls of radius a sufficiently small multiple of r/K, use the bi-Lipschitz bounds to bound both its length and its clearance from the unwanted sides, and insert overlapping rectangles along this chain. RSW and FKG therefore give a uniform lower bound for the connected white crossing configuration.

For the two stems ending at a wired boundary, continue the corridor a distance comparable to r across that boundary and give its exterior vertices, including the actual wired-row vertices, independent auxiliary ghost colors; actual interior free colors are kept. A white crossing of this extended corridor, followed from the interior, has an actual white free path up to the vertex immediately preceding its first wired-boundary contact. This last free vertex is adjacent to the wired vertex. Thus at the black stem one adds precisely one white–black edge, and at the white stem one adds a white–white edge. The continuation crosses only the intended colored boundary side, since the stems are a distance comparable to r from the source and the opposite lip. The $O(\delta)$ boundary tube is absorbed when $r \gg \delta$. All middle portions are interior and use actual common colors. Consequently the ghost crossing event implies the existence of an actual common white arch, and its positive probability lower bound is a lower bound for the latter event. No ghost colors enter the selected actual shield.

Trim at the last white-boundary contact and erase loops as in the straight proof. This gives a simple common primal crosscut with exactly one heterochromatic edge. It separates the entire altered source patch from the target. Its source-side region lies in $H(B(0, 6r))$, hence in $B(p, 6Kr)$, after enlarging constants for lattice rounding. Its supporting free vertices lie in the physical annular shell

$$A(p, r/(2K) - c'\delta, 6Kr + c'\delta).$$

The probability of such a shield is at least a constant $\vartheta_K > 0$.

Start at $r_0 = c''K(\epsilon + \delta)$, with c'' sufficiently large, and use $r_j = \lambda^j r_0$, where $\lambda = 32K^2$ can be enlarged for the fixed discretization constants. The preceding physical shells are disjoint. Shield-existence events use disjoint sets of actual shared free vertices, hence are independent. Every shield with $6Kr_j \le R$ gives the common directed suffix by Lemma 25.1; the two prefixes stay in $B(p, R)$. The probability of no shield is at most $(1 - \vartheta_K)^N$ for a logarithmic number N of available shells. This proves (SH2), for example with $\alpha_K = -\log(1 - \vartheta_K)/\log\lambda$ after changing C_K. □

Application to flattening a regular source. If a Jordan domain has a C^1 embedded boundary near a regular source, an ambient bi-Lipschitz chart H of the stated kind exists after shrinking the neighborhood. Modify only inside $B(p, \epsilon)$ to make a smaller source neighborhood exactly a straight half-plane, preserving the two boundary colors and the target. Choose the two lattice approximations to agree outside the patch. They then satisfy the proposition in the common curved annuli. The estimate compares the original source germ with the flat replacement, despite their different germs. An analogous statement holds for a regular C^1 slit tip with a two-sided tubular chart, provided the common discrete lips have the compatibility stated above. Neither operation presumes conformal invariance.

Scope of the extension. If two slits differ along an entire thin tube of macroscopic length, a tip shield does not isolate all discrepancies: its target-side domain still contains the altered tube farther back. Exact suffix coalescence does not follow from the present argument. In particular (SH2) does not prove straight-tip stability under arbitrary inverse-chart convergence.

25.5 A quantitative input for perturbing a whole straight slit

One can at least exclude three-arm propagation along a finite straight boundary segment. Write $A_3^+(x; r, R)$ for three disjoint alternating free-color arms across the half-annulus based at x on a fixed straight line. The external RSW-and-arm-separation estimate

$$\mathbb{P}(A_3^+(x; r, R)) \le C(r/R)^2 \tag{BA1}$$

holds in critical triangular Bernoulli percolation; see P. Nolin, *Near-critical percolation in two dimensions*, Theorem 23(2), together with half-plane arm separation/quasi-multiplicativity in Section 4.6, https://arxiv.org/abs/0711.4948. It is an external theorem, not a new exponent computation here.

Cover a straight segment of length L by at most $C(1 + L/\epsilon)$ balls of radius ϵ, and let $\mathcal{B}$ be the event that some such ball has three alternating arms in one specified side of the line to distance η, with $\delta \le \epsilon \ll \eta$. Recentring an event at an arbitrary point of the segment to a covering center only changes the inner and outer radii by fixed factors. A union bound in (BA1) gives

$$\mathbb{P}(\mathcal{B}) \le C\frac{(L + \epsilon)\epsilon}{\eta^2}. \tag{BA2}$$

This tends to zero, for example, with $\eta = \epsilon^\beta$ for any $0 < \beta < 1/2$ at fixed L. The two sides of a slit cost only a factor two. This is a concrete useful estimate for boundary pivotal effects away from the two slit endpoints. To turn it into a whole-tube Fréchet coupling, however, one still needs a deterministic implication from a macroscopic ordered-exploration disagreement to one of these three-arm events (or a specified replacement), while controlling the endpoints. That implication has not been proved here; (BA2) alone is not a curve-stability theorem.

26 An affine tensor is readable from a regular endpoint germ

Status. The argument below extends the initial half-plane trace-germ decoder to analytic regular source germs. By ordinary SLE_6 reversibility, it gives a decoder from the ordered terminal trace germ at any analytic regular target, even when the other endpoint and the rest of the domain are rough. It does not decode the unlabelled accessible frontier at a random tip. All almost-sure assertions

are made separately under each fixed affine tensor; no uncountable intersection of full-probability events is used.

Let $\mathcal{X}$ be the usual Polish space of oriented unparameterized continuous finite curves, with the uniform Fréchet metric. Curves are identified under increasing reparameterization, including deletion of pauses. For a continuous path Y starting at 0 and exiting every disk, write $\tau_a(Y)$ for its first hit of $\{|z| = a\}$ and $\mathrm{Stop}_a Y = Y[0, \tau_a(Y)]$. The operations used here are Borel in standard continuous-curve coding; first hits can equivalently be defined by rational-time approximation. For a rough opposite prime end the full curve need not land continuously there. Only finite prefixes near the decoded regular endpoint are coded in $\mathcal{X}$; the unbounded half-plane curves exit every finite disk. This avoids any assumption of a singleton impression at the other mark.

Lemma 26.1 (Unit-radius stopping is almost surely continuous)**.** *Let Y have a law invariant under positive scalar spatial dilation and suppose it is continuous, starts at 0, and exits every disk. Then almost surely*

$$\tau_{1-\delta}(Y) \uparrow \tau_1(Y), \qquad \tau_{1+\delta}(Y) \downarrow \tau_1(Y) \quad (\delta \downarrow 0).$$

Consequently the diameter of $Y[\tau_{1-\delta}, \tau_{1+\delta}]$ tends to zero.

Proof. The first assertion on the left holds pathwise by continuity. The nondecreasing function $a \mapsto \tau_a(Y)$ has only countably many right jumps. Thus the set of radii with $\tau_{a+}(Y) > \tau_a(Y)$ has Lebesgue measure zero for every sample. Spatial dilation invariance implies that the probability of such a jump is the same at every fixed radius. Fubini's theorem on $[1, 2]$ makes this probability zero. Continuity of Y on a compact time interval containing these exit times then gives the diameter assertion. The argument is independent of the chosen continuous strictly increasing curve clock. □

For an orientation-preserving real linear map L carrying $\mathbb{H}$ onto itself, let μ_L be the law of $L\eta$, where η is chordal SLE_6 in $\mathbb{H}$ from 0 to ∞. Scalar multiples of L give the same law. Normalize

$$L = \begin{pmatrix} 1 & u \\ 0 & v \end{pmatrix}, \qquad (u, v) \in \Theta := \mathbb{R} \times (0, \infty). \tag{ED1}$$

The normalized metric represented by this law is $[(L^{-1})^{\mathsf{T}} L^{-1}]$.

Choose a countable bounded Lipschitz determining class $(f_m)_{m \geq 1}$ on $\mathcal{X}$, with values in $[0, 1]$ and Lipschitz constant at most one, allowing constant rescaling of an arbitrary countable bounded Lipschitz determining class. Such a class can be built from truncated distances to a countable dense set and finite rational combinations. Set

$$v_m(L) = \mathbb{E}_{\mu_L} f_m(\mathrm{Stop}_1 Y). \tag{ED2}$$

The vector $v(L) = (v_m(L))_m$ is Borel in L. Indeed, linear image and first-exit stopping are jointly Borel, so integration against the fixed ordinary SLE law preserves Borel measurability.

Brownian spatial scaling is mixing: under the Lamperti transform $Z_s = e^{-s/2} B_{e^s}$ it becomes a shift of a stationary Gaussian process with covariance $e^{-|s-u|/2}$. Its SLE trace factor, and each fixed linear image, are therefore dilation mixing. This gives the ergodicity used below without a separate path-space independence assumption.

The map $L \mapsto v(L)$ is injective. If two unit-radius stopped laws agreed, scaling invariance would identify their stopped laws at every radius. Those restrictions determine the entire ordered unbounded curve, contradicting the half-plane affine identifiability theorem unless the normalized matrices agree. Therefore the Lusin–Souslin theorem makes $v(\Theta)$ Borel and its inverse Borel.

Fix $\rho > 1$ and $r_n = \rho^{-n}$. For a curve γ starting at p define the local observable

$$X_n(\gamma, p) = r_n^{-1}(\gamma[0, \tau_{r_n}(\gamma - p)] - p). \tag{ED3}$$

If a finite curve fails to exit one of the large initial disks, assign a cemetery value to that term. It exits all sufficiently small disks, and finitely many cemetery values do not affect the limits below. Define

$$\ell_m(\gamma, p) = \lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} f_m(X_n(\gamma, p)), \tag{ED4}$$

using a cemetery vector if any limit fails to exist. Each limit is a function of every sufficiently small ordered initial trace prefix: deleting finitely many summands changes no Cesàro limit. In particular this is a Borel ordered-trace-germ statistic.

Lemma 26.2 (The decoder survives a quadratic local error)**.** *Let η be ordinary half-plane SLE_6 and let Φ be a local C^1 embedding at 0, defined on a half-neighborhood of 0, with*

$$\Phi(0) = 0, \qquad \Phi(z) = Lz + O(|z|^2), \tag{ED5}$$

where L is invertible. Assume after a fixed rotation that L maps $\mathbb{H}$ onto itself. Let γ agree with $\Phi(\eta)$ up to the first exit from this local chart. Then almost surely

$$\ell_m(\gamma, 0) = v_m(L) \quad \textit{for every } m. \tag{ED6}$$

Scalar normalization of L is understood in (ED2).

Proof. Put $Y_n = r_n^{-1} L\eta$. Spatial dilation of the trace includes the induced capacity reparameterization. Brownian scaling is mixing; its trace factor and every fixed affine image are therefore dilation ergodic, as proved in the initial half-plane decoder theorem. Hence

$$\frac{1}{N} \sum_{n=1}^{N} f_m(\mathrm{Stop}_1 Y_n) \longrightarrow v_m(L) \quad \text{almost surely.} \tag{ED7}$$

One must control the error in this average; pointwise stopping continuity at all the scaled random samples is not asserted.

For $0 < \delta < 1/4$ define

$$h_\delta(Y) = \min\{2, \operatorname{diam} Y[\tau_{1-\delta}(Y), \tau_{1+\delta}(Y)]\}. \tag{ED8}$$

The linearization error on a fixed bounded scaled chart is at most Cr_n, uniformly over the sample path until its linear image exits the disk of radius $1+\delta$. This uniformity is deterministic: the inverse linear map bounds the relevant preimage disk, and (ED5) bounds the error there. For all sufficiently large n, the first time that $r_n^{-1}\Phi(\eta)$ reaches radius one lies between the corresponding Y_n exit times at radii $1-\delta$ and $1+\delta$. Thus the two stopped curves consist of matched portions with pointwise error at most Cr_n, together with an unmatched terminal portion whose diameter is bounded by the diameter in (ED8). Increasing reparameterizations, with pauses allowed, give

$$|f_m(X_n(\gamma, 0)) - f_m(\mathrm{Stop}_1 Y_n)| \le Cr_n + h_\delta(Y_n). \tag{ED9}$$

The bounded range of f_m justifies capping the diameter by two.

Take a countable sequence $\delta_k \downarrow 0$. Birkhoff's theorem applied to every h_{δ_k} gives

$$\limsup_N \frac{1}{N} \sum_{n=1}^{N} |f_m(X_n(\gamma, 0)) - f_m(\mathrm{Stop}_1 Y_n)| \le \mathbb{E}_{\mu_L} h_{\delta_k}(Y) \tag{ED10}$$

simultaneously for all m, k. Lemma 26.1 and dominated convergence make the right side tend to zero as $k \to \infty$. Combine this with (ED7). Only the initial segment of η enters the argument, since all sufficiently small physical exits occur inside the fixed local chart. This proves (ED6). □

Theorem 26.3 (A universal analytic-endpoint tensor decoder)**.** *There is a Borel rule, depending only on the ordered initial trace germ, the endpoint, and its oriented tangent half-plane, that recovers the projective affine tensor from a single affine-SLE$_6$ sample in every bounded simply connected marked domain with analytic regular source. The rest of its boundary and its target may be rough prime-end data. There is also a terminal version of the rule for every domain with analytic regular target; the source may then be rough. The rule is the same for two domains agreeing near that regular endpoint.*

Proof. Let the law be $(A^{-1})_* \mathrm{SLE}_6(AD, Ap, Aq)$ and assume p is an analytic regular source. A conformal map $f : \mathbb{H} \to AD$ with marks $0, \infty$ extends conformally across 0 by analytic boundary reflection, with $f(0) = Ap$ and $f'(0) \neq 0$. In a physical tangent frame Q carrying the interior tangent half-plane to $\mathbb{H}$, the local map is

$$\Phi(z) = Q(A^{-1} f(z) - p) = Lz + O(|z|^2), \qquad L = QA^{-1} M_{f'(0)}, \tag{ED11}$$

where M_c denotes multiplication by the nonzero complex number c. The linear map L carries $\mathbb{H}$ onto itself and has positive diagonal entries in the boundary frame. Apply Lemma 26.2. The Borel inverse of (ED2) recovers the scalar-normalized L, hence the projective tensor

$$[(L^{-1})^{\mathsf{T}} L^{-1}] = [QA^{\mathsf{T}} AQ^{\mathsf{T}}]. \tag{ED12}$$

Conjugating back by the known orthogonal frame Q recovers $[A^{\mathsf{T}} A]$. The derivative of the conformal map contributes only a positive scalar to this tensor. This rule uses the endpoint germ and tangent half-plane, not the rest of the conformal chart.

Ordinary SLE$_6$ reversibility, already an external theorem used in the research record, says the reversed curve has the same affine-SLE form with marks swapped. Apply the source decoder to that reversed curve at the regular target. Reversibility on prime-end domains follows by conformal transport from the half-plane theorem; only the regular target germ needs a Euclidean analytic boundary extension. □

Corollary 26.4 (Terminal singularity and identifiable mixtures)**.** *For one fixed marked domain with analytic regular target, affine-SLE$_6$ laws with distinct projective affine tensors are mutually singular on the ordered terminal trace-germ sigma-field. More strongly, if π is any Borel probability on normalized affine tensors and $P_\pi = \int P_T \, \pi(dT)$, then the terminal decoder has law π under P_π. Consequently $\pi \mapsto P_\pi$ is injective, and no nontrivial affine mixture can equal one pure affine-SLE law.*

Proof. The same Borel decoder equals T almost surely under P_T. Integrating that equality proves the distribution statement. Distinct pure laws concentrate on disjoint decoder fibers. □

Additional initial-germ obstruction for random mixtures. At an analytic regular source, the initial decoder likewise returns the latent tensor almost surely. Therefore a nontrivial mixture of affine metrics has a nontrivial initial trace-germ sigma-field. At every positive exploration time the metric posterior has already collapsed to a point, even though its prior was nontrivial. This rules out such mixtures under an additional initial-germ zero-one premise. It does not rule out a deterministic source-germ metric assignment; that mechanism still asks whether the current unlabelled frontier recovers the tensor.

Proposition 26.5 (DMP forces restarted mixing measures to be pure)**.** *Suppose an all-state family has the mixed-affine representation $P_S = \int \nu_S^T \, \pi_S(dT)$, with Borel mixing kernel π_S. Start it in a domain with analytic regular source and target. If it has strong DMP, then at each admissible stopping time $\tau > 0$ before the target is reached,*

$$\pi_{S_\tau} = \delta_{D_s(\gamma)} \qquad \text{almost surely,} \tag{ED13}$$

where D_s is the initial tensor decoder. In particular a rule that uses a nondegenerate prior at every restarted state cannot satisfy DMP.

Proof. Sample the latent tensor T with law π_S and then sample the fixed-T affine SLE. The source decoder gives $D_s(\gamma) = T$ almost surely, and this variable is measurable from every positive prefix, including the prefix up to τ. Conditional on this prefix, ordinary fixed-T SLE DMP therefore gives the pure continuation law $\nu_{S_\tau}^{D_s(\gamma)}$. The family's DMP gives the mixed law with mixing measure π_{S_τ}. Since τ precedes reaching the target, the surviving target end has the same analytic regular germ: the compact prefix avoids a sufficiently small target neighborhood. Corollary 26.4 applied to this current state makes the two mixing measures equal, proving (ED13). □

27 Exact suffix coalescence survives curve limits

The source-shield estimate contains more information than a small Fréchet distance. Its exact common suffix is stable under taking a joint curve limit, provided that the prefixes stay in a fixed closed set away from the target. This allows the discontinuous terminal metric decoder to be used without asserting that it is a continuous observable.

Let $K \subset \mathbb{C}$ be closed and consider two oriented continuous curves. Define E_K to be the relation that there exist cut times s_1, s_2 in continuous representatives on $[0,1]$ such that

$$\gamma_i[0, s_i] \subset K \quad (i = 1, 2), \qquad \gamma_1[s_1, 1] = \gamma_2[s_2, 1] \quad \text{as ordered unparameterized curves.} \tag{CB1}$$

The relation is independent of the chosen representatives and allows pauses, as does the usual uniform Fréchet curve space. It implies a common endpoint, which may belong to K in this definition. All applications below have limiting common target $q \notin K$.

Lemma 27.1 (A closed relation of common ordered suffixes)**.** *E_K is closed in the product uniform-Fréchet curve space.*

Proof. Suppose $(\gamma_1^n, \gamma_2^n) \to (\gamma_1, \gamma_2)$ and the pairs belong to E_K. Choose representatives of each convergent sequence converging uniformly to fixed continuous representatives of its limit. This is possible by the definition of the Fréchet metric; reparameterize the cut times along with the curves. Pass to a subsequence on which both cut times s_i^n converge to $s_i \in [0,1]$.

Closedness of K and uniform convergence give $\gamma_i[0, s_i] \subset K$. Linear rescaling of $[s_i^n, 1]$ to $[0,1]$ makes the restricted curves converge uniformly to the restrictions on $[s_i, 1]$, by uniform continuity

of γ_i (a cut at $s_i = 1$ gives a constant curve). The two restrictions at stage n have Fréchet distance zero. Their limits therefore also have distance zero and are the same ordered curve. Thus the limiting pair belongs to E_K. □

Proposition 27.2 (Coalescence transfer and comparison of metric priors)**.** *Suppose pairs of discrete curves with a common target converge marginally to laws P_1, P_2 on the oriented curve space, and their couplings satisfy*

$$\mathbb{P}(E_K) \geq 1 - e_n, \qquad \limsup_n e_n \leq e < 1, \tag{CB2}$$

where K is a fixed closed ball not containing the target. Each limiting domain is analytic and regular near that target, and the two colored target germs agree. Suppose

$$P_i = \int P_{i,T}\, \pi_i(dT), \tag{CB3}$$

where $P_{i,T}$ is the affine-SLE$_6$ law with normalized tensor T in domain i, and π_i are Borel probability measures on tensors. Then, with total variation defined as $\sup_B |\pi_1(B) - \pi_2(B)|$,

$$\|\pi_1 - \pi_2\|_{\mathrm{TV}} \leq e. \tag{CB4}$$

In particular if both marginals are pure affine-SLE laws, any positive uniform lower bound on the discrete coalescence probability forces their tensors to be equal.

Proof. Tightness of the marginals makes the coupled pairs tight. Take a joint subsequential limit. Lemma 27.1 and the closed-set Portmanteau inequality give $\mathbb{P}(E_K) \geq 1 - e$ in the limit. Because the target is outside K, the common suffix on E_K includes a nonconstant ordered terminal germ at that target. The universal terminal decoder of Theorem 26.3 is the same for the two domains, since their colored target germs agree. Let T_i be its value on curve i. The laws of these random variables are π_i, and on E_K they are equal. Hence

$$\sup_B |\mathbb{P}(T_1 \in B) - \mathbb{P}(T_2 \in B)| \leq \mathbb{P}(T_1 \neq T_2) \leq e.$$

For two distinct point masses the left-hand side equals one, proving the last assertion. The proof uses exact suffix equality, not continuity of the decoder and not just small curve distance. □

For the source-shield coupling, take $K = \overline{B}(p, R)$ with the regular target outside this ball. Common discrete targets may vary with mesh, provided $q_n \to q \notin K$; the closed-relation proof already allows this. Equation (SH1) then gives

$$\|\pi_1 - \pi_2\|_{\mathrm{TV}} \leq C(\epsilon/R)^{\alpha} \tag{CB5}$$

for two source-patch perturbations with common flat or straight-slit annuli. This is a quantitative stability statement for mixing measures, and yields exact tensor equality for pure marginals once the right side is less than one. It is conditional on the affine-mixture marginal ansatz; no such ansatz is derived from RSW here.

28 Independent color resampling, initial germs, and a common affine metric

Theorem 28.1 (Initial-germ zero-one law from independent colors and shielding)**.** *Let P be a subsequential exploration-curve limit of the independent color model in a fixed domain with flat*

source germ. Assume the Polish oriented Fréchet curve topology and the flat-source shield estimate of Theorem 25.2. Then the initial ordered trace-germ sigma-field is P-trivial. Precisely, if $\sigma_r\gamma$ denotes the ordered prefix stopped on the radius-r circle about the source, this field is $\bigcap_{r>0}\overline{\sigma(\sigma_r\gamma)}^{P}$. It is the usual ordered positive-prefix germ, with no clock chosen. No affine ansatz, terminal tensor decoder, limiting DMP theorem, or continuum configuration-measurability theorem is required.

More quantitatively, for each sufficiently small r, every bounded observable F of the prefix stopped at radius r and every bounded Lipschitz observable G of the entire curve satisfy

$$|\mathbb{E}_P(FG) - \mathbb{E}_P F\,\mathbb{E}_P G| \le C\|F\|_\infty(\operatorname{Lip} G + \|G\|_\infty)r^{\alpha/(1+\alpha)}, \tag{EDG}$$

first at almost every stopping-continuity radius r. Constants and length units are fixed by the source geometry. The zero-one assertion itself holds on the full initial-germ sigma-field.

Proof. Fix a source-patch radius ϵ and couple two discrete configurations with independent colors inside $B(p,\epsilon)$ and shared independent colors outside. For any fixed $r < \epsilon/3$ and all small enough meshes, the first exploration stopped at radius r is determined entirely by its own interior colors. It is therefore independent of the *entire second exploration*, which uses only the shared outer colors and the second interior color set.

Take a joint subsequential curve limit. Both marginals equal P. Choose an almost-sure stopping-continuity radius $r < \epsilon/3$. The continuous mapping theorem preserves the product law, so the stopped prefix of γ_1 is independent of the full curve γ_2. Consequently any bounded initial-germ observable $F(\gamma_1)$ is independent of $G(\gamma_2)$ for every bounded Lipschitz G on the full curve space. This step concerns the limiting product law; it does not attempt to pass an arbitrary discontinuous germ statistic through a discrete limit.

The shield estimate supplies, in this same coupling, for a fixed small R and $\epsilon \ll R$,

$$\mathbb{P}\{d_{\mathrm{F}}(\gamma_1,\gamma_2) > 2R\} \le C(\epsilon/R)^\alpha.$$

Thus

$$\begin{aligned}|\mathbb{E}_P(FG) - \mathbb{E}_P F\,\mathbb{E}_P G| &= |\mathbb{E}F(\gamma_1)(G(\gamma_1) - G(\gamma_2))| \\ &\le \|F\|_\infty\left(2\operatorname{Lip}(G)R + 2C\|G\|_\infty(\epsilon/R)^\alpha\right).\end{aligned} \tag{EDG1}$$

The left side depends only on the marginal law P. Let $\epsilon \downarrow 0$ with R fixed and then $R \downarrow 0$. It vanishes. Bounded Lipschitz functions determine finite signed measures on a Polish space, so every bounded initial-germ observable is independent of the full curve sigma-field. Since it belongs to that sigma-field, it is almost surely constant. This proves initial trace-germ triviality, including its completion.

For the quantitative prefix assertion, start with a fixed good radius r, use $\epsilon = 4r$, and take $R = r^{\alpha/(1+\alpha)}$ after fixing the reference length unit. For sufficiently small r this is an admissible radius with $\epsilon \ll R$. Formula (EDG1) gives (EDG). □

Proposition 28.2 (Pure affine limit at a flat source)**.** *Let one independent site-percolation model satisfy the flat-source shield coupling of Theorem 25.2. Let D have a flat source germ. Suppose along one mesh subsequence its exploration converges to*

$$P = \int \nu^T_{D,p,q}\,\pi(dT), \tag{ED14}$$

where π is a Borel probability on projective affine tensors. Assume the standard Polish oriented curve topology. Then π is a point mass. No limiting domain Markov theorem or continuum configuration-measurability theorem is needed.

Proof. The regular source endpoint decoder has distribution π under (ED14) and is measurable in the initial trace-germ sigma-field. This sigma-field is trivial by Theorem 28.1. Thus the decoder is almost surely one deterministic tensor, and π is a point mass. The target need only meet the curve-convergence assumptions here; analytic target regularity enters the source-comparison theorem below. □

Lemma 28.3 (Splicing a regular colored source germ)**.** *Fix a bounded reference domain D_0 with flat regular source $p = 0$, analytic regular target q, and one fixed boundary-color convention. Given any analytic regular colored source germ G at 0 with that convention, and any sufficiently small $\epsilon > 0$, there is a bounded marked domain $D_{G,\epsilon}$ agreeing with D_0 outside $B(0, \epsilon)$ and having exactly the germ G at 0. Its target germ at q agrees with D_0. The boundary can be chosen smooth except for any corners already present in D_0; only its two marked germs need to be analytic.*

Proof. For any analytic regular colored source germ G at p and any small ϵ, an orientation-preserving smooth diffeomorphism supported in $B(p, \epsilon)$ can map the source germ of one fixed flat-source reference domain to G, fixing p and the distant regular target. For an explicit local construction, use a radial twist to rotate a small core to the required tangent, followed by a cutoff graph shear to realize the analytic curvature in a still smaller core. The graph derivative can be made arbitrarily small, so the cutoff shear remains a diffeomorphism. Call the resulting domain $D_{G,\epsilon}$. It agrees with the reference outside the source patch and agrees with G in a still smaller neighborhood. The support avoids q, and ambient orientation preserves the boundary color order. Applying the diffeomorphism to D_0 proves the claim. □

Theorem 28.4 (One deterministic metric from a coherent affine mixture)**.** *Consider a translation-covariant source-local exploration family arising from one independent Bernoulli nonbranching cellular percolation model admitting the flat-source shield coupling of Theorem 25.2. Suppose each bounded-domain marginal with analytic regular marks is a Borel mixture of fixed affine-SLE$_6$ laws. Require the family to include the source splices of Lemma 28.3, and require coherent subsequential limits for these comparisons: for every such splice and the reference, compatible common-color discrete approximations along one common mesh subsequence realize the two specified marginal limits.*

Then one deterministic projective affine tensor is shared by every such marginal, and each marginal is the pure affine-SLE$_6$ law for that tensor. Strong DMP and marked inverse-chart continuity are not premises of this theorem.

Proof. The reference mixing measure π_0 is a point mass by Proposition 28.2. Fix another original domain with analytic regular source germ G. Translation covariance allows its source to be placed at 0 without changing its physical projective tensor. For every small ϵ, source locality identifies the original initial prefix law with that in $D_{G,\epsilon}$ through a common source neighborhood. The source tensor decoder is a function of this prefix and the common germ. Its distribution is the mixing measure in either state, so both have one common mixing measure π_G, independent of ϵ.

Coherence and the common-color shield coupling give a coupling of the two limiting curves in $D_{G,\epsilon}$ and D_0 such that their ordered terminal segments agree outside an event of probability at most $C(\epsilon/R)^\alpha$, for one fixed small R in the shared flat outer geometry. Their target germs coincide. Apply the two terminal decoders; the resulting tensor variables have respective laws π_G, π_0 and are equal on the coalescence event. The coupling inequality for total variation therefore gives

$$\|\pi_G - \pi_0\|_{\mathrm{TV}} \le C(\epsilon/R)^\alpha. \tag{ED16}$$

Here total variation is $\sup_B |\pi_G(B) - \pi_0(B)|$. Letting $\epsilon \downarrow 0$ gives $\pi_G = \pi_0$. Since π_0 is a point mass, the original marginal is pure and shares the same tensor. The argument applies separately to each original regular-source state; no simultaneous uncountable full-probability assertion is required. □

Thus, subject to the explicitly stated coherent independent-color scaling and shielding premises, the mixed-affine ansatz on regular domains already collapses to one deterministic common affine metric. This does not derive the mixed-affine ansatz for an arbitrary curve law, and does not classify arbitrary abstract three-axiom families.

Initial characteristic limits become deterministic. At any source to which the preceding zero-one argument applies, a finite initial limit in probability of an adapted characteristic A_t/t or C_t/t is deterministic. To see germ measurability, stop the characteristic at the first exit from an arbitrary source ball. That exit has strictly positive capacity time; the stopped and unstopped limits therefore agree. A limit in probability of variables measurable from the stopped prefix has a measurable version in its completed sigma-field. This holds for every small radius, so the limit belongs to the trivial initial germ. This removes the deterministic-value premise at these starts once existence of a tangent and natural-trace-filtration compatibility are known; neither tangent existence nor the required slit-uniform bound is obtained here.

29 Total-variation decoupling of an initial prefix and a distant suffix

The closed exact-coalescence relation gives a stronger consequence than bounded-Lipschitz covariance. It controls arbitrary measurable observables of the distant terminal suffix in total variation. This uses the actual independent-color source-shield coupling, not just abstract locality or an unforced-crossing bound.

Fix a bounded domain with source p, target q, flat source geometry, and a marginal curve scaling limit P satisfying the hypotheses of Theorem 28.1. Let $\sigma_r\gamma$ be the initial ordered prefix stopped on the radius-r circle about p. For $R < |q-p|$ define $\lambda_R\gamma$ to be the ordered terminal suffix beginning at the *last* visit to $\overline{B}(p,R)$. For any continuous representative on $[0,1]$ this means restriction from

$$t_R(\gamma) = \max\{t \in [0,1] : |\gamma(t) - p| \le R\}$$

to 1, followed by linear reparameterization. The resulting ordered curve does not depend on the representative or pauses.

Both stopping operations are Borel maps on the assumed Polish oriented Fréchet curve space. For completeness, on the continuous-parameter curve space, t_R is Borel because, for fixed a,

$$\{t_R \ge a\} = \left\{ \min_{a \le t \le 1} |\gamma(t) - p| \le R \right\}$$

is closed. Restriction and rescaling are jointly continuous in a curve and its cut parameter. Hence the terminal-suffix map to the quotient curve space is Borel on representatives and invariant under reparameterization. It descends to a Borel map on the Polish quotient: the preimages of Borel sets are saturated Borel sets, whose images and complements under the continuous quotient map are analytic; both are therefore Borel. First-exit stopping has the same Borel property.

Theorem 29.1 (Prefix–suffix decoupling)**.** *There exist $C < \infty$ and $\alpha > 0$, determined by the source-shield constants, such that for all sufficiently small R and all $0 < r < R/64$,*

$$\|\mathcal{L}_P(\sigma_r\gamma, \lambda_R\gamma) - \mathcal{L}_P(\sigma_r\gamma) \otimes \mathcal{L}_P(\lambda_R\gamma)\|_{\mathrm{TV}} \le C(r/R)^\alpha. \tag{SB1}$$

Here total variation is $\sup_A |\mu(A) - \nu(A)|$ *on the product space. In particular this estimate holds at* every *small radius* r*, not merely almost every stopping-continuity radius.*

Proof. Set $\epsilon = 8r$. At the discrete level take two configurations with independent colors in $B(p, \epsilon)$ and identical independent colors outside that ball. Both marginal explorations have the same scaling limit P. Their joint laws are tight. Choose a radius $\rho \in (r, 2r)$ at which first-exit stopping is P-almost surely continuous; such radii have full Lebesgue measure by Lemma 11.1. The first discrete exploration up to radius ρ uses only the first configuration's interior colors, once the mesh is sufficiently small. It is consequently independent of the entire second exploration. Continuous mapping along a joint subsequential limit gives

$$\sigma_\rho \gamma_1 \perp\!\!\!\perp \gamma_2.$$

Stopping the limiting first prefix again, now at radius r, is a Borel operation and requires no continuity assertion at r. Thus $\sigma_r \gamma_1$ is independent of the full curve γ_2.

Apply the source-shield estimate with shield radius $R/2$. Outside an event of probability at most $C(r/R)^\alpha$, both prefixes before coalescence stay in $K = \overline{B}(p, R/2)$ and the ordered suffixes are identical. Lemma 27.1 and the closed-set Portmanteau inequality preserve this statement in the same joint subsequential limit. Since $q \notin \overline{B}(p, R)$, each full curve's last visit to $\overline{B}(p, R)$ is on the common suffix. Therefore

$$\mathbb{P}\{\lambda_R \gamma_1 \neq \lambda_R \gamma_2\} \leq C(r/R)^\alpha.$$

The pairs $(\sigma_r \gamma_1, \lambda_R \gamma_1)$ and $(\sigma_r \gamma_1, \lambda_R \gamma_2)$ form a coupling of the joint law in (SB1) with the displayed product law. The usual coupling bound for total variation proves (SB1). □

Corollary 29.2 (Expected conditional total-variation stability)**.** *Let* $K_r(x, \cdot)$ *be a regular conditional law of* $\lambda_R \gamma$ *given* $\sigma_r \gamma = x$*. Then*

$$\int \|K_r(x, \cdot) - \mathcal{L}_P(\lambda_R \gamma)\|_{\mathrm{TV}} \, \mathcal{L}_P(\sigma_r \gamma)(dx) \leq C(r/R)^\alpha. \tag{SB2}$$

Consequently for bounded Borel functions F, G *of the two respective variables,*

$$|\operatorname{Cov}_P(F(\sigma_r \gamma), G(\lambda_R \gamma))| \leq 2C \|F\|_\infty \|G\|_\infty (r/R)^\alpha. \tag{SB3}$$

Proof. On standard Borel spaces, total variation between a joint law and the product of its marginals is the integral of the total variations of the corresponding regular conditional laws. One may verify the identity using a countable generating algebra to compute the variation of each signed conditional measure. This gives (SB2). Integrating the bounded product observable against the difference of the two probability laws costs at most twice its supremum norm times the convention for total variation used in (SB1), proving (SB3). □

Variants and limitation. The same proof applies to any standard-Borel-valued observable Y_R that agrees whenever two curves belong to $E_{\overline{B}(p,R/2)}$. For example one may use the compact trace outside the open radius-R ball instead of the ordered distant suffix. Measurability of a given variant should be checked separately. Completed prefix sigma-fields are harmless, since versions may be chosen modulo their common marginal null sets.

This is quantitative spatial loss of memory at a deterministic regular source. It does not identify a conditional continuation kernel in an arbitrary random slit domain: the proof has no shield estimate uniform over rough explored frontiers. In particular it neither proves strong DMP in a curve limit nor the whole-slit stability premise of the driving-rigidity results. The whole-tube approach based on boundary three-arm estimates still needs the deterministic implication from ordered curve disagreement to an appropriate arm event.

30 Annular crossing statistics self-average at a regular source

The same source resampling argument applies to a macroscopic observable whenever the closed exact-coalescence relation forces its equality. It therefore controls annular crossings made *before the first outer exit*, even if the full curve later returns arbitrarily close to the source. No source nonreturn assumption is needed.

Proposition 30.1 (Decoupling for coalescence-invariant observations)**.** *Under the hypotheses of Theorem 29.1, let Y_R be a Borel map from the oriented curve space into a standard Borel space. Assume*

$$(\gamma, \widetilde{\gamma}) \in E_{\overline{B}(p,R/2)} \quad \Longrightarrow \quad Y_R(\gamma) = Y_R(\widetilde{\gamma}). \tag{AS1}$$

Then, for $0 < r < R/64$,

$$\big\| \mathcal{L}_P(\sigma_r\gamma, Y_R(\gamma)) - \mathcal{L}_P(\sigma_r\gamma) \otimes \mathcal{L}_P(Y_R(\gamma)) \big\|_{\mathrm{TV}} \le C(r/R)^{\alpha}. \tag{AS2}$$

The same conclusion holds if the first variable is any Borel function of $\sigma_r\gamma$.

Proof. Use exactly the independent-inner/shared-outer coupling in the proof of Theorem 29.1, including a good stopping-continuity radius between r and $2r$. Its limiting curves obey $\sigma_r\gamma_1 \perp\!\!\!\perp \gamma_2$ and

$$\mathbb{P}(E_{\overline{B}(p,R/2)}) \ge 1 - C(r/R)^{\alpha}.$$

Assumption (AS1) implies $\mathbb{P}\{Y_R(\gamma_1) \ne Y_R(\gamma_2)\} \le C(r/R)^{\alpha}$. Couple the two pairs with common first variable $\sigma_r\gamma_1$, as in the earlier proof. The second pair has the product law by independence. This proves (AS2). Applying a Borel map to the first variable cannot increase total variation. □

Fix $a > 0$ sufficiently small that the source-shield construction applies up to scale a, and $|q - p| > a$. Put

$$r_n = a2^{-n}, \qquad n = 1, 2, \dots.$$

For a continuous representative of γ define

$$\tau_n = \inf\{t : |\gamma(t) - p| = 2r_n\}, \qquad \eta_n = \max\{t \le \tau_n : |\gamma(t) - p| \le r_n\}. \tag{AS3}$$

Let $B_n(\gamma)$ be the ordered restriction to $[\eta_n, \tau_n]$, translated by $-p$ and dilated by r_n^{-1}. This is an oriented curve in $\{1 \le |z| \le 2\}$ from its inner to its outer boundary. It is the last inner-to-outer crossing before the first outer hit. Only this crossing is recorded; later returns of the full curve are immaterial.

The map B_n is Borel and invariant under changes of representative. Indeed τ_n is a Borel first-hitting operation. Restrict first to $[0, \tau_n]$, linearly reparameterized on $[0, 1]$. On this stopped curve, η_n is a Borel last visit to the closed inner ball, by the last-visit argument in Section 29. Successive restriction, translation, dilation, and passage to the oriented quotient are Borel. In particular B_n is a Borel function of $\sigma_{2r_n}\gamma$.

There is a second property crucial to the proof: if $(\gamma, \widetilde{\gamma}) \in E_{\overline{B}(p,r_n/2)}$, then $B_n(\gamma) = B_n(\widetilde{\gamma})$. The prefixes before the common suffix remain strictly inside the inner circle, and the common suffix starts there. Both first outer hits thus occur on that suffix, and so do their last inner-ball visits before those hits. The corresponding ordered crossing segments agree exactly.

Theorem 30.2 (Exponential absolute regularity across dyadic source scales)**.** *Under the same source-shield and independent-color limit hypotheses, the annular crossing sequence $(B_n)_{n\ge1}$ satisfies*

$$\begin{aligned} &\big\| \mathcal{L}_P((B_1, \dots, B_m), (B_{m+k}, B_{m+k+1}, \dots)) \\ &\qquad - \mathcal{L}_P(B_1, \dots, B_m) \otimes \mathcal{L}_P(B_{m+k}, B_{m+k+1}, \dots) \big\|_{\mathrm{TV}} \le C_0 2^{-\alpha k}, \qquad m, k \ge 1. \end{aligned} \tag{AS4}$$

The countable product on the fine-scale side is a standard Borel space. Constants are independent of m and k. No stationarity or dilation covariance of the marginal curve limit is assumed.

Proof. For $k \geq 8$, apply Proposition 30.1 with $R = r_m$ and $r = 2r_{m+k}$. The vector $Y_R = (B_1, \ldots, B_m)$ satisfies (AS1): every one of its inner radii is at least r_m, so coalescence inside the radius-$r_m/2$ ball occurs before all the relevant inner and outer visits. The whole countable vector $(B_{m+k}, B_{m+k+1}, \ldots)$ is a Borel function of $\sigma_{2r_{m+k}}\gamma$, because every corresponding outer radius is at most $2r_{m+k}$. Thus (AS2) gives the upper bound

$$C(2r_{m+k}/r_m)^\alpha = C2^\alpha 2^{-\alpha k}.$$

Enlarge C_0 to cover $1 \leq k < 8$ by the trivial total-variation bound one. Interchanging the two coordinates of a joint law preserves total variation, giving (AS4) in the displayed order. □

Corollary 30.3 (A strong law for bounded annular observations)**.** *Let f_n be any deterministic sequence of Borel real functions of an annular oriented crossing, with $|f_n| \leq 1$. Put $Z_n = f_n(B_n)$. Then*

$$\frac{1}{N}\sum_{n=1}^{N}\{Z_n - \mathbb{E}_P Z_n\} \longrightarrow 0 \qquad P\text{-almost surely.} \tag{AS5}$$

In particular, for a fixed bounded Borel f, if the deterministic Cesàro means

$$\frac{1}{N}\sum_{n=1}^{N}\mathbb{E}_P f(B_n)$$

converge to c_f, then the empirical means $N^{-1}\sum_{n\leq N} f(B_n)$ converge almost surely to c_f. These assertions hold simultaneously for any prescribed countable collection of choices of the sequence (f_n).

Proof. The total-variation estimate (AS4) and $|Z_n| \leq 1$ imply

$$|\operatorname{Cov}_P(Z_n, Z_j)| \leq 2C_0 2^{-\alpha|n-j|} \quad (n \neq j).$$

Consequently, for $S_N = \sum_{n\leq N}(Z_n - \mathbb{E}_P Z_n)$,

$$\mathbb{E}_P S_N^2 \leq C_1 N.$$

Chebyshev's inequality and Borel–Cantelli give $S_{\ell^2}/\ell^2 \to 0$ almost surely, since for every $\zeta > 0$,

$$\sum_{\ell=1}^{\infty}\mathbb{P}\{|S_{\ell^2}| > \zeta\ell^2\} \leq \frac{C_1}{\zeta^2}\sum_{\ell=1}^{\infty}\ell^{-2} < \infty.$$

For $\ell^2 \leq N < (\ell+1)^2$, boundedness of the centered summands gives $|S_N - S_{\ell^2}| \leq 2(2\ell+1)$. Dividing by N yields (AS5). Add the deterministic expectations for the second assertion, and intersect countably many probability-one events for the last one. □

Meaning and remaining gap. Annular crossing statistics of one limiting exploration self-average over logarithmically many source scales. The centering sequence is deterministic, even when the scaling limit itself has not been shown dilation invariant. Neither the convergence of those deterministic expectations nor their values is asserted. A source law that is invariant under dyadic dilation would make the expectations constant, but such invariance is an additional premise and is not inferred from precompactness of the collection of scaling limits. The result supplies a rigorous scale-statistics diagnostic without an affine-SLE ansatz; it does not classify those deterministic statistics or produce random slit-domain exploration kernels.

30.1 A martingale approximation and a variance criterion for Gaussian scale fluctuations

The exponential absolute regularity in (AS4) yields more than the centered strong law. The following statements require no stationarity or dilation covariance. Their time variable counts source scales, rather than Loewner capacity.

Let $Z_j = f_j(B_j)$ with deterministic Borel $|f_j| \le 1$, put $Y_j = Z_j - \mathbb{E}Z_j$, and write $\mathcal{A}_j = \sigma(B_1, \dots, B_j)$, with $\mathcal{A}_0$ trivial. Then $|Y_j| \le 2$. For constants $C, c > 0$ depending only on (AS4),

$$\|\mathbb{E}[Y_{j+k} \mid \mathcal{A}_j]\|_1 \le Ce^{-ck}, \qquad j \ge 0,\ k \ge 1. \tag{AM1}$$

Indeed, test the joint law of the coarse vector and Y_{j+k} against a bounded sign of this conditional expectation times Y_{j+k}; the product-law expectation vanishes. Boundedness then gives, for every real $p \ge 1$,

$$\|\mathbb{E}[Y_{j+k} \mid \mathcal{A}_j]\|_p \le C_1 e^{-ck/p}. \tag{AM2}$$

Here C_1 can be chosen independent of p, by $\|V\|_p^p \le 2^{p-1}\|V\|_1$ when $|V| \le 2$.

Proposition 30.4 (An exponentially integrable annular corrector)**.** *Under (AS4), define in L^p, for any finite $p \ge 1$,*

$$U_j = \sum_{k=1}^{\infty} \mathbb{E}[Y_{j+k} \mid \mathcal{A}_j], \qquad U_0 = 0, \qquad D_j = Y_j + U_j - U_{j-1}. \tag{AM3}$$

The definitions give common versions for all finite p and all integer j. The process $M_N = \sum_{j=1}^{N} D_j$ is a square-integrable $\mathcal{A}_N$-martingale and

$$S_N := \sum_{j=1}^{N} Y_j = M_N - U_N. \tag{AM4}$$

For constants $a, C_2 > 0$,

$$\sup_j \mathbb{E}e^{a|U_j|} < \infty, \qquad \sup_j \mathbb{E}e^{a|D_j|} < \infty, \qquad \max_{0 \le j \le N} |U_j| = O(\log N) \quad \textit{almost surely.} \tag{AM5}$$

Proof. Minkowski's inequality and (AM2) imply

$$\|U_j\|_p \le C_1 \sum_{k \ge 1} e^{-ck/p} = \frac{C_1}{e^{c/p} - 1} \le \frac{C_1}{c} p.$$

Thus the series converges in each L^p, with the same L^1 limit. The tower property and termwise conditioning, justified by L^1 convergence, give

$$\mathbb{E}[Y_j + U_j \mid \mathcal{A}_{j-1}] = U_{j-1},$$

so the D_j are martingale differences. Telescoping gives (AM4). The displayed L^p bound and $\|D_j\|_p \le 2 + \|U_j\|_p + \|U_{j-1}\|_p$ give uniform bounds $\|U_j\|_p + \|D_j\|_p \le C_3 p$. Expanding the exponential into its power series and using $m! \ge (m/e)^m$ proves the two exponential-moment assertions for sufficiently small $a > 0$. In particular $\mathbb{P}(|U_j| > x) \le C_4 e^{-ax}$ uniformly in j. The union bound gives

$$\mathbb{P}\left(\max_{j \le N} |U_j| > A \log N\right) \le C_4 N^{1-aA}.$$

Choose $aA > 2$ and sum over integers $N \ge 2$. Borel–Cantelli proves the last assertion of (AM5). □

Deterministic and conditional variance are distinct. Orthogonality of martingale differences and (AM5) show

$$\left|\operatorname{Var}(S_N) - \sum_{j\le N} \mathbb{E}D_j^2\right| = O(\sqrt{N}). \tag{AM6}$$

A martingale central limit theorem would normally ask for convergence of conditional variance averages. For the original bounded annular sequence, absolute regularity allows a direct blocking proof using only a deterministic variance limit, as follows.

Theorem 30.5 (A deterministic variance criterion for an annular invariance principle)**.** *Suppose (AS4) holds and, for a deterministic $\sigma^2 \ge 0$,*

$$\frac{1}{N}\operatorname{Var}(S_N) \longrightarrow \sigma^2. \tag{AM7}$$

Let $\mathcal{S}_N$ be the continuous polygonal interpolation with $\mathcal{S}_N(j/N) = S_j/\sqrt{N}$. On every fixed compact time interval,

$$\mathcal{S}_N \implies \sigma B \quad \textit{in the uniform topology.} \tag{AM8}$$

Consequently $S_N/\sqrt{N}$ tends in law to a centered Gaussian of variance σ^2. Condition (AM7) is an additional deterministic premise; it is not inferred from RSW or (AS4).

Proof. We give a blocking and moment proof to separate (AM7) from a hidden stationarity or conditional-variance assumption. The covariance estimate from (AS4) gives

$$|\operatorname{Cov}(Y_i, Y_j)| \le Ce^{-c|i-j|}, \qquad i \ne j. \tag{AM9}$$

Thus the variance of a sum over any finite subset of indices is at most a constant times its cardinality. The covariance of sums over any two disjoint consecutive intervals, in either order, is uniformly bounded: sum the geometric bound over the two distances from their common boundary.

First consider a centered bounded triangular array whose row is a deterministic bounded weighting of $O(N)$ successive variables Y_j and whose row-sum variance divided by N has a limit v. The absolute-regularity and covariance bounds remain uniform. Split the row into big blocks of length $p_N = \lfloor N^{1/3} \rfloor$, separated by gaps of length $q_N = \lceil A \log N \rceil$, and omit the final remainder. The number of omitted indices is $O(Nq_N/p_N + p_N + q_N) = o(N)$, so their sum divided by $\sqrt{N}$ tends to zero in L^2.

The total-variation distance between the joint big-block law and the product of its block marginals is at most $C(N/p_N)e^{-cq_N} = o(1)$ when A is sufficiently large. To see this, separate the final block from the entire preceding block vector using (AS4), and iterate; total variation adds under this telescoping argument. For independent blocks with these marginals, the sum of block variances divided by N tends to v. Indeed replacing the omitted variables changes the original variance by $o(N)$, and the sum of cross-block covariances is at most $CNp_Ne^{-cq_N} = o(N)$, by (AM9). Each independent block divided by $\sqrt{N}$ is deterministically bounded in magnitude by $Cp_N/\sqrt{N} = o(1)$. Taylor expansion of its characteristic function through second order has total remainder bounded by this maximum times the sum of the normalized block variances. Taking the product of characteristic functions therefore gives the limit $e^{-v\theta^2/2}$. The total-variation approximation and the omitted L^2 remainder prove the same Gaussian limit for the original weighted row. This argument also covers $v = 0$; alternatively that case follows from Chebyshev's inequality.

Apply this triangular-array statement to every fixed linear combination of increments of $S_{\lfloor Nt \rfloor}/\sqrt{N}$. Assumption (AM7) gives $N^{-1}\operatorname{Var}(S_{\lfloor Nt \rfloor}) \to \sigma^2 t$. The uniform bound on covariances between consecutive interval sums then shows that a linear combination with increment

coefficients a_ℓ and time intervals $(t_{\ell-1}, t_\ell]$ has variance limit $\sigma^2 \sum_\ell a_\ell^2 (t_\ell - t_{\ell-1})$. Cramér–Wold thus gives the Brownian finite-dimensional limits.

For tightness, write an ordered quadruple of indices as $i \le i+a \le i+a+b \le i+a+b+d$. Its centered-product expectation is bounded in absolute value by

$$Ce^{-c\max(a,d)}. \tag{AM10}$$

When $a > 0$, separate the first centered factor from the other three using (AS4); when $d > 0$, separate the last centered factor instead. Use the stronger of these two estimates, or the trivial boundedness estimate if both gaps vanish. This argument permits repeated middle indices and repeated endpoint pairs. Summing (AM10) over quadruples in any interval of m indices gives

$$\mathbb{E}\left|\sum_{j=r+1}^{r+m} Y_j\right|^4 \le C_5 m^2, \tag{AM11}$$

because $\sum_{a,d\ge 0} e^{-c\max(a,d)} < \infty$ while the starting index and the middle gap have $O(m^2)$ choices. For the polygonal process, (AM11) gives $\mathbb{E}|\mathcal{S}_N(t) - \mathcal{S}_N(s)|^4 \le C_6 |t-s|^2$ when $|t-s| \ge N^{-1}$: the two fractional endpoint pieces have total magnitude at most $4/\sqrt{N}$, which is absorbed in this fourth-moment bound. For shorter intervals use the deterministic slope bound $2\sqrt{N}$, giving the same estimate. The usual Kolmogorov tightness criterion, together with $\mathcal{S}_N(0) = 0$, proves tightness in the uniform topology. Combining this with the finite-dimensional limits proves (AM8). □

A two-scale criterion for the variance limit. As a more explicit sufficient target, suppose every fixed lag $k \ge 0$ has a deterministic Cesàro covariance limit

$$c_k = \lim_{N\to\infty} \frac{1}{N} \sum_{j=1}^{N-k} \operatorname{Cov}(Y_j, Y_{j+k}).$$

Then (AM9) and dominated summation imply (AM7), with the absolutely convergent formula

$$\sigma^2 = c_0 + 2\sum_{k=1}^{\infty} c_k. \tag{AM12}$$

This provides a condition in terms of two-scale expectations, without requiring exact stationarity. No convergence of these expectations is asserted for the percolation scaling limits under study.

The variance premise cannot be dropped. Even independent bounded observations can have no asymptotic variance rate. For example let $Y_j = a_j \varepsilon_j$, with independent symmetric signs ε_j and deterministic $a_j \in \{0, 1\}$. Choose alternating 0 and 1 blocks whose successive lengths dominate all previous lengths. Then (AS4) holds with zero error, but $N^{-1}\operatorname{Var}(S_N) = N^{-1}\sum_{j\le N} a_j$ has subsequential limits 0 and 1. Thus the new invariance principle isolates another deterministic scale-statistics limit; it does not identify universal expectations or fluctuation variances and does not control Loewner-capacity increments.

31 An isotropic whole-plane law identifies its affine tensor

Lemma 31.1 (Only similarities stabilize an isotropic scale-invariant curve law)**.** *Let μ be the law of an almost surely nonconstant oriented continuous curve starting at 0, with compact continuous*

prefixes up to every finite spatial exit. Suppose its law is invariant under every rotation and every positive scalar dilation. If $L \in GL^+(2,\mathbb{R})$ preserves μ, then L is a similarity. Consequently

$$A_*\mu = B_*\mu \quad \Longleftrightarrow \quad AA^{\mathsf{T}} = cBB^{\mathsf{T}} \text{ for some } c > 0. \tag{WP1}$$

Proof. If L is not a similarity, singular value decomposition, together with the rotation and scalar symmetries, makes $D = \operatorname{diag}(1,a)$, $0 < a < 1$, another symmetry of μ. For $Y \sim \mu$, the law of $X = \sup|\operatorname{Re} Y|$ is invariant under all positive scalar multiplications; hence $X \in \{0,\infty\}$ almost surely. The event $X = 0$ says the nonconstant curve is contained in one specified line through 0. Its images under countably many distinct rotations are disjoint and have equal probabilities, so each has probability zero. Thus $X = \infty$ almost surely. In particular the first time T with $|\operatorname{Re} Y(T)| = 1$ is finite almost surely, and $M = \sup_{s\le T}|\operatorname{Im} Y(s)| < \infty$.

The first unit-radius exit of D^nY occurs by T; its initial trace stays within distance a^nM of the real axis. Since D^nY has law μ, for every $\epsilon > 0$ the probability that the unit-radius stopped trace leaves the ϵ-strip is at most $\mathbb{P}(a^nM > \epsilon) \to 0$. Thus the unit-radius initial trace lies on the real axis almost surely. Rotation invariance also puts it on the imaginary axis almost surely, a contradiction. Finally $A_*\mu = B_*\mu$ is equivalent to $B^{-1}A$ stabilizing μ, and a direct matrix calculation gives (WP1). □

Proposition 31.2 (A conditional whole-plane initial-germ decoder)**.** *Suppose additionally that the curve space is Polish, affine image and first-radius stopping are jointly Borel, and spatial dilation by one fixed factor $\rho > 1$ is ergodic under μ. Then there is one Borel initial ordered trace-germ decoder D such that, for every fixed $A \in GL^+(2,\mathbb{R})$,*

$$A_*\mu\left\{D(Y) = [(A^{-1})^{\mathsf{T}}A^{-1}]\right\} = 1. \tag{WP2}$$

No tangent half-plane or explicit SLE passage formula is needed. This first version concerns a fixed deterministic affine map. Theorem 31.6 below gives a stronger simultaneous statement for sample-dependent local differentiable maps.

Proof. Parameterize the tensors by positive definite determinant-one matrices T, choosing $A_T = T^{-1/2}$. Rotation and scalar invariance imply that every affine-image law has exactly this representation. Take a countable bounded determining family (f_m) for unit-radius stopped curves, and set $v_m(T) = \mathbb{E}_{(A_T)_*\mu} f_m(\operatorname{Stop}_1 Y)$. This vector is Borel by the assumed joint coding. It is injective: equal unit-radius laws give equal stopped laws at every radius by dilation invariance, then equal full laws, and hence equal T by Lemma 31.1. Lusin–Souslin gives a Borel inverse on the image of v.

Under any fixed $(A_T)_*\mu$, Birkhoff averages of $f_m(\operatorname{Stop}_1(\rho^nY))$ tend almost surely to $v_m(T)$. Dilation ergodicity transfers because affine image commutes with scalar dilation. Each sufficiently late term uses only an arbitrarily small initial prefix of Y. The vector of limiting averages is therefore germ measurable. Apply the Borel inverse of v when the vector lies in its image, and otherwise use a cemetery value. This proves (WP2). □

31.1 A decoder stable under random local differentiable maps

The fixed-affine decoder can be strengthened so that its exceptional set does not depend on the local coordinate map. The useful change is to average whole curves in the Riemann sphere before extracting the tensor; first-radius stopping is then used only for Borel germ coding, not for continuity of an averaging observable.

We make an additional explicit properness and coding assumption. Let $\mathcal{C}$ be a Polish space of oriented continuous curves in the Riemann sphere from 0 to ∞, modulo increasing reparameterization

and pauses, with bounded spherical Fréchet metric $d_{\mathcal{C}}$, induced by the chordal metric on the sphere. Assume the affine action and scalar dilation are continuous on this space and first-radius stopping is Borel. Require the proper and source-avoiding curve sets to be Borel in this coding. These are properties of the coding, not consequences of the exploration axioms. A curve is *proper* here if it has a continuous planar representative $y : [0, \infty) \to \mathbb{C}$ with $|y(t)| \to \infty$ as $t \to \infty$.

Lemma 31.3 (No later visit to the source)**.** *Let μ be a dilation-invariant law supported on proper curves from 0 to ∞. Then almost surely the curve has no visit to 0 after it first leaves 0.*

Proof. For $R > 0$, let E_R be the event of a visit to 0 after the first radius-R exit. Dilation invariance makes $\mu(E_R)$ independent of R. The events decrease as R increases. For every proper continuous curve, all visits to 0 occur in a finite time interval, whose initial trace has finite radius. Thus $\bigcap_{R>0} E_R$ is empty on the proper-curve set, and $\mu(E_R) = 0$ for every R. A nontrivial later visit to 0 belongs to E_R for some positive rational R. □

Lemma 31.4 (Uniform spherical linearization)**.** *Let ψ be an orientation-preserving C^1 diffeomorphism on a neighborhood of 0, with $\psi(0) = 0$ and derivative $L \in GL^+(2, \mathbb{R})$. It has a global bi-Lipschitz extension Ψ agreeing with ψ on a smaller neighborhood and fixing ∞, such that*

$$\sup_{z \in \widehat{\mathbb{C}}} d_{\mathrm{sph}}\big(\lambda \Psi(z), \lambda L z\big) \longrightarrow 0 \quad (\lambda \to \infty). \tag{WP3}$$

Consequently the same upper bound controls $d_{\mathcal{C}}(\lambda \Psi Y, \lambda L Y)$ for every curve Y.

Proof. Write $\psi(z) = Lz + r(z)$, where $r(0) = 0$ and $Dr(z) \to 0$. Choose a smooth radial cutoff χ_δ, equal to one in $B(0, \delta)$ and zero outside $B(0, 2\delta)$, and set $\Psi(z) = Lz + \chi_\delta(z) r(z)$, with the perturbation zero outside the chart. For sufficiently small δ the Lipschitz norm of the perturbation is less than half the smallest singular value of L. Hence Ψ is globally bi-Lipschitz. Surjectivity follows, for example, from the contraction mapping applied to $z = L^{-1}(w - \chi_\delta(z) r(z))$ for each w.

There is $c > 0$ with $|\Psi(z)| \geq c|z|$ and $|Lz| \geq c|z|$. Given $\eta > 0$, choose $a > 0$ such that $|\Psi(z) - Lz| \leq \eta |z|$ for $|z| \leq a$. The chordal formula (our convention for d_{sph})

$$d_{\mathrm{sph}}(u, v) = \frac{|u - v|}{\sqrt{1 + |u|^2}\sqrt{1 + |v|^2}}$$

then bounds the distance in (WP3), for $|z| \leq a$, by

$$\frac{\eta \lambda |z|}{1 + c^2 \lambda^2 |z|^2} \leq \frac{\eta}{2c}.$$

For $|z| \geq a$, the triangle inequality through ∞ bounds the distance by $2/(c\lambda a)$. First let $\lambda \to \infty$ and then $\eta \downarrow 0$. Pairing identical curve parameters gives the Fréchet bound. □

Lemma 31.5 (Changing a distant tail is invisible after magnification)**.** *Suppose two proper source-avoiding curves share a nonconstant initial ordered segment. Then*

$$d_{\mathcal{C}}(\lambda Y, \lambda \widetilde{Y}) \longrightarrow 0 \quad (\lambda \to \infty). \tag{WP4}$$

Here source avoidance means that neither curve visits 0 after its initial departure.

Proof. Synchronize parameters along a common initial segment ending at a nonzero point. Each remaining tail is a compact subset of the sphere disjoint from 0; its finite planar points therefore have modulus at least some positive b. After multiplication by λ, both tails lie within chordal distance $1/(\lambda b)$ of ∞. Match their remaining parameters arbitrarily. The common prefixes agree exactly and the two tails have mutual distance at most $2/(\lambda b)$. □

Theorem 31.6 (A robust affine decoder on initial ordered germs)**.** *Let μ be a nonconstant rotation- and dilation-invariant law on $\mathcal{C}$, supported on proper curves, and suppose dilation by one factor $\rho > 1$ is ergodic. There is one Borel tensor-valued decoder D, measurable from every positive-radius initial prefix on the proper source-avoiding curve set, with the following simultaneous property. There is a single μ-conull set $\mathcal{G}$ such that, for every $Y \in \mathcal{G}$, every $A \in GL^+(2,\mathbb{R})$, and every local orientation-preserving C^1 diffeomorphism ψ fixing 0, with derivative L,*

$$D(\psi(AY)) = [((LA)^{-1})^{\mathsf{T}}(LA)^{-1}]. \tag{WP5}$$

The argument of D can be any proper source-avoiding completion of the indicated local curve segment; the value is independent of the completion. In particular A and ψ may depend arbitrarily on Y and on additional randomness, provided the displayed local regularity holds for each realization.

Proof. Let T act by multiplication by ρ, and form empirical orbit measures in the Polish probability-measure space $\mathcal{P}(\mathcal{C})$:

$$\Pi_N(Y) = \frac{1}{N}\sum_{n=1}^{N} \delta_{T^n Y}.$$

The ergodic theorem applied to a countable bounded continuous convergence-determining family gives a single conull Borel set $\mathcal{G}$ on which $\Pi_N(Y) \Rightarrow \mu$. Intersect with the proper source-avoiding set, using Lemma 31.3.

For every $Y \in \mathcal{G}$ and every fixed linear B, commutation of B with T and continuity of its action give

$$\Pi_N(BY) = B_*\Pi_N(Y) \Rightarrow B_*\mu. \tag{WP6}$$

This is a deterministic consequence of weak convergence for the given Y, so it holds simultaneously for all B; it does not intersect uncountably many probability-one events.

Extend ψ to Ψ as in Lemma 31.4. Applying that lemma to $\Psi \circ A$ gives $d_{\mathcal{C}}(T^n \Psi AY, T^n LAY) \to 0$. For every bounded Lipschitz test function, the difference of its empirical averages consequently tends to zero by Cesàro averaging. Together with (WP6), this gives

$$\Pi_N(\Psi AY) \Rightarrow (LA)_*\mu. \tag{WP7}$$

Again this is a pathwise assertion for every admissible A, ψ.

To decode the limiting law, parameterize projective positive definite tensors by determinant-one matrices Q and set $A_Q = Q^{-1/2}$. The map

$$Q \longmapsto (A_Q)_*\mu \in \mathcal{P}(\mathcal{C})$$

is continuous by the continuous affine action and is injective by Lemma 31.1. Its image is Borel and its inverse is Borel by Lusin–Souslin. Apply that inverse to the weak limit of Π_N when the limit exists and belongs to this image; otherwise assign a cemetery value. This defines a Borel decoder. Rotation and scalar invariance identify the tensor of the law on the right side of (WP7) as the class in (WP5).

It remains to check germ measurability. For a proper source-avoiding curve Y, replace the tail after its first radius-r exit by the radial ray to ∞, obtaining $E_r Y$. This is a Borel function of the initial prefix: use Borel stopping followed by continuous concatenation with the radial ray in the spherical curve coding. Lemma 31.5 gives $d_{\mathcal{C}}(T^n Y, T^n E_r Y) \to 0$. Their empirical measures therefore have weak limits simultaneously, with the same limit when it exists. More precisely, their bounded-Lipschitz distance is at most $N^{-1}\sum_{n\le N} d_{\mathcal{C}}(T^n Y, T^n E_r Y) \to 0$. Hence $D(Y) = D(E_r Y)$ for every $r > 0$, including when the decoder has the cemetery value. The same argument makes the decoder independent of the completion of a local curve segment and completes the proof. □

Scope for random tip charts. If L is conformal, the decoded tensor changes by the usual covariant rule $Q \mapsto [L^{-\mathsf{T}} Q L^{-1}]$. If a random physical local curve is obtained by first applying a random conformal local map to Y and then a fixed affine map A, its derivative is $A(aR)$ and (WP5) decodes $[(A^{-1})^{\mathsf{T}} A^{-1}]$, independently of the random similarity aR. Thus dependence of the local map on the sampled curve is not by itself an obstruction once this proper, dilation-ergodic whole-plane model and a C^1 local chart are known. This theorem still supplies an ordered-germ decoder. It does not select a canonical point of an unlabelled frontier, prove an intrinsic typical-tip theorem, establish compatibility at arbitrary exploration stopping times, or derive the required whole-plane law and dilation ergodicity from the three exploration axioms.

31.2 Full scaling ergodicity from the stationary cylinder driver

Zhan's construction in Section 3 of [5] gives more than an ergodic one-dimensional angular statistic once the full cylinder process is retained. The following argument supplies the full discrete-dilation ergodicity premise; it uses the classical Hörmander hypoellipticity theorem for the smoothing step.

Proposition 31.7 (Every stationary cylinder skeleton is ergodic)**.** *Let $\kappa > 0$, $\varrho \geq \kappa/2 - 2$, and $a = \varrho/2 + 1 > 0$. Let $Z_t \in (0, 2\pi)$ be the two-sided stationary diffusion*

$$dZ_t = \sqrt{\kappa}\, dB_t + a \cot(Z_t/2)\, dt, \qquad \pi_Z(dz) = c \sin(z/2)^{4a/\kappa}\, dz. \tag{WP8}$$

Let q_0 be uniform modulo 2π, independent of the entire stationary path Z, and set

$$q_t = q_0 - \int_0^t \cot(Z_s/2)\, ds \pmod{2\pi}. \tag{WP9}$$

Then the cylinder process (q_t, Z_t) is stationary, and its two-sided stationary path law is ergodic under every fixed time shift $h \neq 0$. Consequently any Borel whole-plane Loewner trace constructed from $e^{i\lambda_t}$, $\lambda_t = q_t + Z_t$, is ergodic under multiplication by every fixed scalar $e^h \neq 1$ as an unparameterized curve, on a conull invariant set where the trace is defined.

Proof. The dimension of the radial Bessel diffusion is $1 + 4a/\kappa \geq 2$, so its endpoints are not hit; this is the stationary construction in [5, Section 3 and Appendix B]. Uniform independent q_0 makes q_t uniform conditional on the entire path Z, proving stationarity and giving the invariant probability $\pi = (dq/(2\pi))\pi_Z(dz)$, with full support.

The Markov generator on the open cylinder is

$$\mathcal{L} = \frac{\kappa}{2}\partial_z^2 + a \cot(z/2)\partial_z - \cot(z/2)\partial_q. \tag{WP10}$$

Write $V_1 = \sqrt{\kappa}\,\partial_z$ and $V_0 = a \cot(z/2)\partial_z - \cot(z/2)\partial_q$. Then

$$[V_1, V_0] = \frac{\sqrt{\kappa}}{2} \csc^2(z/2)(-a\partial_z + \partial_q). \tag{WP11}$$

Thus V_1 and $[V_1, V_0]$ span the spatial tangent space at every interior point. The parabolic Hörmander theorem applied to $\mathcal{L} - \partial_t$ gives the strong Feller property at every positive time. Here is the localization needed for the singular endpoints. On a relatively compact subcylinder, the killed backward semigroup applied to a bounded Borel function solves the homogeneous backward Kolmogorov equation in the interior at positive times, in the distributional sense. Hörmander's theorem [16, Theorem 1.1] makes this solution smooth there. For any initial interior point and finite horizon, endpoint nonattainment gives a compact subcylinder containing the entire path with

probability at least $1-\varepsilon$. Choose a slightly larger subcylinder. Coupling nearby starting points with the same Brownian motion, local Lipschitz continuity and Gronwall's inequality show that their paths remain in the larger subcylinder on this same event, for all sufficiently close starting points. The full semigroup thus differs from that killed semigroup by at most $\|f\|_\infty\varepsilon$, uniformly near the given point. Let $\varepsilon \downarrow 0$ to obtain continuity for every bounded Borel f. Hörmander smoothing is an external analytic input, not a consequence of stationarity alone.

For clarity, topological irreducibility at every fixed time $h > 0$ can be checked directly, without a general support theorem. Given two cylinder points (q, z) and (q', z'), choose a smooth controlled path $z(s)$, $0 \le s \le h$, from z to z', remaining in a compact subinterval of $(0, 2\pi)$. It can be chosen so that

$$-\int_0^h \cot(z(s)/2)\, ds = q' - q \pmod{2\pi}.$$

Indeed let the path move to a variable interior level, remain there for a positive time, and return to z'. As that level ranges from 0 to 2π, the integral varies continuously from $-\infty$ to $+\infty$. The required deterministic Brownian control is

$$w(s) = \kappa^{-1/2}\left(z(s) - z - a\int_0^s \cot(z(u)/2)\, du\right).$$

It is a Cameron–Martin path. Brownian motion has positive probability of lying in every uniform neighborhood of w. Local Lipschitz dependence of the additive-noise SDE and its integrated q coordinate shows that the uncontrolled process ends in any prescribed neighborhood of (q', z') with positive probability. Hence each skeleton kernel P_h is topologically irreducible.

A strong Feller irreducible Markov kernel has at most one invariant probability, and its stationary probability is an extreme invariant probability. Equivalently its two-sided stationary Markov path law is shift-ergodic. Apply this standard Markov-kernel fact to P_h for every fixed $h > 0$. The same conclusion holds on the full continuous path space under time shift h. Indeed consider the Markov chain of path segments $(X_{nh+s})_{0\le s\le h}$, where $X = (q, Z)$. Its transition law depends only on the previous endpoint. The endpoint law of any invariant segment law must be invariant for P_h, hence must equal π. The segment law is then necessarily the law obtained by starting the diffusion from π and recording its next length-h segment. Thus the segment chain has a unique invariant law and is ergodic; concatenation intertwines its shift with the full continuous-path shift by h. Negative shifts have the same invariant events.

Finally the deterministic whole-plane Loewner correspondence intertwines shifting λ_t to λ_{t+h} with mapping the capacity-parameterized trace $\gamma(t)$ to $e^{-h}\gamma(t+h)$. After forgetting the parameter this is scalar multiplication by e^{-h}. A Borel factor of an ergodic transformation is ergodic, proving the final assertion. □

Application and remaining premise. For $0 < \kappa < 4$ and $\varrho = \kappa + 2$, the stationary whole-plane SLE$(\kappa; \kappa+2)$ driver of [5, Section 3] therefore has full dilation ergodicity, not only the ergodicity of its coordinate Z used in Section 6 of that paper. Applying Theorem 31.6 still requires its stated Borel spherical curve coding and proper continuous trace from 0 to ∞; this proposition supplies the ergodicity input but does not prove those geometric inputs.

Corollary 31.8 (An affine tensor at a deterministic simple-SLE tip)**.** *Fix $0 < \kappa < 4$ and assume the proper spherical curve coding premises of Theorem 31.6 for the whole-plane SLE$(\kappa; \kappa+2)$ law. Let β be a chordal SLE$_\kappa$ trace, let $t_0 > 0$ be a deterministic half-plane capacity time, and let $A \in GL^+(2, \mathbb{R})$ be fixed. The same decoder applied to the reversed ordered germ of $A\beta$ at $A\beta(t_0)$, translated to start at 0, almost surely returns*

$$[(A^{-1})^{\mathsf{T}} A^{-1}]. \tag{WP12}$$

For a general simply connected domain, use a conformal parameter chart and its corresponding deterministic chordal capacity time.

Proof. Theorem 5.3 of [5] maps the reversed past of β at t_0 by a random conformal map to an initial segment of a whole-plane SLE$(\kappa;\kappa+2)$ trace. Its local inverse at 0 has a random nonzero conformal derivative aR. Proposition 31.7 supplies the whole-plane discrete-dilation ergodicity. Apply Theorem 31.6 to this random inverse chart followed by A. The derivative is $A(aR)$, whose tensor class is exactly (WP12). The decoder needs only the initial ordered germ, so the whole-plane segment can be completed arbitrarily. □

This corollary holds simultaneously at any prescribed countable set of deterministic capacity times. It does not assert correctness at every tip, at a tip chosen by an arbitrary intrinsic rule, or at a stopping time selected by the trace.

31.3 An intrinsic measure class without affine Minkowski convergence

There is a way to address the measure-selection part of the frontier problem without first proving existence of Minkowski content after every affine deformation. This does not supply a typical-tip theorem or a Borel choice of the relevant local frontier arc.

Fix $0 < d < 2$. For a nonempty compact planar set E and $\epsilon > 0$ put

$$m_{E,\epsilon}(dx) = \epsilon^{d-2}\mathbf{1}_{\{\operatorname{dist}(x,E)<\epsilon\}}\,dx. \tag{IM1}$$

Call E content regular if these finite measures converge weakly, as $\epsilon \downarrow 0$, to a finite nonzero measure μ_E. This is an explicit local Minkowski-content hypothesis. The theorem below is deterministic; no SLE or exploration assumption enters its proof.

Theorem 31.9 (A Euclidean canonical affine-stable measure class)**.** *There is a Borel map $E \mapsto \nu_E$ from compact planar sets to finite positive measures, supported on E, with the following properties. It has exact d-dimensional similarity covariance: for $F(x) = sRx + t$, $s > 0$ and R orthogonal, $\nu_{FE} = s^d F_*\nu_E$. For every content-regular E, $\nu_E = \mu_E$. If A is invertible real-linear, with smallest and largest singular values $a, b > 0$, then*

$$|\det A|\, b^{d-2} A_*\mu_E \;\le\; \nu_{AE} \;\le\; |\det A|\, a^{d-2} A_*\mu_E \tag{IM2}$$

as measures. In particular, ν_{AE} and $A_*\mu_E$ have precisely the same null sets. No convergence of the tube measures of AE is assumed. Exact affine covariance of the selected measure is not claimed.

Proof. Consider the Borel set of compact sets

$$\mathcal{E}_d = \left\{ E : 0 < \liminf_{\epsilon\downarrow 0} m_{E,\epsilon}(\mathbb{R}^2) \le \limsup_{\epsilon\downarrow 0} m_{E,\epsilon}(\mathbb{R}^2) < \infty \right\}.$$

The tube-measure map in (IM1) is Borel in the Hausdorff hyperspace of compact sets: the distance is jointly continuous in (E, x), so integrals against bounded continuous test functions are Borel by measurable integration. The tube measure also depends continuously on positive ϵ. Indeed every positive distance level has zero area: if $\operatorname{dist}(x, E) = r > 0$, a nearest point of E gives, in each sufficiently small ball about x, a ball of one quarter its radius where the distance is strictly below r. Thus the level set has no Lebesgue density points. Dominated convergence now proves the asserted continuity. All small-radius limit operations may consequently be coded using rational radii, proving that $\mathcal{E}_d$ is Borel. For $E \in \mathcal{E}_d$, the sufficiently small-radius tube measures have uniformly bounded

masses and a common compact support, namely the closed unit neighborhood of E. Their weak cluster set $K(E)$ as $\epsilon \downarrow 0$ is therefore nonempty compact. Every cluster measure is supported on E, and its mass lies between the displayed positive liminf and finite limsup. Let $C(E)$ be the closed convex hull of $K(E)$; it is compact, since it lies in the compact set of measures supported on E with the same uniform mass bounds.

For clarity, these compact-valued constructions admit Borel coding. Choose a complete metric for weak convergence of finite positive measures. For every open set O in that measure space,

$$K(E) \cap O \neq \varnothing \quad \Longleftrightarrow \quad \begin{array}{c} \text{for some } k, \text{ for every } m, \text{ some } \epsilon \in \mathbb{Q} \cap (0, 1/m) \\ \text{has } \operatorname{dist}(m_{E,\epsilon}, O^c) > 1/k. \end{array} \tag{IM3}$$

Precompactness proves both implications. The right side is Borel. Using a countable basis of balls and (IM3), choose a point of $K(E)$ in each basis open set which meets it: successively take the first smaller basis ball meeting $K(E)$ whose closure lies in the preceding open ball, with radii tending to zero. The centers form a Cauchy sequence; its limit lies in $K(E)$ and in the original basis open set. All choices are least-integer choices of Borel predicates. This gives a Borel countable dense family in $K(E)$, and its rational convex combinations give a Borel countable dense family in $C(E)$.

Select ν_E to minimize the Gaussian energy

$$J_E(\nu) = \iint e^{-|x-y|^2/\operatorname{diam}(E)^2}\, \nu(dx)\nu(dy) \tag{IM4}$$

over $C(E)$. The diameter is positive on $\mathcal{E}_d$, since a singleton has tube mass of order $\epsilon^d \to 0$. The energy is continuous for weak convergence of finite positive measures, and its dependence on E is Borel. It is strictly convex: for a nonzero finite signed measure η, the Gaussian Fourier identity gives

$$\iint e^{-|x-y|^2/\operatorname{diam}(E)^2}\, \eta(dx)\eta(dy) = c_E \int e^{-\operatorname{diam}(E)^2|\xi|^2/4} |\widehat{\eta}(\xi)|^2 d\xi > 0,$$

where $c_E > 0$. The last inequality uses continuity and uniqueness of the Fourier transform of a finite measure. Thus the minimizer exists and is unique. It is Borel in E: the infimum over the countable dense family in $C(E)$ is Borel; choose its first member within $1/j$ of that infimum. Compactness and uniqueness force these approximate minimizers to converge to ν_E. Set $\nu_E = 0$ outside $\mathcal{E}_d$, also for the empty set.

For a similarity $F(x) = sRx + t$, the tube change of variables gives $m_{FE,\epsilon} = s^d F_* m_{E,\epsilon/s}$. The full-radius cluster construction therefore gives $C(FE) = s^d F_* C(E)$. Also $J_{FE}(s^d F_* \nu) = s^{2d} J_E(\nu)$. Uniqueness gives the stated exact similarity covariance. If E is content regular, $K(E) = C(E) = \{\mu_E\}$.

For the affine comparison, the singular-value bounds give tube inclusions

$$A(E_{\epsilon/b}) \subset (AE)_\epsilon \subset A(E_{\epsilon/a}),$$

where E_r is the open radius-r neighborhood. Hence for every bounded nonnegative continuous f,

$$|\det A|\, b^{d-2} m_{E,\epsilon/b}(f \circ A) \leq m_{AE,\epsilon}(f) \leq |\det A|\, a^{d-2} m_{E,\epsilon/a}(f \circ A). \tag{IM5}$$

Full tube convergence for E shows that every weak cluster measure of AE satisfies (IM2). The total masses are bounded above and away from zero, so $AE \in \mathcal{E}_d$. The measure inequalities are preserved by convex combinations and weak limits. They therefore hold for every element of $C(AE)$, in particular its selected minimizer. Testing against nonnegative continuous functions proves the inequalities as Borel measures and finishes the proof. □

Corollary 31.10 (Removing the randomness of an intrinsic-point decoder)**.** *Suppose a compact trace is an affine image $F = AE$ of a content-regular trace, and $\Theta(F, x)$ is a Borel local-arc tensor observation, with a cemetery value permitted. If for one tensor Q,*

$$\Theta(F, x) = Q \quad \text{for } A_*\mu_E\text{-almost every } x, \tag{IM6}$$

then the same equality holds for ν_F-almost every x. Moreover Q is recoverable as a Borel function of F alone whenever (IM6) holds, without choosing a capacity time or a random point.

Proof. The null-set assertion is (IM2). On the Borel set where $\nu_F(\mathbb{R}^2) > 0$, push $\nu_F/\nu_F(F)$ forward by $x \mapsto \Theta(F, x)$; outside it return the cemetery value. This is a Borel probability kernel in F and, under (IM6), equals δ_Q. The Dirac embedding of the Polish tensor space into its probability-measure space is Borel injective and has a Borel inverse on its image. Apply that inverse if the pushed measure is Dirac, and otherwise return the cemetery value. This constructs the claimed deterministic Borel decoder. □

Known natural-length sampling and the gap that remains. Lawler–Rezaei [11] proves existence of SLE Minkowski content and identifies natural parametrization. Zhan's decomposition theorem [12], Theorem 4.1 and Corollaries 4.2–4.3, identifies the chordal-SLE curve–point measure obtained from Minkowski-content sampling with the integral of two-sided radial SLE through z against its Green-function density. Thus a natural-length Palm decomposition is available, and is different from the deterministic-capacity-tip theorem in [5]. To use (IM6) one still needs a Borel extraction of the appropriate ordered frontier arc and a decoder valid at its natural-length typical point. A capacity-almost-everywhere statement alone supplies neither. The robust random-chart decoder reduces the remaining model question to its explicit properness, isotropy, scale-ergodicity, and local-chart premises for that intrinsic-point law. No such frontier identification or all-stopping-time compatibility is asserted here.

31.4 The tensor at a natural-length typical point of a simple SLE

The natural-length Palm law uses a different whole-plane model from the deterministic-capacity tip law. The stationary cylinder argument applies also to this model, with force parameter $\varrho = 2$. We restrict to $0 < \kappa < 4$ so that the relevant traces are simple. Let μ_2 be the whole-plane SLE$(\kappa; 2)$ law from 0 to ∞. Continuous proper traces for this explicit SLE model are established in [13, Theorem 1.12 and Proposition 2.5]. Retain the spherical curve coding and Borel local stopping/completion premises of Theorem 31.6. Proposition 31.7, with $\varrho = 2$, supplies its full discrete-dilation ergodicity.

Call a source-avoiding ordered germ at 0 μ_2-generic if a proper source-avoiding completion Y has

$$\frac{1}{N}\sum_{n=1}^{N} \delta_{\rho^n Y} \implies \mu_2 \quad \text{for one fixed } \rho > 1. \tag{NP1}$$

By tail replacement this does not depend on the completion and is a Borel germ property. It is stronger than merely assigning a tensor value to an arbitrary decoder. The spherical linearization argument shows pathwise that this property is preserved by every local conformal map fixing 0 with nonzero derivative: the derivative is a similarity, and μ_2 is rotation and dilation invariant. More generally applying any fixed affine A makes the limit $A_*\mu_2$. These assertions hold simultaneously for sample-dependent local conformal maps, without another exceptional-set intersection.

Proposition 31.11 (The terminal germ of radial SLE$(\kappa; 2)$)**.** *For every $\theta \in (0, 2\pi)$, take radial SLE$(\kappa; 2)$ in $\mathbb{D}$ from $e^{i\theta}$ to 0, with boundary force point 1. Its reversed terminal ordered germ at 0 is*

μ_2-generic almost surely. The same assertion holds in any conformally equivalent marked domain, using the physical Euclidean coordinate at its interior target.

Proof. We use whole-plane SLE(κ; 2) reversibility, proved in [13, Theorem 1.20], and its conformal Markov property; the relation with radial SLE(κ; 2) is recalled explicitly in [15, Section 2.1] and [14, Section 2.2]. These are external SLE inputs.

First construct a full stationary whole-plane SLE(κ; 2) trace γ from 0 to ∞, parametrized by whole-plane capacity, with Loewner maps g_t and cylinder coordinates λ_t, q_t. Let $Z_t = \lambda_t - q_t \in (0, 2\pi)$. Under inversion $J(z) = 1/z$, $\Gamma = J\gamma$ is a whole-plane SLE(κ; 2) from ∞ to 0. At time 0, conditional on its explored past, the remaining path is radial SLE(κ; 2) in the remaining domain, from its current tip to 0, with force point at the prime end corresponding to ∞. The conformal chart

$$f(z) = \frac{e^{iq_0}}{g_0(1/z)} \tag{NP2}$$

maps that domain to the unit disk, fixes 0, takes the force point to 1, and takes the tip to e^{-iZ_0}. It is analytic with nonzero derivative at 0. The normalization at capacity zero gives $f'(0) = e^{iq_0}$.

Reversibility makes the full reversed Γ a whole-plane SLE(κ; 2) from 0 to ∞. Its initial germ is therefore μ_2-generic almost surely. Applying the sample-dependent chart f preserves (NP1). The resulting reversed terminal germ belongs, conditionally on the past, to the radial law in the statement with angle $\theta = -Z_0$ modulo 2π. The stationary gap density is

$$\pi_Z(dz) = c \sin(z/2)^{8/\kappa}\, dz, \tag{NP3}$$

which is positive on $(0, 2\pi)$. Disintegration therefore proves the proposition for Lebesgue-almost every starting angle θ.

To obtain every angle, start the radial law at an arbitrary $\theta_0 \in (0, 2\pi)$ and run it for a deterministic radial time $s > 0$. After mapping by its radial Loewner map and rotating to put the current force point at 1, its future is a radial law with gap X_s, where

$$dX_t = \sqrt{\kappa}\, dB_t + 2\cot(X_t/2)\, dt, \qquad X_0 = \theta_0. \tag{NP4}$$

The endpoint Bessel dimension is $1 + 8/\kappa > 2$, so neither endpoint is reached. The distribution of X_s is absolutely continuous with respect to Lebesgue measure: on each compact subinterval of $(0, 2\pi)$, the stopped drift is bounded, and Girsanov compares its endpoint law with Brownian motion on the event of no exit. These no-exit events increase to probability one as the compact intervals exhaust $(0, 2\pi)$. Thus X_s lies in the already proved good-angle set almost surely. Conditional DMP gives (NP1) for the future terminal germ in its new chart. The inverse radial chart is analytic near 0 with nonzero conformal derivative, so pathwise conformal invariance of (NP1) transfers the property back to the original terminal germ. For a general marked domain use an interior conformal chart at the target and the same pathwise invariance. No continuity of the good event as a function of θ is assumed. □

Theorem 31.12 (A natural-length almost-everywhere affine tensor decoder)**.** *Let β be chordal SLE$_\kappa$, $0 < \kappa < 4$, in a simply connected domain, with its Minkowski-content measure μ_β. Let $A \in GL^+(2, \mathbb{R})$ be fixed. Use the decoder of Theorem 31.6 constructed from the model μ_2. Almost surely, for μ_β-almost every interior point $z = \beta(t)$, the reversed incoming ordered germ of $A\beta$ at Az, translated to start at 0, has decoded tensor*

$$[(A^{-1})^{\mathsf{T}} A^{-1}]. \tag{NP5}$$

The same holds for the outgoing germ. This is a statement about the intrinsic length measure, not capacity-almost every time.

Proof. Zhan's natural-length Palm decomposition [12, Theorem 4.1 and Corollaries 4.2–4.3] identifies chordal SLE with a point sampled using its Minkowski content with two-sided radial SLE through that point, integrated against the SLE Green-function density. For every marked interior point z, the incoming arm of the two-sided radial curve is radial SLE$(\kappa;2)$ from the source to z, with force point at the remote boundary target. Proposition 31.11 makes its reversed terminal germ μ_2-generic. Hence the affine-image empirical law is $A_*\mu_2$, which the robust decoder identifies as (NP5). The bad-point indicator is jointly Borel in (β, z): on a simple representative the point z has a unique hit (up to pauses), whose first-hit cut is Borel; restriction, reversal, translation, and the germ decoder with its Borel completion are Borel. One may first cut a branch inside $B(z, \operatorname{dist}(z, \partial D)/2)$, so rough source or target boundaries do not enter this observation. Integrating the Palm identity over compact interior sets, followed by a countable exhaustion, proves that the expected Minkowski-content mass of bad points is zero. That mass is nonnegative, so it vanishes almost surely. Reversibility of two-sided radial SLE, recalled in [14, Section 2.2], gives the outgoing assertion. □

What this removes and what remains. For compact simple-SLE pieces satisfying the content-regularity premise of Theorem 31.9, (NP5) supplies (IM6). Consequently a tensor can be recovered from their unlabelled trace by the intrinsic-measure construction, provided the local incoming or outgoing branch is Borel extracted from that trace. A simple compact arc has two possible orientations; (NP5) works for both, so choosing an orientation does not change its almost-everywhere tensor. This does not yet identify the laws of all accessible frontier arcs of an exploration, establish the content premise on a chosen frontier exhaustion, or prove compatibility at every random slit state and every exploration stopping time. It does not derive an affine-SLE representation from the three axioms.

31.5 A tensor from an unlabelled compact simple-SLE arc

The natural-point theorem and the intrinsic measure can be combined without leaving an unproved choice of orientation for a compact simple arc. We give the coding and local-content details explicitly. This result concerns arcs already known to be simple SLE; it does not identify arbitrary accessible boundary pieces of an unknown exploration.

Lemma 31.13 (Borel orientation and cutting of simple compact arcs)**.** *Use the space $\mathcal{C}_c$ of arbitrary compact planar oriented curves, with pauses identified and the Fréchet metric. This is a separate space from the fixed-endpoint spherical 0-to-∞ space. Use its continuous quotient map from $C([0,1],\mathbb{R}^2)$. The nonconstant simple-arc subspace is Borel. Its trace map to the Hausdorff hyperspace of compact planar sets has exactly two elements in each nonempty fiber. Consequently its image is Borel and there is a Borel choice of an orientation for every compact simple arc. Given an oriented simple arc and a point in its trace, cutting there and reversing either resulting branch are jointly Borel operations.*

Proof. The compact planar curve space is Polish: images of rational polygonal paths give separability. For completeness, align the representatives of a Cauchy subsequence successively by increasing homeomorphisms so that their successive uniform distances are summable. Their uniform limit is continuous and represents the Fréchet limit; the original Cauchy sequence has the same limit. The quotient map is continuous.

For continuous representatives, failure of simplicity modulo pauses is the existence of $r \le s \le t$ with

$$\gamma(r) = \gamma(t), \qquad |\gamma(s) - \gamma(r)| \ge 1/n \tag{UA1}$$

for some integer $n \geq 1$. For fixed n this is a closed condition: the three times range over a compact set and subsequential limits preserve both relations. These conditions are invariant under the identification of pauses. Their images and their complements under the continuous Polish quotient are analytic, hence Borel. Excluding their countable union and the constant curves gives the claimed Borel subspace. The absence of a nonconstant closed subpath is equivalent to being an embedded arc after its constant intervals are collapsed.

The trace map is continuous for Fréchet and Hausdorff metrics. An embedded compact arc has exactly two orientations, so its restriction to this Borel subspace is two-to-one. The countable-fiber Borel uniformization theorem gives both a Borel image and a Borel orientation selector. For cutting, the unique point hit (or its constant interval in a representative with pauses) is a Borel first hitting time. Indeed its sublevel events are given by the minimum distance on a compact initial interval being zero. Restriction and reversal are Borel on representatives, invariant after quotienting, and therefore descend by the same analytic/complement argument. Endpoints or a constant branch may be assigned a cemetery observation. □

Lemma 31.14 (Segment content implies weak local tube convergence)**.** *Let $y : [s,t] \to \mathbb{R}^2$ be a simple continuous curve, and let Θ be continuous and strictly increasing. Suppose for every $s \leq u < v \leq t$ the d-dimensional total Minkowski content of $y([u,v])$ exists and equals $\Theta(v) - \Theta(u)$. Then $E = y([s,t])$ is content regular in the sense of (IM1), with*

$$\mu_E = y_*(d\Theta|_{[s,t]}). \tag{UA2}$$

In particular the simultaneous segment-content theorem of Lawler–Rezaei [11, Theorem 1.1] supplies this property for every compact nontrivial interior time segment of a simple chordal SLE in $\mathbb{H}$, on one probability-one set.

Proof. For a finite partition $s = t_0 < \cdots < t_m = t$, set $E_j = y([t_{j-1}, t_j])$. The tube measures satisfy $m_{E,\epsilon} \leq \sum_j m_{E_j,\epsilon}$. The difference is nonnegative, and its total mass tends to

$$\sum_j \{\Theta(t_j) - \Theta(t_{j-1})\} - \{\Theta(t) - \Theta(s)\} = 0.$$

Thus replacing $m_{E,\epsilon}$ by the sum costs $o(1)$ against any bounded continuous test f. Choose the partition fine enough that each E_j has diameter at most δ. On its radius-ϵ tube, f differs from $f(y(t_{j-1}))$ by at most its modulus of continuity at $\delta + 2\epsilon$ on a fixed compact neighborhood of E. The tube masses converge to the corresponding increments of Θ. Riemann–Stieltjes sums give (UA2) by first taking $\epsilon \downarrow 0$ and then $\delta \downarrow 0$. □

Theorem 31.15 (Simultaneous affine decoding from every compact simple-SLE subarc)**.** *Fix $0 < \kappa < 4$, put $d = 1 + \kappa/8$, and retain the stated Polish spherical curve and Borel completion coding. There is a Borel map $\mathcal{T}_\kappa$ from compact planar sets to projective positive definite tensors, with a cemetery value, such that the following holds. If β is chordal SLE_κ in $\mathbb{H}$, then on one probability-one set, simultaneously for every $A \in GL^+(2,\mathbb{R})$ and every $0 < s < t < \infty$,*

$$\mathcal{T}_\kappa(A\beta([s,t])) = [(A^{-1})^{\mathsf{T}} A^{-1}]. \tag{UA3}$$

The input is only the unlabelled compact trace. Neither its orientation, capacity time, source, nor target is supplied. In particular A, s, t may be selected as functions of the sample on this conull set.

Proof. The proof of Theorem 31.12 gives more than correctness for a single fixed affine map: on one conull set, at content-almost every point of β, both local ordered branches are μ_2-generic before affine deformation. For each such germ, (NP1) and commutation with linear maps give the affine empirical law $A_*\mu_2$ simultaneously for all A. No uncountable intersection of probability-one events is needed. Intersect with the simultaneous segment-content event of Lawler–Rezaei.

For any compact simple arc F, choose its orientation by Lemma 31.13. At each interior point x, apply the μ_2 tensor decoder to its reversed incoming branch translated to start at 0, using a Borel radial completion. Call this observation $\Theta(F, x)$. A branch can be cut inside a small ball about x before completing it, so only its germ enters. This is a Borel observation by the lemma and the previously established Borel germ decoder.

Set $E = \beta([s, t])$ and $F = AE$ on the conull set above. Lemma 31.14 makes E content regular, with positive finite atomless measure μ_E. By simplicity and the natural parametrization identity, μ_E is exactly the restriction of the full natural measure μ_β to E. At $A_*\mu_E$-almost every point of F, the chosen incoming branch is one of the two already good branches of the full curve. Its observation is therefore the right side of (UA3), regardless of the sample-dependent orientation choice. Theorem 31.9 makes ν_F equivalent to $A_*\mu_E$. Apply the Dirac-pushforward construction of Corollary 31.10 to $\Theta(F, \cdot)$. Return the cemetery value off the Borel simple-arc set or when the selected measure vanishes or the pushforward is not a single non-cemetery Dirac mass. This defines the required Borel map and proves (UA3).

All comparisons after the two conull events were deterministic, and their statements hold for every affine map and every subinterval. This proves the simultaneous assertion. □

Limit of this advance. This provides an intrinsic measure and orientation-independent decoder on known affine simple-SLE arcs, including sample-dependent subarcs. It does not assert that an arbitrary accessible frontier decomposes into such arcs with Borel extraction, identify those arc laws for the unknown exploration family, or supply continuation kernels at every random slit state. No classification follows without those steps.

A relevant tip theorem and the unresolved selection problem. Zhan's Theorem 5.3 in [5] maps the reversed past of a simple chordal SLE_κ, $0 < \kappa < 4$, at a deterministic capacity time conformally to an initial whole-plane $\mathrm{SLE}(\kappa; \kappa + 2)$ segment. The conformal map is random. The continuation above supplies full dilation ergodicity from the stationary cylinder driver and handles sample-dependent local C^1 maps, under the explicit proper trace and Borel coding premises. This gives the deterministic-time ordered-germ conclusion (WP12), not an intrinsic frontier decoder.

There is a further issue even if fixed-time decoding is obtained. Correctness at a dense set of rational original capacity times, or at almost every such time, does not give a canonical decoder on an unlabelled open frontier arc. Exceptional points could give other values, and an affine change of the capacity clock need not preserve null sets. Zhan's remarks after Theorem 6.6 explicitly distinguish the local models at capacity-typical and natural-length-typical tips. Thus a frontier theorem still needs a canonical local arc extraction and a decoder valid at intrinsic typical points, with compatibility at all exploration stopping times. The tube-cluster construction above provides an affine-stable measure class for compact content-regular model traces, and the cited natural-length Palm decomposition is available for SLE. The natural-point transfer and unlabelled-subarc theorem now give the decoder for known simple-SLE arcs. Identifying the appropriate frontier pieces and their model laws remains open.

32 Restart states and direct driving bounds for nonsimple traces

Scope. The exploration itself is not assumed to contain any embedded time interval. It is an ordered, possibly densely self-touching curve. A decoder theorem for simple SLE arcs is a separate statement about different objects. At a finite capacity time τ, its state is the remaining target component with its inherited colored boundary and its distinguished source and target *prime ends*, not just the two Euclidean endpoint locations. If $f : \mathbb{H} \to D$ is the initial inverse chart, its restarted inverse chart is

$$f_\tau(z) = f(g_\tau^{-1}(z + W_\tau)). \tag{NS1}$$

The centered future driver is $W_{\tau+s} - W_\tau$. This identity uses Loewner composition and does not assert conformal covariance of the unknown physical family. The marked-state space is required to be Borel, and genuine strong DMP is a hypothesis on this state space. The chart is coordinate bookkeeping, not an extra physical state variable: the law in the physical domain does not depend on its choice.

Uniform approximation of a deterministic ordered curve by simple interior paths does not by itself assert strict growth of its filled hulls. The topological characterization in Yuan [7], Theorem 1.1 and Remark 1.4, additionally requires strictly increasing filled hulls. For the results below continuous Loewner generation and the usual capacity clock are standing admissibility hypotheses. No stochastic semimartingale property is included in admissibility. Initially the capacity lifetime may be $\zeta \le \infty$; strong DMP and all restart assertions are imposed on $\{\tau < \zeta\}$ for stopping times τ, allowing $\tau = \infty$ elsewhere. The full curve tends to its target prime end, so the charted trace tends to infinity as capacity tends to ζ.

A continuum crossing assumption. Fix $M > 0$ and $0 < q < 1$. At every admissible marked state, suppose that every nonseparating boundary quadrilateral of modulus at least M has exploration crossing probability at most q. Here the two side arcs of the quadrilateral lie on the domain boundary, the two crossed arcs lie in the domain, and nonseparating means that the source and target can be connected while avoiding the quadrilateral. All crossing events, admissible pairs of states and quads, and evaluation kernels are assumed jointly Borel. The constants are uniform over the state class, including its random slit states. This is a precise upper unforced-crossing input; it is not a substitute for a two-sided configuration-level RSW hypothesis. We use a boundary-robust crossing convention: the joining subcurve may lie in the closed quadrilateral and may touch its two boundary side arcs between its crosscut hits. Closures at boundary sides are interpreted with their inherited prime-end sides. In particular, a nonsimple trace which joins the two semicircles of a half-annulus while touching the real line is a crossing. An estimate stated only for subcurves strictly inside the open quad is not silently substituted for this assumption.

Lemma 32.1 (Time-zero crossing bounds become conditional bounds)**.** *Suppose a Borel family satisfies genuine strong DMP on the stated restart class and the preceding time-zero crossing assumption at every state. For every stopping time τ, on $\{\tau < \zeta\}$, and every $\mathcal{F}_\tau$-measurable admissible nonseparating quad Q_τ of modulus at least M in the remaining state,*

$$\mathbb{P}\{\textit{future exploration crosses } Q_\tau \mid \mathcal{F}_\tau\} \le q \quad \textit{almost surely.} \tag{NS2}$$

The same bound holds in the restarted half-plane chart, with the same constants.

Proof. Let X_τ be the marked state and let P_x be the exploration kernel. Strong DMP identifies the conditional crossing probability with $P_{X_\tau}(\text{cross } Q_\tau)$. The numerical time-zero bound holds for every admissible pair (x, Q), so evaluation at the random pair gives (NS2). This does not intersect

an uncountable collection of almost-sure statements. Conformal maps preserve the quad modulus, its separation status, and the crossing event. Consequently its pullback to the physical state has exactly the same bound. We have transported a geometric inequality, not identified the law of a conformal image with a law in another physical domain. □

Lemma 32.2 (A deterministic driver localization bound)**.** *Let a continuous chordal Loewner trace have filled hulls K_s and driver W_s, with* $\operatorname{hcap}(K_s) = 2s$. *If $K_t \subset \{z : |z| \le R\}$, then*

$$\sup_{0\le s\le t} |W_s| \le 3R. \tag{NS3}$$

Also $\sup\{\operatorname{Im} z : z \in K_t\} \le 2\sqrt{t}$. *These facts do not require a simple trace.*

Proof. For fixed $s \le t$, Schwarz-reflect the hydrodynamic map g_s across the real line outside $[-R, R]$. On $|z| > R$ it is univalent and has expansion $g_s(z) = z + \sum_{n\ge1} b_n z^{-n}$. The area theorem applied after rescaling by R gives

$$\sum_{n\ge1} \frac{n|b_n|^2}{R^{2n+2}} \le 1.$$

Consequently Cauchy–Schwarz gives, at $z = \pm 2R$,

$$|g_s(z) - z| \le R\left(\sum_{n\ge1} \frac{1}{n4^n}\right)^{1/2} < R.$$

The real interval which is the image of the two sides of the hull, and in particular its current tip image W_s, lies between $g_s(-2R)$ and $g_s(2R)$ by boundary order. This proves (NS3). For the height bound, while z is unswallowed its image height Y_s satisfies

$$\partial_s Y_s = -\frac{2Y_s}{|g_s(z) - W_s|^2}, \qquad \partial_s Y_s^2 \ge -4.$$

A point of original height greater than $2\sqrt{t}$ therefore cannot be swallowed by time t. The closed-hull bound follows by closure. □

Proposition 32.3 (Conditional Brownian-size driving bounds without simplicity)**.** *For a continuous-driver family satisfying the hypotheses of Lemma 32.1, whose full traces tend to the target prime end (and hence to infinity in their half-plane chart), the capacity lifetime is infinite almost surely. There are $c, C > 0$, depending only on M, q, such that, for every finite stopping time τ, $h > 0$, and $x \ge 0$,*

$$\mathbb{P}\left\{\sup_{0\le s\le h} |W_{\tau+s} - W_\tau| > x\sqrt{h} \mid \mathcal{F}_\tau\right\} \le Ce^{-cx}. \tag{NS4}$$

In particular, for some $a, C' > 0$,

$$\mathbb{E}\left[\exp\left\{a \sup_{0\le s\le h} |W_{\tau+s} - W_\tau|/\sqrt{h}\right\}\middle| \mathcal{F}_\tau\right] \le C'. \tag{NS5}$$

For every $p > 0$ this gives a constant C_p with conditional p-th moment at most $C_p h^{p/2}$. The driver is almost surely locally α-Hölder for every $\alpha < 1/2$.

Proof. Work first at time zero in the half-plane chart, before the possible capacity lifetime. Fix $u > 0$ and choose $d = 4e^{\pi M}$. To bound horizontal travel before reaching height u, use the successive first hitting times of the lines $\operatorname{Re} z = kdu$, $k = 1, 2, \ldots$, before upper exit. At the first hit of line kdu, all the earlier trace and its filled hull lie to its left. The half-annulus centered at $(k+1)du$, of inner radius $2u$ and outer radius $du/2$, lies entirely to the right of that line and is untouched. As a boundary quad its modulus for crossing between the semicircles is

$$\pi^{-1} \log \frac{du/2}{2u} = M.$$

It does not separate the current source prime end from infinity. Indeed, a source access arc eventually lies near the tip, to the left of this quad. Any path from such an access point to infinity first enters the empty right half-plane to the left of the quad, where it can be rerouted around its outer semicircle. This verifies the prime-end separation statement without equating prime ends which have the same physical impression. Reaching the next line at height at most u requires crossing this quad from its outer to its inner semicircle. Its conditional probability is therefore at most q by (NS2). The same construction starts at $k = 0$ with the source at zero.

Iterating gives probability at most q^n for hitting line ndu before upper exit. Reflection of this geometric construction treats left exit, without assuming reflection invariance of the laws. If the capacity lifetime were at most a fixed $T < \infty$, the height bound would confine the full trace to height $2\sqrt{T}$, while convergence to infinity would require unbounded horizontal extent. Use the preceding estimate with $u = 3\sqrt{T}$, so that no upper exit is possible on this event. The iterated crossing estimate makes that event have probability zero. A countable union over integer T proves infinite lifetime.

Now put $u = 2\sqrt{h}$. While the trace stays in $[-L, L] \times [0, u]$, each filled hull lies in the disk of radius $\sqrt{L^2 + u^2}$. Lemma 32.2 implies that, for $L \geq u$, a driver supremum greater than $5L$ requires a horizontal trace exit before time h; the height bound excludes upper exit. The probability of reaching either side $\pm L$ before upper exit is at most $2q^{\lfloor L/(du) \rfloor}$. Taking $L = x\sqrt{h}/5$ when $x \geq 10$, and enlarging the prefactor for $x < 10$, proves (NS4) at zero. The proof only used target-visible Loewner hulls and first hits of vertical lines, so dense self-touching does not affect it.

Apply the identical argument to the centered future driver in (NS1). Lemma 32.1 supplies the same crossing constants after conditioning at τ. This proves (NS4) for all finite stopping times. Tail integration gives (NS5) for $0 < a < c$ and the moment bounds. Kolmogorov continuity, using arbitrarily large p, also gives pathwise Hölder exponents $\alpha < 1/2$ on every compact capacity interval. □

Relation to the existing literature and limits of the estimate. This is a direct version of the rectangle argument underlying Kemppainen–Smirnov [3], Section 3.3, Propositions 3.8–3.9. Their Theorem 1.5 starts with families of simple curves and obtains nonsimple limits. One cannot apply that formulation merely by choosing deterministic simple approximations of each sample: the approximating *laws* need not retain the conditional crossing bound. The direct argument above instead assumes continuous-driver admissibility for the continuum family and proves its quantitative estimate there.

These estimates control unsigned increments. They do not bound $|\mathbb{E}[W_{t+h} - W_t \mid \mathcal{F}_t]|$ by $O(h)$, establish semimartingality, or identify an affine conformal structure. Uniform quad bounds are stable under a fixed quasiconformal change with altered constants; Brownian-size bounds therefore also hold in coordinates which need not be the correct affine coordinates for stochastic driving rigidity.

33 Unsigned regularity and the missing continuum signed budget

33.1 Unsigned Brownian-scale regularity does not supply a semimartingale

The distinction between topological Loewner admissibility and stochastic regularity is substantive even after strong quantitative estimates are available. The following process example isolates the missing signed cancellation. It is not asserted to obey source locality, domain Markov for an undecorated slit domain, or a curve-level unforced-crossing estimate.

Proposition 33.1 (A driver-level obstruction to an unsigned regularity argument)**.** *For every $\kappa > 0$, $H \in (1/2, 1)$, and $a \neq 0$, there is a continuous process X on $[0,1]$ with all the following properties in its usual natural filtration:*

1. *Its paths are locally α-Hölder for every $\alpha < 1/2$.*
2. *For every $c > 0$ there is $C_c < \infty$ such that, for every stopping time τ and deterministic $h > 0$ with $\tau + h \le 1$ almost surely,*

$$\mathbb{E}\left[\exp\left(c\,\frac{\sup_{0\le u\le h}|X_{\tau+u} - X_\tau|}{\sqrt{h}}\right)\,\middle|\,\mathcal{F}_\tau\right] \le C_c. \tag{UR1}$$

3. *Its sums of squared increments along dyadic partitions have the same limit as Brownian motion of variance rate κ.*
4. *It is not a semimartingale.*

Proof. Let $\varphi(x) = \operatorname{dist}(x, \mathbb{Z})$ and define

$$g_H(t) = \sum_{n=0}^{\infty} 2^{-nH}\varphi(2^n t), \qquad X_t = \sqrt{\kappa}B_t + a g_H(t), \tag{UR2}$$

where B is standard Brownian motion. Uniform convergence defines a continuous deterministic function. The bounds $|\varphi(x) - \varphi(y)| \le \min\{|x-y|, 1/2\}$ show

$$|g_H(t) - g_H(s)| \le C_H |t-s|^H. \tag{UR3}$$

Indeed split the series where $2^n|t-s|$ first exceeds one, and sum the finite geometric series of Lipschitz bounds and the remaining geometric tail separately. Both sums are bounded by a constant times $|t-s|^H$. The pathwise Hölder assertion follows.

The natural filtrations of X and B are identical, since the shift ag_H is deterministic. Brownian strong Markov and (UR3) give, for $h \le 1$, the bound in (UR1) by

$$e^{c|a|C_H}\,\mathbb{E}\exp\left(c\sqrt{\kappa}\sup_{0\le u\le 1}|B_u|\right) < \infty.$$

Reflection bounds for Brownian maxima make the last expectation finite for every $c > 0$.

Here is an explicit verification of the variation obstruction. Set $q = 2^{1-H} > 1$, and use the mesh $t_j = j2^{-m}$, $0 \le j \le 2^m$. All terms with $n \ge m$ vanish at the mesh points. On each mesh interval all the earlier tent functions are linear, so

$$\Delta_j g_H = 2^{-m}\sum_{n=0}^{m-1} q^n r_{n+1}(j), \tag{UR4}$$

where the sign vectors r_{n+1} are the dyadic Rademacher vectors. Their orthogonality on this mesh gives

$$Q_m := \sum_j (\Delta_j g_H)^2 = 2^{-m}\frac{q^{2m}-1}{q^2-1}, \qquad \max_j |\Delta_j g_H| \le 2^{-m}\frac{q^m-1}{q-1}. \tag{UR5}$$

Consequently

$$\sum_j |\Delta_j g_H| \ge \frac{Q_m}{\max_j |\Delta_j g_H|} \ge \frac{q^m+1}{q+1} \longrightarrow \infty. \tag{UR6}$$

Thus g_H has infinite variation. On the other hand $Q_m \to 0$ because $H > 1/2$. More generally (UR3) bounds the sums of squared increments on any subinterval of the dyadic mesh by $C_H^2 2^{-m(2H-1)}$, so they tend to zero uniformly over mesh endpoints. Cauchy–Schwarz bounds the mixed Brownian/g_H sums by the square root of the product of their squared increment sums. The standard Brownian dyadic quadratic-variation limit therefore proves the asserted Brownian quadratic variation for X.

For completeness, the failure of semimartingality can also be seen without relying only on the deterministic infinite-variation criterion. The bounded deterministic elementary predictable strategies $L_t^m = \operatorname{sign}(a\Delta_j g_H)$ on $(t_j, t_{j+1}]$ give the elementary sums

$$\sum_j L^m_{t_{j+1}}(X_{t_{j+1}} - X_{t_j}) = |a| \sum_j |\Delta_j g_H| + Z_m, \tag{UR7}$$

where Z_m is centered Gaussian with variance at most κ. They diverge to $+\infty$ in probability by (UR6). For a continuous semimartingale, its elementary integrals with predictable integrands bounded by one are bounded in probability on a fixed finite horizon: localize the martingale bracket and finite-variation total variation, then use the martingale L^2 inequality. This contradicts (UR7). □

What RSW regularity can legitimately supply. Kemppainen–Smirnov [3], Proposition 2.6, Theorem 1.5, and Section 3.3, give conformally stable conditional unforced-crossing criteria, continuous driving with every Hölder exponent below 1/2, and Brownian-scale exponential driving bounds in their stated setup. Their Section 1.4 explicitly permits nonsimple curves already having continuous Loewner drivers; it does not alone give that property for an arbitrary law of uniformly approximable nonsimple curves. These are unsigned estimates; they do not assert semimartingality. The example above proves that even the listed stronger process-level conclusions, including conditional estimates at stopping times and Brownian quadratic variation, cannot replace the missing signed cancellation argument. It is not a counterexample to the full curve axioms.

33.2 A continuum signed budget on visited slit states

A useful first target is weaker than the uniform linear conditional-bias bound (BI1). It asks for finite aggregate predictable variation, not for a pointwise drift density. Its probabilistic content is classical quasimartingale theory; the formulation below makes the exact continuum restart observable and its stopping rule explicit.

Assume only the Borel continuous-law and restart-identity premises of (KR1), not the semimartingale premises of Proposition 18.1. Let $\{P_f\}$ be this family, with drivers starting at zero and defined on $[0,\infty)$. Fix $L, T > 0$, set

$$\sigma_L = \inf\{t \ge 0 : |W_t| \ge L\}, \qquad X_t = W_{t\wedge\sigma_L}, \quad 0 \le t \le T, \tag{QV1}$$

and use the usual natural filtration of the original W. For $\ell < 0 < r$, define the bounded restarted mean

$$m_h(f;\ell,r) = \mathbb{E}_{P_f} W_{h\wedge\inf\{u:W_u\notin(\ell,r)\}}. \tag{QV2}$$

This is a Borel function of its arguments: the stopped evaluation of a continuous path at its first exit from an open interval is Borel and is bounded between the two endpoints. No claim of continuity in f is made.

Proposition 33.2 (An exact signed restart budget)**.** *Suppose (KR1) holds in the inherited centered capacity charts. For every deterministic partition $\pi = (0 = t_0 < \cdots < t_n = T)$,*

$$\mathbb{E}[X_{t_{j+1}} - X_{t_j} \mid \mathcal{F}_{t_j}] = \mathbf{1}_{\{t_j < \sigma_L\}} m_{t_{j+1}-t_j}(f_{t_j}; -L - W_{t_j}, L - W_{t_j}). \tag{QV3}$$

Consequently the aggregate budget

$$\mathcal{V}_{L,T}(f) = \sup_{\pi} \sum_{j=0}^{n-1} \mathbb{E}_{P_f}\left[\mathbf{1}_{\{t_j < \sigma_L\}} |m_{t_{j+1}-t_j}(f_{t_j}; -L - W_{t_j}, L - W_{t_j})|\right] \tag{QV4}$$

is exactly the conditional variation of X. It is finite if and only if $X = M + A$, $A_0 = 0$, has a continuous canonical decomposition with M a uniformly integrable martingale on $[0,T]$ and $\mathbb{E}\operatorname{Var}_{[0,T]} A < \infty$. In that case

$$\mathcal{V}_{L,T}(f) = \mathbb{E}\operatorname{Var}_{[0,T]} A. \tag{QV5}$$

If it is finite for every L, T, every driver is a continuous semimartingale locally. No absolute continuity of A, initial tangent, or Brownian bracket follows just from this finiteness.

Proof. On $\{t_j \geq \sigma_L\}$ the stopped increment vanishes. Otherwise freeze the past at t_j. The conditional future driver is $P_{f_{t_j}}$ by (KR1). The original stopping boundary $(-L, L)$ becomes exactly $(-L - W_{t_j}, L - W_{t_j})$ for that centered future driver. This proves (QV3), hence the conditional-variation identity (QV4).

Since X is bounded and continuous, the quasimartingale decomposition theorem gives precisely the stated continuous canonical decomposition when its conditional variation is finite; see Rao [4]. Conversely, given this decomposition, martingale increments have conditional mean zero, so each partition sum is at most $\mathbb{E}\operatorname{Var} A$. For the reverse inequality, bounded predictable elementary processes on deterministic time partitions are dense among bounded predictable integrands for the finite measure $H \mapsto \mathbb{E}\int_0^T H_s\, d|A|_s$. Approximate the predictable sign of the signed measure dA by such processes, all bounded by one. For an elementary process with value $H_j \in \mathcal{F}_{t_j}$ on $(t_j, t_{j+1}]$,

$$\mathbb{E}\int_0^T H_s\, dA_s = \sum_j \mathbb{E}[H_j(X_{t_{j+1}} - X_{t_j})] \leq \sum_j \mathbb{E}|\mathbb{E}[X_{t_{j+1}} - X_{t_j} \mid \mathcal{F}_{t_j}]|.$$

The approximation gives (QV5). The local conclusion follows by taking $L \uparrow \infty$ and $T \uparrow \infty$, using continuity and uniqueness of continuous canonical decompositions. □

Why this is a weaker target, and why it is still missing. The bound (BI1) gives $\mathcal{V}_{L,T} \leq KT$ and thus suffices. Finiteness of (QV4) alone also permits singular finite-variation drift. For example, Brownian motion plus a deterministic Cantor function has finite conditional variation on compact intervals but its canonical drift is not absolutely continuous. This example illustrates the distinction between the two probabilistic conditions, not a proposed exploration family. An unsigned estimate of Brownian order gives only a bound of order $\sum_j \sqrt{t_{j+1} - t_j}$, which diverges on finer equal partitions. The budget must be proved from interactions of locality and DMP, in a suitable common affine structure; none of those additional cancellation estimates has been established here. One cannot replace the offsets in (QV3) by the fresh symmetric interval $(-L, L)$: the localization boundary is inherited from the past.

33.3 The shrinking-slit anomaly fits inside Brownian-size bounds

Proposition 33.3 (Uniform unsigned bounds do not control initial bias rates)**.** *Let $b_n \neq 0$ with $|b_n| \to \infty$, and let $\kappa > 0$. Put*

$$a_n(t) = b_n(t \wedge b_n^{-2}), \qquad X_t^n = \sqrt{\kappa} B_t + a_n(t), \quad t \geq 0. \tag{SP1}$$

Then $\sup_t |X_t^n - \sqrt{\kappa} B_t| \leq |b_n|^{-1} \to 0$ under the displayed coupling. Uniformly in n, the processes satisfy the conditional Brownian-scale exponential bound (UR1), at every bounded stopping time and every deterministic $h > 0$. Every X^n is a semimartingale with bracket κt, but its initial drift is b_n. The optimal constant in its unstopped linear conditional-bias bound is $K_n = |b_n|$.

Proof. The deterministic shifts obey

$$|a_n(t) - a_n(s)| \leq \min\{|b_n||t-s|, |b_n|^{-1}\} \leq \sqrt{|t-s|}. \tag{SP2}$$

Brownian strong Markov therefore gives (UR1), uniformly in n, by adding one inside the normalized supremum. The uniform-convergence and bracket claims are immediate. For $0 < h < b_n^{-2}$ the initial mean increment is $b_n h$, proving the optimal lower bound on K_n; the Lipschitz bound $|b_n|$ proves the upper bound. □

For the mirrored slit derivatives in (DR9), a variance anomaly produces

$$b_\epsilon^\pm = (\kappa/2 - 3)\frac{e_\epsilon^\pm}{d_\epsilon^2}, \qquad \frac{e_\epsilon^\pm}{d_\epsilon^2} = \mp \frac{4}{\epsilon^3 \sqrt{\epsilon^2 + 4}}. \tag{SP3}$$

For $\kappa \neq 6$ these are of order ϵ^{-3}. Substituting either sequence into (SP1) gives pulses of duration $\asymp \epsilon^6$ and height $\asymp \epsilon^3$. Thus even uniform convergence to Brownian motion, exact Brownian bracket, and uniform conditional Brownian-size bounds allow precisely the inverse-cubic initial-bias scale which the slit test needs to exclude. Choosing $b_\epsilon = b/d_\epsilon$ similarly realizes the inverse-linear drift scale. The sharp corner at the end of a pulse can be smoothed without changing these orders.

These processes are not claimed to be the locality-related drivers of one exploration family. The point is narrower and rigorous: neither uniform unsigned driving estimates nor weak continuity of the driver laws implies the rate $\epsilon K_\epsilon \to 0$ required in (BI5). A successful shrinking-slit proof must use additional structure of the locality comparison, not only the size and continuity conclusions of RSW.

34 What the continuum crossing estimates give once the signed budget is finite

The finite aggregate signed budget supplies a continuous semimartingale, but by itself it supplies no regularity of its canonical characteristics. The unsigned continuum estimate nevertheless supplies one additional conclusion at that point: the bracket is absolutely continuous with a bounded density. This does not require a tangent of the drift, or any regularity of its total variation.

Proposition 34.1 (Brownian-order conditional moments dominate the bracket)**.** *Let X be a continuous semimartingale in a usual filtration, with canonical decomposition $X = X_0 + M + A$, where M is a continuous local martingale starting at zero and A is continuous predictable finite variation. Suppose that for every $0 \leq s < t \leq T$,*

$$\mathbb{E}\big[(X_t - X_s)^2 \mid \mathcal{F}_s\big] \leq K(t-s) \quad \textit{almost surely} \tag{BR1}$$

with one deterministic finite K. Then, pathwise,

$$d\langle M\rangle_t \le K\,dt. \tag{BR2}$$

Consequently M is square integrable on $[0,T]$ and its bracket has a predictable density v with $0 \le v \le K$ almost everywhere. No bound on the variation or initial tangent of A is asserted.

Proof. Write $C = \langle M\rangle$. For fixed deterministic $s < t$, the quadratic increment sums along refining deterministic partitions of $[s,t]$ converge in probability to $C_t - C_s$. For any $E \in \mathcal{F}_s$, the tower property and (BR1) give

$$\mathbb{E}\left[\mathbf{1}_E \sum_i (X_{t_{i+1}} - X_{t_i})^2\right] \le K(t-s)\mathbb{P}(E).$$

Choose an almost-surely converging subsequence of the nonnegative sums and apply Fatou. Thus

$$\mathbb{E}[\mathbf{1}_E(C_t - C_s)] \le K(t-s)\mathbb{P}(E). \tag{BR3}$$

In particular $\mathbb{E}C_T \le KT$, and $C_t - Kt$ is a continuous supermartingale of class D: all of its stopped absolute values are dominated by $C_T + KT$.

A continuous predictable finite-variation supermartingale of class D is nonincreasing: its existing special-semimartingale decomposition has zero martingale part, and uniqueness identifies its drift with the negative increasing Doob–Meyer compensator. Apply this fact to $C_t - Kt$ to obtain (BR2). Finite expected bracket makes M square integrable, and the predictable Radon–Nikodym density of dC gives the final assertion. □

Application to the continuum classification problem. The conditional exponential estimate already proved in the nonsimple curve class implies (BR1) with a uniform numerical constant depending only on the crossing constants. If the aggregate signed restart budget is finite for a bounded stopped driver, then the quasimartingale criterion supplies its canonical decomposition. Such a stopped driver retains (BR1): on $\{\sigma_L > s\}$ its increment is bounded by the original driver supremum over $[s,t]$, and otherwise it vanishes. The proposition then gives bounded absolutely continuous bracket automatically. Thus the first signed-budget target, if established, leaves drift regularity as the characteristic bottleneck; it does not leave both bracket regularity and drift regularity independent.

Corollary 34.2 (The remaining singular drift is transported without an anomaly)**.** *Assume the proposition for all relevant chart drivers. Before contact with a locality deletion, let h_t be the evolving deletion map and $u'(t) = h_t'(W_t)^2 > 0$. Write $dA = b_t\,dt + dA_t^{\perp}$ for the decomposition relative to capacity Lebesgue measure. Then the restarted deleted-chart characteristics satisfy*

$$d\widehat{A}_{u(t)} = h_t'(W_t)\,dA_t + \tfrac{1}{2}h_t''(W_t)(v_t - 6)\,dt, \qquad d\widehat{A}^{\perp}_{u(t)} = h_t'(W_t)\,dA_t^{\perp}. \tag{BR4}$$

In particular the deletion cannot create or cancel a singular drift component by the variance-six anomaly. The singular part transforms as a one-form, with its capacity clock changed as stated.

Proof. The deletion Itô identity is

$$d\widehat{W}_{u(t)} = h_t'(W_t)\,dM_t + h_t'(W_t)\,dA_t + \tfrac{1}{2}h_t''(W_t)\,dC_t - 3h_t''(W_t)\,dt.$$

Insert $dC = v\,dt$ and use uniqueness of canonical decomposition. The strictly positive continuous capacity-clock derivative makes u locally bi-Lipschitz before deletion contact, so it preserves the distinction between absolutely continuous and singular measures. All terms except the indicated pushforward of $dA^{\perp}$ are absolutely continuous. This proves (BR4). □

A failed uniform-variation shortcut. An unscaled bound on total drift variation in mirrored shrinking deletion charts does not by itself give the local first-order rates needed by the existing rigidity theorem. The anomaly can be supported on capacity windows much shorter than the slit scale. The previously proved pulse example shows why bounded variation budgets alone cannot be replaced silently by drift-density bounds. Conversely, imposing a uniform $o(\sqrt{t})$ initial variation modulus over *all* positive scalings of every chart would be far too strong: if $A^{(\lambda)}_{\lambda^2 t} = \lambda A_t$, then

$$\lambda\, \mathbb{E}\, \mathrm{Var}(A; [0,T]) \le \omega(\lambda^2 T)\lambda\sqrt{T}, \qquad \omega(r) \longrightarrow 0,$$

and $\lambda \downarrow 0$ already forces $A = 0$, without locality. Any useful added variation premise must therefore specify its chart normalization and its domain dependence rather than hide an all-scalings martingale assumption in a uniform modulus.

Remaining research question. Can source locality, physical translation covariance, and the precise crossing hypothesis control the finite signed budget, or rule out the one-form-valued singular drift in (BR4), in one fixed unknown affine structure? The present proposition does not answer either question. It narrows the required characteristic input once semimartingality has been obtained.

Why an interior-hitting shortcut would be invalid. The boundary-unforced crossing estimate alone does not imply a positive lower driving tail: it includes deterministic SLE_0 among its examples. In the Kemppainen–Smirnov definition the unforced annulus part is empty when the inner circle misses the current domain boundary; see [3, Definition 2.3 and Section 1.4]. Thus an argument treating first entry into an interior ball as an already controlled unforced crossing would add an unproved premise. The next ray-deletion comparison repairs the particular driving-confinement argument by using corrected locality; it does not assert general interior small-ball nonconcentration.

34.1 Source locality and a ray deletion give a lower driving tail

The upper unforced-quad estimate alone does not control first entry into an interior ball. The following argument uses source locality essentially, turning the relevant interior point into a boundary slit tip in a comparison domain. It proves a lower *driving* tail, not a general interior small-ball nonconcentration estimate.

Admissible comparison states. For a physical inverse chart $f : \mathbb{H} \to D$ and $r > 0$, require that the domain

$$D' = f(\mathbb{H} \setminus \{iy : y \ge r\}) \tag{RL1}$$

belongs to the family, with source $f(0)$ and either of the two appropriate target prime ends over the original target infinity. It is enough to use the right-hand target end. A canonical inverse chart for this comparison domain is $f \circ F_r$, where

$$F_r(w) = \frac{r}{2}\left(\sqrt{w+1} - \frac{1}{\sqrt{w+1}}\right), \tag{RL1a}$$

with the principal square root. Indeed the square root maps the half-plane onto a quadrant, and $s \mapsto (r/2)(s - 1/s)$ maps that quadrant onto the ray-deleted half-plane, taking 1 to 0. This also provides a jointly Borel comparison state and target choice. Choose boundary colors so they agree with the original colors on the two real sides near zero. Source locality is assumed to compare the ordered prefixes up to first encounter with the removed ray. For regular distinct source and

target points this is a usual source-neighborhood deletion. At rough states, this is an explicit prime-end/source-deletion admissibility requirement, not a consequence of Euclidean neighborhood locality at coincident impressions.

Lemma 34.3 (A lower driving tail from locality and boundary quads)**.** *Assume continuous Loewner generation with capacity $2t$, corrected source locality for (RL1), and the uniform boundary-robust unforced quad bound with constants $M > 0$, $q < 1$ on all these states. Put*

$$\delta = \frac{1}{4} e^{-2\pi M}. \tag{RL2}$$

Then, for every $h > 0$,

$$\mathbb{P}\left\{ \sup_{0 \le s \le h} |W_s| > \delta\sqrt{h} \right\} \ge 1 - q. \tag{RL3}$$

Strong DMP gives the same conditional bound in the inherited centered chart after any finite stopping time.

Proof. In the original half-plane chart set $z = i\sqrt{h}$ and delete the upward ray $A = \{iy : y \ge \sqrt{h}\}$. The comparison domain is $U = \mathbb{H} \setminus A$. It is simply connected. Its two ends at infinity are distinguished prime ends; choosing either one as target allows the boundary colors on the real line to match the original source colors. Corrected source locality couples the original trace and the comparison trace through their first hit of A.

Let E be the event $\sup_{s \le h} |W_s| \le \delta\sqrt{h}$. On E, the original trace and hull up to time h lie in the vertical strip $|\operatorname{Re} w| \le \delta\sqrt{h}$ by the Loewner ODE. If its trace height were at most $\sqrt{h}$, its hull would lie in a disk of radius $\sqrt{(1+\delta^2)h}$. The area theorem would give $2h \le (1+\delta^2)h$, impossible because $\delta < 1$. Consequently the trace reaches height $\sqrt{h}$ by time h and there enters $B(z, 2\delta\sqrt{h})$.

Entry into the open ball occurs strictly before the first encounter with A. If the first ray encounter before h is outside this ball, its height is greater than $\sqrt{h}$, so an earlier height-$\sqrt{h}$ point was already in the ball. If the first encounter itself lies in the ball, continuity puts an earlier point in that open ball. If there is no ray encounter before h, the asserted visit already precedes it. Thus the coupled comparison trace visits the ball on E, using only the common prefix; E itself need not be a common-prefix event.

In U, consider the annulus centered at z of inner radius $2\delta\sqrt{h}$ and outer radius $\sqrt{h}/2$, slit along the upward ray. It is a boundary quad of opening angle 2π, with both side arcs on the two prime-end sides of A. Its modulus is

$$\frac{1}{2\pi} \log \frac{\sqrt{h}/2}{2\delta\sqrt{h}} = M.$$

It is unforced: source zero and either target end at infinity can be connected outside its outer circle, on the chosen target side. The comparison trace starts outside its outer circle, so visiting the inner ball requires a crossing. The boundary-robust convention counts side contacts and a visit ending at the slit tip as well. The quad bound gives comparison visit probability at most q. Hence $\mathbb{P}(E) \le q$, proving (RL3).

The same comparison can be made in every inherited restart chart. Genuine strong DMP therefore supplies the conditional version with the same constants. No conformal covariance of the physical family was used. □

What this does and does not establish. Together with the upper bracket theorem and the stochastic lower-tail criterion, this gives uniformly elliptic bracket for a semimartingale driver from the current crossing bound and the stated ray-deletion locality. It does not identify the density

as six, remove singular drift, or prove semimartingality. In particular it does not infer a general interior hitting estimate from the boundary-quad condition. Deterministic vertical SLE_0 fails the source-locality comparison used here, so it does not contradict this argument.

34.2 The driving variation index is two before semimartingality

The lower driving tail has a consequence which does not assume a finite signed budget, a bracket, or a semimartingale decomposition. For a continuous real path x on $[a,b]$, write

$$V_r(x;[a,b]) = \sup_{\pi} \sum_j |x(t_{j+1}) - x(t_j)|^r, \qquad r > 0, \tag{VI1}$$

where the supremum is over finite partitions of that interval.

Proposition 34.4 (Uniform diffusive lower tails force variation index two)**.** *Let X be continuous and adapted, and suppose there are fixed $\delta, p > 0$ such that for all deterministic $t \geq 0$ and $h > 0$,*

$$\mathbb{P}\{ \sup_{0 \leq u \leq h} |X_{t+u} - X_t| > \delta\sqrt{h} \mid \mathcal{F}_t \} \geq p. \tag{VI2}$$

Then, on one probability-one event, $V_r(X;[a,b]) = \infty$ for every nontrivial interval and every $0 < r < 2$. If its paths are also locally Hölder for every exponent below $1/2$, then on that same event $V_r(X;[a,b]) < \infty$ on every compact interval for every $r > 2$. Thus the variation index is exactly two on every nontrivial interval. No assertion about V_2 or a quadratic-variation limit is made.

Proof. Fix first a deterministic compact interval $[a,b]$, put $n = 2^m$, $h = (b-a)/n$, $t_j = a + jh$, and define

$$I_j = \mathbf{1}_{\{\sup_{0 \leq u \leq h} |X_{t_j+u} - X_{t_j}| > \delta\sqrt{h}\}}.$$

These indicators are measurable at t_{j+1}, and $\mathbb{E}[I_j \mid \mathcal{F}_{t_j}] \geq p$. The martingale differences $D_j = I_j - \mathbb{E}[I_j \mid \mathcal{F}_{t_j}]$ are orthogonal in L^2 and satisfy $\mathbb{E}(\sum_j D_j)^2 \leq n$. Therefore

$$\mathbb{P}\{\textstyle\sum_j I_j < pn/2\} \leq \frac{4}{p^2 n}. \tag{VI3}$$

This is summable over dyadic n. Borel–Cantelli gives $\sum_j I_j \geq pn/2$ for all sufficiently large m, almost surely. For every successful block, choose a time where its displacement from the left endpoint exceeds $\delta\sqrt{h}$. The ordered union of these times and the block endpoints is a finite partition, so simultaneously for every $r > 0$,

$$V_r(X;[a,b]) \geq \frac{pn}{2} \delta^r h^{r/2} = \frac{p\delta^r}{2} (b-a)^{r/2} n^{1-r/2}. \tag{VI4}$$

No adapted selection is needed: this is a pathwise lower bound for a supremum over partitions. It diverges for every $r < 2$ on the same event. Intersect over rational intervals, and place a rational subinterval inside any other nontrivial interval.

If $r > 2$, choose $1/r < \alpha < 1/2$. A Hölder-α constant K on a compact interval bounds every partition sum by $K^r \sum_j |t_{j+1} - t_j|^{\alpha r} \leq K^r (b-a)^{\alpha r}$. The simultaneous upper assertion follows from the countable collection of rational Hölder exponents. □

By the ray-locality lemma and the already proved upper driving bounds, the continuum family has this variation index under the stated ray-deletion admissibility. In particular its drivers cannot have finite variation on any nontrivial interval. The lower proof controls block oscillations, not terminal increments of the blocks; replacing it by a claim of positive quadratic variation along deterministic partitions would be unjustified. Nor does variation index two imply semimartingality: the earlier Brownian-plus-Takagi example already shows the latter conclusion needs signed information.

34.3 A bounded bracket becomes uniformly elliptic

Proposition 34.5 (Conditional lower tails make a bounded bracket elliptic)**.** *Let $X = X_0 + M + A$ be a continuous semimartingale with $d\langle M\rangle = v_t dt$, $0 \le v_t \le K$, and A continuous finite variation. Suppose the conditional lower-tail bound (RL3), with lower probability $p > 0$, holds with X in place of W, at least at every deterministic $\tau = t$ and for all $h > 0$ in a finite horizon. Then*

$$v_t \ge \frac{p\delta^2}{32} \quad dt \otimes d\mathbb{P}\text{-almost everywhere.} \tag{NC3}$$

No density or initial tangent of A is assumed.

Proof. Choose a bounded predictable version of v. For Lebesgue-almost every deterministic t, Fubini and Lebesgue differentiation give, almost surely,

$$\frac{1}{h}\int_t^{t+h} v_s ds \longrightarrow v_t, \qquad \frac{\mathrm{Var}(A;[t,t+h])}{\sqrt{h}} \longrightarrow 0. \tag{NC4}$$

The second assertion follows because the finite Stieltjes variation measure has a finite one-sided Lebesgue derivative almost everywhere. Fix such a t, set $\eta = p\delta^2/32$, and let $E = \{v_t < \eta\} \in \mathcal{F}_t$. The lower tail assumption implies

$$p\mathbb{P}(E) \le \mathbb{P}\{E, \sup_{u\le h} |X_{t+u} - X_t| > \delta\sqrt{h}\}.$$

The corresponding drift supremum is at most $\mathrm{Var}(A;[t,t+h])$, so its probability of exceeding $\delta\sqrt{h}/2$ tends to zero by (NC4). Since $v \le K$, M is square integrable on finite horizons and Doob's inequality gives

$$\mathbb{P}\{E, \sup_{u\le h} |M_{t+u} - M_t| > \delta\sqrt{h}/2\} \le \frac{16}{\delta^2 h}\mathbb{E}[\mathbf{1}_E(C_{t+h} - C_t)].$$

Boundedness of v and (NC4) permit dominated convergence of the last normalized expectation to $\mathbb{E}[\mathbf{1}_E v_t]$. Thus $p\mathbb{P}(E) \le 16\eta\mathbb{P}(E)/\delta^2 = (p/2)\mathbb{P}(E)$, forcing $\mathbb{P}(E) = 0$. Apply Fubini again. □

Combining this proposition with the upper bracket theorem and the ray-locality lemma gives $c\,dt \le dC \le K\,dt$, with $c = (1-q)\delta^2/32 > 0$, once the finite signed budget supplies semimartingality. The density is not yet identified as six. The lower conclusion uses corrected ray-deletion locality, not the boundary-quad estimate alone.

34.4 Bounded Brownian bracket does not remove a singular drift

The reflected Brownian process $X_t = |B_t|$ is a sharp process-level warning at the next stage. In its usual natural filtration it has canonical decomposition $X = N + \Lambda$, where N is Brownian motion and Λ is the continuous Skorokhod regulator, increasing only on $\{X = 0\}$. That set has zero Lebesgue measure almost surely, so $d\Lambda$ is singular and nonzero. Tanaka's formula gives the decomposition in the Brownian filtration; the regulator is recovered from the occupation local time of X, so both summands are adapted to its smaller natural filtration, where the martingale identity continues to hold by conditioning.

For every stopping time τ of X,

$$\sup_{u\le h}\big||B_{\tau+u}| - |B_\tau|\big| \le \sup_{u\le h}|B_{\tau+u} - B_\tau|. \tag{RB1}$$

Brownian strong Markov and conditioning therefore give uniform conditional Gaussian-size increment bounds for X, even though its finite signed budget is carried by a singular regulator. In particular its bracket density is exactly one. This is not a curve-family counterexample: neither source locality nor the domain-indexed exploration axioms are asserted for this example. It shows why bounded bracket and unsigned driving estimates do not finish the drift argument.

35 Analytic locality and signed characteristic measures

Scope and matched localization. This section uses source locality only to identify two initial ordered trace laws and their natural filtrations. On their common trace, let h_t be the evolving analytic deletion map, recentered at the source, and write

$$p_t = h_t'(W_t) > 0, \quad q_t = h_t''(W_t), \quad u(t) = \int_0^t p_s^2\, ds, \quad Y_t = \widehat{W}_{u(t)} = h_t(W_t) - h_0(0). \tag{LB1}$$

The exact recentering convention has no effect on the derivatives or characteristics. The two capacity clocks and their inverses are stopping-time clocks in the corresponding natural trace filtrations, by geometric reconstruction of a stopped trace. Stop on a strictly positive interval where the deletion has not collided with the trace, its field and inverse are absolutely continuous in time as C^2-valued fields on compact analytic source neighborhoods, with bounded spatial derivatives, and

$$|W| \le L, \quad |Y| \le \widehat{L}, \quad 0 < m \le p \le P, \quad |q| \le Q, \quad \theta \le T. \tag{LB2}$$

All identities below concern the processes stopped at this same physical trace time θ, not two independently selected exit rules. The inherited stop $u(\theta)$ is a stopping time in deleted capacity, and $u(\theta) \le P^2 T$. The deletion PDE gives the stated time regularity before collision. In particular, the time dependence of the field is pathwise finite variation; no stochastic regularity of W is assumed in writing (LB1).

Proposition 35.1 (Local semimartingality and finite signed budgets are analytic-locality invariants)**.** *Under this setup, the original driver is locally a continuous semimartingale if and only if the deleted driver is, on the matched initial intervals. Suppose they have canonical decompositions $W = M + A$ and $\widehat{W} = \widehat{M} + \widehat{A}$, with brackets C and $\widehat{C}$. Set $D_t = \widehat{A}_{u(t)}$. Then, before the matched stop,*

$$dD = p\, dA + \frac{q}{2}(dC - 6dt), \qquad d\widehat{C}_{u(t)} = p^2\, dC. \tag{LB3}$$

For the bounded stopped processes let $V = \mathbb{E}\operatorname{Var} A$ and $\widehat{V} = \mathbb{E}\operatorname{Var} D$, equivalently their conditional-variation budgets when finite. Then

$$\widehat{V} \le (P + LQ)V + \tfrac{1}{2}QL^2 + 3QT, \tag{LB4}$$

and

$$V \le \left(m^{-1} + Q\widehat{L}m^{-3}\right)\widehat{V} + \tfrac{1}{2}Q\widehat{L}^2 m^{-3} + 3QT/m. \tag{LB5}$$

In particular finite expected canonical variation, hence finite aggregate signed budget, transfers both ways after these matched bounded localizations. Neither estimate supplies a first-order time rate.

Proof. Assume first W is a semimartingale. The finite-variation-field Itô–Wentzell formula applied to $h_t(W_t)$ gives

$$dY_t = p_t\, dM_t + p_t\, dA_t + \tfrac{1}{2}q_t\, dC_t - 3q_t dt.$$

Here the final term is the Loewner deletion field identity $(\partial_t h_t)(W_t) = -3h_t''(W_t)$, which requires no stochastic calculus to derive. The geometric stopping-time clocks transfer the semimartingale and its natural filtration to deleted capacity. Conversely, if the deleted driver is a semimartingale, its pullback Y is one in the common trace filtration. Apply the same finite-variation-field formula to $h_t^{-1}(Y_t + h_0(0))$. The inverse field is adapted, spatially analytic, and of finite variation in time on this interval; its derivatives are bounded under localization. Thus W is a semimartingale. Uniqueness of canonical continuous decompositions now gives (LB3), without assuming any drift density or initial tangent.

We use a useful elementary budget/bracket estimate. If a continuous semimartingale $Z = N + F$ starts at zero, is bounded by K, and has $\mathbb{E}\operatorname{Var} F < \infty$, then

$$\mathbb{E}\langle N\rangle_T \le K^2 + 2K\mathbb{E}\operatorname{Var} F. \tag{LB6}$$

To prove it, stop to bound the bracket and total variation, take expectations in Itô's formula for Z^2, and then use Fatou. The resulting finite expected bracket also makes the martingale square integrable, so the usual unstopped identity is justified. Apply (LB6) first to W and use (LB3) and (LB2):

$$\widehat{V} \le PV + \tfrac{1}{2}Q\mathbb{E}C_\theta + 3QT,$$

which gives (LB4). In the opposite direction put $\widetilde{C}_t = \widehat{C}_{u(t)}$ and solve (LB3):

$$dA = p^{-1}dD - \frac{q}{2p^3}d\widetilde{C} + 3qp^{-1}dt.$$

Apply (LB6) to Y and use (LB2) to obtain (LB5). These estimates are invoked with the source-side budget finite; the corresponding conclusion then guarantees finiteness on the other side. The equality between budgets and expected variation is the earlier exact conditional-variation identity, applied to the matched stopped processes. A budget in the original capacity and one in the deleted capacity must retain their inherited stopping rules. □

Proposition 35.2 (Two deletion directions eliminate singular characteristics)**.** *Take two admissible deletion maps h_t^+, h_t^- on the same initial trace. Suppose their drivers are continuous semimartingales. Define $p^\pm, q^\pm, u_\pm$ as in (LB1), let $D_t^\pm = \widehat{A}^\pm_{u_\pm(t)}$, and suppose*

$$\Delta_t = p_t^+ q_t^- - p_t^- q_t^+ \neq 0 \tag{LB7}$$

on a positive initial interval. Then the original driver is a semimartingale and its characteristic measures satisfy

$$\boxed{\begin{aligned} dA &= \frac{q^-\,dD^+ - q^+\,dD^-}{\Delta}, \\ dC - 6dt &= \frac{2(p^+\,dD^- - p^-\,dD^+)}{\Delta}. \end{aligned}} \tag{LB8}$$

No initial derivatives or drift densities are required. In particular, if both deleted drift measures are absolutely continuous in their capacity clocks, then the original drift and bracket are absolutely continuous on this interval. The original drift need not have been assumed absolutely continuous.

Proof. The preceding proposition transfers semimartingality from either deleted chart to the original one. Apply (LB3) to both, and invert the two-by-two system of signed measures. This gives (LB8). The clocks $du_\pm = (p^\pm)^2 dt$ are locally bi-Lipschitz after localization, so absolute continuity of a deleted drift in its own capacity is exactly absolute continuity of its pullback $D^\pm$ in t. Equation (LB8) then proves the assertion. □

The singular-part alternative is exact. Take pathwise Lebesgue decompositions with respect to dt. Writing superscript $\perp$ for singular Stieltjes parts, (LB3) says

$$dD^{\perp} = p\,dA^{\perp} + \tfrac{1}{2}q\,dC^{\perp}, \qquad d\widetilde{C}^{\perp} = p^2\,dC^{\perp}. \tag{LB9}$$

Thus absence of a singular bracket is invariant under analytic local coordinate changes. Once the bracket is absolutely continuous, absence of a singular drift is invariant as well. If the bracket has a singular part, an Itô correction can cancel that part from a particular chart's drift; two distinct directions with (LB7) cannot both cancel it. Formula (LB8) also gives the pathwise measure bounds

$$\begin{aligned} |dA| &\le \frac{|q^-|\,|dD^+| + |q^+|\,|dD^-|}{|\Delta|}, \\ |dC - 6dt| &\le \frac{2p^+|dD^-| + 2p^-|dD^+|}{|\Delta|}. \end{aligned} \tag{LB10}$$

These also hold for singular parts, with the second left side replaced by the positive measure $dC^{\perp}$.

For the mirrored shrinking slits of (DR6), initially $p_0^{\pm} = d_\epsilon \asymp \epsilon$ and $q_0^- = -q_0^+ = -e_\epsilon^+$ with $|e_\epsilon^+| \asymp \epsilon^{-1}$. The determinant is nonzero. By stopping when the coefficients depart by a sufficiently small relative amount from these initial values, (LB10) gives, with an absolute numerical K independent of small ϵ,

$$|dA| \le K\epsilon^{-1}(|dD^+| + |dD^-|), \qquad |dC - 6dt| \le K\epsilon(|dD^+| + |dD^-|). \tag{LB11}$$

In particular singular bracket mass costs at least inverse-linear singular drift mass across these two deleted charts. The matched positive stopping interval may shrink with ϵ. Therefore finiteness of each deleted budget does not permit passing to $\epsilon = 0$ in (LB11) on a fixed interval. A quantitative uniform budget estimate is still needed; no regularity or classification is being deduced from bare locality by this calculation.

What this changes in the research target. Finiteness of a localized signed budget is a property of the local physical source law within a fixed complex structure, rather than an artifact of choosing one of its analytic deletion charts. Its numerical size is not invariant: (LB3)–(LB5) exhibit the bracket correction and small first derivative which can inflate it. Analytic coordinate changes are being used throughout; an arbitrary real affine normalization is not such a change. Thus this result neither selects the unknown affine structure nor asserts semimartingality in all affine structures. The new positive regularization is the two-direction elimination of singular characteristics. Obtaining finite budgets and the quantitative scale control required for Brownian rigidity remains open.

35.1 Target-invariant force-point branches still fail source locality

A useful test is the familiar target-invariant branching $\mathrm{SLE}_\kappa(\kappa - 6)$ construction. Target invariance holds within one physical domain; it does not assert invariance under deleting remote pieces of that domain. The following calculation excludes the standard one-sided force-point family from the present conjecture, including the singular drift at an initially coincident force point. It does not assume a finite initial drift tangent.

Use the convention

$$dW_t = \sqrt{\kappa}\,dB_t + \frac{\rho}{W_t - V_t}\,dt, \qquad dV_t = \frac{2}{V_t - W_t}\,dt, \qquad (W_0, V_0) = (0, 0^+), \quad \rho > -2. \tag{FP1}$$

The right force point is the boundary-prime-end limit immediately to the right of the source, transported by the Loewner map. The gap $(V-W)/\sqrt{\kappa}$ is a Bessel process of dimension $\delta = 1+2(\rho+2)/\kappa > 1$. In particular, its zero set has Lebesgue measure zero, and the force drift in (FP1) is locally integrable. These standard statements follow from the Bessel construction; see Sheffield [2], Section 3.1, Propositions 3.1 and 3.3, and Section 3.3.

Proposition 35.3 (Only the ordinary SLE_6 member is source-local)**.** *Let $\kappa > 0$. Consider the canonical conformally covariant family with one force point immediately on a specified side of its source, defined by (FP1). If it satisfies source-neighborhood locality under remote hull deletions, then $\kappa = 6$ and $\rho = 0$. In particular, for $4 < \kappa < 8$, the target-invariant $\mathrm{SLE}_\kappa(\kappa-6)$ family fails that locality whenever $\kappa \neq 6$.*

Proof. Let A be a fixed half-plane hull away from 0, let $h : \mathbb{H} \setminus A \to \mathbb{H}$ be its hydrodynamic map, and compare the exploration in $\mathbb{H}$ with that in $\mathbb{H} \setminus A$. Center the new source at $h(0)$. Work up to a positive stopping time before the growing hull encounters A, chosen so that the comparison maps are analytic across a fixed interval containing W_t and V_t. Let h_t be the evolving deletion map and $ds = h_t'(W_t)^2 dt$. The centered transformed driver and force point are $h_t(W_t) - h(0)$ and $h_t(V_t) - h(0)$. The force point is transported exactly: the deletion-map PDE and (FP1) give

$$d\{h_t(V_t)\} = \frac{2h_t'(W_t)^2}{h_t(V_t) - h_t(W_t)}\, dt. \tag{FP2}$$

The driving change-of-coordinate calculation yields the Brownian part $\sqrt{\kappa}\, d\widehat{B}_s$ and drift, in the original dt clock,

$$\frac{\rho h_t'(W_t)}{W_t - V_t} + \left(\frac{\kappa}{2} - 3\right) h_t''(W_t).$$

For the force-point law in the deleted domain, the required drift in that same clock is

$$\frac{\rho h_t'(W_t)^2}{h_t(W_t) - h_t(V_t)}.$$

Therefore locality would require

$$R_t = \rho \left\{ \frac{h_t'(W_t)}{W_t - V_t} - \frac{h_t'(W_t)^2}{h_t(W_t) - h_t(V_t)} \right\} + \frac{\kappa - 6}{2} h_t''(W_t) = 0 \tag{FP3}$$

for Lebesgue-almost every sufficiently small t. This follows from uniqueness of the canonical semimartingale drift: the force point is the canonical one-sided source-boundary Loewner flow, with its instant-reflecting continuation at collisions. It is adapted to the driver and is transported as the specified boundary side, not an independently reassignable mark. The capacity change is invertible on the stopped comparison interval. All drift terms are integrable since $\delta > 1$.

For an analytic H with $H'(w) > 0$, put $d = w - v$. Taylor expansion gives

$$\frac{H'(w)}{d} - \frac{H'(w)^2}{H(w) - H(v)} = -\frac{1}{2} H''(w) + \frac{1}{6} H'(w) \mathcal{S}H(w)\, d + O(d^2), \tag{FP4}$$

where $\mathcal{S}H = H'''/H' - \frac{3}{2}(H''/H')^2$ is the Schwarzian. The remainder is uniform in a compact analytic neighborhood on which H' is bounded away from zero. Consequently

$$R_t = \frac{\kappa - 6 - \rho}{2} h_t''(W_t) + \frac{\rho}{6} h_t'(W_t) \mathcal{S}h_t(W_t)(W_t - V_t) + O((W_t - V_t)^2). \tag{FP5}$$

Choose, for example, the vertical slit $A = \{x + iy : 0 \le y \le a\}$ with $x, a > 0$. Its map is $h(z) = x + \sqrt{(z-x)^2 + a^2}$, with the hydrodynamic branch. At 0 both $h''(0) \neq 0$ and $\mathcal{S}h(0) \neq 0$. More explicitly, writing $u = z - x$, the Schwarzian on the real analytic source interval is

$$\mathcal{S}h(z) = -\frac{3a^2(u^2 + a^2/2)}{u^2(u^2 + a^2)^2} < 0. \tag{FP6}$$

As $t \downarrow 0$, $W_t, V_t \to 0$ and the analytic maps h_t converge locally to h, with their derivatives. If $\kappa - 6 - \rho \neq 0$, (FP5) therefore makes R_t bounded away from zero on a positive initial interval, contradicting (FP3). Thus $\rho = \kappa - 6$.

If this common value is nonzero, the second term of (FP5), divided by $W_t - V_t$, tends to $\rho h'(0)\mathcal{S}h(0)/6 \neq 0$. After decreasing the positive comparison time, $R_t \neq 0$ whenever $W_t \neq V_t$. The latter holds for Lebesgue-almost every time by the Bessel zero-set statement, again contradicting (FP3). Hence $\rho = 0$ and $\kappa = 6$.

Although written in half-plane coordinates, the test applies to bounded physical domains: map the half-plane and the remote slit into a disk by one fixed Möbius map. The resulting physical domains agree near their common source. The same calculation in their chordal charts then contradicts source locality there. □

Why target invariance does not contradict this exclusion. For a Möbius H one has the exact identity

$$\frac{H'(w)}{w - v} - \frac{H'(w)^2}{H(w) - H(v)} = -\frac{1}{2}H''(w). \tag{FP7}$$

Thus the force-point drift with $\rho = \kappa - 6$ cancels the entire coordinate anomaly for Möbius changes of target. A remote domain hull deletion instead has nonzero Schwarzian, leaving the second term of (FP5). Schramm–Wilson [1], Theorem 3, establishes the Möbius coordinate-change theorem with sum of force weights $\kappa - 6$; it does not assert remote-domain locality: https://nyjm.albany.edu/nyjm/j/2005/11-31.pdf. Sheffield's Proposition 3.14 and Section 4.2 use target invariance to construct branching radial explorations within one fixed domain. Those statements are consistent with the calculation above.

There is also an extra-state issue for strong DMP. A force point started at 0^+ is usually separated from the tip at positive times. Conditional continuation then carries its current boundary-prime-end image V_t, rather than restarting with a fresh force point at W_t^+. The standard force-point Markov property is therefore a property of the domain with this additional mark (and, in side-swapping variants, the appropriate excursion state). It is not by itself the required strong-DMP identity on a domain with only the two colored boundary arcs and marked source/target. A proposed intrinsic recovery of that mark from the colored domain would require its own proof. This issue is independent of the locality failure already proved.

Consequence for the present research direction. Even if the desired RSW estimates were granted to these branches, they would not be counterexamples to the current conjecture: the locality test excludes them before RSW enters. They are nevertheless a useful singular-drift test. Their Bessel force drift is integrable, so merely obtaining semimartingality or finite aggregate signed variation would not eliminate them. The exact locality comparison supplies information beyond initial drift tangents; in this explicit ansatz it is the Schwarzian remainder that distinguishes target independence from full source locality.

The failure of source locality persists after any one fixed affine change applied to the entire family: such a change preserves common source neighborhoods and transports their stopped-law equality.

36 Unsuccessful routes, revised objectives, and next experiments

Why several tempting arguments fail. Independent site colors give exact discrete domain Markov and locality at every occupation probability; these axioms do not by themselves encode criticality. Near-critical scaling limits are therefore natural candidates, but configuration convergence and a Markov theorem in the near-critical parameter do not establish the required ordered exploration kernels at random slit domains. We do not use this incomplete transfer as the main counterexample. Nor does singularity relative to isotropic SLE alone exclude an anisotropic affine image. A genuine dilation defect would do so.

A random mixture over fixed affine metrics is also not automatically domain Markov: observing the initial curve can update the metric posterior. The exact example avoids that failure through deterministic terminal information that remains visible in every current domain. This should be distinguished from unsupported claims that every stationary random environment yields a domain-Markov annealed interface.

Applying Schramm's usual Brownian-driving argument directly would silently assume that a conformally mapped slit domain has the same law as the original domain. Physical translations do not imply that premise. The deletion-chart identity above identifies the precise missing drift control. We have not proved that the original axioms imply semimartingale driving, driving martingales, or even a fixed underlying conformal structure.

Historical proposed strengthening, retained for comparison. This earlier proposal is not the current assumption package or research priority. It is retained to distinguish it from the continuum focus below. Restrict to nondegenerate continuous Loewner exploration laws with strong domain Markov, fixed-target locality, physical translation covariance, consistency under changing distant boundary conditions and target, and continuity under controlled Jordan-domain approximation, including changes near the terminal point. Require a genuine criticality condition, such as nondegenerate crossing estimates uniform over all scales, formulated for a common continuum coloring or a compatible exploration system. Does this force one deterministic conformal metric, and then affine SLE_6? Even the sufficiency of this strengthened collection is open here. One must not promote its plausibility to a theorem. Full similarity covariance alone is already defeated by the exact construction above.

Current priorities after the 28 October scope clarification. The continuum classification under uniform RSW is the immediate problem. Strong DMP, corrected source-neighborhood locality, and translation covariance are assumed for the continuum family; proving them for lattice limits is deferred. The old discrete-transfer and independent-color coupling routes remain recorded as tools for that later stage.

Section 32 makes an explicit upper unforced-quad input sufficient for continuous admissible non-simple traces to have infinite capacity lifetime and conditional exponential driving bounds. This uses no embedded exploration interval or conformal covariance of the unknown family. Continuous-driver admissibility is separate, and has not been inferred from arbitrary samplewise simple approximations.

The corrected ray-locality comparison now supplies a uniform diffusive lower tail. Proposition 34.4 consequently forces variation index two before any semimartingale premise. The next target at

that checkpoint was finite aggregate signed restart variation (QV4) in a common unknown affine structure. If obtained, the upper and lower tail bounds automatically give a bounded uniformly positive bracket density, without drift tangents. Thus the remaining characteristic tasks are eliminating the intrinsically transported singular drift and controlling the absolutely continuous bias strongly enough to identify variance six.

The exact locality-budget formulas show why finiteness transports through analytic deletions but its numerical size may inflate. The matched comparison intervals can shrink with the deleted slit; no uniform first-order budget rate follows. The one-force-point calculation gives an example where a full Schwarzian comparison rules out singular initial bias without tangent assumptions. Investigate whether that non-Möbius information can control general signed budgets, rather than assuming a finite-dimensional force-point closure or all-scalings drift modulus.

The decoder remains a supporting model result. Its simple arcs must not be interpreted as time segments of the densely self-touching exploration. No frontier-model law, source-germ tensor recovery at every stopping time, or nonconstant conserved-label construction is proved here.

Reproducibility and preservation. The concise handoff is `STATUS.md`. The polished driving, endpoint-decoding, and independent-color coupling results are in `final.tex`; the historical fixed-target counterexample remains under `archive/`. The detailed record contains all substantive arguments, including useful failed approaches, without relying on the conversation or ignored scratch files. Both current reports are compiled under `results/`. Independent checks provide evidence of correctness; successful compilation checks presentation only.

37 Priority change: retain the unknown affine structure

The 29 October discussion identifies a gap in treating finite signed restart variation as the next unconditional target. We do not know whether ordinary driving in the wrong affine structure has a finite signed budget, even for a known affine image of SLE_6. We must first understand that model and choose regularity targets that encompass it. Existence of one common affine structure with tame driving is a major part of the classification problem, not an available normalization.

The immediate program has two linked parts: determine the stochastic regularity and capacity changes of known affine-SLE_6 in arbitrary coordinates, and extract constraints from the abstract continuum axioms that remain valid in such coordinates. Retain one unknown constant physical Beltrami coefficient as part of the analysis; a conformal Loewner chart transports it to a history-dependent field. Neither an ordinary Itô formula at the tip nor absolute continuity of the affine capacity clock is available without proof. The conditional rigidity criteria remain useful after the appropriate regularity is obtained. The results below provide coordinate-compatible exit clocks and a necessary clock test for wrong-coordinate semimartingality.

38 Affine deformation without assuming regular driving characteristics

Scope. The statements in this section are deterministic statements about bounded half-plane hulls and continuously generated Loewner chains. No simple-curve, zero-area, or semimartingale hypothesis is made. Boundary values of conformal maps are interpreted at the distinguished tip prime end. A real-linear map of the half-plane preserves filled hulls, and the transformed chain is assumed to have its continuous capacity parametrization, as supplied by the affine admissibility results for the model under consideration.

Let $g_t : \mathbb{H}\setminus K_t \to \mathbb{H}$ be hydrodynamically normalized, $g_t(z) = z + 2t/z + O(z^{-2})$, with continuous driver W_t, and put $f_t = g_t^{-1}$. Let $L(z) = az + b\bar{z}$ be an orientation-preserving real-linear self-map of $\mathbb{H}$. Write

$$s(t) = \tfrac{1}{2}\,\mathrm{hcap}(LK_t), \qquad F_t = \widehat{g}_{s(t)} \circ L \circ f_t, \qquad U_t = F_t(W_t), \tag{AB1}$$

where $\widehat{g}$ uniformizes the transformed hulls. Then

$$\mu_{F_t}(z) = \frac{b}{a}\,\frac{\overline{f_t'(z)}}{f_t'(z)}. \tag{AB2}$$

For a model in which $L\gamma$ is SLE_6, $U_t = \sqrt{6}B_{s(t)}$. This is an identity for a known model, not a classification consequence of the exploration axioms.

Proposition 38.1 (An interior Stieltjes evolution)**.** *At every $z \in \mathbb{H}$, locally in finite time,*

$$dF_t(z) = \frac{2\,ds(t)}{F_t(z) - U_t} - \left\{ \frac{2(F_t)_z(z)}{z - W_t} + \frac{2(F_t)_{\bar{z}}(z)}{\bar{z} - W_t} \right\} dt. \tag{AB3}$$

The equation is an equality of finite-variation measures, locally uniformly on compact subsets of $\mathbb{H}$. It does not require that s be absolutely continuous. The field $\mu_t = \mu_{F_t}$ satisfies, in the interior,

$$\partial_t \mu_t = -v_t \partial_z \mu_t - \overline{v_t}\partial_{\bar{z}}\mu_t + (v_t' - \overline{v_t'})\mu_t, \qquad v_t(z) = \frac{2}{z - W_t}. \tag{AB4}$$

Proof. Inverse Loewner evolution gives $\partial_t f_t = -v_t f_t'$. Apply the ordinary chain rule to (AB1), using the Loewner equation for $\widehat{g}$ in its clock s and the real-linear derivative of L. The two contributions from $\partial_t f_t$ are exactly the two dt terms in (AB3). On any compact spatial set and finite localized time interval, the images stay a positive distance from the real line; the spatial derivatives and denominators are bounded. Since s is continuous increasing, the displayed terms have finite total variation. For (AB4), differentiate $\partial_t f_t' = -v_t f_t'' - v_t' f_t'$ and its conjugate in (AB2). This calculation is entirely in the interior. □

Interior finite variation of F does not justify applying Itô's formula to $F_t(W_t)$. Neither boundary differentiability of F_t at W_t nor the necessary uniform boundary time regularity has been proved.

38.1 A normalization-safe first variation at an arbitrary hull

We use the boundary-fixed affine perturbation

$$L_\epsilon(w) = w + \epsilon\nu(\bar{w} - w), \qquad \nu \in \mathbb{C}, \quad \epsilon \in \mathbb{R}. \tag{AV1}$$

For small ϵ it maps $\mathbb{H}$ onto itself and fixes every real boundary point. These perturbations span the two real anisotropy directions modulo similarities.

Fix a bounded hull $K = K_t$ contained in $\{|w| \le R\}$ with $R > 0$. Take a real smooth radial cutoff χ_R equal to one on $|w| \le 2R$ and zero on $|w| \ge 4R$, with derivatives of order j bounded by $C_j R^{-j}$. Set

$$V(w) = \chi_R(w)\nu(\bar{w} - w), \qquad \Phi_\epsilon(w) = w + \epsilon V(w), \qquad f = g_t^{-1}, \quad W = W_t. \tag{AV2}$$

For sufficiently small $|\epsilon|$ this is a plane diffeomorphism, fixes $\mathbb{R}$ pointwise, preserves $\mathbb{H}$, and equals L_ϵ on K. Define the compactly supported bounded coefficient on $\mathbb{H}$

$$\eta(z) = V_{\bar{w}}(f(z))\,\frac{\overline{f'(z)}}{f'(z)}. \tag{AV3}$$

Extend it to the lower half-plane by $\eta(\bar{z}) = \overline{\eta(z)}$.

Proposition 38.2 (First variation indexed by the original hull prefix)**.** *Let*

$$s_\epsilon = \tfrac{1}{2}\,\mathrm{hcap}(L_\epsilon K), \qquad U_\epsilon = \widehat{g}^{\,\epsilon}_{s_\epsilon}(L_\epsilon\gamma(t)), \tag{AV4}$$

where the second expression is interpreted by the tip prime end. Both quantities are differentiable at $\epsilon = 0$*, and*

$$\boxed{\begin{aligned} \partial_\epsilon U_\epsilon|_0 &= -\frac{2}{\pi}\,\mathrm{Re}\int_{\mathbb{H}} \frac{\eta(z)}{z-W}\,d^2z, \\ \partial_\epsilon s_\epsilon|_0 &= \frac{1}{\pi}\,\mathrm{Re}\int_{\mathbb{H}} \eta(z)\,d^2z. \end{aligned}} \tag{AV5}$$

The integrals are absolutely convergent, including at the tip. Their values are independent of the choice of the stated cutoff. There are absolute cutoff-dependent constants such that

$$|\partial_\epsilon U_\epsilon|_0 \le C|\nu|R, \qquad |\partial_\epsilon s_\epsilon|_0 \le C|\nu|R^2. \tag{AV6}$$

Proof. The derivative of Φ_ϵ is uniformly close to the identity, so this map is a global diffeomorphism for sufficiently small ϵ. Since $\mathrm{Im}\,\Phi_\epsilon(x+iy) = y(1-2\epsilon\,\mathrm{Re}(\nu)\chi_R(x+iy))$, it preserves both half-planes. Its image of K agrees with $L_\epsilon K$.

Put $F_\epsilon = \widehat{g}^{\,\epsilon}_{s_\epsilon} \circ \Phi_\epsilon \circ f$ and reflect it across $\mathbb{R}$. It is a plane quasiconformal map with principal normalization

$$F_\epsilon(z) = z + \frac{2(s_\epsilon - t)}{z} + O(z^{-2}) \quad (z \to \infty). \tag{AV7}$$

Indeed Φ_ϵ is the identity outside a fixed disk, so the composition there is the composition of the two hydrodynamic conformal maps. Its Beltrami coefficient is, in the upper half-plane,

$$m_\epsilon(z) = \frac{\epsilon V_{\bar w}(f(z))}{1+\epsilon V_w(f(z))}\,\frac{\overline{f'(z)}}{f'(z)} = \epsilon\eta(z) + O(\epsilon^2), \tag{AV8}$$

with the reflected coefficient in the lower half-plane. The remainder bound is in L^∞, supported on one bounded disk independent of ϵ. Boundedness of the support follows also directly from hydrodynamic univalence: the image under g_t of the portion of $\mathbb{H}\setminus K$ in a fixed disk is bounded. With the above cutoff scales, the support is contained in a disk of radius CR.

Here is an explicit version of the required differentiability argument. Let

$$\mathcal{C}h(z) = \frac{1}{\pi}\int_{\mathbb{C}} \frac{h(\zeta)}{z-\zeta}\,d^2\zeta, \qquad \mathcal{S} = \partial_z\mathcal{C}.$$

For fixed $p > 2$, the Beurling transform is bounded on L^p. For sufficiently small ϵ the principal solution has

$$F_\epsilon = z + \mathcal{C}h_\epsilon, \qquad h_\epsilon = (I - m_\epsilon\mathcal{S})^{-1}m_\epsilon. \tag{AV9}$$

The Neumann series and (AV8) imply $h_\epsilon/\epsilon \to \eta$ in L^p, and also in L^1 since all these functions have the same bounded support. On functions supported in this disk, the Cauchy transform maps L^p, $p > 2$, continuously to bounded continuous functions, by Hölder's inequality at its integrable singularity and the elementary far-field bound. Consequently

$$\frac{F_\epsilon(z) - z}{\epsilon} \longrightarrow \mathcal{C}\eta(z) \tag{AV10}$$

uniformly in z, including $z = W \in \mathbb{R}$. The boundary identity $U_\epsilon = F_\epsilon(W)$ gives the first formula (AV5); reflection combines the two half-plane integrals. The coefficient of $1/z$ in (AV9) is $\pi^{-1}\int h_\epsilon$, so (AV7) and L^1 convergence give the second formula. Neither step differentiates the physical tip.

All cutoffs produce exactly the same transformed hull $L_\epsilon K$ and distinguished prime end, so their driver and capacity variations coincide. Finally $\|\eta\|_\infty \le C|\nu|$, its support lies in $|z| \le CR$, and $|W| \le 3R$. Integrating $1/|z-W|$ over this disk gives the first bound (AV6); its area gives the second. □

Corollary 38.3 (An explicit constant-anisotropy response)**.** *For the boundary-fixed affine deformation (AV1), the driver response can also be written without a cutoff:*

$$\boxed{\partial_\epsilon U_\epsilon|_0 = -\frac{2}{\pi}\,\mathrm{Re}\int_{\mathbb{H}} \frac{\nu}{z-W}\left(\frac{\overline{f'(z)}}{f'(z)} - 1\right) d^2z.} \tag{AV11}$$

The integral is absolutely convergent. This formula remains indexed by the original hull prefix.

Proof. Use the actual affine map in place of the cutoff and write $F_\epsilon = \widehat{g}^{\epsilon}_{s_\epsilon} \circ L_\epsilon \circ f$ and $Q_\epsilon = F_\epsilon - L_\epsilon$. Set $\phi = \overline{f'}/f'$ and $\mu_\epsilon = \epsilon\nu\phi/(1-\epsilon\nu)$. The Beltrami equation gives exactly

$$(Q_\epsilon)_{\bar z} - \mu_\epsilon (Q_\epsilon)_z = \epsilon\nu(\phi - 1). \tag{AV12}$$

The map Q_ϵ has real boundary values since L_ϵ fixes $\mathbb{R}$, so reflect Q_ϵ and the two coefficients across the real line. Its normalization at infinity is $Q_\epsilon = O(|z|^{-1})$. The right side divided by ϵ, denoted m, is bounded and satisfies $m(z) = O(|z|^{-2})$ at infinity, because $f'(z) = 1 + O(z^{-2})$. In particular $m \in L^p \cap L^q$ for fixed $p > 2 > q > 1$.

Writing $\xi_\epsilon = (Q_\epsilon)_{\bar z}$, the normalized solution of (AV12) is

$$\xi_\epsilon = \epsilon(I - \mu_\epsilon \mathcal{S})^{-1} m, \qquad Q_\epsilon = \mathcal{C}\xi_\epsilon. \tag{AV13}$$

For small ϵ, the two inverse operators are bounded on both L^p and L^q, so $\xi_\epsilon/\epsilon \to m$ in their intersection. Near the Cauchy kernel singularity use $p > 2$; at infinity use $q < 2$. This proves uniform convergence $Q_\epsilon/\epsilon \to \mathcal{C}m$, including boundary points. The normalized solution in (AV13) agrees with the geometrically defined Q_ϵ: their difference solves the homogeneous Beltrami equation and vanishes at infinity, so factorization through the normalized quasiconformal homeomorphism reduces it to a bounded entire function vanishing at infinity. Finally $L_\epsilon(W) = W$, and reflection yields (AV11). □

Proposition 38.4 (Uniform Taylor expansion along matched prefixes)**.** *Suppose $K_t \subset \{|z| \le R\}$ for $0 \le t \le T$, and use the same cutoff V from (AV2) for every prefix. Write D_t and E_t for the two first derivatives in (AV5). There are absolute constants $c, C > 0$, depending only on the fixed cutoff profile, such that for $|\epsilon\nu| \le c$,*

$$\begin{aligned} \sup_{t\le T} |U_\epsilon(t) - W_t - \epsilon D_t| &\le C\epsilon^2|\nu|^2 R, \\ \sup_{t\le T} |s_\epsilon(t) - t - \epsilon E_t| &\le C\epsilon^2|\nu|^2 R^2, \\ \sup_{t\le T} |D_t| \le C|\nu|R, \qquad \sup_{t\le T} |E_t| &\le C|\nu|R^2. \end{aligned} \tag{AV14}$$

In particular the matched-prefix pair is differentiable in the uniform path norm at $\epsilon = 0$, with continuous first derivative paths D, E. This remains a statement in the original prefix clock.

Proof. For every $t \le T$, the coefficients $m_{\epsilon,t}$ in (AV8) are supported in the same disk of radius CR. The bounds $\|\eta_t\|_\infty \le C|\nu|$ and $\|m_{\epsilon,t} - \epsilon\eta_t\|_\infty \le C\epsilon^2|\nu|^2$ hold with constants independent of t, the

hulls, and R. Fix one $p > 2$. Choosing c so that the Neumann-series inverses on L^p have norm at most two gives

$$\|h_{\epsilon,t} - \epsilon\eta_t\|_p \le C\epsilon^2|\nu|^2 R^{2/p}. \tag{AV15}$$

The difference is supported in the same disk. For a function k supported in that disk, Hölder's inequality gives $\|\mathcal{C}k\|_\infty \le CR^{1-2/p}\|k\|_p$. Thus (AV15) gives the first remainder in (AV14), even before evaluating at the moving real point W_t. The coefficient at infinity is estimated in L^1: $\|h_{\epsilon,t} - \epsilon\eta_t\|_1 \le C\epsilon^2|\nu|^2R^2$, proving the second remainder. The two first-derivative bounds follow by the same estimates. Continuity of $U_\epsilon(t)$ and $s_\epsilon(t)$, together with uniform convergence of the difference quotients, gives continuity of D_t and E_t. □

Corollary 38.5 (Temporal one-half Hölder regularity of the linear clock response)**.** *Under the preceding proposition,*

$$|E_t - E_s| \le C|\nu|R\sqrt{t-s}, \qquad 0 \le s < t \le T. \tag{AV17}$$

This is a deterministic statement in the original capacity clock. It does not assert absolute continuity of E or of s_ϵ.

Proof. Write $\Delta = t - s$. Monotonicity of transformed hull capacity gives $s_\epsilon(t) - s_\epsilon(s) \ge 0$ for both signs of ϵ. Choose its sign opposite to that of $E_t - E_s$ and apply the two Taylor remainder bounds. For $0 < a \le c/|\nu|$ this gives

$$|E_t - E_s| \le \frac{\Delta}{a} + 2Ca|\nu|^2R^2.$$

For Δ below a sufficiently small constant times R^2, choose a proportional to $\sqrt{\Delta}/(|\nu|R)$. For larger Δ, use $|E_t - E_s| \le C|\nu|R^2$ from (AV14). The case $\nu = 0$ is immediate. □

Uniform stochastic parameter regularity. For a continuous Loewner driver on $[0,T]$, the strip and height bounds give the deterministic hull-radius estimate

$$R_T \le \sup_{t\le T}|W_t| + 2\sqrt{T}. \tag{AV16}$$

Consequently the conditional driving tail (NS4), applied in a restarted centered half-plane chart, implies conditional exponential tails for $\sup_{t\le h}|D_t|/(|\nu|\sqrt{h})$ and for $\sqrt{\sup_{t\le h}|E_t|/(|\nu|h)}$, with changed numerical constants. The two uniform Taylor remainders have respectively the same Brownian and quadratic size bounds multiplied by $\epsilon^2|\nu|^2$. These conclusions hold for the unknown crossing-controlled driver as well as the affine-SLE model. They give regularity in the *deformation parameter*. Apart from the separate one-half Hölder bound on E, they give no temporal regularity estimate for D and no finite signed variation of W or D.

38.2 Temporal regularity of the first affine driver response

Use the fixed-cutoff matched-prefix setup above, with $K_t \subset B_R$ for $t \le T$. Let $D_t = \partial_\epsilon U_\epsilon(t)|_0$ in affine direction ν. For $0 \le s < t \le T$ set

$$r_{s,t} = \sup_{s\le u\le t}|W_u - W_s| + 2\sqrt{t-s}.$$

Proposition 38.6 (The linear affine driver response retains temporal roughness bounds)**.** *For every integer $N \ge 1$, there is C_N depending only on N and the fixed cutoff profile such that*

$$|D_t - D_s| \le C_N|\nu|R^{1/(N+1)}r_{s,t}^{N/(N+1)}. \tag{TD1}$$

Consequently if W is Hölder for every exponent below $1/2$, then so is its matched-prefix first affine response D.

Proof. Fix $p \geq 2(N+1)$, taking $p > 2$ in the case $N = 1$. The principal Beltrami solution at prefix s is $F_s^\epsilon = \mathrm{id} + \mathcal{C}h_{\epsilon,s}$, with coefficient support in B_{CR} and

$$\|h_{\epsilon,s}\|_p \leq C_p|\epsilon\nu|R^{2/p}, \qquad |\epsilon\nu| \leq c_p.$$

The Cauchy transform maps $L^p(\mathbb{C})$ to the homogeneous Hölder class with exponent $1 - 2/p$, by splitting its difference kernel at the separation of the two evaluation points. Thus

$$|F_s^\epsilon(z) - F_s^\epsilon(w)| \leq |z - w| + C_p|\epsilon\nu|R^{2/p}|z - w|^{1-2/p}. \tag{TD2}$$

This holds on the closed half-plane, since the Cauchy representation is continuous there.

The old-chart increment hull between s and t is contained in $B(W_s, r_{s,t})$: its centered driving oscillation bounds horizontal extent and its height is at most $2\sqrt{t-s}$. The comparison map F_s^ϵ maps it to the increment hull in the deformed chart. The map agrees on the actual physical hull with the affine deformation, since the fixed cutoff is one on all prefixes. Equation (TD2) and the deterministic driver localization estimate applied to this increment give

$$|U_\epsilon(t) - U_\epsilon(s)| \leq 3r_{s,t} + C_p|\epsilon\nu|R^{2/p}r_{s,t}^{1-2/p}. \tag{TD3}$$

No stochastic Markov property is used here.

The rational expression for $m_{\epsilon,t}$ and the uniformly convergent Beurling Neumann series are analytic in the real parameter ϵ on one interval $|\epsilon\nu| < c_p$, uniformly over prefixes in B_R. More explicitly, their complex analytic extension is obtained by extending the upper and lower reflected rational coefficient formulas with the same complex parameter; reflection symmetry is needed only for real parameters. Their L^p Taylor coefficients are bounded by $R^{2/p}(C_p|\nu|)^j$. The Cauchy transform estimate gives a uniform order-N remainder $C_{N,p}|\nu|^{N+1}|\epsilon|^{N+1}R$ for the matched-prefix driving value.

Choose finite-difference weights $a_1, \ldots, a_N$ with $\sum_{k=1}^N a_k k^j = \mathbf{1}_{\{j=1\}}$ for $1 \leq j \leq N$. They give

$$D_t = \frac{1}{\epsilon}\sum_{k=1}^{N} a_k(U_{k\epsilon}(t) - W_t) + O_N(|\epsilon|^N|\nu|^{N+1}R), \tag{TD4}$$

uniformly in t. Subtract this formula at s and apply (TD3), obtaining

$$|D_t - D_s| \leq C_N\left\{\frac{r_{s,t}}{|\epsilon|} + |\nu|R^{2/p}r_{s,t}^{1-2/p} + |\epsilon|^N|\nu|^{N+1}R\right\}. \tag{TD5}$$

For $r_{s,t}/R$ below a sufficiently small constant, choose $|\epsilon\nu| = c_N(r_{s,t}/R)^{1/(N+1)}$ so all the finite-difference parameters remain within the analytic interval. The first and third terms give (TD1); the second is no larger because $2/p \leq 1/(N+1)$. For larger $r_{s,t}/R$, (TD1) follows from $\sup|D| \leq C|\nu|R$. The case $\nu = 0$ is immediate.

If W is Hölder-α on a fixed horizon with $\alpha < 1/2$, then $r_{s,t} \leq C(\omega)(t-s)^\alpha$. Equation (TD1) gives exponent $\alpha N/(N+1)$ for D. Given any $\beta < 1/2$, first choose $\beta < \alpha < 1/2$ and then N sufficiently large. Countable choices of α, N give the simultaneous conclusion. □

Corollary 38.7 (The linear clock response has every temporal exponent below one)**.** *For every $0 < \theta < 2$, the same bounded-hull setup gives*

$$|E_t - E_s| \leq C_\theta|\nu|R^{2-\theta}r_{s,t}^\theta. \tag{TD6}$$

If W is Hölder for every exponent below $1/2$, then E is Hölder for every exponent below 1. In particular it has finite q-variation for every $q > 1$, its sums of squared increments tend to zero along every sequence of partitions with mesh tending to zero, and the same is true of mixed sums $\sum \Delta W \Delta E$. Neither bounded variation nor absolute continuity is asserted.

Proof. Choose arbitrarily large $p > 2$, and restrict real parameters to a small interval $|\epsilon\nu| \le \delta_p$ on which the preceding L^p bounds hold. Put $\alpha = 1 - 2/p$. For $r = r_{s,t} \le R$, (TD2) bounds the radius of the transformed incremental hull by $C_p R (r/R)^\alpha$. Its half-plane capacity is at most the square of this radius. Consequently, uniformly in those real parameters,

$$0 \le s_\epsilon(t) - s_\epsilon(s) \le C_p R^2 (r/R)^{2\alpha}.$$

For capacity, the holomorphic parameter argument bounds the uniform order-N Taylor remainder by

$$C_{N,p} |\epsilon|^{N+1} |\nu|^{N+1} R^2.$$

Apply the finite-difference formula for its first parameter derivative at both s and t. For $a = |\epsilon\nu|$ small enough to accommodate all sample points this yields

$$|E_t - E_s| \le C_{N,p} |\nu| R^2 \{ (r/R)^{2\alpha}/a + a^N \}.$$

Optimize a proportional to $(r/R)^{2\alpha/(N+1)}$, capping it at the allowed parameter radius. Small r/R gives exponent $2\alpha N/(N+1)$; larger r/R uses the uniform bound on E. Choose p, N so this exponent exceeds the prescribed θ. This proves (TD6). The temporal conclusion follows exactly as for D, using $r_{s,t} \le C(\omega)(t-s)^\gamma$ for each $\gamma < 1/2$.

For the final assertions choose a temporal Hölder exponent for E greater than $1/q$ to obtain finite q-variation, or greater than $1/2$ to obtain vanishing squared-increment sums. For mixed sums, choose exponents $\gamma < 1/2$ for W and $\beta < 1$ for E with $\gamma + \beta > 1$; the absolute mixed sum is bounded by a constant times $T \operatorname{mesh}(\pi)^{\gamma+\beta-1}$. □

This is temporal path regularity of the linear affine response under the original-prefix clock. It proves neither finite signed variation nor semimartingality, and does not justify differentiating the own-capacity wrong-coordinate driver with respect to anisotropy.

A useful check. For the deterministic vertical slit driven by $W_t = 0$, the pure-shear direction $\nu = i$ gives the response $16\sqrt{t}/\pi$; this agrees with the derivative of the standard square-root driver of a slanted straight ray. Indeed the straight-ray driver $c\sqrt{s}$ has ray angle $\theta(c) = \frac{\pi}{2}(1 - c/\sqrt{c^2+16})$, so $\theta'(0) = -\pi/8$, whereas the shear changes the vertical ray angle by $-2\epsilon + O(\epsilon^2)$. This checks the sign and constant geometrically. For a pure vertical stretch, symmetry gives zero driver response and the geometric capacity derivative is $-4t$.

Indexing warning. The derivative in (AV5) compares the same original ordered prefix. It is not the derivative at fixed numerical capacity in the transformed curve. Inverting $s_\epsilon(t)$ changes the time argument of the original rough driver. Even a differentiable clock perturbation can destroy differentiability in ϵ for a Brownian path. No fixed-capacity pathwise first variation is claimed.

A model-independent consequence of the crossing radius bound. For a continuum family satisfying the conditional crossing assumptions used earlier, the filled hull during a restarted capacity interval of length h lies in a disk of radius R_h with $\mathbb{P}(R_h > x\sqrt{h} \mid \mathcal{F}_\tau) \le Ce^{-cx}$, after changing constants. Apply (AV6) to the restarted half-plane hull. The affine response of its endpoint driver

has conditional moments $\mathbb{E}(|\delta U_h|^p \mid \mathcal{F}_\tau) \le C_p|\nu|^p h^{p/2}$; the capacity response has moments bounded by $C_p|\nu|^p h^p$. This concerns affine perturbation of the *restarted chart hull*, not a fixed physical affine perturbation of a rough slit domain. The latter is encoded by (AB2), rather than a constant chart coefficient. Neither estimate establishes temporal semimartingality of the driver or of its first variation.

Literature boundary. Sung–Wang [21, Theorem 1.1], treats a simple chord and deformation supported away from it. The proof above, using the principal-solution method of Ahlfors–Bers [20], instead keeps the bounded filled hull fixed, pulls a compactly supported smooth physical deformation into its uniformizing half-plane, and uses the principal Beltrami solution. It does not use a theorem about simple approximants or omit swallowed positive-area sets from a physical-domain change of variables.

39 Mean affine response through a reverse Loewner flow

Let $W_t = \sqrt{\kappa}B_t$, $\kappa > 0$, be the driver of ordinary chordal SLE, and let $f_t = g_t^{-1}$. The matched-prefix affine driver response in direction ν is

$$D_t = -\frac{2}{\pi}\operatorname{Re}\nu I_t, \qquad I_t = \int_{\mathbb{H}} \frac{\overline{f_t'(W_t+z)}/f_t'(W_t+z) - 1}{z}\, dA(z).$$

The same original hull prefix, rather than the same deformed capacity, is held fixed.

Absolute integrability and Fubini. For $K_t \subset B_R$, the inverse map is hydrodynamically univalent and $f_t'(z) = 1 + O(R^2/|z|^2)$ uniformly for $|z| > CR$. Since $|W_t| \le 3R$,

$$\int_{\mathbb{H}} \frac{|\overline{f_t'(W_t+z)}/f_t'(W_t+z) - 1|}{|z|}\, dA(z) \le CR.$$

Indeed the near part is bounded by $2/|z|$ over a disk of radius CR, and the far part by $CR^2/|z|^3$. For Brownian driving we can take $R = \sup_{s\le t}|W_s| + 2\sqrt{t}$, so $\mathbb{E}R \le C_\kappa\sqrt{t}$. Thus expectation and the displayed integral may be interchanged. This uses neither a simple trace nor a zero-area hull.

Reverse-flow representation. For fixed t and $0 \le r \le t$, put

$$H_r(z) = g_{t-r}(f_t(z + W_t)) - W_{t-r}.$$

Then $H_0(z) = z$, $H_t(z) = f_t(z + W_t)$, and pathwise the reverse Loewner equation is driven by $W_t - W_{t-r}$. This reversed increment process has Brownian law. Hence the derivative field at time t has the same law as the spatial derivative of

$$dh_r(z) = -\frac{2}{h_r(z)}dr + \sqrt{\kappa}\, d\widetilde{B}_r, \qquad h_0(z) = z.$$

The sign of the Brownian term does not affect its law. Its spatial derivative satisfies $\partial_r h_r' = 2h_r'/h_r^2$. Therefore

$$\frac{\overline{h_t'(z)}}{h_t'(z)} = \exp\left\{\int_0^t 2\left(\frac{1}{\overline{h_r(z)}^2} - \frac{1}{h_r(z)^2}\right) dr\right\}.$$

For fixed initial height $y > 0$, the reverse image height increases, so all the coefficients are bounded away from their boundary singularity on finite time intervals.

Define

$$u(t,z) = \mathbb{E}\frac{\overline{h_t'(z)}}{h_t'(z)}.$$

It is the bounded Feynman–Kac solution, in the stochastic semigroup sense, of

$$\partial_t u = \mathcal{L}u + qu, \quad u(0,z) = 1,$$

where

$$\mathcal{L} = \frac{\kappa}{2}\partial_x^2 - \frac{2x}{x^2+y^2}\partial_x + \frac{2y}{x^2+y^2}\partial_y, \qquad q(z) = \frac{8ixy}{(x^2+y^2)^2}.$$

No boundary condition at $y = 0$ is inserted: the stochastic representation from each interior starting point specifies the solution. Uniqueness among bounded classical solutions, or within the stated stochastic semigroup class, follows by stopped Itô/Feynman–Kac on $y \geq y_0 > 0$ and localization in x, since the phase potential is purely imaginary. Consequently

$$\mathbb{E}D_t = -\frac{2}{\pi}\operatorname{Re}\left\{\nu\int_{\mathbb{H}}\frac{u(t,z)-1}{z}\,dA(z)\right\}.$$

The integral is absolutely convergent by the preceding Fubini bound.

Symmetry and scaling. Reflection of Brownian driving gives $u(t,-\bar{z}) = \overline{u(t,z)}$, so the last integral is purely imaginary. Brownian scaling makes it $iJ_\kappa\sqrt{t}$ for a real constant J_κ. Thus

$$\mathbb{E}D_t = c_\kappa \operatorname{Im}(\nu)\sqrt{t}, \qquad c_\kappa = 2J_\kappa/\pi.$$

The first-order mean vertical-stretch response vanishes, whereas a shear response is allowed by symmetry. Determining whether c_6 is zero, and more generally computing c_κ, is a concrete next model task. This calculation does not determine temporal semimartingality of D or of the finite-affine wrong-coordinate driver.

An interior cancellation, not a boundary conclusion. Let $\mathcal{L}^*$ denote the formal transpose under the bilinear integral pairing, so the multiplication potential remains q, not its complex conjugate. Direct differentiation gives the pointwise identity

$$(\mathcal{L}^* + q)\frac{1}{z} = \frac{\kappa-6}{z^3}, \qquad z \in \mathbb{H}.$$

At $\kappa = 6$ the interior term vanishes. This does not justify integrating by parts over the full half-plane: the test $1/z$ is singular at the boundary and has a noncompact tail. Boundary and infinity fluxes have not been controlled. In particular no vanishing or nonvanishing of c_6 is deduced. Studying those fluxes is a precise next question for the affine-SLE_6 model.

40 Coordinate-compatible exit clocks before semimartingality

The upper and lower conditional oscillation bounds already proved for the continuum family have a stronger clock consequence. It is compatible with working in a wrong affine structure: it assumes neither a signed budget nor a canonical semimartingale decomposition. All constants may depend on the fixed affine distortion through the crossing constants.

Let X be a continuous adapted process on $[0,\infty)$ in a usual filtration. Suppose constants $C, c, \delta, p > 0$, with $0 < p < 1$, satisfy, for every finite stopping time τ, every $h > 0$, and every $x \geq 0$,

$$\mathbb{P}\{\sup_{u \leq h} |X_{\tau+u} - X_\tau| > x\sqrt{h} \mid \mathcal{F}_\tau\} \leq Ce^{-cx}, \tag{7}$$

$$\mathbb{P}\{\sup_{u \leq h} |X_{\tau+u} - X_\tau| > \delta\sqrt{h} \mid \mathcal{F}_\tau\} \geq p. \tag{8}$$

For the current exploration assumptions, these follow from the upper crossing estimate, corrected ray-deletion locality, and strong DMP. The rough-state admissibility qualifications of those results remain in force. Physical translation covariance is not needed for this deduction.

Proposition 40.1 (Uniform Brownian-order exit-time control)**.** *For $r > 0$, put*

$$\xi_r(\tau) = \inf\{u \geq 0 : |X_{\tau+u} - X_\tau| \geq r\}.$$

There are constants $a_i > 0$, depending only on the input constants, such that, conditionally on $\mathcal{F}_\tau$,

$$\mathbb{P}\{\xi_r(\tau) \leq h \mid \mathcal{F}_\tau\} \leq C\exp(-cr/\sqrt{h}), \tag{9}$$

$$\mathbb{P}\{\xi_r(\tau) > h \mid \mathcal{F}_\tau\} \leq (1-p)^{\lfloor \delta^2 h/(4r^2)\rfloor}, \tag{10}$$

$$a_1 r^2 \leq \mathbb{E}[\xi_r(\tau) \mid \mathcal{F}_\tau] \leq a_2 r^2, \qquad \mathbb{E}[\exp(a_3 \xi_r(\tau)/r^2) \mid \mathcal{F}_\tau] \leq a_4. \tag{11}$$

In particular, exits occur in finite time almost surely.

Proof. For the fast bound, the exit event implies that the supremum on $[0,h]$ is at least r. Apply (7) first at thresholds $r - \varepsilon$ and then let $\varepsilon \downarrow 0$. For the slow bound, split time into blocks of length dr^2, where $d = 4/\delta^2$. If the exit has not occurred by the end of a block, both the path and its value at the start of that block lie in the original interval $[X_\tau - r, X_\tau + r]$. Their displacement throughout the block is at most $2r = \delta\sqrt{dr^2}$. The conditional lower bound at the block's starting stopping time therefore bounds the conditional survival probability by $1 - p$. Iteration proves (10). Integrating that tail gives the upper mean and an exponential moment. Choose $a > 0$ small enough that $Ce^{-c/\sqrt{a}} \leq 1/2$. The fast bound gives $\mathbb{P}\{\xi_r(\tau) > ar^2 \mid \mathcal{F}_\tau\} \geq 1/2$, hence the lower mean bound $ar^2/2$. □

Proposition 40.2 (Adaptive quadratic activity is nondegenerate)**.** *Define $\sigma_0^r = 0$ and successive exits by*

$$\sigma_{k+1}^r = \inf\{t \geq \sigma_k^r : |X_t - X_{\sigma_k^r}| \geq r\}, \qquad \nu_r = r^2 \sum_{k \geq 1} \delta_{\sigma_k^r}.$$

The exit times do not accumulate on a finite interval. There are fixed $0 < a < b < \infty$ and $A, \alpha > 0$ such that for every deterministic interval $I = [s,t]$, $h = t - s > 0$,

$$\mathbb{P}\{\nu_r(I) < ah \text{ or } \nu_r(I) > bh\} \leq A\exp(-\alpha h/r^2).$$

Consequently, on one probability-one event, the measures $\nu_{2^{-m}}$ are relatively compact on every finite interval, and every subsequential weak limit ν satisfies

$$a\,dt \leq d\nu \leq b\,dt$$

(after decreasing a and increasing b if necessary). These measures are realized squared increments on the corresponding exit partitions, not a claimed canonical quadratic variation.

Proof. Finite-time nonaccumulation follows from continuity: infinitely many oscillations of size r on a compact interval contradict uniform continuity. Every exit time is finite by the preceding proposition, and every completed exit increment has square exactly r^2.

Write $D_k = (\sigma^r_{k+1} - \sigma^r_k)/r^2$. For some $\lambda > 0$ and $K < \infty$, the preceding proposition gives

$$\mathbb{E}[e^{\lambda D_k} \mid \mathcal{F}_{\sigma^r_k}] \le K.$$

Also its fast-tail proof supplies $a_0 > 0$ with $\mathbb{P}\{D_k > a_0 \mid \mathcal{F}_{\sigma^r_k}\} \ge 1/2$. Thus, for every $\eta > 0$,

$$\mathbb{E}[e^{-\eta D_k} \mid \mathcal{F}_{\sigma^r_k}] \le \rho := \tfrac{1}{2}(1 + e^{-\eta a_0}) < 1.$$

Iteration and exponential Markov inequalities give, for sums of n successive full exit durations, the bounds

$$\mathbb{P}\{\textstyle\sum_{j=1}^n D_j > H \mid \text{initial past}\} \le K^n e^{-\lambda H}, \qquad \mathbb{P}\{\textstyle\sum_{j=1}^n D_j \le H \mid \text{initial past}\} \le \rho^n e^{\eta H}.$$

No independence of the durations is used.

To count exits in a fixed interval starting at s, first wait for the first global exit $\theta \ge s$. The residual time $R = \theta - s$ also has a uniform conditional exponential moment at scale r^2. Indeed, before θ, the process remains within r of the previous exit value, hence within $2r$ of X_s. Repeating the slow-tail proof with that larger confinement band gives $\mathbb{P}\{R > u \mid \mathcal{F}_s\} \le (1-p)^{\lfloor \delta^2 u/(16r^2)\rfloor}$. Consequently for a sufficiently small common λ, $\mathbb{E}[e^{\lambda R/r^2} \mid \mathcal{F}_s] \le K_0$. If an interval has at most n exits, its residual duration plus n full durations is at least h. If it has at least $n+1$ exits, n full durations after its first exit fit inside h. Thus

$$\mathbb{P}\{\nu_r(I)/r^2 \le n\} \le K_0 K^n e^{-\lambda h/r^2}, \qquad \mathbb{P}\{\nu_r(I)/r^2 \ge n+1\} \le \rho^n e^{\eta h/r^2}.$$

Choose n respectively as a sufficiently small and sufficiently large multiple of h/r^2. Rounding and bounded values of h/r^2 are absorbed in A. This proves (40.2).

For dyadic r, the right side of that estimate is summable for every fixed rational interval. Borel–Cantelli and a countable intersection give eventual upper and lower mass bounds on every rational interval. The upper bounds give relative compactness of finite measures on compact intervals. For any weak subsequential limit, use rational open intervals for the upper mass bound and rational closed intervals for the lower bound, applying the Portmanteau inequalities. Approximation by rational endpoints gives (40.2); in particular the limits have no atoms. Finite endpoint conventions in the counts do not affect the conclusion, since each atom of ν_r has mass r^2. □

What is gained and what is still missing. The continuum axioms already supply nonzero finite quadratic activity along observable stopping-time partitions, with subsequential activity clocks comparable to capacity. This is stronger information than merely stating a variation index. It does not establish uniqueness of the clock, convergence along deterministic partitions, semimartingality, or variance six. Even uniqueness of a quadratic-activity clock could not by itself establish semimartingality: the earlier Brownian-plus-infinite-variation Hölder example remains a warning about missing signed cancellation.

At exit times the displacement is $\varepsilon_k r$, with $\varepsilon_k \in \{-1, 1\}$. The remaining observable signed quantity is

$$r \sum_{k<N} \left| \mathbb{E}[\varepsilon_k \mid \mathcal{F}_{\sigma^r_k}] \right|.$$

The new clock estimate controls the number of summands by order T/r^2; it does not improve the resulting crude bound of order T/r on this signed budget. A useful next question is how the

predictable exit-sign bias changes under a candidate constant affine normalization, or whether locality yields cancellations for a corrected observable depending on the transported Beltrami field. Finite signed variation of the raw driver in every affine coordinate system is not being assumed or recommended.

41 Affine capacity clocks and the limits of scaling arguments

41.1 Global affine capacity comparison does not control clock increments

The affine clock at least has a deterministic global comparison. This uses only the geometric capacity theorem of Lalley–Lawler–Narayanan [19, Theorem 1], not any regularity of the trace or driver.

Proposition 41.1 (Capacity distortion by a fixed affine map)**.** *Let $L(x+iy) = \alpha x + \beta y + i\delta y$, where $\alpha, \delta > 0$ and $\beta \in \mathbb{R}$. There are constants $0 < c_L \le C_L < \infty$ such that every bounded half-plane hull K satisfies*

$$c_L \operatorname{hcap}(K) \le \operatorname{hcap}(LK) \le C_L \operatorname{hcap}(K). \tag{GC1}$$

Consequently, if $\operatorname{hcap}(K_t) = 2t$, its affine clock $s(t) = \operatorname{hcap}(LK_t)/2$ satisfies $c_L t \le s(t) \le C_L t$. The constants can be chosen uniformly when L ranges over a compact set of invertible half-plane-preserving linear maps.

Proof. For a family of disks $\{B_j\}$ with bounded radii, the elementary Vitali selection argument gives, for every $\lambda \ge 1$,

$$\left| \bigcup_j \lambda B_j \right| \le (2\lambda + 3)^2 \left| \bigcup_j B_j \right|. \tag{GC2}$$

Indeed one can select disjoint disks such that each original disk meets a selected disk of at least half its radius. Its λ-dilate then lies in the $(2\lambda+3)$-dilate of that selected disk. Sum their areas; arbitrary families reduce to countable subfamilies of the open union.

Let m, M be the smallest and largest singular values of L, and $d = \det L$. Put

$$H(K) = \bigcup_{z \in K} B(z, \operatorname{Im} z), \qquad \operatorname{hsiz}(K) = |H(K)|.$$

For $z = x + iy$, the ellipse $LB(z, y)$ contains $B(Lz, my)$ and is contained in $B(Lz, My)$, whereas the corresponding disk in $H(LK)$ is $B(Lz, \delta y)$. Since $m \le \delta \le M$, (GC2) yields

$$\frac{d}{(2M/\delta + 3)^2} \operatorname{hsiz}(K) \le \operatorname{hsiz}(LK) \le d(2\delta/m + 3)^2 \operatorname{hsiz}(K). \tag{GC3}$$

The cited theorem gives universal positive a, b with $a \operatorname{hsiz}(K) \le \operatorname{hcap}(K) \le b \operatorname{hsiz}(K)$. Apply it to both K and LK. This proves (GC1), the clock assertion, and local uniformity in L. □

This is a comparison from the original source time zero, not an increment estimate. After mapping out K_t, the transported affine map is quasiconformal with a history-dependent field; it is no longer the original linear map. Applying (GC1) to that increment would be invalid. Even for increasing continuous functions, $s(t) \asymp t$ does not imply absolute continuity: $s(t) = t + \epsilon C(t)$, with C a Cantor staircase supported on $[1, 2]$, has a singular part and still satisfies such a comparison. Determining whether the actual affine-SLE clock has a singular component remains open here.

41.2 A necessary clock regularity condition for wrong-coordinate semimartingality

Let η_s be chordal SLE_6 in $\mathbb{H}$, parametrized by $\mathrm{hcap}(K_s) = 2s$, and let B_s be its ordinary Brownian driver divided by $\sqrt{6}$. Let L be an orientation-preserving real-linear self-map of $\mathbb{H}$. Put

$$\tau(s) = \tfrac{1}{2}\,\mathrm{hcap}(LK_s), \qquad \gamma_{\tau(s)} = L\eta_s,$$

and denote the ordinary Loewner driver of γ by W_t. We assume continuous Loewner generation, strictly growing hulls, and infinite lifetimes for both parametrizations. Thus τ and its inverse are continuous increasing clocks. Completed natural filtrations are used.

Proposition 41.2 (Semimartingality forces absolute continuity of the affine clock)**.** *Suppose $W = M + A$ is a continuous semimartingale in its natural filtration and its martingale bracket $C = \langle M \rangle$ satisfies*

$$c\,dt \le dC_t \le C_0\,dt \tag{AC1}$$

on the relevant localized intervals, with deterministic constants $0 < c \le C_0 < \infty$. Then τ is absolutely continuous in s. There exists a Brownian-predictable H such that, locally,

$$M_{\tau(s)} = \int_0^s H_u\,dB_u, \qquad c\,d\tau(s) \le H_s^2 ds \le C_0\,d\tau(s). \tag{AC2}$$

In particular, a nonzero singular component of $d\tau$ would exclude semimartingality of W, whenever the crossing/locality hypotheses needed for (AC1) hold.

Proof. An initial segment of W reconstructs the corresponding ordered Loewner trace and hull. Applying L^{-1} reconstructs the original SLE segment, its original capacity, and its driver. Conversely the Brownian driver up to s reconstructs the transformed segment and its driver up to $\tau(s)$. Thus, with the usual completions,

$$\mathcal{F}^W_{\tau(s)} = \mathcal{F}^B_s. \tag{AC3}$$

This assertion retains the ordered trace and distinguished tip; an unlabelled hull or frontier alone is not substituted for that state.

For every fixed s, $\tau(s)$ is a stopping time in the W filtration: at observed time t one can reconstruct the original capacity $\tau^{-1}(t)$, and $\{\tau(s) \le t\} = \{\tau^{-1}(t) \ge s\}$. Consequently $N_s = M_{\tau(s)}$ is a continuous local martingale in the time-changed filtration, which is the Brownian filtration by (AC3). Localize both clocks and the martingale in the usual way. The Brownian martingale representation theorem gives $N_s = \int_0^s H_u\,dB_u$, and therefore $C_{\tau(s)} = \int_0^s H_u^2 du$.

The measure inequality (AC1), applied pathwise to each clock interval, gives

$$c\,d\tau \le d(C \circ \tau) \le C_0\,d\tau.$$

Its middle measure equals $H_s^2 ds$. The lower inequality forces $d\tau \ll ds$ and proves all assertions. □

The already established upper conditional driving moment bound yields the upper inequality in (AC1) once semimartingality is assumed. The ray-deletion locality lower tail yields the lower inequality. Thus (AC1) uses precisely those explicit continuum crossing and prime-end admissibility hypotheses; it is not a consequence of semimartingality alone. No singularity of the actual affine clock has been proved here. Absolute continuity of τ is only necessary, not sufficient for semimartingality. It also does not imply absolute continuity of its inverse: τ' might vanish on a positive-measure set. Global comparisons $c_L s \le \tau(s) \le C_L s$ do not remove this possibility or give an increment comparison.

41.3 Why scaling and a trivial germ do not regularize the bracket

Here is a process-level obstruction to strengthening the previous scaling lemma without its initial-bracket premise. It is not a proposed exploration family satisfying locality and DMP.

Proposition 41.3 (A self-similar elliptic martingale without an initial bracket tangent)**.** *Let B be standard Brownian motion and define*

$$X_t = \int_0^t \sqrt{2 + \sin(B_s/\sqrt{s})}\, dB_s, \qquad t \geq 0. \tag{AC4}$$

The integrand at zero may be chosen arbitrarily. Then X is a continuous martingale in its own natural filtration, its initial germ is trivial, and it has exact Brownian scaling. Its bracket satisfies

$$dt \leq d\langle X\rangle_t \leq 3dt. \tag{AC5}$$

Nevertheless $\langle X\rangle_t/t$ has no limit in probability as $t \downarrow 0$. It satisfies conditional Brownian-scale upper oscillation tails and a positive conditional lower oscillation tail uniformly over stopping times.

Proof. The integrand is bounded and progressive, so the integral is a square-integrable martingale on each finite horizon. A martingale remains a martingale in its own smaller natural filtration; its bracket is the pathwise quadratic variation and hence remains

$$Q_t = \int_0^t [2 + \sin(B_s/\sqrt{s})]ds.$$

This proves (AC5). The natural X filtration is contained in the Brownian filtration, so its completed initial germ is trivial. For every $a > 0$, substitution $\widetilde{B}_u = B_{au}/\sqrt{a}$ in the stochastic integral proves $(a^{-1/2}X_{at})_{t\geq 0} \overset{\text{law}}{=} (X_t)_{t\geq 0}$. The same substitution shows Q_t/t has the fixed distribution of Q_1.

This distribution is nondegenerate. Fix small $\epsilon > 0$. On $[\epsilon, 1]$, Brownian paths have positive probability of lying in a sufficiently small uniform neighborhood of either continuous function $s \mapsto \pi\sqrt{s}/2$ or its negative. This follows from Brownian full support after extending each function continuously to zero. On these two events the integral of the sine over $[\epsilon, 1]$ is close respectively to $1 - \epsilon$ and $-(1 - \epsilon)$, whereas its contribution on $[0, \epsilon]$ has absolute value at most ϵ. Thus Q_1 is nonconstant.

If Q_t/t converged in probability at zero, an almost surely converging subsequence would make its limit measurable in the Brownian initial germ. The limit would be deterministic, contradicting the fixed nondegenerate distribution. Hence there is no such limit.

For completeness, the upper bracket bound and the exponential martingale inequality give conditional Gaussian upper tails for $\sup_{u\leq h} |X_{\sigma+u} - X_\sigma|/\sqrt{h}$ at every stopping time σ. For the lower tail, restart the martingale at σ and apply Dambis–Dubins–Schwarz. Its bracket over $[\sigma, \sigma + h]$ is at least h, so its maximum dominates the maximum of a Brownian motion over $[0, h]$. This gives any fixed positive Brownian lower-tail probability, conditionally on the past. Passing to the natural X filtration preserves both bounds by conditional expectation. □

This example rules out the shortcut "exact scaling + Brownian germ triviality + bounded elliptic bracket implies Brownian driving." It does not exclude the possibility that the actual wrong-coordinate affine-SLE driver is a semimartingale whose bracket density oscillates on every logarithmic scale near zero.

42 Finite-mesh locality characteristics before semimartingality

The existing characteristic transformation rules presuppose a semimartingale. The conditional crossing moments already allow a finite-mesh version with an error tending to zero in total expected variation. This identifies the signed quantities to study without first choosing a preferred affine structure or asserting a canonical quadratic variation.

42.1 Compactness of realized squared increments

Proposition 42.1 (Bounded cluster clocks on deterministic meshes)**.** *Let X be continuous and adapted on $[0,T]$ in a usual filtration and suppose, for deterministic $s<t$, that*

$$\mathbb{E}[|X_t - X_s|^2 \mid \mathcal{F}_s] \le K_2(t-s), \qquad \mathbb{E}[|X_t - X_s|^4 \mid \mathcal{F}_s] \le K_4(t-s)^2. \tag{FM1}$$

For a deterministic partition $\pi = (t_i)$ write

$$a_i = \mathbb{E}[(X_{t_{i+1}} - X_{t_i})^2 \mid \mathcal{F}_{t_i}].$$

Let Q^π be the continuous piecewise linear function starting at zero with increment a_i on $[t_i, t_{i+1}]$, and let

$$R^\pi(t) = \sum_{t_{i+1}\le t} (X_{t_{i+1}} - X_{t_i})^2.$$

Every Q^π is increasing and K_2-Lipschitz. There is an absolute constant C such that

$$\mathbb{E}\sup_{t\le T} |R^\pi(t) - Q^\pi(t)|^2 \le CK_4T\,|\pi| + CK_2^2|\pi|^2, \tag{FM2}$$

where $|\pi|$ denotes the largest mesh interval. In particular the realized square-sum processes are asymptotically tight with continuous increasing cluster limits whose densities are bounded by K_2. For every fixed mesh sequence with $\sum_n |\pi_n| < \infty$, almost surely their uniform cluster set is nonempty and every cluster path is increasing and K_2-Lipschitz. Neither uniqueness nor a positive lower density is asserted.

Proof. The slope claim follows from $0 \le a_i \le K_2(t_{i+1} - t_i)$. At partition endpoints the difference is the discrete martingale

$$Z_j = \sum_{i<j} \{(X_{t_{i+1}} - X_{t_i})^2 - a_i\}.$$

Conditional centering, orthogonality, and (FM1) give $\mathbb{E}Z_{\#\pi}^2 \le K_4 \sum_i (t_{i+1} - t_i)^2 \le K_4 T|\pi|$. Doob's inequality controls the maximum at grid points. Between grid points R^π is constant and Q^π moves by at most $K_2|\pi|$, which proves (FM2). The Q^π lie in one compact subset of $C([0,T])$ by Arzela–Ascoli. Equation (FM2) proves tightness and the assertion about cluster laws. For summable mesh sizes, Markov's inequality and Borel–Cantelli give $\|R^{\pi_n} - Q^{\pi_n}\|_\infty \to 0$ almost surely, hence the pathwise cluster assertion. A cluster selection need not produce one canonical adapted clock, and no such selection is claimed. □

The conditional exponential bounds from the crossing estimate imply (FM1), also for the bounded stopped drivers used in (QV1). Thus this is a further regularity consequence available before any signed budget is proved. It concerns cluster limits along deterministic meshes, whereas the earlier exit-clock theorem also supplies a lower density for its different, adaptive partitions. These facts do not identify the two kinds of cluster clock.

42.2 The variance-six anomaly has a finite-mesh form

Use a locality deletion map h_t on a common physical trace before contact, as in (LB1). Stop at a matched trace time $\theta \le T$ so that the driver stays in a compact real interval, h_t is analytic on a fixed larger neighborhood of that interval, and its spatial derivatives through order three, and the time derivatives needed below through spatial order two, are bounded by deterministic constants. Also require $0 < m \le h_t'(W_t) \le P$. Such bounds are localization conditions on a deletion map strictly away from contact. Write

$$X_t = W_{t\wedge\theta}, \quad Y_t = h_{t\wedge\theta}(X_t) - h_0(W_0), \quad A_t = t \wedge \theta, \quad U_t = \int_0^{t\wedge\theta} h_s'(W_s)^2\, ds. \tag{FM3}$$

Here Y is the deleted driver at the inherited clock U, not at deterministic deleted capacity. Suppose the original driving increments have the conditional supremum moment bounds

$$\mathbb{E}[\sup_{0\le u\le h} |X_{t+u} - X_t|^j \mid \mathcal{F}_t] \le C_j h^{j/2}, \qquad j = 1, \dots, 6, \tag{FM4}$$

for $t + h \le T$. These are supplied by the stopped crossing estimate.

For each deterministic partition interval put

$$\begin{aligned} b_i^X &= \mathbb{E}[\Delta_i X \mid \mathcal{F}_{t_i}], & a_i^X &= \mathbb{E}[(\Delta_i X)^2 \mid \mathcal{F}_{t_i}], \\ b_i^Y &= \mathbb{E}[\Delta_i Y \mid \mathcal{F}_{t_i}], & a_i^Y &= \mathbb{E}[(\Delta_i Y)^2 \mid \mathcal{F}_{t_i}], \\ d_i &= \mathbb{E}[\Delta_i A \mid \mathcal{F}_{t_i}], & e_i &= \mathbb{E}[\Delta_i U \mid \mathcal{F}_{t_i}], \\ p_i &= h'_{t_i\wedge\theta}(X_{t_i}), & q_i &= h''_{t_i\wedge\theta}(X_{t_i}). \end{aligned} \tag{FM5}$$

Proposition 42.2 (Conditional characteristic covariance without Ito calculus)**.** *In this setup there is a deterministic C, depending only on the localized map and moment bounds and T, such that, with $\delta_i = t_{i+1} - t_i$,*

$$\begin{aligned} b_i^Y &= p_i b_i^X + \tfrac{1}{2} q_i (a_i^X - 6d_i) + r_i, \\ a_i^Y &= p_i^2 a_i^X + v_i, \qquad e_i = p_i^2 d_i + w_i, \\ |r_i| + |v_i| + |w_i| &\le C\delta_i^{3/2}. \end{aligned} \tag{FM6}$$

All equalities and bounds hold almost surely. In particular

$$a_i^Y - 6e_i = p_i^2 (a_i^X - 6d_i) + O(\delta_i^{3/2}), \tag{FM7}$$

and the sums of absolute errors in (FM6)–(FM7) are bounded by $CT\sqrt{|\pi|}$. No signed-budget, semimartingale, or canonical quadratic-variation assumption is used.

Proof. Put $S_i = \sup_{u\in[t_i,t_{i+1}]} |X_u - X_{t_i}|$ and $\delta = \delta_i$. Spatial Taylor expansion and time integration of the stopped field give

$$\Delta_i Y = p_i \Delta_i X + \tfrac{1}{2} q_i (\Delta_i X)^2 - 3 q_i \Delta_i A + \mathcal{R}_i, \qquad |\mathcal{R}_i| \le C(S_i^3 + \delta S_i + \delta^2). \tag{FM8}$$

For clarity, the time contribution is $\int_{t_i\wedge\theta}^{t_{i+1}\wedge\theta} (\partial_s h_s)(X_{t_{i+1}}) ds$. The Loewner deletion PDE has the removable boundary value $(\partial_s h_s)(W_s) = -3h_s''(W_s)$. The bounded first spatial derivative of $\partial_s h_s$ permits replacing $X_{t_{i+1}}$ by W_s at cost $C\delta S_i$. The bounded time derivative of h_s'' and bounded h_s''' permit replacing $h_s''(W_s)$ by q_i at cost $C\delta(S_i + \delta)$. This proves (FM8), including intervals straddling θ; after θ everything is frozen. Taking conditional expectations and using (FM4) proves the first line of (FM6).

The simpler pathwise bound $|\Delta_i Y - p_i \Delta_i X| \le C(S_i^2 + \delta)$ follows directly from the bounded field derivatives. Squaring and using (FM4) shows

$$\mathbb{E}[|(\Delta_i Y)^2 - p_i^2(\Delta_i X)^2| \mid \mathcal{F}_{t_i}] \le C\mathbb{E}[S_i^3 + \delta S_i + S_i^4 + \delta^2 \mid \mathcal{F}_{t_i}] \le C\delta^{3/2},$$

after enlarging C on the fixed horizon. Finally $|h_s'(W_s)^2 - p_i^2| \le C(S_i + \delta)$ while the clock is active. Integration and conditional expectation prove the estimate for e_i. Summing $\delta_i^{3/2} \le \sqrt{|\pi|}\delta_i$ gives the last claim. □

What this does and does not bridge. The pair consisting of predictable signed increments and the variance-six defect has a coordinate transformation rule already at finite resolution. Its total error vanishes on fine deterministic meshes. Once a signed budget is finite this recovers the familiar characteristic-measure calculation, but its derivation here does not presuppose that missing regularity. Since $\sum_i |a_i^X - 6d_i| \le (K_2 + 6)T$, it also explains directly why analytic locality changes can add only a bounded amount to an aggregate signed budget under matched bounded localizations. There is still no estimate forcing that budget to be finite.

The inherited times U_{t_i} are stopping times in the deleted natural trace filtration by geometric reconstruction. Their future increments are random, so (FM6) must not be reinterpreted as a fixed-duration transition identity for the deleted driver. Also the maps here are analytic deletion maps. A fixed physical affine deformation of a rough restarted domain carries a nonconstant transported Beltrami coefficient; it is not covered by replacing h_t with a constant affine chart map.

43 Exit signs give exact signed-regularity criteria

The adaptive exits used to measure unsigned activity also suffice to test the missing signed regularity. This gives a concrete grading of the remaining gap, without selecting an affine structure or assuming a semimartingale decomposition. The criteria below are implications and equivalences for processes; no exit-sign estimate is deduced from the curve axioms in this section.

Let W be a continuous adapted process in a usual filtration, with $W_0 = 0$, and suppose its radius-r exits are finite almost surely. Put

$$\sigma_0^r = 0, \qquad \sigma_{k+1}^r = \inf\{t \ge \sigma_k^r : |W_t - W_{\sigma_k^r}| \ge r\}, \qquad G_k^r = \mathcal{F}_{\sigma_k^r}.$$

Write

$$\beta_k^r = \mathbb{E}\left[\frac{W_{\sigma_{k+1}^r} - W_{\sigma_k^r}}{r} \,\middle|\, G_k^r\right], \qquad b_k^r = r\beta_k^r. \tag{EX1}$$

Every displacement divided by r is $+1$ or -1. Fix $L, T > 0$ and

$$\tau = T \wedge \inf\{t : |W_t| \ge L\}, \qquad X_t = W_{t\wedge\tau}, \quad 0 \le t \le T.$$

Let $\mathcal{V}(X)$ be the conditional variation over deterministic time partitions, as in (QV4), and define the pure exit-sign budget

$$\mathcal{B}_r = \mathbb{E} \sum_{k:\, \sigma_k^r < \tau} |b_k^r|. \tag{EX2}$$

Proposition 43.1 (One family of exit skeletons tests the whole signed budget)**.** *One has*

$$\mathcal{V}(X) \le \liminf_{r\downarrow 0} \mathcal{B}_r. \tag{EX3}$$

Thus $\liminf_{r\downarrow 0}\mathcal{B}_r<\infty$ *is equivalent to* $X=M+A$ *with* M *a uniformly integrable continuous martingale and* A *predictable continuous finite variation with* $\mathbb{E}\operatorname{Var}A<\infty$. *In that case*

$$\lim_{r\downarrow 0}\mathcal{B}_r=\mathcal{V}(X)=\mathbb{E}\operatorname{Var}_{[0,T]}A. \tag{EX4}$$

In particular one may test finiteness along any deterministic sequence of radii decreasing to zero. No uniqueness of an unsigned activity clock is required.

Proof. For a deterministic partition $0=t_0<\cdots<t_n=T$, let H_j be $\mathcal{F}_{t_j}$-measurable and bounded by one. Set $u_j=t_j\wedge\tau$ and $K_j=\inf\{k:\sigma^r_k\geq u_j\}$. Each K_j is a stopping index for (G^r_k), since $\{K_j\leq k\}=\{u_j\leq\sigma^r_k\}$. The active variable $H_j\mathbf{1}_{\{t_j<\tau\}}$ is measurable in $\mathcal{F}_{u_j}$ and hence in $G^r_{K_j}$. Consequently

$$L_k=\sum_{j=0}^{n-1}H_j\mathbf{1}_{\{t_j<\tau\}}\mathbf{1}_{\{K_j\leq k<K_{j+1}\}}$$

is G^r_k-measurable, bounded by one, and supported on $\sigma^r_k<\tau$. Thus the expected discrete gain is at most $\mathcal{B}_r$:

$$\mathbb{E}\sum_{j=0}^{n-1}H_j\mathbf{1}_{\{t_j<\tau\}}(W_{\sigma^r_{K_{j+1}}}-W_{\sigma^r_{K_j}})=\mathbb{E}\sum_k L_k b^r_k\leq\mathcal{B}_r. \tag{EX5}$$

For clarity the equality follows first after truncating the sum at a deterministic index. The truncated gain telescopes into at most n differences of driver values prior to the first grid exit after τ; all such values lie in $[-L-r,L+r]$. Dominated convergence applies to the gains, and absolute summability applies to the predictable terms if $\mathcal{B}_r<\infty$. If that budget is infinite the desired upper bound is automatic. This avoids an unjustified optional sampling at the last completed exit before a deterministic time.

The value at the first grid exit after u_j is within $2r$ of W_{u_j}. Hence (EX5) differs from $\mathbb{E}\sum_j H_j(X_{t_{j+1}}-X_{t_j})$ by at most $4nr$. Taking $r\downarrow 0$ and then the supremum over the predictable variables and partitions proves (EX3).

If the liminf is finite, the bounded continuous quasimartingale theorem and Proposition 33.2 give the asserted decomposition and $\mathcal{V}(X)=\mathbb{E}\operatorname{Var}A$. For the converse set, on $\{\sigma^r_k<\tau\}$,

$$\widetilde{b}^r_k=\mathbb{E}[W_{\sigma^r_{k+1}\wedge\tau}-W_{\sigma^r_k}\mid G^r_k].$$

Optional sampling of the uniformly integrable stopped martingale gives

$$\mathbb{E}\sum_{\sigma^r_k<\tau}|\widetilde{b}^r_k|\leq\mathbb{E}\operatorname{Var}A.$$

There is at most one excursion straddling τ, and its displacement between τ and its full exit is at most $2r$. Since $\{\sigma^r_k<\tau\}$ is G^r_k-measurable, conditional expectation and summation give the precise terminal-error bound

$$\mathbb{E}\sum_{\sigma^r_k<\tau}|b^r_k-\widetilde{b}^r_k|\leq 2r. \tag{EX6}$$

Therefore $\mathcal{B}_r\leq\mathbb{E}\operatorname{Var}A+2r$. Combining this with (EX3) proves (EX4). □

Additional hypotheses for the energy criterion. Assume now the uniform diffusive exit estimates from Proposition 40.1. In particular, for constants $a_1, a_2 > 0$ independent of k, r,

$$a_1 r^2 \le d_k^r := \mathbb{E}[\sigma_{k+1}^r - \sigma_k^r \mid G_k^r] \le a_2 r^2. \tag{EX7}$$

The same estimates give exponential residual-exit tails after any bounded stopping time: before the next global exit, the process remains within $2r$ of its current position. The block proof of Proposition 40.1 therefore gives

$$\mathbb{E}[\sigma_{K(\theta)}^r - \theta \mid \mathcal{F}_\theta] \le Cr^2, \qquad K(\theta) = \inf\{k : \sigma_k^r \ge \theta\}. \tag{EX8}$$

The original conditional lower oscillation estimate also excludes all constant nontrivial time intervals almost surely: subdivide any fixed rational interval into n pieces, use the conditional bound on each piece, and let $n \to \infty$. Continuity then implies that the maximum length of an exit-grid interval intersecting a fixed compact time interval tends to zero as $r \downarrow 0$. Otherwise converging endpoints of intervals of length bounded below would exhibit a constant interval.

Proposition 43.2 (Exit-sign energy tests square-integrable drift)**.** *Define*

$$\mathcal{E}_r = \mathbb{E} \sum_{\sigma_k^r < \tau} \frac{(b_k^r)^2}{d_k^r}, \qquad \widetilde{\mathcal{E}}_r = \mathbb{E} \sum_{\sigma_k^r < \tau} \frac{(\widetilde{b}_k^r)^2}{d_k^r}. \tag{EX9}$$

The following conditions are equivalent:

1. $\liminf_{r \downarrow 0} \mathcal{E}_r < \infty$;
2. $\liminf_{r \downarrow 0} \widetilde{\mathcal{E}}_r < \infty$;
3. $X = M + \int_0^{\cdot \wedge \tau} v_s ds$, *where* M *is a uniformly integrable continuous martingale and* v *predictable, with* $\mathbb{E} \int_0^\tau v_s^2 ds < \infty$.

For the truncated means there is the exact limit

$$\lim_{r \downarrow 0} \widetilde{\mathcal{E}}_r = \mathbb{E} \int_0^\tau v_s^2 ds. \tag{EX10}$$

For the raw pure-sign energy only the boundedness equivalence is asserted; the terminal excursion can cause a uniformly bounded energy error. By (EX7), condition 1 is equivalently boundedness along some radii of $\mathbb{E} \sum_{\sigma_k^r < \tau} (\beta_k^r)^2$.

Proof. For a bounded deterministic-time elementary predictable process $H = \sum_j H_j \mathbf{1}_{(t_j, t_{j+1}]}$, use the indices and grid strategy L_k from the preceding proof, now with the bound $\sup |H|$. Set $L^r(t) = L_k$ on $(\sigma_k^r, \sigma_{k+1}^r]$, and zero beyond the first exit after τ. The vanishing grid mesh and (EX8) imply

$$\mathbb{E} \int_0^\infty |L^r(t)|^2 dt \longrightarrow \mathbb{E} \int_0^\tau |H_t|^2 dt. \tag{EX11}$$

Indeed on $[0, \tau]$ there is bounded pointwise convergence away from the finitely many partition endpoints; the expected time of the excess terminal interval is $O(r^2)$. Predictability gives

$$\mathbb{E} \sum_k L_k^2 d_k^r = \mathbb{E} \int_0^\infty |L^r(t)|^2 dt.$$

Both $\mathbb{E}\sum_k L_k b_k^r$ and $\mathbb{E}\sum_k L_k \widetilde{b}_k^r$ tend to $\ell(H) := \mathbb{E}\sum_j H_j(X_{t_{j+1}} - X_{t_j})$; the first uses the $O(nr\sup|H|)$ discretization error, and the second also uses (EX6). Cauchy–Schwarz, followed by (EX11), yields

$$|\ell(H)| \le \sqrt{\liminf_{r\downarrow 0} \mathcal{E}_r}\, \left(\mathbb{E}\int_0^\tau H_t^2 dt\right)^{1/2},$$

and the same inequality with $\widetilde{\mathcal{E}}_r$. Bounded deterministic-time elementary predictable processes are dense in the predictable $L^2(\mathbb{P}\,dt)$ space. Riesz representation therefore gives a predictable v, zero after τ, such that $\ell(H) = \mathbb{E}\int_0^\tau H_t v_t dt$, and its squared norm is bounded by the corresponding liminf energy. Testing a single interval and every bounded $\mathcal{F}_s$ variable shows that $X - \int_0^{\cdot\wedge\tau} v_t dt$ is a martingale. Its integrable dominating variable is $L + \int_0^\tau |v_t| dt$, so it is uniformly integrable; continuity gives the continuous martingale decomposition.

Conversely, assume condition 3. Conditional Cauchy–Schwarz on each excursion, including its possible truncation at τ, gives

$$(\widetilde{b}_k^r)^2 \le d_k^r\, \mathbb{E}\left[\int_{\sigma_k^r}^{\sigma_{k+1}^r\wedge\tau} v_s^2 ds \,\middle|\, G_k^r\right] \quad \text{on } \{\sigma_k^r < \tau\}.$$

Thus $\widetilde{\mathcal{E}}_r \le \mathbb{E}\int_0^\tau v_s^2 ds$. The lower norm bound from the Riesz argument gives the reverse liminf, proving (EX10). Finally, set $p_k^r = \mathbb{P}\{\sigma_k^r < \tau < \sigma_{k+1}^r \mid G_k^r\}$. Then $|b_k^r - \widetilde{b}_k^r| \le 2rp_k^r$ on active starts, and

$$\mathbb{E}\sum_{\sigma_k^r<\tau} \frac{|b_k^r - \widetilde{b}_k^r|^2}{d_k^r} \le \frac{4}{a_1}\mathbb{E}\sum_k p_k^r \le \frac{4}{a_1}.$$

The triangle inequality in the indexed L^2 space proves equivalence of the two bounded-energy conditions. This also shows why replacing the truncated energy by the raw sign energy need not preserve the numerical limit. □

Corollary 43.3 (Any subquadratic sign energy distinguishes AC drift from singular drift)**.** *Fix $1 < p < 2$, and define*

$$\mathcal{E}_{r,p} = \mathbb{E}\sum_{\sigma_k^r<\tau} \frac{|b_k^r|^p}{(d_k^r)^{p-1}}. \tag{EX12}$$

Then $\liminf_{r\downarrow 0}\mathcal{E}_{r,p} < \infty$ is equivalent to $X = M + \int_0^{\cdot\wedge\tau} v_s ds$ with $\mathbb{E}\int_0^\tau |v_s|^p ds < \infty$. In that case the raw *pure-sign energy has the exact limit*

$$\lim_{r\downarrow 0}\mathcal{E}_{r,p} = \mathbb{E}\int_0^\tau |v_s|^p ds. \tag{EX13}$$

By (EX7), boundedness is equivalently boundedness of $r^{2-p}\mathbb{E}\sum_{\sigma_k^r<\tau}|\beta_k^r|^p$.

Proof. Repeat the preceding proof with Hölder conjugate $p' = p/(p-1)$. The grid strategy satisfies

$$\mathbb{E}\sum_k |L_k|^{p'} d_k^r \longrightarrow \mathbb{E}\int_0^\tau |H_t|^{p'} dt.$$

Weighted Hölder's inequality and duality between predictable $L^{p'}$ and L^p therefore identify a predictable density v with $\mathbb{E}\int_0^\tau |v_s|^p ds$ bounded by the liminf energy. For the terminal-truncated means, conditional Hölder gives

$$|\widetilde{b}_k^r|^p \le (d_k^r)^{p-1}\mathbb{E}\left[\int_{\sigma_k^r}^{\sigma_{k+1}^r\wedge\tau} |v_s|^p ds \,\middle|\, G_k^r\right]$$

on active starts. Summation supplies the reverse upper bound, so the truncated-mean energy converges to the asserted L^p norm.

For the full-exit means, the terminal correction obeys

$$\mathbb{E}\sum_{\sigma_k^r<\tau}\frac{|b_k^r-\widetilde{b}_k^r|^p}{(d_k^r)^{p-1}}\le\frac{2^p}{a_1^{p-1}}\,r^{2-p}\mathbb{E}\sum_k(p_k^r)^p\le\frac{2^p}{a_1^{p-1}}\,r^{2-p}\longrightarrow 0. \tag{EX14}$$

The triangle inequality for the corresponding weighted L^p norms therefore transfers the same numerical limit to the raw sign energy. The exponent $2-p>0$ is essential for this last assertion. At $p=1$ the limiting object is an arbitrary finite-variation drift measure, as in Proposition 43.1; for every $p>1$ in the stated range, finiteness forces a density with respect to capacity time. □

Corollary 43.4 (Order-r exit-sign bias supplies bounded drift)**.** *If, for all sufficiently small radii and all active starts, $|\beta_k^r|\le Kr$ with a deterministic constant K, then $X=M+\int_0^{\cdot\wedge\tau}v_s ds$ with $|v_s|\le K/a_1$ almost everywhere.*

Proof. By (EX7), $|b_k^r|\le Kr^2\le(K/a_1)d_k^r$. The predictable test argument therefore bounds $|\ell(H)|$ by $(K/a_1)\mathbb{E}\int_0^\tau|H_t|dt$. The L^1–L^∞ dual representation supplies the claimed bounded predictable density and hence the martingale decomposition. □

The classification gap is now quantitatively explicit. Unsigned estimates give $|\beta_k^r|\le 1$ and order T/r^2 exit terms. The exact finite-variation criterion asks for bounded $r\mathbb{E}\sum|\beta_k^r|$. The square-integrable drift criterion asks for bounded $\mathbb{E}\sum(\beta_k^r)^2$. The stronger pointwise bound $|\beta_k^r|\le Kr$ gives the bounded conditional-bias premise used in Theorem 19.3. To apply that theorem one must still control these constants along mirrored shrinking-slit states: with uniform a_1, the required rate is $\epsilon K_\epsilon\to 0$. Pointwise bounded drift in each unrelated state does not supply that rate.

Even bounded drift alone gives no initial tangent without locality's additional comparison: $v(t)=\sin(\log t)$ has bounded magnitude but $t^{-1}\int_0^t v(s)ds=\frac{1}{2}(\sin\log t-\cos\log t)$ has no limit at zero. Brownian motion plus this deterministic integral is only a process illustration, not a proposed family obeying the curve axioms. All these criteria are in a specified coordinate system. Neither finite raw exit-sign budget nor bounded sign energy is asserted in every affine structure, and no preferred affine tensor has been constructed.

44 Source-linear exit energy is sufficient for locality rigidity

The preceding exit criteria can be linked to classification using a weaker hypothesis than pointwise bounded driving drift. The new input is a bound on *expected initial drift energy per unit capacity*, expressible entirely in exit-sign observables. It still requires a quantitative rate along the locality-related shrinking-slit states. Neither that rate nor the energy bound is derived from RSW here.

Fix $1<p<2$ and one physical conformal structure. Assume the Borel family, continuous Loewner generation, strong natural-filtration DMP (KR1), and source locality for the mirrored tests (DR6), with the invertible capacity-clock and rough-prime-end requirements used previously. Assume also the uniform upper conditional driving moments and diffusive exit estimates already obtained from the explicit crossing/locality inputs. Thus Proposition 34.1 and the exit-energy criteria apply once their stated signed hypotheses hold.

For each state f, write $\mathcal{E}_{r,p}^f(\theta)$ for the raw energy in (EX12), stopped at a bounded stopping time θ at which the stopped driver is bounded. The proofs of Propositions 43.1–43.2 and Corollary 43.3 apply to every such θ: only boundedness of the stopped driver, the stopping-time property, and the

residual-exit estimate are used. In particular θ need not be exactly the first exit from a symmetric driving interval.

Theorem 44.1 (Conditional classification from initial exit energy)**.** *Suppose the following additional signed conditions hold.*

1. *For every deterministic state f and $L, T > 0$, with $\tau_{L,T} = T \wedge \inf\{t : |W_t| \geq L\}$,*

$$\liminf_{r \downarrow 0} \mathcal{E}^f_{r,p}(\tau_{L,T}) < \infty. \tag{ER1}$$

2. *For each deterministic state f, there is a bounded stopping time $\rho_f > 0$ almost surely, with bounded stopped driver, and a finite deterministic constant $K(f)$ such that*

$$\limsup_{h \downarrow 0} \frac{1}{h} \liminf_{r \downarrow 0} \mathcal{E}^f_{r,p}(h \wedge \rho_f) \leq K(f)^p. \tag{ER2}$$

3. *For each state f, along some sequence $\epsilon_n \downarrow 0$, the mirrored deleted states obey*

$$\epsilon_n \max_{\sigma = \pm} K(f^\sigma_{\epsilon_n}) \longrightarrow 0. \tag{ER3}$$

Then every driver is $\sqrt{6}B$ in the specified conformal structure, and the exploration family is chordal SLE_6 in that structure.

Proof. The exit-energy criterion applied to (ER1), and uniqueness on overlapping stopped intervals, supply the global continuous canonical decomposition

$$W = M + A, \qquad A_t = \int_0^t b_s ds,$$

where b is predictable and locally $L^p(\mathbb{P}\, dt)$ after bounded driving localization. The upper conditional moment estimate and Proposition 34.1 give $C = \langle M \rangle = \int_0^t v_s ds$ with a locally bounded predictable density. The exact energy limit (EX13) turns (ER2) into

$$\limsup_{h \downarrow 0} \frac{1}{h} \mathbb{E}_f \int_0^{h \wedge \rho_f} |b_s|^p ds \leq K(f)^p. \tag{ER4}$$

No pointwise bound or right limit of b is assumed.

Fix a deterministic state f and one ϵ. Couple its initial trace to each mirrored deleted-domain trace by locality and use their invertible capacity clocks. Put $d = d_\epsilon$, $e = e^+_\epsilon$, and $K_\epsilon = \max_\sigma K(f^\sigma_\epsilon)$. The canonical characteristic identity (BI3) reads, before contact,

$$p^\sigma_t b_t + \tfrac{1}{2} q^\sigma_t (v_t - 6) = (p^\sigma_t)^2 \widehat{b}^\sigma_{u_\sigma(t)}, \qquad du_\sigma(t) = (p^\sigma_t)^2 dt. \tag{ER5}$$

Here $p^\sigma_0 = d$ and $q^\pm_0 = \pm e$. There is an almost surely positive stopping time λ_ϵ on which the original and both deleted initial localizations remain valid, p^σ_t lies between $d/2$ and $2d$, and each q^σ_t remains within $|e|/2$ of its initial value. The two-by-two system in (ER5) is then uniformly invertible in these relative bounds. With a numerical constant C independent of ϵ,

$$|b_t| \leq C d \sum_{\sigma = \pm} |\widehat{b}^\sigma_{u_\sigma(t)}|, \qquad |v_t - 6| \leq C \frac{d^2}{|e|} \sum_{\sigma = \pm} |\widehat{b}^\sigma_{u_\sigma(t)}| \tag{ER6}$$

for almost every $t < \lambda_\epsilon$.

Changing variables in the first inequality, using $u_\sigma(h \wedge \lambda_\epsilon) \le 4d^2 h$ and stopping before $\rho_{f_\epsilon^\sigma}$, gives

$$\mathbb{E}\,\mathrm{Var}(A; [0, h \wedge \lambda_\epsilon]) \le \frac{C}{d} \sum_{\sigma=\pm} \mathbb{E} \int_0^{4d^2 h \wedge \rho_{f_\epsilon^\sigma}} |\widehat{b}_u^\sigma| du.$$

Hölder's inequality on probability times capacity and (ER4) imply, at fixed ϵ,

$$\mathbb{E} \int_0^{4d^2 h \wedge \rho_{f_\epsilon^\sigma}} |\widehat{b}_u^\sigma| du \le 4d^2 h\big(K_\epsilon + o_\epsilon(1)\big), \qquad h \downarrow 0.$$

The second inequality in (ER6) is treated identically. Thus

$$\limsup_{h \downarrow 0} \frac{1}{h}\, \mathbb{E}\,\mathrm{Var}(A; [0, h \wedge \lambda_\epsilon]) \le C d K_\epsilon, \tag{12}$$

$$\limsup_{h \downarrow 0} \frac{1}{h}\, \mathbb{E}\,\mathrm{Var}(C - 6\,\mathrm{id}; [0, h \wedge \lambda_\epsilon]) \le C \frac{d^2}{|e|} K_\epsilon. \tag{13}$$

In applying (ER4), the deleted drift is evaluated in its own canonical filtration and capacity clock. Its law is the deleted-state law by locality; the time change in (ER5) is not replaced by a deterministic capacity equality.

For any $a > 0$, Markov's inequality and $\mathbb{P}\{\lambda_\epsilon < h\} \to 0$ give

$$\limsup_{h \downarrow 0} \mathbb{P}\left\{ h^{-1} \sup_{s \le h} |A_s| > a \right\} \le C a^{-1} d K_\epsilon,$$

and the corresponding bound for $C_s - 6s$ with dK_ϵ replaced by $d^2 K_\epsilon / |e|$. Since $d_\epsilon \asymp \epsilon$ and $|e_\epsilon| \asymp \epsilon^{-1}$, condition (ER3), first taking $h \downarrow 0$ at fixed ϵ_n and then $n \to \infty$, proves

$$h^{-1} \sup_{s \le h} |A_s| \longrightarrow 0, \qquad h^{-1} \sup_{s \le h} |C_s - 6s| \longrightarrow 0 \quad \text{in probability} \tag{ER7}$$

in every deterministic state.

It remains to propagate the initial conclusion without assuming a tangent of the original density. Proposition 18.1 and strong DMP transport the canonical shifted characteristics at each deterministic time t. Along any fixed sequence $h_n \downarrow 0$, for almost every conditioned history their law is the restarted state's canonical law. Statement (ER7) holds in that state. Conditional probabilities are bounded by one, so dominated convergence under the regular conditional law yields

$$\frac{A_{t+h_n} - A_t}{h_n} \longrightarrow 0, \qquad \frac{C_{t+h_n} - C_t}{h_n} \longrightarrow 6 \quad \text{in probability.} \tag{ER8}$$

On the other hand, absolute continuity and Fubini imply that for Lebesgue-almost every deterministic t, these same quotients converge almost surely to b_t and v_t. Uniqueness of limits in probability gives $b_t = 0$ and $v_t = 6$ almost surely for almost every t. Another Fubini argument gives $A = 0$ and $C_t = 6t$ up to indistinguishability. Lévy's characterization identifies $W = \sqrt{6}B$. □

What is sufficient, and what is not proved. This theorem replaces pointwise order-r exit-sign bias by averaged order-r signed energy. Conditions (ER1)–(ER3) are still missing theorems for the unknown family in an appropriate common structure. Finite exit energy alone supplies AC drift and bounded bracket but no rate in the slit comparison. The drift and variance anomaly rates are respectively ϵK_ϵ and $\epsilon^3 K_\epsilon$. All estimates are conditional classification tools, not a construction of the affine normalizer. In arbitrary initial coordinates an adequate regularity statement may need a correction involving an unknown constant physical Beltrami coefficient before these signed-energy hypotheses can be expected to hold.

45 Analytic clock families can acquire singular time change

The affine-prefix results give analytic dependence of the capacity clock on a small deformation parameter in the uniform path norm, and very high temporal regularity of its first parameter derivative. These properties, even supplemented by strict monotonicity and global capacity comparison, do not alone prove absolute continuity of a finite-deformation clock. The following explicit example separates this analytic issue from the unresolved geometry of the actual affine-SLE clock.

Proposition 45.1 (A singular analytic clock family)**.** *There is a family of continuous strictly increasing clocks $s_\epsilon : [0,\infty) \to [0,\infty)$, $|\epsilon| < 1$, with the following properties.*

1. *$s_0(t) = t$, $s_\epsilon(t) = t$ outside $[1,2]$, and $\frac{1}{2}t \le s_\epsilon(t) \le 2t$ for every $t \ge 0$.*
2. *On every compact time interval the family extends holomorphically in ϵ into the Banach space of continuous paths on the disk $|\epsilon| < 4$. In particular it has uniform Taylor remainders of all orders on any smaller parameter disk.*
3. *Every finite parameter derivative at zero is temporally Hölder of every exponent below one. Its quadratic-increment sums vanish along every sequence of partitions of vanishing mesh. Its mixed sums with any path Hölder of some positive exponent also vanish, after choosing an appropriate exponent for the derivative path.*
4. *For every real $0 < |\epsilon| < 1$, ds_ϵ has a nonzero singular continuous component relative to dt. The first parameter derivative $\partial_\epsilon s_\epsilon|_0$ is not of bounded variation.*

This example has no asserted realization by Loewner hulls.

Proof. Work first on the circle $[0, 2\pi]$. For real $|\epsilon| < 1$, set

$$p_{\epsilon,N}(x) = \prod_{n=1}^{N} (1 + \epsilon \cos(5^n x)), \qquad \mu_{\epsilon,N}(dx) = p_{\epsilon,N}(x)\frac{dx}{2\pi}.$$

Every factor is positive. In the Fourier expansion, frequencies are $k = \sum_{n=1}^{N} \sigma_n 5^n$, $\sigma_n \in \{-1, 0, 1\}$, with coefficient $(\epsilon/2)^{\#\{n:\sigma_n \neq 0\}}$. Such representations are unique: the largest nonzero digit in a difference of two representations has magnitude at least 5^m, whereas all lower difference digits have total magnitude at most $2\sum_{n<m} 5^n < 5^m$. In particular the constant Fourier coefficient is one and these are probability measures. Their Fourier coefficients stabilize at every fixed integer frequency, so compactness and uniqueness of Fourier coefficients give a unique weak limit μ_ϵ.

Let

$$C_{\epsilon,N}(x) = \int_0^x p_{\epsilon,N}(u)\, du.$$

The nonconstant term with highest digit m has frequency of magnitude at least $5^m - \sum_{n<m} 5^n \ge (3/4)5^m$. Summing the absolute coefficients of all terms with this highest digit gives $|\epsilon|(1 + |\epsilon|)^{m-1}$. Since $|\int_0^x e^{iku} du| \le 2/|k|$, the largest-digit block contributes at most

$$C 5^{-m} |\epsilon| (1 + |\epsilon|)^{m-1} \tag{SC1}$$

in uniform norm. Therefore $C_{\epsilon,N}$ converges uniformly in x and locally uniformly for complex $|\epsilon| < 4$ to a holomorphic $C([0, 2\pi])$-valued function C_ϵ. For real $|\epsilon| < 1$ it is increasing, $C_\epsilon(0) = 0$, and $C_\epsilon(2\pi) = 2\pi$. The uniform convergence also shows that the weak measure limit has no atoms and $dC_\epsilon = 2\pi\, d\mu_\epsilon$.

Now define

$$s_\epsilon(t) = \begin{cases} t, & t \notin [1,2], \\ 1 + \dfrac{x + C_\epsilon(x)}{4\pi}, & t = 1 + x/(2\pi), \quad 0 \le x \le 2\pi. \end{cases} \tag{SC2}$$

On the middle interval its increment measure has one-half Lebesgue part, so the clock is strictly increasing; all joins are continuous. It maps $[1,2]$ onto $[1,2]$, giving the stated global comparisons. Analyticity and uniform Taylor bounds follow from (SC1).

The first derivative is zero outside $[1,2]$ and inside equals

$$\partial_\epsilon s_\epsilon(t)|_0 = \frac{1}{4\pi} \sum_{n \ge 1} 5^{-n} \sin(5^n x). \tag{SC3}$$

For a temporal separation h its increment has magnitude at most $C|h|(1 + |\log|h||)$: split the sum at $5^n|h| = 1$ and use $|\sin(a+b) - \sin a| \le \min\{|b|, 2\}$. More generally, in the coefficient of ϵ^j the sum of absolute Fourier coefficients in largest-digit block m is $\binom{m-1}{j-1}$, up to constants depending only on j. The same split gives the modulus

$$C_j|h|(1 + |\log|h||)^j. \tag{SC4}$$

This holds across the joins as well, since all derivative paths vanish at both endpoints. Thus each derivative is Hölder-α for every $\alpha < 1$. An exponent $\alpha > 1/2$ gives vanishing quadratic sums. For mixed sums with a Hölder-β path, choose $\alpha < 1$ with $\alpha + \beta > 1$ and bound their absolute sum by $CT\,\mathrm{mesh}(\pi)^{\alpha+\beta-1}$.

We prove singularity directly. Under μ_ϵ,

$$\mathbb{E}_{\mu_\epsilon} \cos(5^n x) = \epsilon/2, \qquad \mathbb{E}_{\mu_\epsilon}[\cos(5^n x)\cos(5^m x)] = \epsilon^2/4 \quad (n \ne m).$$

These identities are the Fourier coefficients at 5^n and $5^n \pm 5^m$. Hence the centered variables are pairwise orthogonal and bounded. Their Cesaro average converges in $L^2(\mu_\epsilon)$ to $\epsilon/2$, with squared error at most C/N. At $N = k^2$, Chebyshev and Borel–Cantelli give almost sure convergence; boundedness interpolates between consecutive squares. Under normalized Lebesgue measure the same argument gives almost sure limit zero. For $\epsilon \ne 0$ the two full-measure sets are disjoint, proving $\mu_\epsilon \perp dx$. Formula (SC2) therefore has a nonzero singular continuous increment component in $[1,2]$.

Finally put $F(x) = \sum_{n \ge 1} 5^{-n} \sin(5^n x)$. If F were BV, its periodic distributional derivative would be a finite signed measure λ with distributional Fourier series $\sum_{n \ge 1} \cos(5^n x)$. Convolution with the normalized Fejer kernel of degree $M = 2 \cdot 5^N$ would have L^1 norm at most $\|\lambda\|_{\mathrm{TV}}$ and would equal

$$P_N(x) = \sum_{n=1}^{N} \left(1 - \frac{5^n}{M+1}\right) \cos(5^n x).$$

All weights are between $1/2$ and one. Orthogonality gives $\|P_N\|_2 \ge c\sqrt{N}$. The fourth moment is at most CN^2: in a zero signed sum of four powers of five the largest exponent must cancel with an opposite copy, since three lower powers have magnitude less than the largest. Only pairings remain. Therefore $\|P_N\|_4 \le C\sqrt{N}$. Interpolation gives $\|P_N\|_1 \ge \|P_N\|_2^3 / \|P_N\|_4^2 \ge c\sqrt{N}$, a contradiction. This proves the claimed failure of BV. □

Consequence for the research strategy. Even an all-order affine-parameter Taylor theory whose clock derivatives have every temporal exponent below one does not, by itself, settle absolute continuity of the finite affine capacity clock. One needs an additional estimate that controls the increment measures, or new information from the actual Loewner/SLE geometry. The example does not decide the affine-SLE clock question, and is not a counterexample to the exploration classification conjecture.

45.1 An exact increment-measure target for clock regularity

There is a positive measure-theoretic formulation of the missing estimate. It concerns a martingale in the dyadic *time variable*; it makes no semimartingale assumption about a Loewner driver.

Proposition 45.2 (A dyadic clock criterion)**.** *Let s be a continuous increasing function on $[0,T]$, with $s(0)=0$ and $s(T)=m>0$. For dyadic cells I at level n, put*

$$a_n(t)=\frac{s(\sup I)-s(\inf I)}{|I|},\quad t\in I.$$

Then $ds \ll dt$ if and only if (a_n) is uniformly integrable relative to Lebesgue measure on $[0,T]$. A sufficient condition is

$$\sup_n H_n<\infty,\qquad H_n=\int_0^T a_n\log\frac{a_n}{m/T}\,dt. \tag{SC5}$$

Write $m_I=s(\sup I)-s(\inf I)$ and I_-,I_+ for the children of I. Condition (SC5) is equivalent, up to absolute multiplicative constants, to finiteness of

$$\sum_{I\ dyadic}\frac{(m_{I_-}-m_{I_+})^2}{m_I}, \tag{SC6}$$

where a zero-mass summand is defined to be zero. If the clock is random, a uniform bound on $\mathbb{E}H_n$ is sufficient for almost sure absolute continuity.

Proof. The a_n form a nonnegative martingale on the probability space $([0,T],dt/T)$ with the increasing dyadic sigma-fields. They converge almost everywhere to the density of the absolutely continuous part of ds, by differentiation of measures. Uniform integrability is equivalent to preservation of their common integral m at the limit, which is equivalent to absence of singular mass. If $ds=a\,dt$, then a_n are the dyadic conditional averages of the integrable function a and hence are uniformly integrable.

The entropy condition gives uniform integrability by the elementary $x\log x$ tail bound. Jensen's inequality makes H_n nonnegative and increasing. At a split of a positive-mass cell, put $p=m_{I_-}/m_I$. The entropy increment is exactly

$$m_I\{p\log(2p)+(1-p)\log(2(1-p))\}.$$

For $\delta=2p-1$, its bracket equals $\sum_{k\ge1}\delta^{2k}/[2k(2k-1)]$, and is bounded between $\delta^2/2$ and $(\log 2)\delta^2$. Summing the increments proves the equivalence with (SC6). Finally a uniform expected entropy bound and monotone convergence give finite limiting entropy almost surely, which implies the pathwise criterion. □

This supplies a concrete new model task: control the dyadic mass-split energy of the actual affine-SLE capacity clock. Uniform integrability, rather than merely temporal Hölder regularity of clock tangents, is exactly what is needed for clock absolute continuity. The entropy bound is sufficient, not necessary; no such bound is proved here from the exploration axioms or for affine SLE. Clock AC would remain only a necessary condition in the existing wrong-driver semimartingale test.

46 Selecting an unknown affine tensor through finite signed tests

One need not identify the affine tensor before formulating a useful signed regularity estimate. It can be retained as a compact parameter and selected jointly with the estimates. The following conditional principle makes this precise without extending from a dense collection of domains or assuming a domain Feller property.

Fix $1 < p < \infty$ and $q = p/(p-1)$. On $C([0,T],\mathbb{R})$ let $W_t(w) = w(t)$, and suppose all coordinate variables are integrable under the law P. Let $\mathcal{H}_T$ consist of processes

$$H_t = \sum_{i=0}^{n-1} h_i(W|_{[0,t_i]})\mathbf{1}_{(t_i,t_{i+1}]}(t), \qquad 0 = t_0 < \cdots < t_n = T,$$

where every h_i is bounded and continuous on its past-path space. Put

$$\mathcal{J}_{p,T}(P) = \sup_{H\in\mathcal{H}_T} \left\{ p\,\mathbb{E}_P \sum_i h_i(W)(W_{t_{i+1}} - W_{t_i}) - (p-1)\mathbb{E}_P \int_0^T |H_t|^q dt \right\}. \tag{FS1}$$

Proposition 46.1 (A signed-test characterization of L^p drift)**.** *The functional in (FS1) is finite if and only if, in the completed right-continuous natural filtration, there is a predictable $b \in L^p(P\otimes dt)$ such that*

$$W_t = W_0 + M_t + \int_0^t b_s ds \tag{FS2}$$

with M a continuous martingale starting at zero. In that case

$$\mathcal{J}_{p,T}(P) = \mathbb{E}_P \int_0^T |b_s|^p ds. \tag{FS3}$$

The drift is unique up to $P\otimes dt$-null sets.

Proof. Write $L(H) = \mathbb{E}\sum_i h_i \Delta W_i$ and $\|H\|_q = (\mathbb{E}\int |H|^q dt)^{1/q}$. Scaling H by any real constant in (FS1), and optimizing that scalar, shows

$$|L(H)| \le \mathcal{J}_{p,T}(P)^{1/p} \|H\|_q. \tag{FS4}$$

In particular L respects equivalence modulo $P\otimes dt$. Continuous bounded past-path functions are dense in every finite L^q space of past-measurable variables, since path space is Polish. Consequently $\mathcal{H}_T$ is dense in the predictable L^q space. For the usual augmentation one may first approximate any augmented past variable by raw past variables at times $t+\delta$, then delay the start of its elementary interval by δ; the lost interval has vanishing $P\otimes dt$ norm. This gives the same density result.

Extend L continuously and apply L^q duality on the raw predictable sigma-field (completed for $P\otimes dt$) to obtain a predictable $b \in L^p$, choosing a raw predictable version, with $L(H) = \mathbb{E}\int Hb\,dt$ and $\|b\|_p \le \mathcal{J}_{p,T}(P)^{1/p}$. Testing one interval $(s,t]$ against a bounded continuous functional of $W|_{[0,s]}$ gives zero conditional mean for increments of $W - \int b dt$. The identity extends to every bounded past-measurable variable: it is equality of finite signed measures on the Polish past-path space, tested against all bounded continuous functions. Thus the continuous integrable process M in (FS2) is a martingale in the raw natural filtration. It extends to the usual augmentation: it is closed by its integrable terminal value, and right limits of conditional expectations agree with its continuous paths.

Conversely, under (FS2), martingale increments contribute zero, so $L(H) = \mathbb{E}\int Hb\,dt$. Pointwise convex duality gives

$$\sup_{h\in\mathbb{R}}\{phb - (p-1)|h|^q\} = |b|^p.$$

Approximate the optimizing predictable variable $\operatorname{sgn}(b)|b|^{p-1}$ in L^q by members of $\mathcal{H}_T$ to obtain equality (FS3). Uniqueness follows from uniqueness of the continuous semimartingale decomposition. □

Proposition 46.2 (Lower semicontinuity and common-tensor selection)**.** *Let $\mathcal{K}$ be a compact parameter space of normalized affine tensors. For each state f in an arbitrary index set and each horizon T, suppose $A \mapsto P^A_{f,T}$ is weakly continuous on $C([0,T],\mathbb{R})$, and, for each deterministic time t, the family of random variables W_t is uniformly integrable over these laws. Then $A \mapsto \mathcal{J}_{p,T}(P^A_{f,T})$ is lower semicontinuous.*

Fix a deterministic finite envelope $B_{f,T}$ for every desired state/horizon pair. If every finite collection of elementary signed tests, from any finite collection of state/horizon pairs, admits one common $A \in \mathcal{K}$ for which all the individual test expressions in (FS1) are at most their prescribed $B_{f,T}$, then there exists one $A_* \in \mathcal{K}$ such that

$$\mathcal{J}_{p,T}(P^{A_*}_{f,T}) \le B_{f,T} \quad \text{for every desired } (f,T). \tag{FS5}$$

In particular all these drivers simultaneously have absolutely continuous L^p drift with the corresponding envelopes. The same conclusion holds if finite collections can only be satisfied up to an arbitrary common error $\eta > 0$.

Proof. For fixed H, its penalty expectation is a bounded continuous function of the law. Each increment expectation is continuous as well: its integrand is continuous, and its absolute value is at most a fixed multiple of $|W_{t_i}| + |W_{t_{i+1}}|$; uniform integrability allows truncation. Therefore each elementary test expression is continuous in A. Their supremum is lower semicontinuous.

The constraint that one test expression is at most $B_{f,T}$ is closed in $\mathcal{K}$. The simultaneous finite-test hypothesis is exactly the finite-intersection property for these closed sets. Compactness gives a point in their entire intersection, and taking the supremum over tests gives (FS5). In the approximate version, compactness and continuity first let $\eta \downarrow 0$ for each fixed finite collection, giving the exact finite-intersection property. No countability of the state index set and no dense-state extension are used. □

One possible compact tensor class is $\{Q = Q^{\mathsf{T}} > 0 : \det Q = 1,\ \lambda^{-1} I \le Q \le \lambda I\}$. The common ellipticity bound λ must be proved or assumed. The finite envelopes must also be specified in advance; the claim that each finite test collection has some finite but uncontrolled budget is not enough for the compactness conclusion.

46.1 Physical affine-coordinate continuity for a fixed state

The law-continuity premise above can be proved pathwise for changes of affine coordinates. The transported coefficient is essential. This is not a continuity assertion as the physical state itself varies.

Proposition 46.3 (Uniform driver continuity under asymptotically conformal changes)**.** *Let K_t, $0 \le t \le T_0$, be a continuously driven half-plane Loewner chain with driver W_t and $K_t \subset B_R$. Let $F_n : \mathbb{H} \to \mathbb{H}$ be quasiconformal homeomorphisms fixing $0, 1, \infty$, with maximal dilatations tending to one. Suppose the image hull chains admit their continuous capacity parametrizations. Write*

$$s_n(t) = \tfrac{1}{2}\operatorname{hcap}(F_n K_t), \qquad U_n(t) = \widehat{g}^{\,n}_{s_n(t)}(F_n \gamma(t)),$$

where tips are interpreted as prime ends. Then

$$\sup_{t \le T_0} |s_n(t) - t| \longrightarrow 0, \qquad \sup_{t \le T_0} |U_n(t) - W_t| \longrightarrow 0. \tag{FS6}$$

Consequently their own-capacity drivers converge uniformly to W on every $[0,T]$ with $T < T_0$.

Proof. Reflect F_n across the real line. Normalized quasiconformal compactness gives $F_n \to \mathrm{id}$ uniformly in spherical distance, and hence uniformly on compact subsets of the plane. The image hulls all lie in a fixed disk $B_{R'}$. At each fixed t, their complements converge in the Caratheodory kernel sense to $\mathbb{H} \setminus K_t$: any compact subset persists by uniform convergence of F_n^{-1}, and any purported larger kernel neighborhood would contain points $F_n(z_n) \in F_n K_t$ converging to a point of K_t. Hydrodynamic maps converge locally uniformly. Since all hulls are uniformly bounded, the maps reflect conformally across the real line outside a fixed disk; convergence on a sufficiently large fixed circle gives convergence of their $1/z$ coefficients, hence $s_n(t) \to t$. Monotonicity in t and a finite-grid sandwich against the continuous limit give uniform clock convergence.

Set $f_t = g_t^{-1}$ and consider

$$Q_{n,t} = \widehat{g}^{\,n}_{s_n(t)} \circ F_n \circ f_t.$$

This is a quasiconformal self-homeomorphism of the half-plane, fixing infinity, with dilatation tending to one uniformly in t. Uniform hydrodynamic bounds for hulls in a fixed disk give, for every sufficiently large fixed Y,

$$\sup_{t \le T_0} |Q_{n,t}(iY) - iY| \le CR^2/Y + o_n(1), \tag{FS7}$$

enlarging R by a fixed factor if necessary. Indeed $f_t(iY) = iY + O(R^2/Y)$ uniformly in t, F_n converges uniformly on the compact set of these values, and the transformed hydrodynamic map changes its argument by $O(R^2/Y)$.

For any sequence t_n, the images of the three distinct sphere points $\infty, iY, 2iY$, with one sufficiently large fixed Y, remain respectively infinity and in separated compact neighborhoods of $iY, 2iY$. Normalized quasiconformal compactness therefore keeps the reflected Q_{n,t_n} away from degenerate limits. Any subsequential normalized quasiconformal limit is therefore a conformal sphere homeomorphism fixing infinity and preserving $\mathbb{H}$, hence $z \mapsto az + b$ with $a > 0$ and $b \in \mathbb{R}$. The estimate (FS7), for arbitrarily large fixed Y, forces $a = 1, b = 0$. Thus $Q_{n,t} \to \mathrm{id}$ uniformly on bounded closed half-plane sets, uniformly in t. Since W_t remains bounded, $U_n(t) = Q_{n,t}(W_t)$ proves the driver assertion in (FS6). Uniform inverse-clock convergence on $[0,T]$, followed by uniform continuity of W, proves the final statement. □

For a fixed physical state $f_0 : \mathbb{H} \to D_0$, choose one third marked prime end $f_0(1)$ to normalize its chordal charts. If physical affine maps $A_n \to \mathrm{id}$, the comparison $F_n = \varphi_n \circ A_n \circ f_0$, where $\varphi_n : A_n D_0 \to \mathbb{H}$ sends the three corresponding ends to $0, 1, \infty$, has coefficient

$$\mu_{F_n}(z) = \mu_{A_n} \frac{\overline{f_0'(z)}}{f_0'(z)}. \tag{FS8}$$

Its norm tends to zero, so the preceding proposition applies. At a general base affine map A_0, use the reference domain $A_0 D$ and the relative maps $A_n A_0^{-1}$. For each deterministic convergent sequence of affine maps, this gives samplewise convergence of normalized driving paths on horizons strictly inside the available capacity lifetime, on the common probability-one event of admissibility for that countable sequence. Bounded continuous functionals then give weak law continuity by dominated convergence. Admissibility almost surely for each individual affine map suffices for this argument; a universal event for an uncountable tensor class is not asserted. No domain Feller assertion is used. Uniform integrability of increments remains a separate requirement, for example from uniform crossing tails over a compact ellipticity class.

Corollary 46.4 (The selection continuity assumptions under uniform crossings)**.** *Suppose a fixed physical-state law has the uniform avoidable-quad crossing upper bound used earlier, and its curves are continuously Loewner admissible after every affine map in a compact normalized ellipticity class $\mathcal{K}$, with infinite capacity lifetime in each such chart. Normalize each initial chart by three corresponding prime ends. Then the associated own-capacity driver laws are weakly continuous in $A \in \mathcal{K}$ and have uniformly integrable coordinate increments on each finite capacity horizon. Thus the lower semicontinuity premise of Proposition 46.2 holds for that state.*

Proof. Weak continuity is the preceding deterministic argument, coupled through the fixed physical curve sample. If all affine maps have quasiconformal dilatation at most K_*, modulus distortion shows that a transformed quad of modulus at least K_*M pulls back to one of modulus at least M. Avoidability and the crossing event are topological, so the same probability upper bound survives uniformly over the class, with the enlarged modulus threshold. The established driving upper-tail theorem consequently supplies uniform exponential tails for the driving supremum divided by $\sqrt{T}$, with constants depending only on the original crossing constants and K_*. This gives uniform integrability, indeed all finite moments, for the finitely many coordinate increments in each elementary test. □

Status and role. These propositions are compactness and stochastic duality tools. They do not prove the simultaneous finite-test bound or a common ellipticity class from the exploration axioms. They show that a regularity theorem may select its affine tensor as part of its conclusion. It is unnecessary to know the tensor first, but it remains necessary to prove an estimate uniform over finite collections of tests. Mere L^p drift, without the additional locality/slit control in the rigidity theorems, is not claimed to finish classification.

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